

The actual number of ATP molecules generated through cellular respiration varies from cell to cell. In most eukaryotic cells, the NADH that is made in the cytosol during glycolysis cannot diffuse through the inner membrane of the mitochondrion. Instead, it must be actively transported into the mitochondrial matrix. The active transport of NADH consumes ATP. As a result, most eukaryotic cells produce only about 36 ATP molecules per glucose molecule.

The efficiency of cellular respiration can vary depending on conditions in the cell. In general, the efficiency when 38 ATP molecules are generated can be estimated as shown below:

$$\begin{aligned} \text{Efficiency of cellular respiration} &= \frac{\text{Energy required to make ATP}}{\text{Energy released by oxidation of glucose}} \\ &= \frac{38 \times 7 \text{ kcal}}{686 \text{ kcal}} \times 100\% = 39\% \end{aligned}$$

Thus, cellular respiration is nearly 20 times more efficient than glycolysis alone. In fact, the efficiency of cellular respiration is quite impressive compared with the efficiency of machines that humans have designed, such as the car shown in Figure 7-13. An automobile engine, for example, is only about 25 percent efficient in extracting energy from gasoline to move a car. Most of the remaining energy released from gasoline is lost as heat.



**FIGURE 7-13**

Through cellular respiration, cells are more efficient at generating energy than many machines—including cars.

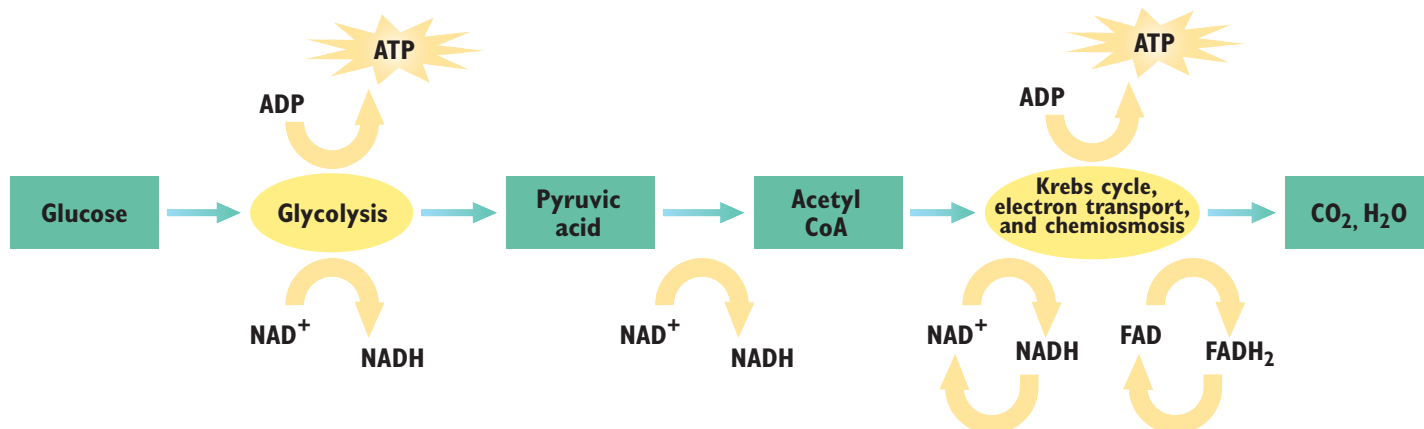
## A SUMMARY OF CELLULAR RESPIRATION

Cellular respiration occurs in two stages, as listed below and shown in Figure 7-14:

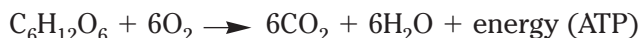
1. Glycolysis—Glucose is converted into pyruvic acid, producing a small amount of ATP and NADH.
2. Aerobic respiration—Pyruvic acid is converted into CO<sub>2</sub> and water in the presence of oxygen, producing a large amount of ATP.

**FIGURE 7-14**

Cellular respiration occurs in two stages: glycolysis and aerobic respiration (which includes the conversion of pyruvic acid to acetyl CoA, the Krebs cycle, the electron transport chain, and chemiosmosis).

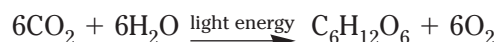


The following equation summarizes the complete oxidation of glucose in cellular respiration:



In addition to glucose, many other compounds can be used as fuel in cellular respiration. Molecules derived from the breakdown of fats, proteins, and carbohydrates can enter glycolysis or the Krebs cycle at various points in order to yield more energy to an organism.

The equation above can be considered the opposite of the overall equation for photosynthesis, if glucose is considered to be a product of photosynthesis:

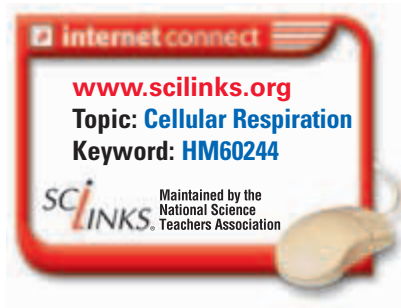


That is, the products of photosynthesis are reactants in cellular respiration, and the products of cellular respiration are reactants in photosynthesis. However, cellular respiration is not the reverse of photosynthesis. These two processes involve different biochemical pathways and occur at different sites inside cells.

## Another Role of Cellular Respiration

Cellular respiration provides the ATP that all cells need to support the activities of life. But providing cells with ATP is not the only important function of cellular respiration. Cells also need specific organic compounds from which to build the macromolecules that compose their own structures. Some of these specific compounds may not be contained in the food that a heterotroph consumes.

The molecules formed at different steps in glycolysis and the Krebs cycle are often used by cells to make the compounds that are missing in food. Compounds formed during glycolysis and the Krebs cycle can be diverted into other biochemical pathways in which the cell makes the molecules it requires. For example, approximately 10 of the amino acids needed by the human body can be made with compounds diverted from the Krebs cycle.



## SECTION 2 REVIEW

1. In what part of a mitochondrion does the Krebs cycle occur?
2. In what part of a mitochondrion is the electron transport chain located?
3. What four-carbon compound is regenerated at the end of the Krebs cycle?
4. What molecule does oxygen become a part of at the end of the electron transport chain?
5. Is cellular respiration more or less efficient than fermentation?
6. List the two processes that together result in cellular respiration.

### CRITICAL THINKING

7. **Predicting Results** Sometimes protons leak out of a cell or are used for purposes other than ATP production. How would this loss of protons affect the production of ATP in aerobic respiration?
8. **Inferring Relationships** How does the arrangement of the cristae in the inner membrane of mitochondria affect the rate of aerobic respiration? Explain your answer.
9. **Making Calculations** Calculate the efficiency of cellular respiration if a cell generates 32 ATP molecules per molecule of glucose.

# CHAPTER HIGHLIGHTS

## SECTION 1

## Glycolysis and Fermentation

- Cellular respiration is the process by which cells break down organic compounds to produce ATP.
- Cellular respiration begins with glycolysis, which takes place in the cytosol of cells. During glycolysis, one glucose molecule is oxidized to form two pyruvic acid molecules. Glycolysis results in a net production of two ATP molecules and two NADH molecules.
- If oxygen is not present, glycolysis may lead to anaerobic pathways in which pyruvic acid is converted into other organic molecules in the cytosol. Glycolysis combined with these anaerobic pathways is called *fermentation*. Fermentation does not produce ATP, but it does regenerate  $\text{NAD}^+$ , which helps keep glycolysis operating.
- In lactic acid fermentation, an enzyme converts pyruvic acid into lactic acid.
- In alcoholic fermentation, other enzymes convert pyruvic acid into ethyl alcohol and  $\text{CO}_2$ .
- Through glycolysis, only about 2 percent of the energy available from the oxidation of glucose is captured as ATP.

### Vocabulary

cellular respiration (p. 131)  
pyruvic acid (p. 131)  
NADH (p. 131)

anaerobic (p. 131)  
aerobic respiration (p. 131)  
glycolysis (p. 132)

$\text{NAD}^+$  (p. 133)  
fermentation (p. 133)  
lactic acid fermentation (p. 134)

alcoholic fermentation (p. 135)  
kilocalorie (p. 135)

## SECTION 2

## Aerobic Respiration

- In eukaryotic cells, the processes of aerobic respiration occur inside the mitochondria. The Krebs cycle occurs in the mitochondrial matrix. The electron transport chain is embedded in the inner mitochondrial membrane.
- In the mitochondrial matrix, pyruvic acid produced in glycolysis is converted into acetyl CoA. Then, acetyl CoA enters the Krebs cycle. Each turn of the Krebs cycle generates three NADH, one  $\text{FADH}_2$ , one ATP, and two  $\text{CO}_2$  molecules.
- NADH and  $\text{FADH}_2$  donate electrons to the electron transport chain in the inner mitochondrial membrane. These electrons are passed from molecule to molecule in the transport chain.
- As electrons pass along the electron transport chain, protons donated by NADH and  $\text{FADH}_2$  are pumped into the space between the inner and outer mitochondrial membranes. This pumping creates a concentration gradient of protons and a charge gradient across the inner mitochondrial membrane. As protons move through ATP synthase, down their concentration and charge gradients, and back into the mitochondrial matrix, ATP is produced.
- During aerobic respiration, oxygen accepts both protons and electrons from the electron transport chain. As a result, oxygen is converted to water.
- Cellular respiration can produce up to 38 ATP molecules from the oxidation of a single molecule of glucose. Thus, up to 39 percent of the energy released by the oxidation of glucose can be transferred to ATP. However, most eukaryotic cells produce only about 36 ATP molecules per molecule of glucose.
- Cellular respiration uses the processes of glycolysis and aerobic respiration to obtain energy from organic compounds.

### Vocabulary

mitochondrial matrix (p. 137)  
acetyl CoA (p. 138)

Krebs cycle (p. 138)  
oxaloacetic acid (p. 138)

citric acid (p. 138)  
FAD (p. 139)

# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain the relationship between the terms.
  - alcoholic fermentation* and *lactic acid fermentation*
  - glycolysis* and *pyruvic acid*
  - mitochondrial matrix* and *Krebs cycle*
- Use the following terms in the same sentence: *acetyl CoA*, *citric acid*, and *oxaloacetic acid*.
- Explain the difference between the terms *fermentation* and *cellular respiration*.
- Word Roots and Origins** The word *glycolysis* is derived from the Greek words *glykys*, which means “sweet,” and *lysis*, which means “loosening.” Using this information, explain why the term *glycolysis* is a good name for the biological process it describes.

## UNDERSTANDING KEY CONCEPTS

- Compare** the two stages of cellular respiration.
- Explain** why the net yield of ATP molecules in glycolysis is two, even though four ATP molecules are produced.
- Describe** what causes your muscles to become fatigued and sometimes develop cramps when you exercise too strenuously.
- Calculate** the efficiency of glycolysis if the number of ATP molecules produced during glycolysis were 5 times greater.
- Name** the two areas of the mitochondrion where the major stages of aerobic respiration occur.
- Explain** the importance of the cyclical nature of the Krebs cycle.
- Define** the specific role that oxygen plays in the electron transport chain.
- Summarize** the process of electron transport and chemiosmosis, including where the electron transport chain is located, what structures make up the electron transport chain, where the electrons come from and their characteristics, what happens to them as they move along the electron transport chain and as they reach the end of the chain, and what is accomplished by this process.
- Explain** why most eukaryotic cells produce fewer than 38—the maximum number possible—ATP molecules for every glucose molecule that is oxidized by cellular respiration.
- Compare** the efficiency of glycolysis alone with the efficiency of cellular respiration.

## 15. Unit 3—Cellular Respiration

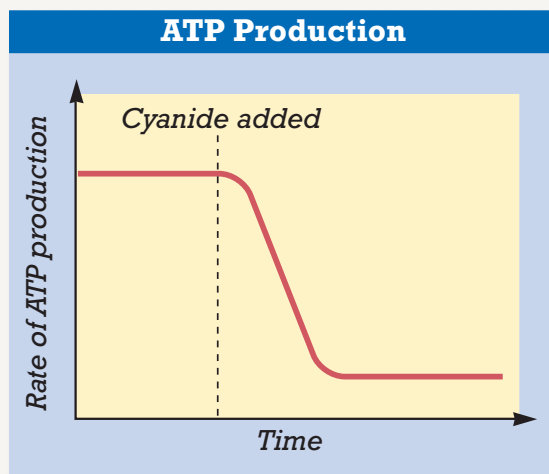


Write a report summarizing how exercise physiologists regulate the diets and training of athletes. Find out how diets vary according to the needs of each athlete. Research the relationship between exercise and metabolism.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that shows the activities of fermentation: *alcoholic fermentation*, *anaerobic pathway*, *fermentation*, *glycolysis*, *lactic acid fermentation*, and *pyruvic acid*.

## CRITICAL THINKING

- Evaluating Results** Some yeast can use fermentation or cellular respiration. If oxygen is present, these yeast cells consume glucose much more slowly than if oxygen is absent. Explain this observation.
- Inferring Relationships** How does cellular respiration ultimately depend on photosynthesis?
- Predicting Results** Some eukaryotic cells must use ATP to move NADH into the mitochondrial matrix. Would you expect cellular respiration to be more or less efficient in prokaryotic cells than in eukaryotic cells? Explain your answer.
- Interpreting Graphics** The graph below shows the rate of ATP production by a culture of yeast cells over time. At the time indicated by the dashed line, cyanide was added to the culture. Cyanide blocks the flow of electrons to  $O_2$  from the electron transport chain in mitochondria. Explain why adding cyanide affects ATP production in the way indicated by the graph.





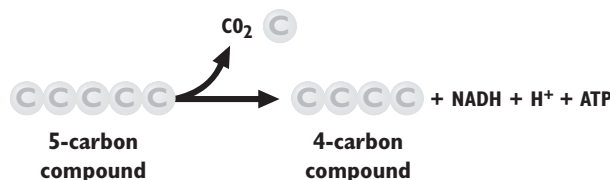


# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- Which of the following must pyruvic acid be converted into before the Krebs cycle can proceed?
  - NADH
  - glucose
  - citric acid
  - acetyl CoA
- Which of the following occurs in lactic acid fermentation?
  - Oxygen is consumed.
  - Lactic acid is converted into pyruvic acid.
  - $\text{NAD}^+$  is regenerated for use in glycolysis.
  - Electrons pass through the electron transport chain.
- Which of the following is not a product of the Krebs cycle?
  - $\text{CO}_2$
  - ATP
  - $\text{FADH}_2$
  - ethyl alcohol
- In which way is cellular respiration similar to photosynthesis?
  - They both make G3P.
  - They both involve ATP.
  - They both involve chemiosmosis.
  - all of the above
- ATP is synthesized in chemiosmosis when which of the following moves across the inner mitochondrial membrane?
  - NADH
  - oxygen
  - protons
  - citric acid

**INTERPRETING GRAPHICS:** The illustration shows part of a biochemical pathway. Use the illustration to answer the question that follows.

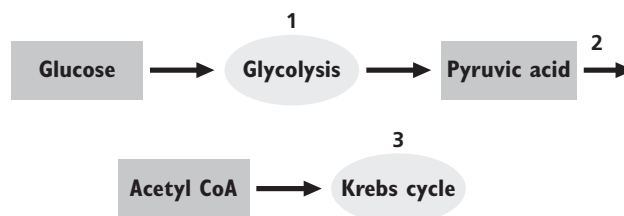


- This reaction occurs during which of the following processes?
  - Krebs cycle
  - acetyl CoA formation
  - alcoholic fermentation
  - lactic acid fermentation

**DIRECTIONS:** Complete the following analogy.

- glycolysis : pyruvic acid :: Krebs cycle :
  - $\text{O}_2$
  - ATP
  - lactic acid
  - acetyl CoA

**INTERPRETING GRAPHICS:** The illustration below shows some stages and reactants of cellular respiration. Use the illustration to answer the question that follows.



- At which of the points is ATP, the main energy currency of the cell, produced?
  - 1 only
  - 2 only
  - 1 and 3
  - 1, 2, and 3

## SHORT RESPONSE

The inner membrane of a mitochondrion is folded; these folds are called *cristae*.

How might cellular respiration be different if the inner mitochondrial membrane were not folded?

## EXTENDED RESPONSE

Oxygen is produced during the reactions of photosynthesis, and it is used in the reactions of cellular respiration.

*Part A* How does oxygen get into or out of chloroplasts and mitochondria?

*Part B* What are the roles of oxygen in the processes of photosynthesis and cellular respiration, and how are the roles similar?

## Test TIP

Make sure you get plenty of rest the night before the test. Eat a healthy breakfast the day of the test and wear comfortable clothing.

# Observing Cellular Respiration

## OBJECTIVES

- Measure the rate of cellular respiration in germinating seeds.
- Compare cellular respiration rates in germinating and nongerminating seeds.

## PROCESS SKILLS

- experimenting
- collecting data
- analyzing data

## MATERIALS

- |                                         |                                  |
|-----------------------------------------|----------------------------------|
| ■ volumeter jar and screw-on lid        | ■ 100 mL graduated cylinder      |
| ■ tap water at room temperature         | ■ glass or plastic beads         |
| ■ ring stand with support ring          | ■ 3 plastic or paper cups        |
| ■ 8.5 in. × 11 in. piece of cardboard   | ■ 3 volumeters                   |
| ■ 8.5 in. × 11 in. sheet of white paper | ■ cotton                         |
| ■ cellophane tape                       | ■ 3 soda-lime packets            |
| ■ 40 germinating corn or pea seeds      | ■ forceps                        |
| ■ 40 nongerminating corn or pea seeds   | ■ Pasteur pipet                  |
|                                         | ■ colored water                  |
|                                         | ■ ruler with millimeter markings |



## Background

1. When glucose is oxidized in cellular respiration, what other substance is consumed and what substances are produced?
2. Write the balanced equation for the complete oxidation of glucose in cellular respiration.

## PART A Setting Up the Apparatus

1. Pour water into the volumeter jar until the jar is about two-thirds full. Screw on the lid.
2. Place the cardboard on top of the ring stand support ring, and tape a sheet of white paper to the cardboard. Adjust the support ring so that the cardboard is level, as shown in the illustration below.
3. **CAUTION** Put on a lab apron, safety goggles, and protective gloves. Keep the seeds, which may have been treated with a fungicide, away from your skin. Place 40 germinating seeds in a graduated cylinder and measure their volume. Do the same for the 40 nongerminating seeds.
4. Add beads to the seeds that have the smaller volume until the combined volume is the same as the volume of the other group of seeds. Then transfer both groups to separate cups. To a third cup, add a volume of beads equal to the volume of each of the other two cups.
5. Remove the stopper assemblies from three volumeter tubes and transfer the contents of the three cups to separate tubes. Place a 2 cm plug of dry cotton into each tube, leaving a gap of about 1 cm between the cotton and the seeds or beads.
6. **CAUTION** Soda lime is corrosive. Do not touch it. If it gets on your skin or clothing, wash it off at the sink. If it gets in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Using forceps, place a packet of soda lime wrapped in gauze on top of the cotton plug in each tube. Soda lime absorbs the CO<sub>2</sub> that is produced as a result of respiration.
7. Gently but firmly press the stopper assembly into each volumeter tube. Insert the tubes into the volumeter jar through the large holes in the lid.

8. Use a Pasteur pipet to place a small drop of colored water into the three capillary tubes. Tilt two of the tubes slightly until the drops are lined up with the outermost calibration mark. Carefully attach these tubes to the latex tubing on the two volumeter tubes containing seeds. Position the drop in the third capillary tube near the middle of the tube. Attach this tube to the volumeter tube that contains only beads. This volumeter is the control volumeter. Tape all three capillary tubes to the paper on the ring stand.
9. Wait 5 min for the temperature to become uniform throughout the volumeter jar. While you wait, make a data table like the one shown below. Then return the drops in the capillary tubes to their original positions by using the syringes to inject air into or withdraw air from the volumeter tubes, if necessary.

## PART B Measuring Respiration Rates

10. On the paper beneath the capillary tubes, mark the position of one end of each drop of colored water. Note the time. Repeat this procedure every 5 min for 20 min. If respiration is rapid, you may have to reposition the drops as you did in Step 9. In which direction would you expect a drop to move if respiration in the volumeter tube were causing it to move?
11. Remove the paper from the ring stand and use a ruler to measure the distance moved by the drops during each time interval. If you repositioned any drops in Step 10, be sure to add this adjustment when you measure the distances. Enter the measurements in the "Uncorrected" columns in your data table.
12.   Clean up your materials and wash your hands before leaving the lab.

## Analysis and Conclusions

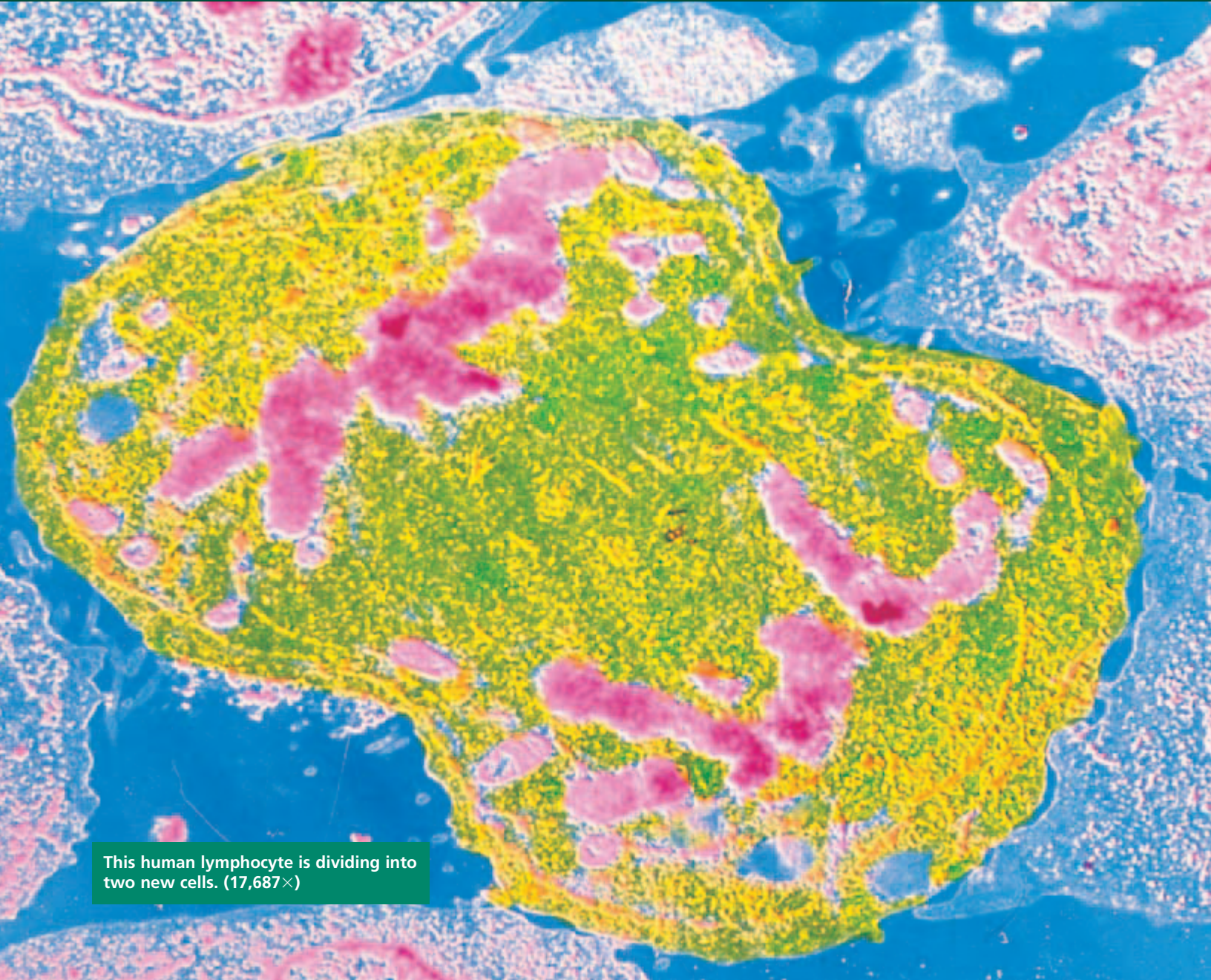
1. No respiration should have occurred in the control volumeter, which contained only beads. Therefore, any movement of the drop in the control volumeter must have been caused by changes in the temperature of the volumeter jar or the air pressure in the classroom. Since these changes would have affected all three volumeters to the same extent, you must subtract the distance you measured for the control volumeter from the distances you measured for the other two volumeters. Do this calculation for each time interval, and enter the results in the "Corrected" columns in your data table.
2. Each capillary tube has a capacity of 0.063 mL between each 1 cm mark on the tube. Use this information to calculate the volume of  $O_2$  consumed by the germinating and nongerminating seeds during each time interval. Enter these results in your data table.
3. Prepare a graph to show the volume of  $O_2$  consumed versus time; use different symbols or colors to distinguish the points for the germinating seeds from those for the nongerminating seeds. Make sure each point represents the cumulative volume of  $O_2$  consumed. For example, the point plotted for the 15–20 min interval should represent the volume consumed during that interval plus the volume consumed during all of the preceding intervals. Draw the best-fit line through the points for each group of seeds. From the slope of this line, calculate the average rate of respiration in milliliters of  $O_2$  per minute for both groups of seeds.
4. Which group of seeds had the higher average rate of respiration? What is the significance of this difference in terms of a seed's ability to survive for long periods?

### MEASUREMENTS OF CELLULAR RESPIRATION

| Time interval (min) | Distance moved by drops in volumeters (mm) |                   |           |                      |           | Volume of $O_2$ consumed (mL) |                |
|---------------------|--------------------------------------------|-------------------|-----------|----------------------|-----------|-------------------------------|----------------|
|                     | Control                                    | Germinating seeds |           | Nongerminating seeds |           | Germinating                   | Nongerminating |
|                     |                                            | Uncorrected       | Corrected | Uncorrected          | Corrected |                               |                |
| 0–5                 |                                            |                   |           |                      |           |                               |                |
| 5–10                |                                            |                   |           |                      |           |                               |                |
| 10–15               |                                            |                   |           |                      |           |                               |                |
| 15–20               |                                            |                   |           |                      |           |                               |                |



# CELL REPRODUCTION



This human lymphocyte is dividing into two new cells. (17,687 $\times$ )

**SECTION 1** *Chromosomes*

**SECTION 2** *Cell Division*

**SECTION 3** *Meiosis*



Unit 4—Cell Reproduction  
Topics 1–6

# CHROMOSOMES

*Recall that DNA is a long, thin molecule that stores genetic information. The DNA in a human cell is estimated to consist of three billion nucleotides. To visualize the enormity of three billion nucleotides, imagine increasing a cell nucleus to the size of a basketball. Then, imagine taking the DNA out of the basketball-sized nucleus and stretching it into a straight line. That line of DNA would stretch for more than 20 miles. How can a nucleus hold so much DNA? Inside the nucleus, the DNA is coiled and packed in a complicated yet organized manner. As a cell prepares to divide, the DNA coils even further into tightly compacted structures.*

## CHROMOSOME STRUCTURE

During cell division, the DNA in a eukaryotic cell's nucleus is coiled into very compact structures called chromosomes. **Chromosomes** are rod-shaped structures made of DNA and proteins. In Figure 8-1, you can see the many levels of DNA coiling required to form a chromosome.

The chromosomes of stained eukaryotic cells undergoing cell division are visible as darkened structures inside the nuclear membrane. Each chromosome is a single DNA molecule associated with proteins. The DNA in eukaryotic cells wraps tightly around proteins called **histones**. Histones help maintain the shape of the chromosome and aid in the tight packing of DNA. Nonhistone proteins are generally involved in controlling the activity of specific regions of the DNA.

## OBJECTIVES

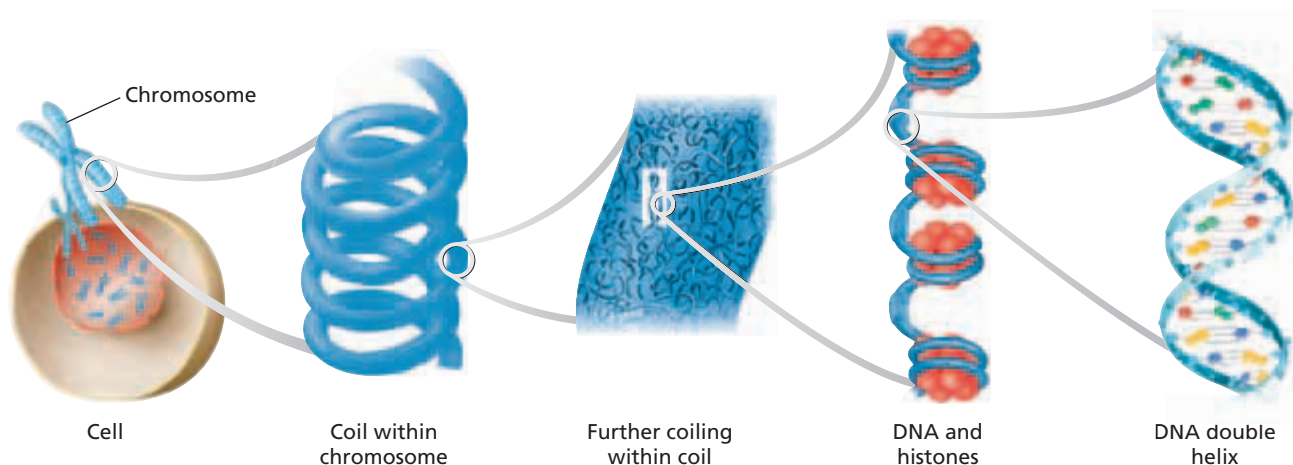
- **Describe** the structure of a chromosome.
- **Identify** the differences in structure between prokaryotic chromosomes and eukaryotic chromosomes.
- **Compare** the numbers of chromosomes in different species.
- **Explain** the differences between sex chromosomes and autosomes.
- **Distinguish** between diploid and haploid cells.

## VOCABULARY

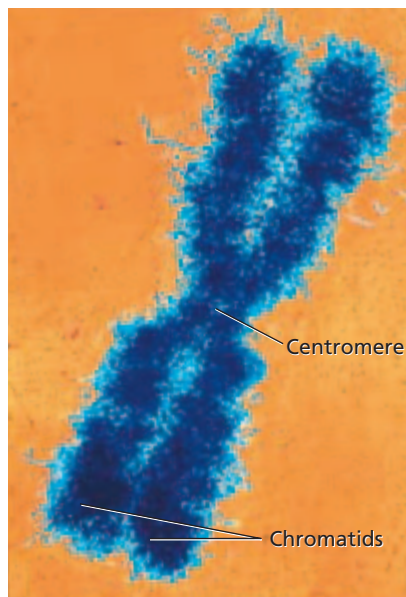
chromosome  
histone  
chromatid  
centromere  
chromatin  
sex chromosome  
autosome  
homologous chromosome  
karyotype  
diploid  
haploid

**FIGURE 8-1**

As a cell prepares to divide, its DNA coils around histones and twists into rod-shaped chromosomes.







**FIGURE 8-2**

Chromosomes, such as this one isolated from a dividing human cell, consist of two identical chromatids. (TEM 12,542 $\times$ )

Figure 8-2 shows a chromosome that was isolated from a dividing cell. Notice that the chromosome consists of two identical halves. Each half of the chromosome is called a **chromatid**. Chromatids form as the DNA makes a copy of itself before cell division. When the cell divides, each of the two new cells will receive one chromatid from each chromosome. The two chromatids of a chromosome are attached at a point called a **centromere**. The centromere holds the two chromatids together until they separate during cell division. As you will learn in the next section, centromeres are especially important for the movement of chromosomes during cell division.

Between cell divisions, DNA is not so tightly coiled into chromosomes. Regions of DNA uncoil in between cell divisions so they can be read and so the information can be used to direct the activities of the cell. The less tightly coiled DNA-protein complex is called **chromatin**.

As you might expect, chromosomes are simpler in prokaryotes than in eukaryotes. The DNA of most prokaryotes consists of only one chromosome, which is attached to the inside of the cell membrane. Prokaryotic chromosomes consist of a circular DNA molecule. As with eukaryotic chromosomes, prokaryotic chromosomes must be very compact to fit into the cell.

## CHROMOSOME NUMBERS

Each species has a characteristic number of chromosomes in each cell. Table 8-1 lists the number of chromosomes found in some organisms. Some species of organisms have the same number of chromosomes. For example, potatoes, plums, and chimpanzees all have 48 chromosomes in each cell.

### Sex Chromosomes and Autosomes

Human and animal chromosomes are categorized as either sex chromosomes or autosomes. **Sex chromosomes** are chromosomes that determine the sex of an organism, and they may also carry genes for other characteristics. In humans, sex chromosomes are either X or Y. Females normally have two X chromosomes, and males normally have an X and a Y chromosome. All of the other chromosomes in an organism are called **autosomes**. Two of the 46 human chromosomes are sex chromosomes, and the remaining 44 chromosomes are autosomes.

Every cell of an organism produced by sexual reproduction has two copies of each autosome. The organism receives one copy of each autosome from each parent. The two copies of each autosome are called **homologous chromosomes**, or *homologues*. Homologous chromosomes are the same size and shape and carry genes for the same traits. For example, if one chromosome in a pair of homologues contains a gene for eye color, so will the other chromosome in the homologous pair.

**TABLE 8-1 Chromosome Numbers of Various Species**

| Organism            | Number of chromosomes |
|---------------------|-----------------------|
| Adder's tongue fern | 1,262                 |
| Carrot              | 18                    |
| Cat                 | 32                    |
| Chimpanzee          | 48                    |
| Dog                 | 78                    |
| Earthworm           | 36                    |
| Fruit fly           | 8                     |
| Garden pea          | 20                    |
| Gorilla             | 48                    |
| Horse               | 64                    |
| Human               | 46                    |
| Lettuce             | 18                    |
| Orangutan           | 48                    |
| Sand dollar         | 52                    |

Figure 8-3 shows a **karyotype**, which is a photomicrograph of the chromosomes in a normal dividing cell found in a human. Notice that the 46 human chromosomes exist as 22 homologous pairs of autosomes and 2 sex chromosomes (XY in males and XX in females).

## Diploid and Haploid Cells

Cells having two sets of chromosomes are **diploid**. Diploid cells have two autosomes for each homologous pair. Diploid cells also have two sex chromosomes in animals, including humans, and in many other organisms that have sex chromosomes. All human cells, except reproductive cells (sperm cells and egg cells), are normally diploid cells. Diploid is commonly abbreviated as  $2n$ . In humans, the diploid, or  $2n$ , number of chromosomes is 46—22 pairs of homologous autosomes and 2 sex chromosomes.

Sperm cells and egg cells are **haploid** cells, which contain only one set of chromosomes. Haploid cells have half the number of chromosomes that are present in diploid cells. Thus, haploid cells have only one autosome of each homologous pair and only one sex chromosome (23 total). Haploid is abbreviated as  $1n$ . When a sperm cell ( $1n$ ) and an egg cell ( $1n$ ) combine to create the first cell of a new organism, the new cell will be diploid ( $2n$ ). If the reproductive cells were diploid, the new cell would have too many chromosomes and would not be functional.



**FIGURE 8-3**

Karyotypes, such as this one, are used to examine an individual's chromosomes. Karyotypes are made from a sample of a person's blood. White blood cells from the sample are treated chemically to stimulate mitosis and to arrest mitosis in metaphase. The chromosomes are then photographed, cut out, and arranged by size and shape into pairs.

## SECTION 1 REVIEW

1. Name the proteins that DNA wraps around to form a chromosome in eukaryotic cells.
2. How do the structure and location of a prokaryotic chromosome differ from that of a eukaryotic chromosome?
3. Does chromosome number indicate whether an organism is a plant or an animal? Explain.
4. Contrast sex chromosomes with autosomes.
5. Using Table 8-1, list the haploid and diploid number of chromosomes for each organism.

### CRITICAL THINKING

6. **Forming Reasoned Opinions** Is there a correlation between the number of chromosomes and the complexity of an organism? Give support for your answer.
7. **Predicting Results** What would be the consequence for future generations of cells if sperm and egg cells were normally diploid?
8. **Interpreting Graphics** What is the sex of the person whose chromosomes are shown in Figure 8-3 above? Explain your answer.

## SECTION 2

### OBJECTIVES

- **Describe** the events of cell division in prokaryotes.
- **Name** the two parts of the cell that are equally divided during cell division in eukaryotes.
- **Summarize** the events of interphase.
- **Describe** the stages of mitosis.
- **Compare** cytokinesis in animal cells with cytokinesis in plant cells.
- **Explain** how cell division is controlled.

### VOCABULARY

binary fission  
mitosis  
asexual reproduction  
meiosis  
gamete  
interphase  
cytokinesis  
prophase  
spindle fiber  
metaphase  
anaphase  
telophase  
cell plate

# CELL DIVISION

*Approximately 2 trillion cells—about 25 million cells per second—are produced by an adult human body every day.*

*All cells come from the division of preexisting cells. Cell division (also called cell reproduction) is the process by which cells produce offspring cells. Cell division differs in prokaryotes and eukaryotes. But cell reproduction in both prokaryotes and eukaryotes produces the same result—two cells from one.*

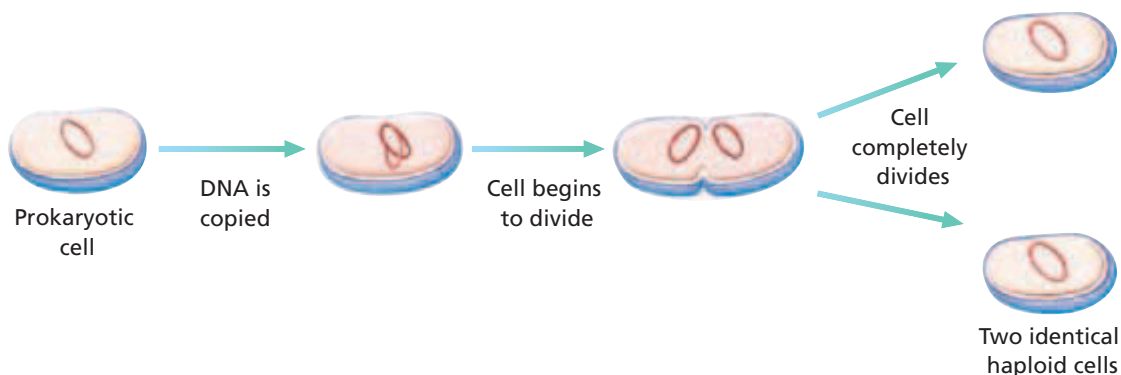
## CELL DIVISION IN PROKARYOTES

Prokaryotes have cell walls but lack nuclei and membrane-bound organelles. A prokaryote's single DNA molecule is not coiled around proteins to form chromosomes. Instead, a prokaryote's DNA is a circular chromosome attached to the inner surface of the plasma membrane like a rope attached to the inner wall of a tent. For most prokaryotes, cell division takes place through a process called binary fission.

**Binary fission** is the division of a prokaryotic cell into two offspring cells, as shown in Figure 8-4. The DNA is copied, resulting in two identical chromosomes attached to the inside of the prokaryote's inner cell membrane. A new cell membrane then begins to develop between the two DNA copies. The cell grows until it reaches approximately twice the cell's original size. As new material is added, the growing cell membrane pushes inward and the cell is constricted in the center, like a balloon being squeezed in the middle. A new cell wall forms around the new membrane. Eventually, the dividing prokaryote is split into two independent cells. Each cell contains one of the identical chromosomes that resulted from the copying of the original cell's chromosome.

**FIGURE 8-4**

Most prokaryotes reproduce by binary fission, in which two identical cells are produced from one cell.



# CELL DIVISION IN EUKARYOTES

In eukaryotic cell division, both the cytoplasm and the nucleus divide. There are two kinds of cell division in eukaryotes. The first type of cell division that you will learn about is called mitosis. **Mitosis** results in new cells with genetic material that is identical to the genetic material of the original cell. Mitosis occurs in organisms undergoing growth, development, repair, or asexual reproduction. **Asexual reproduction** is the production of offspring from one parent.

The second type of cell division that you will learn about (in the next section) is called meiosis. **Meiosis** occurs during the formation of **gametes**, which are haploid reproductive cells. Meiosis reduces the chromosome number by half in new cells. Each new cell has the potential to join with another haploid cell to produce a diploid cell with a complete set of chromosomes.

## The Cell Cycle

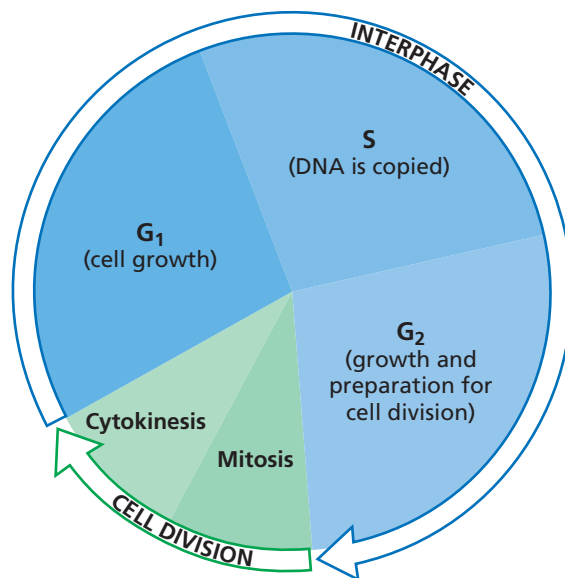
The *cell cycle* is the repeating set of events in the life of a cell. Cell division is one phase of the cycle. The time between cell divisions is called **interphase**. Interphase is divided into three phases, and cell division is divided into two phases, as shown in Figure 8-5.

During cell division, the chromosomes and cytoplasm are equally divided between two offspring cells. Cell division consists of mitosis and cytokinesis. During mitosis, the nucleus of a cell divides. **Cytokinesis** is the division of the cell's cytoplasm.

## Interphase

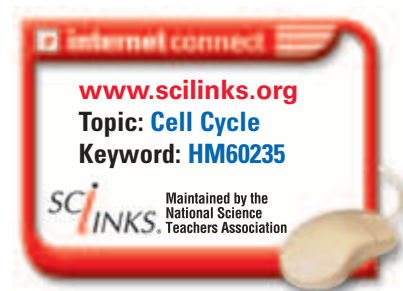
Notice in Figure 8-5 that cells spend most of the cell cycle in interphase. Following cell division, offspring cells are approximately half the size of the original cell. During the first stage of interphase—called the *G<sub>1</sub> phase*—offspring cells grow to mature size. *G<sub>1</sub>* stands for the time gap following cell division and preceding DNA replication. After cells have reached a mature size, many proceed into the next phase of interphase, called the *S phase*. During the *S phase*, the cell's DNA is copied (synthesized). The *G<sub>2</sub> phase* represents the time gap following DNA synthesis (*S phase*) and preceding cell division. The *G<sub>2</sub> phase* is a time during which the cell prepares for cell division.

Cells can also exit the cell cycle (usually from the *G<sub>1</sub> phase*) and enter into a state called the *G<sub>0</sub> phase*. During the *G<sub>0</sub> phase*, cells do not copy their DNA and do not prepare for cell division. Many cells in the human body are in the *G<sub>0</sub> phase*. For example, fully developed cells in the central nervous system stop dividing at maturity and normally never divide again.



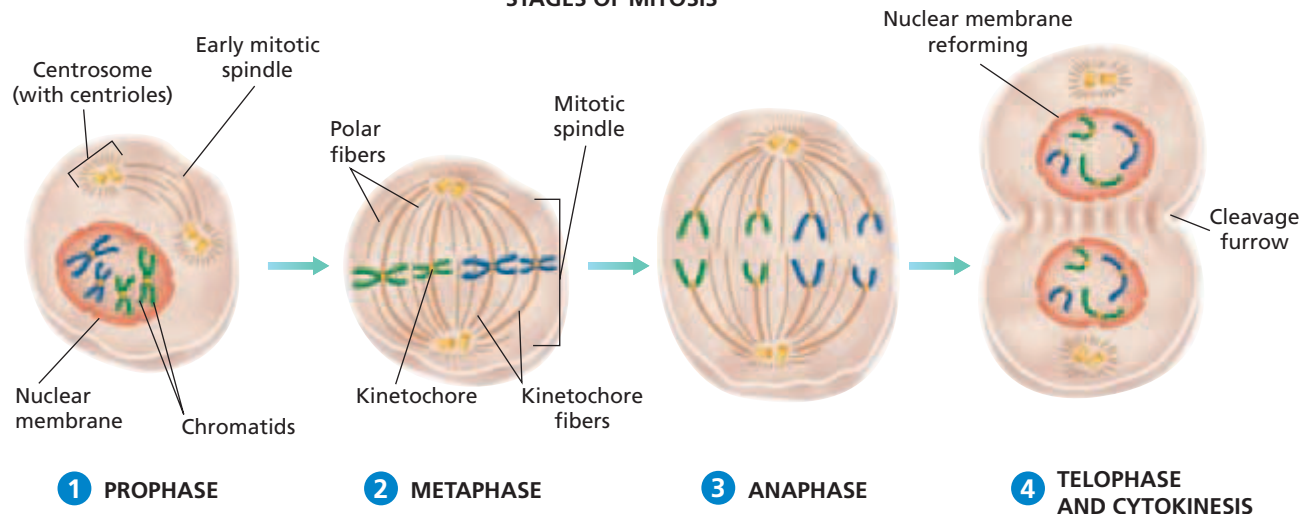
**FIGURE 8-5**

The cell cycle consists of interphase and cell division. Phases of growth, DNA synthesis, and preparation for cell division make up interphase. Cell division is divided into mitosis (division of the nucleus) and cytokinesis (division of the cytoplasm).





## STAGES OF MITOSIS



**FIGURE 8-6**

- 1 During prophase, the copied DNA coils into chromosomes.
- 2 During metaphase, the chromosomes line up along the midline of the dividing cell.
- 3 During anaphase, the chromatids of each chromosome begin moving toward opposite poles of the cell.
- 4 During telophase, the chromosomes reach opposite poles of the cell, and a cleavage furrow is formed. Cytokinesis follows.

## STAGES OF MITOSIS

Mitosis is the division of the nucleus, which occurs during cell division. Mitosis is a continuous process that allows for the organized distribution of a cell's copied DNA to offspring cells. The process of mitosis is usually divided into four phases for ease of understanding: prophase, metaphase, anaphase, and telophase.

### Prophase

**Prophase** is the first phase of mitosis. Prophase, shown in step 1 of Figure 8-6, begins with the shortening and tight coiling of DNA into rod-shaped chromosomes that can be seen with a light microscope. Recall that during the S phase, each chromosome is copied. The two copies of each chromosome—the chromatids—stay connected to one another by the centromere. At this time, the nucleolus and the nuclear membrane break down and disappear.

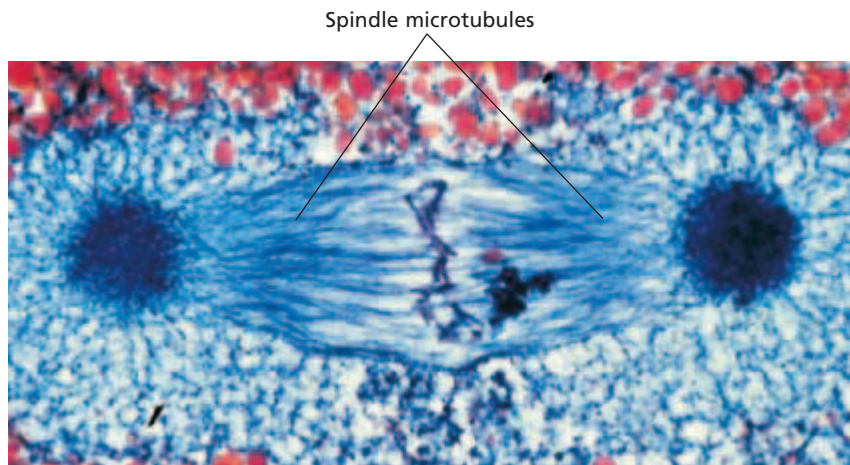
Two pairs of dark spots called *centrosomes* appear next to the disappearing nucleus. In animal cells, each centrosome contains a pair of small, cylindrical bodies called *centrioles*. The centrosomes of plant cells lack centrioles. In both animal and plant cells, the centrosomes move toward opposite poles of the cell during prophase.

As the centrosomes separate, **spindle fibers** made of microtubules radiate from the centrosomes in preparation for metaphase. This array of spindle fibers is called the *mitotic spindle*, which serves to equally divide the chromatids between the two offspring cells during cell division. Two types of spindle fibers make up the mitotic spindle: kinetochore fibers and polar fibers. *Kinetochore fibers* attach to a disk-shaped protein—called a *kinetochore*—that is found in the centromere region of each chromosome. Kinetochore fibers extend from the kinetochore of each chromatid to one of the centrosomes. *Polar fibers* extend across the dividing cell from centrosome to centrosome but do not attach to the chromosomes.

### Word Roots and Origins

#### *kinetochore*

from the Greek *kinetos*, meaning "moving," and *choros*, meaning "place"



**FIGURE 8-7**

This micrograph of the spindle apparatus during metaphase shows the kinetochore fibers moving the chromosomes to the center of the dividing cell. The wormlike structures in the center are the chromosomes. (LM 1,080 $\times$ )

## Metaphase

**Metaphase**, as shown in step ② of Figure 8-6, is the second phase of mitosis. During metaphase, chromosomes are easier to identify by using a microscope than during other phases; thus, karyotypes are typically made from photomicrographs of chromosomes in metaphase. As shown in Figure 8-7 above, the kinetochore fibers move the chromosomes to the center of the dividing cell during metaphase. Once in the center of the cell, each chromosome is held in place by the kinetochore fibers.

## Anaphase

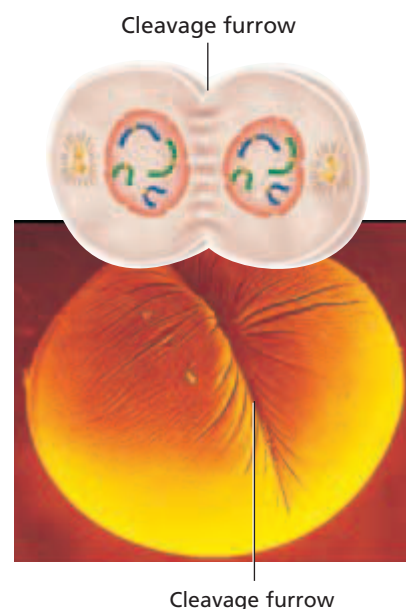
During **anaphase**, shown in step ③ of Figure 8-6 on the previous page, the chromatids of each chromosome separate at the centromere and slowly move, centromere first, toward opposite poles of the dividing cell. After the chromatids separate, they are considered to be individual chromosomes.

## Telophase

**Telophase** is shown in step ④ in Figure 8-6 on the previous page. After the chromosomes reach opposite ends of the cell, the spindle fibers disassemble, and the chromosomes return to a less tightly coiled chromatin state. A nuclear envelope forms around each set of chromosomes, and a nucleolus forms in each of the newly forming cells.

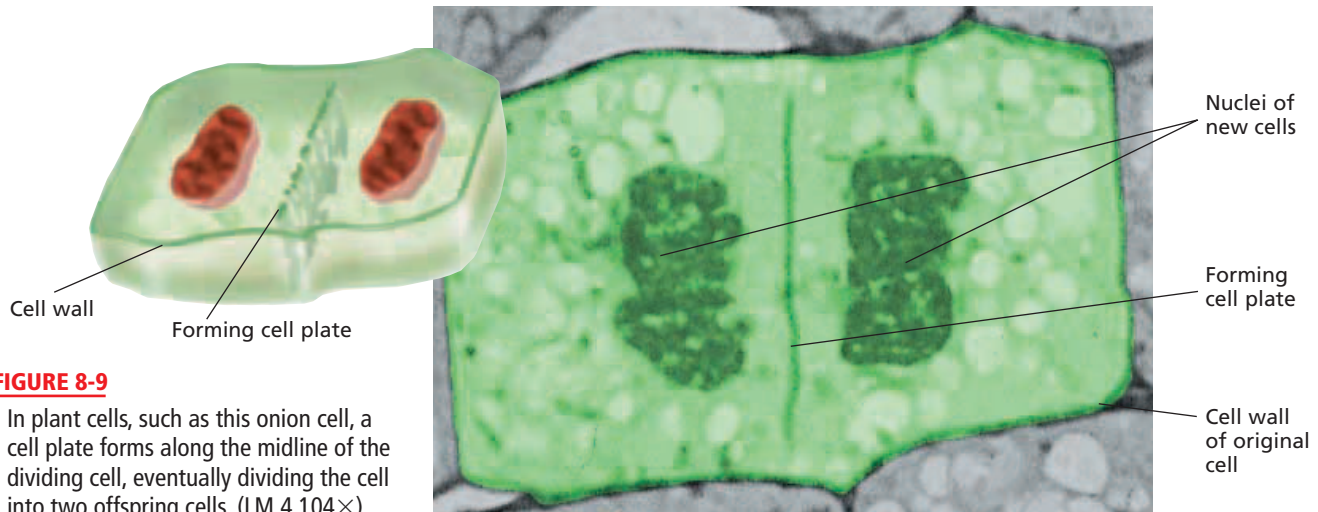
# CYTOKINESIS

During telophase, the cytoplasm begins dividing by the process of cytokinesis. In animal cells, cytokinesis begins with a pinching inward of the cell membrane midway between the dividing cell's two poles, as shown in Figure 8-8. The area of the cell membrane that pinches in and eventually separates the dividing cell into two cells is called the *cleavage furrow*. The cleavage furrow pinches the cell into two cells through the action of microfilaments.



**FIGURE 8-8**

In animal cells, such as this frog cell, the cell membrane pinches in at the center of the dividing cell, eventually dividing the cell into two offspring cells. (SEM 78 $\times$ )



**FIGURE 8-9**

In plant cells, such as this onion cell, a cell plate forms along the midline of the dividing cell, eventually dividing the cell into two offspring cells. (LM 4,104 $\times$ )

Figure 8-9 shows cytokinesis in plant cells. In plant cells, vesicles from the Golgi apparatus join together at the midline of the dividing cell to form a **cell plate**. A cell wall eventually forms from the cell plate at the midline, dividing the cell into two cells.

In both animal cells and plant cells, offspring cells are approximately equal in size. Each offspring cell receives an identical copy of the original cell's chromosomes and approximately one-half of the original cell's cytoplasm and organelles.



## Quick Lab

### Identifying Prefixes and Suffixes

**Materials** dictionary, 3  $\times$  5 in. index cards (18), pencil

#### Procedure

1. Write each of the following prefixes and suffixes on separate cards: *pro-*, *meta-*, *ana-*, *telo-*, *cyto-*, *oo-*, *inter-*, *-kinesis*, and *-genesis*.
2. Use a dictionary to find the definition of each prefix and suffix. Write the definitions on cards.
3. Play "Memory" with a partner. Mix the cards, and place each one face down on the table. Turn over two cards. If the two cards consist of a prefix or suffix and its definition, pick up the cards, and take another turn. If the two cards do not match, turn them face down again, and leave them in the same place.
4. Repeat step 3 until no cards remain on the table. The player with the most pairs wins.

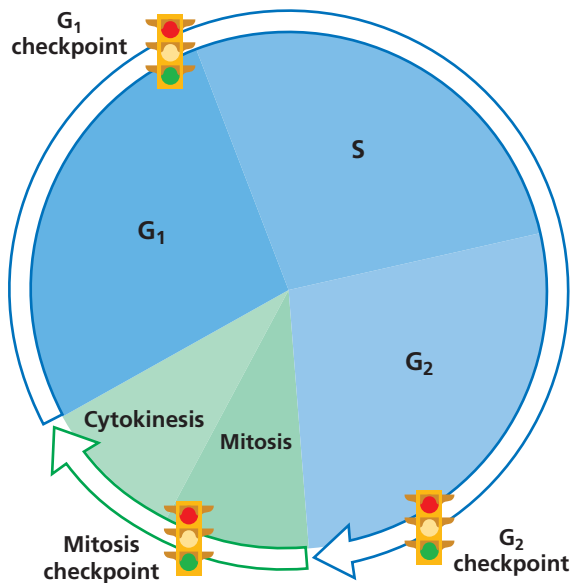
**Analysis** How does knowing the meaning of a prefix or suffix help you understand a word's meaning?

## CONTROL OF CELL DIVISION

Recall that a cell spends most of its time in interphase, the time between cell divisions. What triggers a cell to leave interphase and begin dividing? In eukaryotes, proteins regulate the progress of cell division at certain checkpoints. This system of checkpoints can be thought of as a kind of "traffic signal" for the cell. Certain feedback signals from the cell can trigger the proteins to initiate the next phase of the cell cycle, much as a green light signals traffic to move forward. Other feedback signals from the cell can trigger the proteins to halt the cycle, just as a red light signals traffic to stop.

Control occurs at three main checkpoints. These checkpoints are illustrated in Figure 8-10 on the next page.

**1. Cell growth ( $G_1$ ) checkpoint.** Proteins at this checkpoint control whether the cell will divide. If the cell is healthy and has grown to a suitable size during the  $G_1$  phase, proteins will initiate DNA synthesis (the S phase). The cell copies its DNA during this phase. If conditions are not favorable for DNA synthesis, the cell cycle will stop at this point. The cell cycle may also stop at this checkpoint if the cell needs a rest period. Certain cells pass into the  $G_0$  phase at this checkpoint. Many cells that have passed into the  $G_0$  phase will never divide again.



**FIGURE 8-10**

The cell cycle in eukaryotes is controlled at three inspection points, or checkpoints. Many proteins are involved in the control of the cell cycle.

- 2. DNA synthesis (G<sub>2</sub>) checkpoint.** At this point in the G<sub>2</sub> phase, DNA repair enzymes check the results of DNA replication. If this checkpoint is passed, proteins will signal the cell to begin the molecular processes that will allow the cell to divide mitotically.
- 3. Mitosis checkpoint.** If a cell passes this checkpoint, proteins signal the cell to exit mitosis. The cell then enters into the G<sub>1</sub> phase, the major growth phase of the cell cycle, once again.

## When Control Is Lost: Cancer

The proteins that regulate cell growth and division are coded for by genes. If a mutation occurs in one of these genes, the proteins may not function properly. Cell growth and division may be disrupted as a result. Such a disruption could lead to *cancer*, the uncontrolled growth of cells. Cancer cells do not respond normally to the body's control mechanisms. Some mutations cause cancer by overproducing growth-promoting molecules, which can lead to increased cell division. Other mutations may interfere with the ability of control proteins to slow or stop the cell cycle.

## SECTION 2 REVIEW

1. Name the process by which prokaryotic cells divide.
2. What is the name of the process by which the cell's cytoplasm divides?
3. During which of the phases of interphase does an offspring cell grow to mature size?
4. During which phase of mitosis do chromatids separate to become chromosomes?
5. Explain the main difference between cytokinesis in animal cells and cytokinesis in plant cells.
6. Which type of molecule controls the cell cycle?

### CRITICAL THINKING

7. **Predicting Results** What would happen if cytokinesis took place before mitosis?
8. **Applying Information** What would result if chromosomes did not replicate during interphase?
9. **Evaluating Information** Why are individual chromosomes more difficult to see during interphase than during mitosis?



## STEM CELLS: Promise and Difficulty

**S**tem cells are unspecialized cells that give rise to the different types of cells that make up the human body. Scientists researching stem cells hope someday to be able to use them to replace damaged or diseased cells in the body. There are two general types of stem cells: embryonic stem cells and adult stem cells.

### Embryonic Stem Cells

Embryonic stem cells seem to show more promise in medical treatment than do adult stem cells. Embryonic stem cells are easier to find than are adult stem cells. Embryonic stem cells can reproduce indefinitely in culture and have the potential to grow into any cell type. However, embryonic stem cells would be genetically different from the cells of a transplant recipient. The recipient's immune system could reject the cells, causing transplant failure.

### Adult Stem Cells

Some stem cells remain in the body into adulthood. These adult stem cells naturally produce just one or a few types of cells. For example, bone marrow stem cells give rise only to new blood cells. Studies have shown some success in coaxing adult stem cells into becoming other cell types. Using a person's own stem cells for cell transplant would avoid a possible immune response. However, because there are so few adult stem cells, they can be difficult to find. Adult stem cells also have a limited life span in the lab, which gives rise to questions about the life span of any transplant done with adult stem cells.

### Stem Cell Controversy

Despite the possibilities of stem cell use, research on them has been controversial. Embryonic stem cells are harvested from human embryos

that are unused for fertility treatment. When embryonic stem cells are harvested, the embryo is destroyed. Many people believe it is unethical to destroy embryos that have the potential to develop into babies. The harvesting of adult stem cells causes no lasting harm to the donor, but the potential for the cells may be limited.

Stem cell transplants might one day be used routinely to treat diseases and disorders such as Alzheimer's disease, diabetes, cancer, and spinal cord injuries. We may even be able to grow complete new organs from stem cells. Funding will be one factor that affects the direction of research. Federal, state, and private funding set the landscape for the future of stem cell research. While the promise of stem cells seems unlimited, major advances will only be achieved through years of intensive research.

## REVIEW

1. How do adult stem cells differ from embryonic stem cells?
2. Why is stem cell research controversial?
3. **Critical Thinking** What do you think are the strongest reasons for and against further research? Based on these two points, would you propose stem cell research be regulated? If so, how?



*In 2004, New York Mets coach Don Baylor (center) enjoyed his first day back with the team after receiving stem cell replacement therapy for cancer.*



# MEIOSIS

*Meiosis is a process of nuclear division that reduces the number of chromosomes in new cells to half the number in the original cell. The halving of the chromosome number counteracts a joining of cells later in the life cycle of the organism.*

## FORMATION OF HAPLOID CELLS

In animals, meiosis produces gametes, which are haploid reproductive cells. Human gametes are sperm cells and egg cells. Sperm and egg cells each contain 23 ( $1n$ ) chromosomes. The fusion of a sperm and an egg results in a zygote that contains 46 ( $2n$ ) chromosomes.

Cells preparing to divide by meiosis undergo the  $G_1$ , S, and  $G_2$  phases of interphase. During interphase, the cell grows to a mature size and copies its DNA. Thus, cells begin meiosis with a duplicate set of chromosomes, just as cells beginning mitosis do. Because cells undergoing meiosis divide twice, diploid ( $2n$ ) cells that divide meiotically result in four haploid ( $1n$ ) cells rather than two diploid ( $2n$ ) cells. The stages of the first cell division are called *meiosis I*, and the stages of the second cell division are called *meiosis II*.

## MEIOSIS I

While reading about each phase of meiosis I, shown in Figure 8-11 on the next page, notice how these phases compare with the corresponding phases that occur in mitosis.

### Prophase I

In prophase I (step ①), DNA coils tightly into chromosomes. As in the prophase of mitosis, spindle fibers appear. Then, the nucleolus and nuclear membrane disassemble. Notice that every chromosome lines up next to its homologue. The pairing of homologous chromosomes, which does not occur in mitosis, is called **synapsis**. Each pair of homologous chromosomes is called a **tetrad**. In each tetrad, chromatids of the homologous chromosomes are aligned lengthwise so that the genes on one chromosome are adjacent to the corresponding genes on the other chromosome.

## OBJECTIVES

- **Compare** the end products of meiosis with those of mitosis.
- **Summarize** the events of meiosis I.
- **Explain** crossing-over and how it contributes to the production of unique individuals.
- **Summarize** the events of meiosis II.
- **Compare** spermatogenesis and oogenesis.
- **Define** *sexual reproduction*.

## VOCABULARY

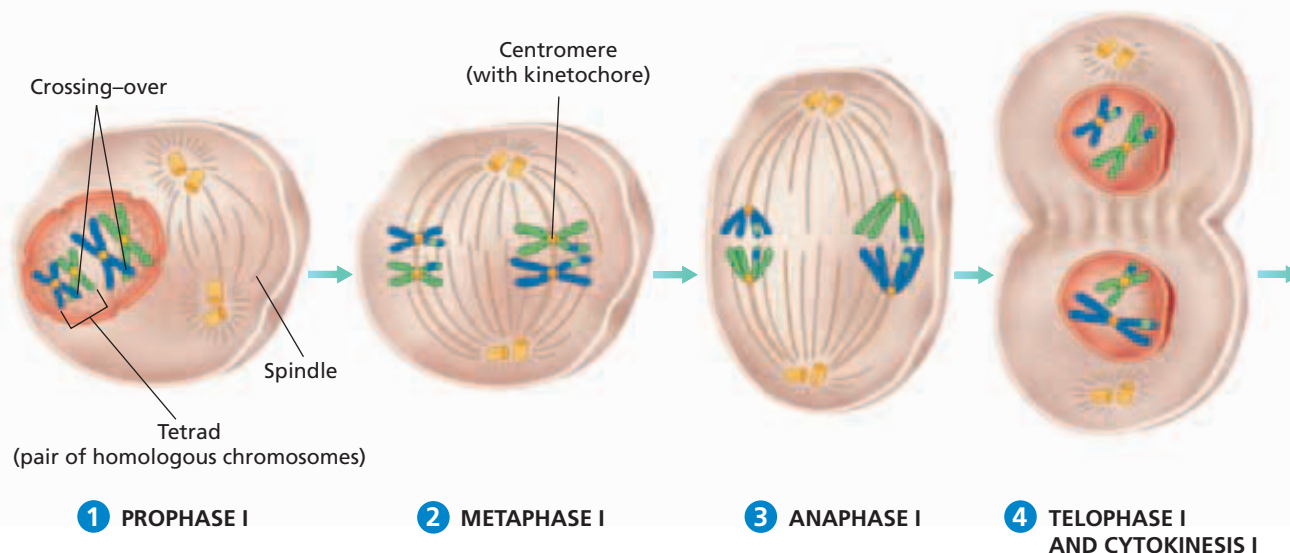
synapsis  
tetrad  
crossing-over  
genetic recombination  
independent assortment  
spermatogenesis  
oogenesis  
polar body  
sexual reproduction

## Word Roots and Origins

### tetrad

from the Greek *tetras*,  
meaning "four"

## MEIOSIS I



**FIGURE 8-11**

Meiosis occurs in diploid reproductive cells. Before meiosis begins, the DNA of the diploid reproductive cells is copied. Meiosis I results in two haploid cells. Meiosis II results in four haploid offspring cells.

During synapsis, the chromatids within a homologous pair twist around one another, as shown in Figure 8-12. Portions of chromatids may break off and attach to adjacent chromatids on the homologous chromosome—a process called **crossing-over**. This process permits the exchange of genetic material between maternal and paternal chromosomes. Thus, **genetic recombination** results, because a new mixture of genetic material is created.

### Metaphase I

During metaphase I (step ②), the tetrads line up randomly along the midline of the dividing cell, as shown in Figure 8-11. The orientation of the pair of chromosomes is random with respect to the poles of the cell. Spindle fibers from one pole attach to the centromere of one homologous chromosome. Spindle fibers from the opposite pole attach to the other homologous chromosome of the pair.

### Anaphase I

During anaphase I (step ③), each homologous chromosome (consisting of two chromatids attached by a centromere) moves to an opposite pole of the dividing cell. The random separation of the homologous chromosomes is called **independent assortment**. Independent assortment results in genetic variation.

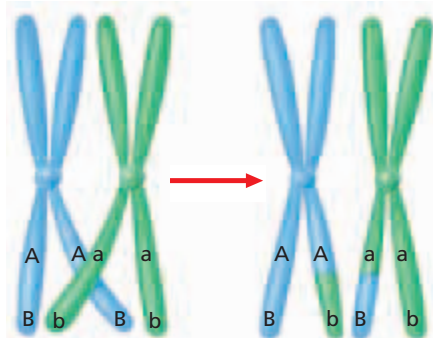
### Telophase I and Cytokinesis I

During telophase I (step ④), the chromosomes reach the opposite ends of the cell, and cytokinesis begins. Notice that the new cells contain a haploid number of chromosomes.

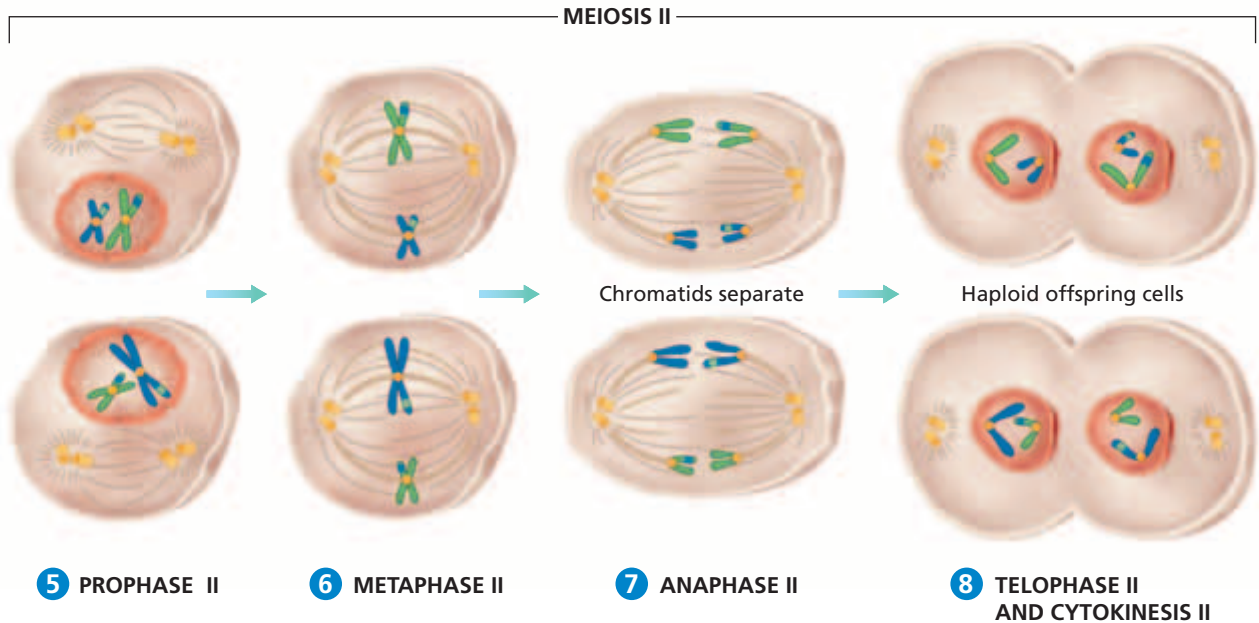
During meiosis I, the original cell produces two new cells, each containing one chromosome from each homologous pair. The new cells contain half the number of chromosomes of the original cell. However, each new cell contains two copies (as chromatids), because the original cell copied its DNA before meiosis I.

**FIGURE 8-12**

Crossing-over occurs when chromosomes that make up a tetrad exchange portions of their chromatids during synapsis. Crossing-over results in an exchange of genes and in new combinations of genes.







## MEIOSIS II

Meiosis II occurs in each cell formed during meiosis I and is not preceded by the copying of DNA. The events of meiosis II are shown above. In some species, meiosis II begins after the nuclear membrane re-forms in the new cells. In other species, meiosis II begins immediately following meiosis I.

### Prophase II, Metaphase II, and Anaphase II

During prophase II (step **5**), spindle fibers form and begin to move the chromosomes toward the midline of the dividing cell. In metaphase II (step **6**), the chromosomes move to the midline of the dividing cell, with each chromatid facing opposite poles of the dividing cell. In anaphase II (step **7**), the chromatids separate and move toward opposite poles of the cell.

### Telophase II and Cytokinesis II

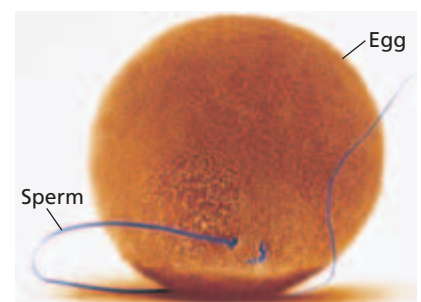
In telophase II (step **8**), a nuclear membrane forms around the chromosomes in each of the four new cells. Cytokinesis II then occurs, resulting in four new cells, each of which contains half of the original cell's number of chromosomes.

## DEVELOPMENT OF GAMETES

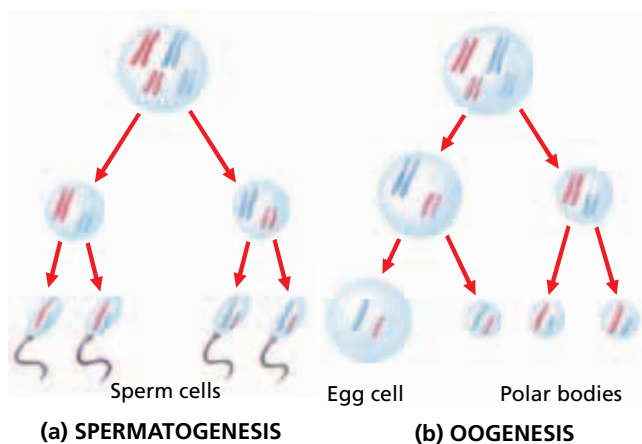
In animals, the only cells that divide by meiosis are those that produce gametes within the reproductive organs. However, organisms vary in timing and structures associated with gamete formation. In humans, meiosis occurs in the testes (males) and the ovaries (females). Figure 8-13 shows a male human gamete joining with a female human gamete.

**FIGURE 8-13**

When the female gamete (the egg) joins with a male gamete (sperm), the genetic instructions from the male and female are combined, and a new individual is formed. (SEM 1,225×)







**FIGURE 8-14**

(a) In the formation of male gametes, the original cell produces four sperm cells by meiosis. (b) In the formation of egg cells, the original cell produces one egg and three polar bodies by meiosis. The egg cell receives most of the original cell's cytoplasm.

In the testes, meiosis is involved in the production of male gametes known as *sperm cells* or *spermatozoa*. In the development of sperm cells, a diploid reproductive cell divides meiotically to form four haploid cells called *spermatids*. Each spermatid develops into a mature sperm cell. The production of sperm cells is called **spermatogenesis**, shown in Figure 8-14a.

**Oogenesis** (OH-oh-JEN-uh-sis) is the production of mature egg cells, or ova. During oogenesis, a diploid reproductive cell divides meiotically to produce one mature egg cell (ovum). During cytokinesis and cytokinesis II of oogenesis, the cytoplasm of the original cell is divided unequally between new cells. As Figure 8-14b shows, one cell, which develops into a mature egg cell, receives most of the cytoplasm of the original cell. As a result, one egg cell is produced by meiosis. The other three products of meiosis, called **polar bodies**, eventually will degenerate.

## SEXUAL REPRODUCTION

**Sexual reproduction** is the production of offspring through meiosis and the union of a sperm and an egg. Offspring produced by sexual reproduction are genetically different from the parents because genes are combined in new ways in meiosis. In fact, except in the case of identical twins, sexually produced offspring contain unique combinations of their parents' genes. The evolutionary advantage of sexual reproduction is that it enables species to adapt rapidly to new conditions. For example, if disease strikes a crop of grain, a few plants may have genetic variations that make them resistant to the disease. Although many individuals may die, these few resistant plants survive and reproduce.



## SECTION 3 REVIEW

1. How do the end products of meiosis differ from the end products of mitosis?
2. How does anaphase I in meiosis differ from anaphase in mitosis?
3. Explain the role of crossing-over in ensuring genetic variation.
4. During which stage of meiosis is the diploid number of chromosomes reduced to the haploid number of chromosomes?
5. Describe the differences between spermatogenesis and oogenesis.
6. Why is meiosis essential to sexual reproduction?

### CRITICAL THINKING

7. **Applying Information** Explain why the chromosomes in the haploid cells that are produced by meiosis I look different from those produced by meiosis II.
8. **Relating Concepts** Explain how it might happen that a human offspring with 47 chromosomes could be produced.
9. **Distinguishing Relevant Information** In humans, the egg is larger than the sperm. Explain how it is possible that a child inherits equally from its mother and father.

# CHAPTER HIGHLIGHTS

## SECTION 1

## Chromosomes

- Chromosomes are tightly coiled DNA molecules. In eukaryotes, proteins called *histones* help maintain the compact structure of chromosomes.
- Chromosomes in prokaryotes are simpler than chromosomes in eukaryotes.
- Each species has a characteristic number of chromosomes in each cell.
- Sex chromosomes are chromosomes that determine the sex of an organism. All of the other chromosomes in an organism are autosomes.
- Cells having two sets of chromosomes are diploid ( $2n$ ). Haploid cells ( $1n$ ) have half the number of chromosomes that are present in diploid cells.

### Vocabulary

chromosome (p. 151)

histone (p. 151)

chromatid (p. 152)

centromere (p. 152)

chromatin (p. 152)

sex chromosome (p. 152)

autosome (p. 152)

homologous

chromosome (p. 152)

karyotype (p. 153)

diploid (p. 153)

haploid (p. 153)

## SECTION 2

## Cell Division

- Cell division is the process by which cells reproduce themselves. Binary fission is the process of cell division in prokaryotes.
- The cell cycle is the repeating set of events in the life of a cell. The cell cycle consists of cell division and interphase.
- Cell division in eukaryotes includes nuclear division (mitosis) and the division of cytoplasm (cytokinesis).
- Interphase consists of growth ( $G_1$ ), DNA replication ( $S$ ), and preparation for cell division ( $G_2$ ).
- Mitosis is divided into prophase, metaphase, anaphase, and telophase. Mitosis results in two offspring cells that are genetically identical to the original cell.
- During cytokinesis in animal cells, a cleavage furrow pinches in and eventually separates the dividing cell into two cells. In plant cells, a cell plate separates the dividing cell into two cells.
- Cell division in eukaryotes is controlled by many proteins. Control occurs at three main checkpoints. Cancer may result if cells do not respond to control mechanisms.

### Vocabulary

binary fission (p. 154)

mitosis (p. 155)

asexual reproduction (p. 155)

meiosis (p. 155)

gamete (p. 155)

interphase (p. 155)

cytokinesis (p. 155)

prophase (p. 156)

spindle fiber (p. 156)

metaphase (p. 157)

anaphase (p. 157)

telophase (p. 157)

cell plate (p. 158)

## SECTION 3

## Meiosis

- Meiosis is a process of nuclear division that reduces the number of chromosomes in new cells to half the number in the original cell. Meiosis produces gametes.
- Cells undergoing meiosis divide twice. Diploid cells that divide meiotically result in four haploid cells rather than two diploid cells as in mitosis.
- Meiosis I includes prophase I, metaphase I, anaphase I, and telophase I. Crossing-over during prophase I results in genetic recombination.
- Meiosis II includes prophase II, metaphase II, anaphase II, and telophase II. Four new haploid cells result.
- Spermatogenesis is the process by which sperm cells are produced. Oogenesis is the process that produces egg cells.
- Sexual reproduction is the formation of offspring through the union of a sperm and an egg. Offspring produced by sexual reproduction are genetically different from the parents.

### Vocabulary

synapsis (p. 161)

tetrad (p. 161)

crossing-over (p. 162)

genetic recombination (p. 162)

independent assortment (p. 162)

spermatogenesis (p. 164)

oogenesis (p. 164)

polar body (p. 164)

sexual reproduction (p. 164)

# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
  - autosome* and *sex chromosome*
  - synapsis* and *crossing-over*
  - haploid* and *diploid*
- Explain the relationship between the terms *chromosome* and *homologous chromosomes*.
- Use the following terms in the same sentence: *mitosis*, *meiosis*, and *cytokinesis*.
- Word Roots and Origins** The word *oogenesis* is derived from the Greek *oion*, which means “egg,” and *geneia*, which means “birth.” Using this information, explain why the term *oogenesis* is a good name for the biological process it describes.

## UNDERSTANDING KEY CONCEPTS

- Describe** the structure of a chromosome.
- Compare** prokaryotic chromosomes with eukaryotic chromosomes.
- State** the number of chromosomes in normal human cells.
- Identify** the type of chromosome that determines the sex of an organism.
- Distinguish** between haploid cells and diploid cells.
- Define** the term *binary fission*. In what type of organism does this type of cell division occur?
- Differentiate** mitosis from cytokinesis.
- Describe** the events that occur in the  $G_0$ ,  $G_1$ , and  $G_2$  phases.
- Summarize** the phases of mitosis.
- Describe** the process of cytokinesis in plant cells.
- List** the three main checkpoints at which the cell cycle is controlled.
- Identify** the type of nuclear division that results in haploid cells.
- Compare** the phases of meiosis I with those of meiosis II.
- Name** the meiotic process that results in genetic recombination.
- Summarize** the major characteristics of spermatogenesis and oogenesis.

- Identify** the relationship between sexual reproduction and genetic variation.

### 21. Unit 4—Cell Reproduction

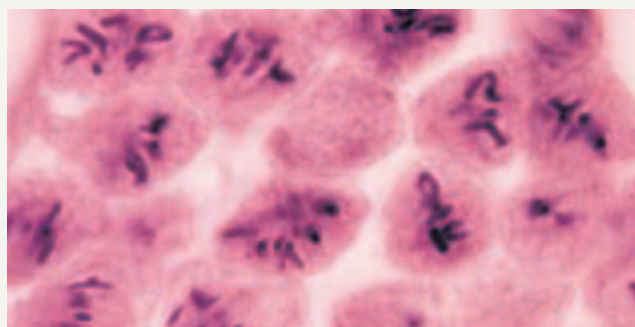


Write a report summarizing how different cancer-fighting drugs kill cancer cells by interrupting the events of the cell cycle.

- CONCEPT MAPPING** Use the following terms to create a concept map that shows the connection between cellular division in prokaryotes and eukaryotes: *binary fission*, *cell cycle*, *cytokinesis*, *eukaryote*, *mitosis*, and *prokaryote*.

## CRITICAL THINKING

- Predicting Results** Can mitosis occur in the absence of cytokinesis? Support your answer. If your answer is yes, describe how the new cell would appear in the  $G_1$  phase.
- Making Calculations** If you consider the mass of DNA in a sperm (a haploid cell) to be 1, what would the relative value be for the DNA mass of a cell in the  $G_2$  phase of the cell cycle?
- Applying Information** Does a cell in metaphase II have the same mass of DNA as a diploid cell in the  $G_1$  phase? Assume that both cells are from the same animal. Explain your answer.
- Analyzing Information** The events of mitosis in plants and animals are very similar, with the exception of the absence of centrioles in plants. How has the absence of centrioles in plant cells influenced scientists' thinking about the function of centrioles in mitosis?
- Interpreting Graphics** The photograph below shows cell division in a grasshopper testis. The offspring cells are gametes. Do you think the photograph shows mitosis or meiosis? Explain your answer.



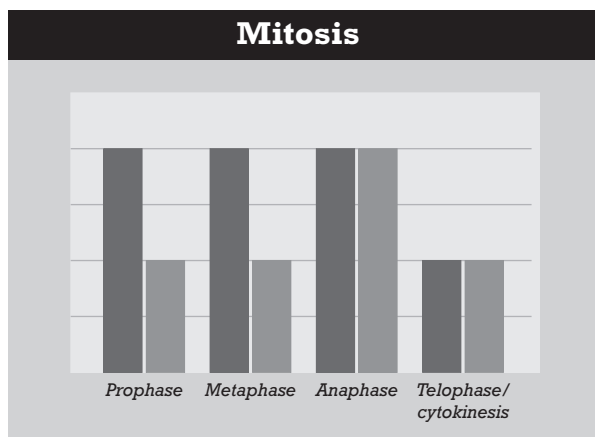


# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- Which of the following statements about prokaryotic chromosomes is true?
  - Prokaryotes have at least two chromosomes.
  - Prokaryotic chromosomes consist of a circular DNA molecule.
  - Prokaryotic chromosomes include histone and nonhistone proteins.
  - Prokaryotic chromosomes are made of DNA wrapped tightly around histone proteins.
- Crossing-over occurs during which process?
  - mitosis
  - meiosis I
  - meiosis II
  - interphase

**INTERPRETING GRAPHICS:** The graph below shows the relative mass of DNA and chromosome number for a cell undergoing mitosis. Use the graph to answer the questions that follow.



■ DNA mass

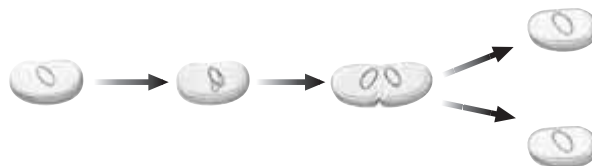
■ Chromosome number

- In which phase of mitosis do chromatids separate and become individual chromosomes?
  - prophase
  - metaphase
  - anaphase
  - telophase/cytokinesis
- What process occurs that leads to the decrease in the cell's DNA mass?
  - prophase
  - metaphase
  - anaphase
  - telophase/cytokinesis

**DIRECTIONS:** Complete the following analogy.

5. prokaryote : binary fission :: eukaryote :
  - mitosis
  - cytokinesis
  - crossing-over
  - genetic recombination

**INTERPRETING GRAPHICS:** The diagram below shows a model of cell division. Use the diagram to answer the question that follows.



6. Which type of cell division is shown in the diagram?
  - mitosis
  - meiosis
  - binary fission
  - sexual reproduction

## SHORT RESPONSE

Human cells are either diploid or haploid. All human cells with 46 chromosomes are diploid.

Would a human cell with *any* 23 chromosomes be haploid? Explain your answer.

## EXTENDED RESPONSE

For a cell to function efficiently, its surface area must exceed that of its volume.

**Part A** Explain how cell division maintains the relationship between surface area and volume.

**Part B** How does a stable ratio between surface area and volume help maintain proper cell functioning?

### Test TIP

To help you learn the stages of mitosis and meiosis, make a note card describing each stage, mix the cards up, and practice reordering the stages.



## Observing Mitosis in Plant Cells

### OBJECTIVES

- Examine the dividing root-tip cells of an onion.
- Identify the phase of mitosis that different cells in an onion root tip are undergoing.
- Determine the relative length of time each phase of mitosis takes in onion root-tip cells.

### PROCESS SKILLS

- observing
- classifying
- collecting
- organizing
- analyzing data
- calculating

### MATERIALS

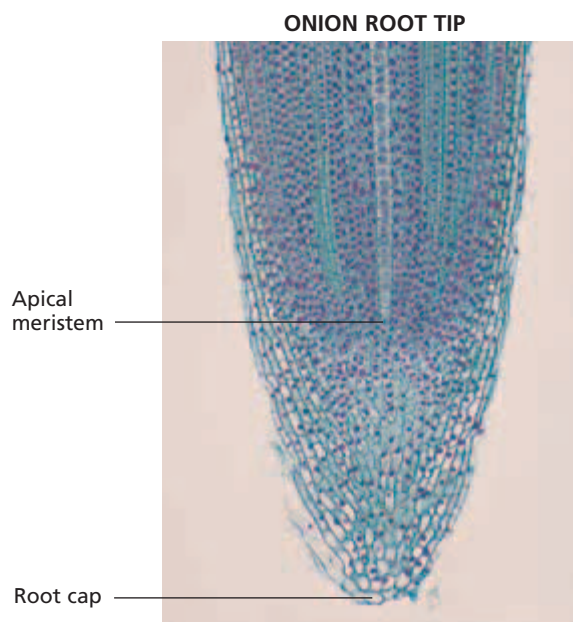
- compound light microscope
- prepared microscope slide of a longitudinal section of *Allium* (onion) root tip

### Background

1. Mitosis is divided into four phases: prophase, metaphase, anaphase, and telophase.
2. Interphase is not considered a part of mitosis.
3. List the visible characteristics of each phase of mitosis.
4. In many plants, there are growth regions called *meristems* where mitosis is ongoing. Meristems are found in the tips of plant roots and shoots.

### PART A Identifying the Phases of Mitosis

1. Look at the photograph below of a longitudinal section of an onion root tip. Find the meristem on the photograph. As you can see, the meristem is located just behind the root cap.



2. **CAUTION** Slides break easily. Use caution when handling them. Using low power on your microscope, bring the meristem region on your slide into focus.

**TABLE A RELATIVE DURATION OF EACH PHASE OF MITOSIS**

| Phase of mitosis | Tally marks | Count | Percentage | Time (in minutes) |
|------------------|-------------|-------|------------|-------------------|
| Prophase         |             |       |            |                   |
| Metaphase        |             |       |            |                   |
| Anaphase         |             |       |            |                   |
| Telophase        |             |       |            |                   |

**TABLE B DATA COLLECTED BY THE ENTIRE CLASS**

| Phase of mitosis | Count | Percentage | Time (in minutes) |
|------------------|-------|------------|-------------------|
| Prophase         |       |            |                   |
| Metaphase        |       |            |                   |
| Anaphase         |       |            |                   |
| Telophase        |       |            |                   |

3. In your lab report, prepare a data table like Table A.
4. Examine the meristem carefully. Choose a sample of about 50 cells. Look for a group of cells that appear to have been actively dividing at the time that the slide was made. The cells will appear in rows, so it should be easy to keep track of them. The dark-staining bodies are the chromosomes.
5. For each of the cells in your sample, identify the stage of mitosis and place a mark in the "Tally marks" column beside the appropriate phase.
9. Record the actual time for each phase in your data table.
10. Make another data table, similar to Table B, shown above. Collect and record the count for each phase of mitosis for the entire class. Fill in the percentage and time information using the data collected by the entire class.
11.  Clean up your materials before leaving the lab.

## **PART B** Calculating the Relative Length of Each Phase

6. When you have classified each cell in your sample, count the tally marks for each phase and fill in the "Count" column. In which phase of mitosis were the greatest number of cells? In which phase were the fewest number of cells?
7. Calculate the percentage of cells found in each phase. Divide the number of cells in a phase by the total number of cells in your sample, and multiply by 100 percent. Enter the figures under "Percentage."
8. The percentage of cells found in each phase can be used as a measure of how long each phase lasts. For example, if 25 percent of the cells are in prophase, then prophase takes 25 percent of the total time it takes for a cell to undergo mitosis. Mitosis in onion cells takes about 80 minutes. Calculate the actual time for each phase using this information and the percentage you have just determined.

Duration of phase (in minutes) =

$$\frac{\text{percentage}}{100} \times 80 \text{ minutes}$$

## Analysis and Conclusions

1. What color are the chromosomes stained?
2. How can you distinguish between early and late anaphase?
3. According to your data table, which phase of mitosis lasts the longest? Why might this phase require more time than other phases of mitosis?
4. According to your data table, which phase takes the least amount of time?
5. How do your results compare with those of the entire class?
6. In this investigation, you assumed that the percentage of the total time that any given phase takes is equal to the percentage of cells in that phase at any moment. Why might this not be true for very small samples of cells?

## Further Inquiry

1. Given the rate of mitosis in a type of animal cells, how could you determine how long each phase of mitosis takes in those cells?
2. Cancerous tissue is composed of cells undergoing uncontrolled, rapid cell division. How could you develop a procedure to identify cancerous tissue by counting the number of cells undergoing mitosis?

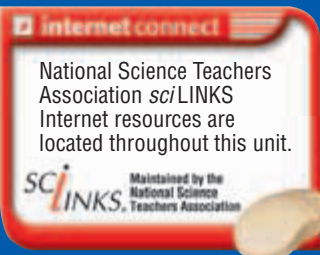
# UNIT 3

## CHAPTERS

- 9 Fundamentals of Genetics
- 10 DNA, RNA, and Protein Synthesis
- 11 Gene Expression
- 12 Inheritance Patterns and Human Genetics
- 13 Gene Technology



References to *Scientific American* project ideas are located throughout this unit.



# GENETICS AND BIOTECHNOLOGY

“... [G]enetics has come to occupy an important position at the center of the sciences of life . . . genetics became allied with biochemistry; it revolutionized bacteriology, played a major role in the emergence of the molecular biology of the fifties, resisted the challenge of ecology, took hold of cancer research and is even now reaching out to revolutionize taxonomy and its old rival embryology.”

From “Mendel, Mendelism and Genetics,” by Robert C. Olby. Online, World Wide Web, July 2, 1997.  
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These tiger cub siblings look very similar to one another because they have inherited characteristics from their parents.

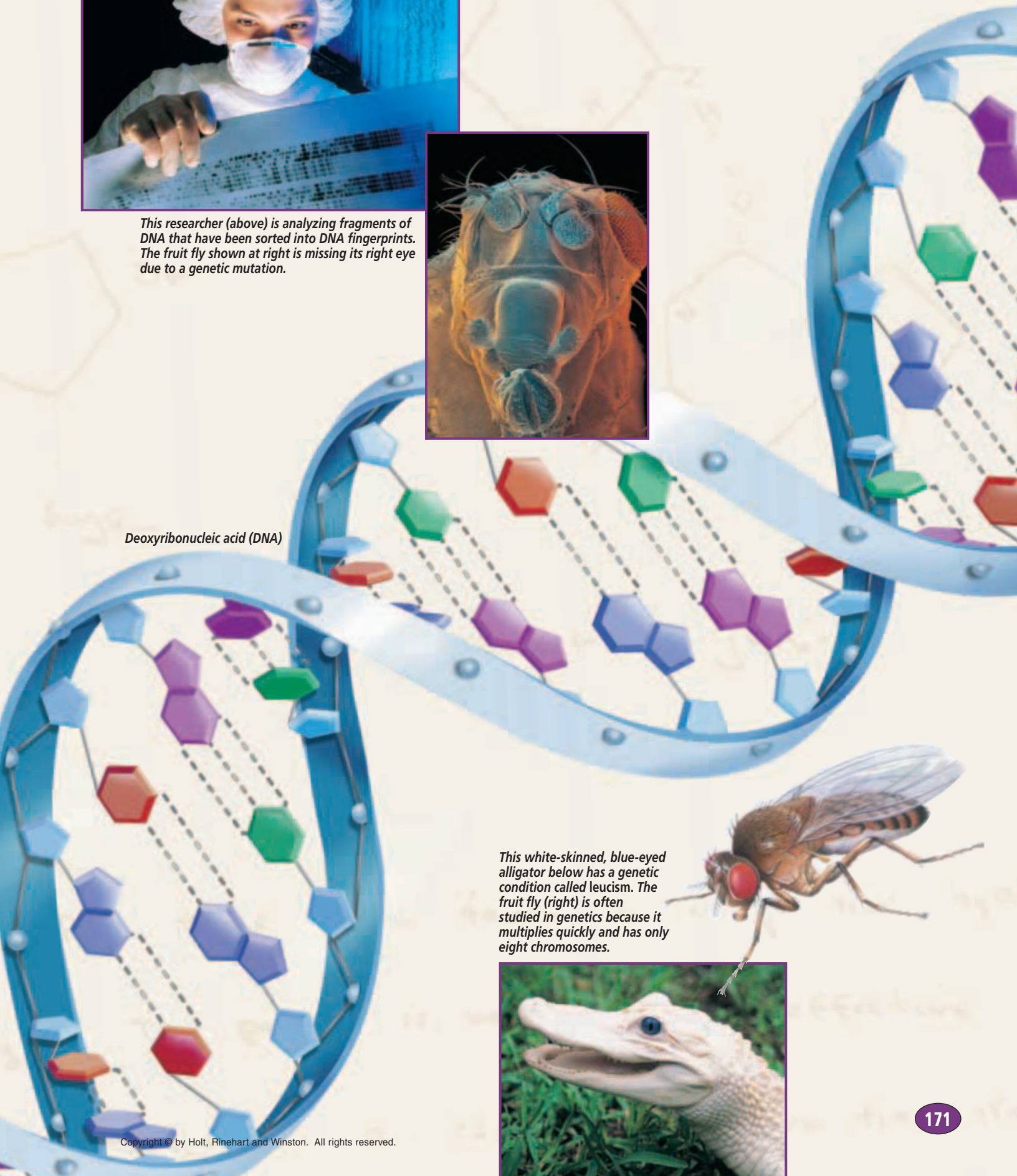




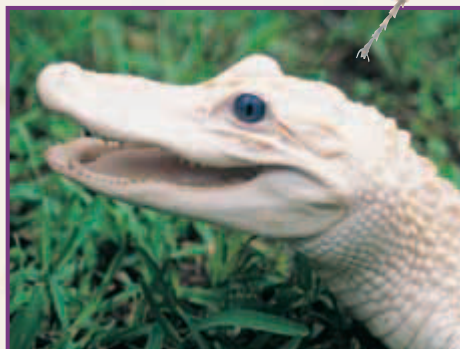
*This researcher (above) is analyzing fragments of DNA that have been sorted into DNA fingerprints. The fruit fly shown at right is missing its right eye due to a genetic mutation.*



Deoxyribonucleic acid (DNA)



*This white-skinned, blue-eyed alligator below has a genetic condition called leucism. The fruit fly (right) is often studied in genetics because it multiplies quickly and has only eight chromosomes.*





# FUNDAMENTALS OF GENETICS



The unique appearance of this white-skinned, blue-eyed alligator is the result of a genetic condition.

SECTION 1 *Mendel's Legacy*

SECTION 2 *Genetic Crosses*



Unit 5—Heredity  
Topics 1–6

# MENDEL'S LEGACY

*Genetics is the field of biology devoted to understanding how characteristics are transmitted from parents to offspring. Genetics was founded with the work of Gregor Johann Mendel. This section describes Mendel's experiments and the principles of genetics that resulted from them.*

## GREGOR MENDEL

In 1843, at the age of 21, Gregor Mendel, shown in Figure 9-1, entered a monastery in Brunn, Austria. His task of tending the garden gave him time to think and to observe the growth of many plants. In 1851, he entered the University of Vienna to study science and mathematics. His mathematics courses included training in the then-new field of statistics. Mendel's knowledge of statistics later proved valuable in his research on **heredity**—the transmission of characteristics from parents to offspring. When Mendel returned to the monastery, he taught in a high school and also kept a garden plot. Although he studied many plants, he is remembered most for his experiments with *Pisum sativum*, a species of garden peas.

### Mendel's Garden Peas

Mendel observed seven characteristics of pea plants. A characteristic is a heritable feature, such as flower color. Each characteristic occurred in two contrasting traits. A **trait** is a genetically determined variant of a characteristic, such as yellow flower color. The pea characteristics that Mendel observed were *plant height* (traits: long and short), *flower position along stem* (traits: axial and terminal), *pod color* (traits: green and yellow), *pod appearance* (traits: inflated and constricted), *seed texture* (traits: round and wrinkled), *seed color* (traits: yellow and green), and *flower color* (traits: purple and white). Mendel used his knowledge of statistics to analyze his observations of these seven characteristics.



**FIGURE 9-1**

Gregor Johann Mendel lived from 1822 to 1884. Mendel's experiments with garden peas led to his discovery of the basic principles of genetics.

## OBJECTIVES

- **Describe** how Mendel was able to control how his pea plants were pollinated.
- **Describe** the steps in Mendel's experiments on true-breeding garden peas.
- **Distinguish** between dominant and recessive traits.
- **State** two laws of heredity that were developed from Mendel's work.
- **Describe** how Mendel's results can be explained by scientific knowledge of genes and chromosomes.

## VOCABULARY

genetics  
heredity  
trait  
pollination  
self-pollination  
cross-pollination  
true-breeding  
P generation  
F<sub>1</sub> generation  
F<sub>2</sub> generation  
dominant  
recessive  
law of segregation  
law of independent assortment  
molecular genetics  
allele

**FIGURE 9-2**

Mendel controlled the breeding of his pea plants and tracked the inheritance of traits by transferring pollen from the anthers of one plant to the stigma of another plant.



Mendel collected seeds from his pea plants and carefully recorded each plant's traits and seeds. The next year, he planted the seeds. He observed that purple-flowering plants grew from most of the seeds obtained from purple-flowering plants but that white-flowering plants grew from some of the seeds of purple-flowering plants. And when experimenting with the characteristic of plant height, he observed that while tall plants grew from most of the seeds obtained from tall plants, short plants grew from some of the seeds obtained from tall plants. Mendel wanted to find an explanation for such variations.

## Mendel's Methods

Mendel was able to observe how traits were passed from one generation to the next by carefully controlling how pea plants were pollinated. **Pollination** occurs when pollen grains produced in the male reproductive parts of a flower, called the *anthers*, are transferred to the female reproductive part of a flower, called the *stigma*.

**Self-pollination** occurs when pollen is transferred from the anthers of a flower to the stigma of either that flower or another flower on the same plant. **Cross-pollination** occurs between flowers of two plants. Pea plants normally reproduce through self-pollination.

Self-pollination can be prevented by removing all of the anthers from the flowers of a plant. Then, cross pollination can be performed by manually transferring pollen from the flower of a second plant to the stigma of the antherless plant, as Figure 9-2 shows. By preventing self pollination and manually cross-pollinating pea plants, Mendel selected parent plants that had specific traits and observed the traits that appeared in the offspring.





# MENDEL'S EXPERIMENTS

Mendel initially studied each characteristic and its contrasting traits individually. He began by growing plants that were true-breeding for each trait. Plants that are **true-breeding**, or pure, for a trait always produce offspring with that trait when they self-pollinate. For example, pea plants that are true-breeding for the trait of yellow pods self-pollinate to produce offspring that have yellow pods. Mendel produced true-breeding plants by self-pollinating the pea plants for several generations, as Figure 9-3 shows. He eventually obtained 14 true-breeding plant types, one for each of the 14 traits observed.

Mendel cross-pollinated pairs of plants that were true-breeding for contrasting traits of a single characteristic. He called the true-breeding parents the **P generation**. He cross-pollinated by transferring pollen from the anthers of one plant to the stigma of another plant. For example, if he wanted to cross a plant that was true-breeding for the trait of yellow pods with one that was true-breeding for the trait of green pods, he first removed the anthers from the plant that produced green pods. Then, he dusted the pollen from a yellow-podded plant onto the stigma of a green-podded plant and allowed the seeds to develop.

When the plants matured, Mendel recorded the number of each type of offspring produced by each cross. He called the offspring of the P generation the first filial generation, or **F<sub>1</sub> generation**. He then allowed the flowers from the F<sub>1</sub> generation to self-pollinate and collected the seeds. Mendel called the plants in this generation the second filial generation, or **F<sub>2</sub> generation**. Following this process, Mendel performed hundreds of crosses and documented the results of each by counting and recording the observed traits of every cross. Table 9-1 summarizes the results of many of Mendel's crosses.

## Word Roots and Origins

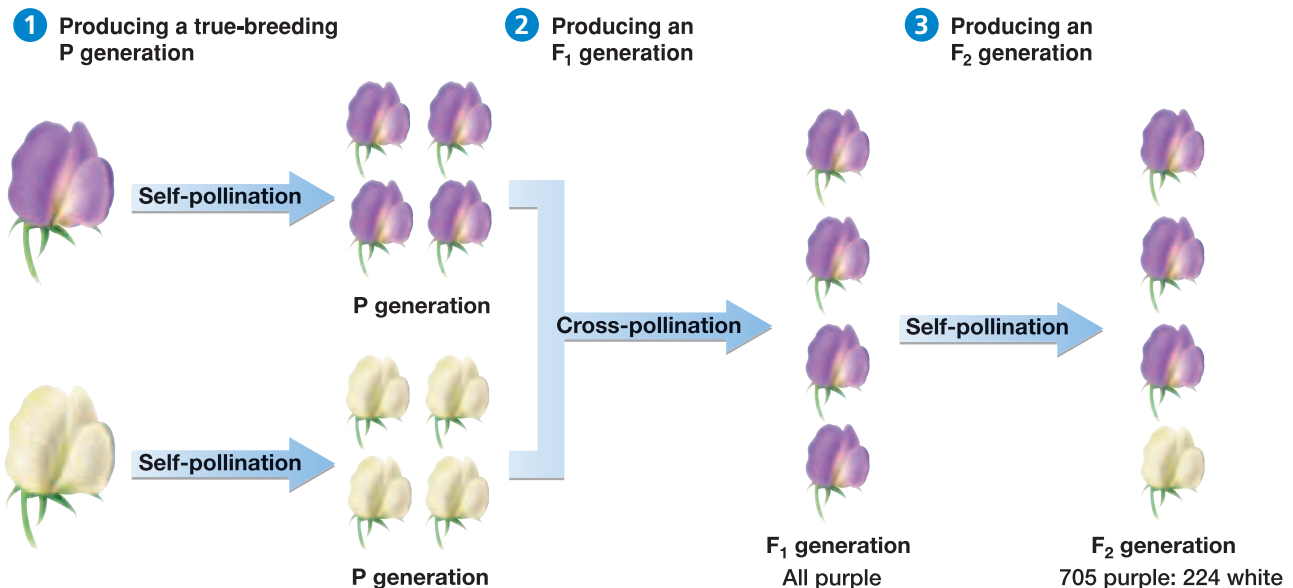
### *filial*

from the Latin *filialis*, meaning  
"of a son or daughter"

**FIGURE 9-3**

Mendel bred plants for several generations that were true breeding for specific traits. These plants he called the P generation. He then observed the passage of these specific traits through successive generations.

### THREE STEPS OF MENDEL'S EXPERIMENTS





**TABLE 9-1** *Mendel's Crosses and Results*

| Characteristic                 | P generation                                                                                                | F <sub>1</sub> generation                                                                         | F <sub>2</sub> generation       | Observed ratio | Predicted ratio |
|--------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------|----------------|-----------------|
| Position of flowers along stem | <br>axial × terminal       | <br>all axial    | 651 axial<br>207 terminal       | 3.14:1         | 3:1             |
| Height of plant                | <br>tall × short           | <br>all tall     | 787 tall<br>277 short           | 2.84:1         | 3:1             |
| Pod appearance                 | <br>inflated × constricted | <br>all inflated | 882 inflated<br>299 constricted | 2.95:1         | 3:1             |
| Pod color                      | <br>green × yellow        | <br>all green   | 428 green<br>152 yellow         | 2.82:1         | 3:1             |
| Seed texture                   | <br>round × wrinkled     | <br>all round  | 5,474 round<br>1,850 wrinkled   | 2.96:1         | 3:1             |
| Seed color                     | <br>yellow × green       | <br>all yellow | 6,022 yellow<br>2,001 green     | 3.01:1         | 3:1             |
| Flower color                   | <br>purple × white       | <br>all purple | 705 purple<br>224 white         | 3.15:1         | 3:1             |

## MENDEL'S RESULTS AND CONCLUSIONS

In one of his experiments, Mendel crossed a plant true-breeding for green pods with one true-breeding for yellow pods, as shown in Figure 9-4. The resulting seeds produced an  $F_1$  generation that had only green-podded plants. No yellow pods developed even though one parent had been true-breeding for yellow pods. Only one of the two traits found in the P generation appeared in the  $F_1$  generation.

Next, Mendel allowed the  $F_1$  plants to self-pollinate and planted the resulting seeds. When the  $F_2$  generation plants grew, he observed that about three-fourths of the  $F_2$  plants had green pods and about one-fourth had yellow pods.

His observations and careful records led Mendel to hypothesize that something within the pea plants controlled the characteristics observed. He called these controls *factors*. Mendel hypothesized that each trait was inherited by means of a separate factor. Because the characteristics studied had two alternative forms, he reasoned that a *pair* of factors must control each trait.

### Recessive and Dominant Traits

Whenever Mendel crossed strains, one of the P traits failed to appear in the  $F_1$  plants. In every case, that trait reappeared in a ratio of about 3:1 in the  $F_2$  generation. This pattern emerged in thousands of crosses and led Mendel to conclude that one factor in a pair may prevent the other from having an effect. Mendel hypothesized that the trait appearing in the  $F_1$  generation was controlled by a **dominant** factor because it masked, or dominated, the factor for the other trait in the pair. He thought that the trait that did not appear in the  $F_1$  generation but reappeared in the  $F_2$  generation was controlled by a **recessive** factor.

Thus, a trait controlled by a recessive factor had no observable effect on an organism's appearance when that trait was paired with a trait controlled by a dominant factor.

### The Law of Segregation

Mendel concluded that the paired factors separate during the formation of reproductive cells. That is, each reproductive cell, or gamete, receives one factor of each pair. When two gametes combine during fertilization, the offspring have two factors for each characteristic. The **law of segregation** states that a pair of factors is segregated, or separated, during the formation of gametes.

### The Law of Independent Assortment

Mendel also crossed plants that differed in two characteristics, such as flower color and seed color. The data from these more-complex crosses showed that traits produced by dominant factors do not necessarily appear together. A green seed pod produced by a dominant factor could appear in a white-flowering pea plant.

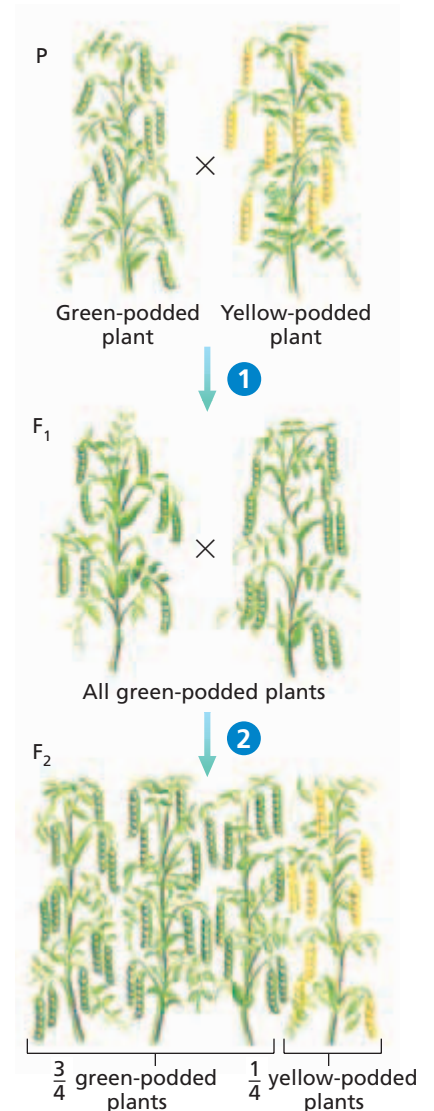


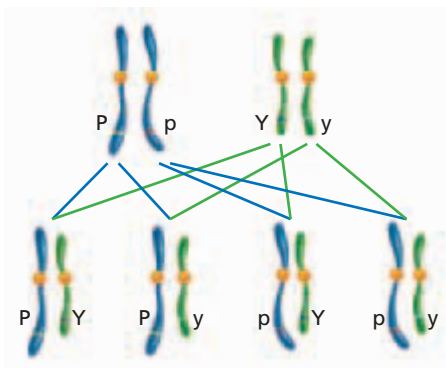
FIGURE 9-4

① True-breeding green-podded pea plants crossed with true-breeding yellow-podded pea plants produce only green-podded plants. ② Yet when the  $F_1$  generation is permitted to self-pollinate, about one-fourth of the plants of the  $F_2$  generation are yellow-podded plants.

### Word Roots and Origins

#### recessive

from the Latin *recessus*, meaning "to recede"



**FIGURE 9-5**

Independent assortment of these two pairs of homologous chromosomes (Pp and Yy) would result in gametes that contain the allele combinations shown above. *P* denotes the dominant purple flower color, and *p* denotes the recessive white flower color. *Y* denotes yellow seed color, and *y* denotes green seed color.

Mendel concluded that the factors for individual characteristics are not connected. Recall that the random separation of homologous chromosomes is called *independent assortment*. The **law of independent assortment** states that factors separate independently of one another during the formation of gametes.

## SUPPORT FOR MENDEL'S CONCLUSIONS

Most of Mendel's findings agree with what biologists now know about molecular genetics. **Molecular genetics** is the study of the structure and function of chromosomes and genes. A chromosome is a threadlike structure made up of DNA. A gene is the segment of DNA on a chromosome that controls a particular hereditary trait. Because chromosomes occur in pairs, genes also occur in pairs. Each of two or more alternative forms of a gene is called an **allele**. Mendel's *factors* are now called *alleles*.

Letters are used to represent alleles. Capital letters refer to dominant alleles, and lowercase letters refer to recessive alleles. For example, the dominant allele for the trait of purple flower color may be represented by *P*, and the recessive allele for the trait of white flower color may be represented by *p*, as is shown in Figure 9-5. Whether a letter is capitalized or lowercased is important. The actual letter selected to represent an allele is typically the first letter of the dominant trait. During meiosis, gametes receive one chromosome from each homologous pair of chromosomes. Thus, when the gametes combine in fertilization, the offspring receives from each parent one allele for a given trait.

Mendel's law of independent assortment is supported by the independent segregation of chromosomes to gametes during meiosis. Therefore, the law of independent assortment is observed only for genes located on separate chromosomes or located far apart on the same chromosome.

## SECTION 1 REVIEW

1. Describe what a *true-breeding* plant is.
2. Outline how Mendel produced plants that had genes for both contrasting traits of a characteristic.
3. Define the terms *dominant* and *recessive*.
4. State in modern terminology the two laws of heredity that resulted from Mendel's work.
5. Differentiate genes from alleles.

### CRITICAL THINKING

6. **Evaluating Results** How did Mendel's  $F_1$  generation plants differ from his  $F_2$  generation plants?
7. **Recognizing Relationships** Many inherited disorders of humans appear in children of parents who do not have the disorder. How can you explain this?
8. **Applying Information** During meiosis, what allows genes located on the same chromosome to separate independently of one another?

# Science in Action

## Do Genes Jump?

The rediscovery of Mendel's work in 1900 is said to mark the birth of the science of genetics. Barbara McClintock, born in 1902, devoted her life to this new science. Ironically, certain assumptions about genetics—which were strongly believed but wrong—prevented early acceptance of McClintock's exciting conclusions.



Barbara McClintock

### **HYPOTHESIS:** Genes Can Move

The prevailing opinion among most geneticists in Barbara McClintock's time was that genes were lined up on chromosomes in an unchanging way, much like beads on a string. However, observations from her experiments with maize (*Zea mays*) told McClintock otherwise.

As a graduate student at Cornell, McClintock developed new staining techniques and discovered that each of the 10 maize chromosomes could be distinguished under the microscope. She noted that some changes in the appearance of the corn kernels and plants were coupled with changes in the shape of one or more of the maize chromosomes. She also noted that kernels that had been exposed to X rays germinated and grew into seemingly normal plants but that the shapes of some of their chromosomes had changed. She proposed that the maize genome had a dynamic chromosome repair system that allowed for growth even after major chromosome damage that X rays had caused. Geneticists at the time assumed that a mutated gene was dead and could not be reactivated. McClintock's findings, however, challenged this belief.

### **METHODS:** Analyze Maize Chromosomes

McClintock wanted to examine more closely the results of growing maize that contained broken chromosomes. So during the winter of 1944–1945, she planted self-pollinated kernels that were the product of many generations of inbreeding and self-fertilization. She hoped to track chromosome repairs by observing changes in chromosome shapes.

### **RESULTS:** Unexpected Changes

When the plants germinated, McClintock was astounded at the results. The leaves had odd patches that lacked the normal green coloration. These patches occurred regularly along the leaf blades. She compared chromosomes of these plants with those of the parent plants under the microscope and concluded that parts of the offspring plants' chromosomes had changed position.

### **CONCLUSION:** Genes Have the Ability to "Jump"

The changes that McClintock saw in these plants and their chromosomes led her to conclude that genes are not stable within the chromosome but can move to a new place on a chromosome or to a new chromosome entirely. McClintock called these movable genes *controlling elements*. They were later called *transposons*.

McClintock observed two kinds of transposons: dissociators and activators. Dissociators could jump to a new chromosomal location when signaled by activators. The dissociators would then cause changes in nearby genes on the chromosome and in the color of the kernels and leaves in the maize. McClintock verified her conclusions through repeated experiments.

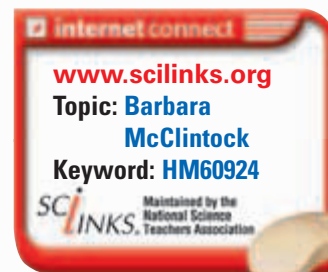
McClintock summarized her findings in 1951 at a Cold Spring Harbor symposium, but her work was not well received. After many years, recognition of the value of her work grew and she was awarded the Nobel Prize in medicine or physiology in 1983.

*The colorful maize plant Zea mays was harvested at Cornell University during the late 1920s and early 1930s to study its genetics. The varied colors of its kernels worked as a multi-colored spreadsheet of genetic data.*



### **REVIEW**

1. What was the purpose of exposing the maize kernels to X rays?
2. What is a transposon?
3. **Critical Thinking** Why was it important that McClintock used self-fertilized kernels in her experiments?





## SECTION 2

### OBJECTIVES

- **Differentiate** between the genotype and the phenotype of an organism.
- **Explain** how probability is used to predict the results of genetic crosses.
- **Use** a Punnett square to predict the results of monohybrid and dihybrid genetic crosses.
- **Explain** how a testcross is used to show the genotype of an individual whose phenotype expresses the dominant trait.
- **Differentiate** a monohybrid cross from a dihybrid cross.

### VOCABULARY

genotype  
phenotype  
homozygous  
heterozygous  
probability  
monohybrid cross  
Punnett square  
genotypic ratio  
phenotypic ratio  
testcross  
complete dominance  
incomplete dominance  
codominance  
dihybrid cross

# GENETIC CROSSES

*Today, geneticists rely on Mendel's work to predict the likely outcome of genetic crosses. In this section, you will learn how to predict the probable genetic makeup and appearance of offspring resulting from specified crosses.*

## GENOTYPE AND PHENOTYPE

An organism's genetic makeup is its **genotype** (JEEN-uh-TIEP). The genotype consists of the alleles that the organism inherits from its parents. For example, the genotype of the white-flowering pea plant in Figure 9-6 consists of two recessive alleles for white flower color, represented as  $pp$ . The genotype of a purple-flowering pea plant may be either  $PP$  or  $Pp$ . Either of these two genotypes would result in a pea plant that has purple flowers because the  $P$  allele is dominant.

An organism's appearance is its **phenotype** (FEE-noh-TIEP). The phenotype of a  $PP$  or a  $Pp$  pea plant is purple flowers, whereas the phenotype of a  $pp$  pea plant is white flowers. As this example shows, a phenotype does not always indicate genotype. In addition to recessive alleles, certain environmental factors can affect phenotype. For example, lack of proper nutrition can cause a genetically tall plant to remain short.



**FIGURE 9-6**

The flower color genotype of the pea plant on the left is  $pp$ . The plant's phenotype is white flowers. The flower color phenotype of the pea plant on the right is purple flowers. The plant's genotype is either  $Pp$  or  $PP$ .

When both alleles of a pair are alike, the organism is said to be **homozygous** (HOH-moh-ZIE-guhs) for that characteristic. An organism may be homozygous dominant or homozygous recessive. For example, a pea plant that is homozygous dominant for flower color has the genotype *PP*. A pea plant that is homozygous recessive for flower color has the genotype *pp*. When the two alleles in the pair are different, the organism is **heterozygous** (HET-uhr-OH-ZIE-guhs) for that characteristic. A pea plant that is heterozygous for flower color has the genotype *Pp*.

## PROBABILITY

**Probability** is the likelihood that a specific event will occur. A probability may be expressed as a decimal, a percentage, or a fraction. Probability is determined by the following equation:

$$\text{Probability} = \frac{\text{number of times an event is expected to happen}}{\text{number of times an event could happen}}$$

For example, in Mendel's experiments, the dominant trait of yellow seed color appeared in the  $F_2$  generation 6,022 times. The recessive trait of green seed color appeared 2,001 times. The total number of individuals was 8,023 (6,022 + 2,001). Using the probability equation above we can determine that the probability that the dominant trait will appear in a similar cross is

$$\frac{6,022}{8,023} = 0.75$$

Expressed as a percentage, the probability is 75 percent. Expressed as a fraction, the probability is  $3/4$ .

The probability that the recessive trait will appear in an  $F_2$  generation is

$$\frac{2,001}{8,023} = 0.25$$

Expressed as a percentage, the probability is 25 percent. Expressed as a fraction, the probability is  $1/4$ . Fractions can also be expressed as ratios. For example, the ratio 1:3 represents the same probability that  $1/4$  does. Probability tells us that there are three chances in four that an offspring of two heterozygous individuals will have the dominant trait and one chance in four that the offspring will have only the recessive trait.

The results predicted by probability are more likely to occur when there are many trials. For example, many coin tosses should yield a result of heads 50 percent of the time and tails 50 percent of the time. However, if you toss a coin only a few times, you might not get this result. But *each* time a coin is tossed, the probability of landing tails is 50 percent. Only after many, many tries would you be *likely* to get the percentage of heads predicted on the basis of probability, that is, 50 percent heads and 50 percent tails.



### Quick Lab

#### Calculating Probability

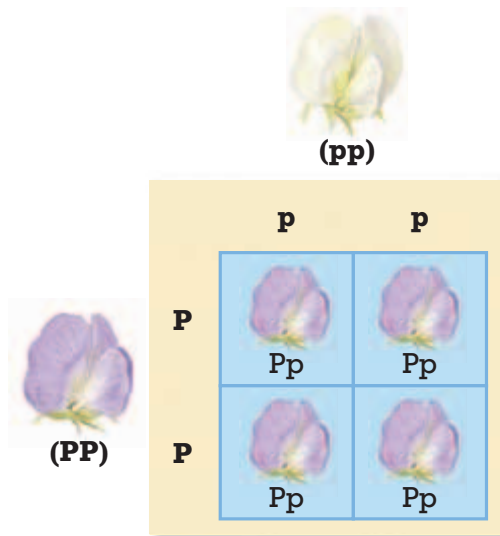
**Materials** paper sack containing 20 jelly beans of three different colors (with an unknown number of each color)

#### Procedure

1. Obtain a sack of 20 jelly beans from your teacher. Do not look into the sack. Do not eat the jelly beans. There are three possible colors of jelly beans that can be pulled from the sack. Pull one jelly bean out, and record the color. Return the jelly bean to the sack, and shake the bag to mix the jelly beans.
2. Repeat step 1 until you have examined 20 jelly beans.
3. Determine the probability of getting a jelly bean of a specific color with a single draw. Do this for each of the three colors of jelly beans. Compare your results with those of the rest of the class.

**Analysis** Does anyone have the same probabilities that you do? Are any probabilities very close to yours? Are any probabilities very different from yours? From these observations, determine how many jelly beans of each color are in your sack.

## PREDICTING RESULTS OF MONOHYBRID CROSSES



**FIGURE 9-7**

A pea plant homozygous for purple flowers that is crossed with a pea plant homozygous for white flowers will produce only purple-flowering offspring. Note that all of the offspring, called *monohybrids*, are heterozygous for flower color.

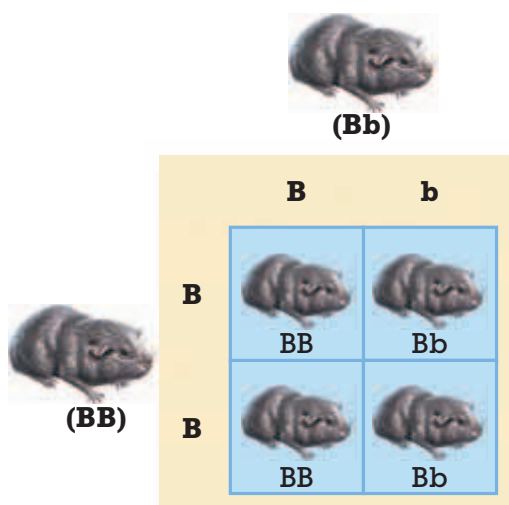
A cross in which only one characteristic is tracked is a **monohybrid** (MAHN-oh-HIE-brid) **cross**. The offspring of a monohybrid cross are called *monohybrids*. A cross between a pea plant that is true-breeding for producing purple flowers and one that is true-breeding for producing white flowers is an example of a monohybrid cross. Biologists use a diagram called a **Punnett** (PUHN-uh) **square**, such as the one shown in Figure 9-7, to aid them in predicting the probable distribution of inherited traits in the offspring. The following examples show how a Punnett square can be used to predict the outcome of different types of crosses.

### Example 1: Homozygous × Homozygous

Figure 9-7 shows a cross between a pea plant homozygous for purple flower color (PP) and a pea plant homozygous for white flower color (pp). The alleles carried in gametes of the homozygous dominant parent are represented by P's on the left side of the Punnett square. The alleles carried in gametes of the homozygous recessive parent are represented by p's across the top of the Punnett square. Each box within the Punnett square is filled in with the letters, or alleles, that are above it and at left of it outside the square. The combinations of alleles in the four boxes indicate the possible genotypes that can result from the cross. The predicted genotype is Pp in every case. Thus, there is a 100 percent probability that the offspring will have the genotype Pp and thus the phenotype purple flower color.

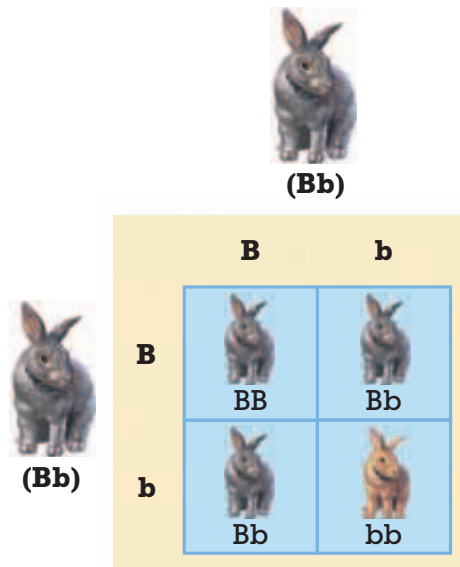
**FIGURE 9-8**

Crossing a guinea pig homozygous for black coat color with one heterozygous for black coat color produces all black-coated monohybrid offspring. Note that half of the monohybrid offspring are predicted to be homozygous for coat color.



### Example 2: Homozygous × Heterozygous

Figure 9-8 shows a cross between a guinea pig that is homozygous dominant for the trait of black coat color (BB) and a guinea pig that is heterozygous for this trait (Bb). The letter b stands for the recessive allele. Genotype bb results in a brown coat. Notice that two possible genotypes can result from this cross: BB or Bb. The probability of an offspring having the genotype BB is 2/4, or 50 percent. The probability of an offspring having the genotype Bb is also 2/4, or 50 percent. You could expect about 50 percent of the offspring resulting from this cross to be homozygous dominant for the black coat and about 50 percent to be heterozygous dominant for a black coat. The probable phenotype is black coat color in every case; thus, 4/4, or 100 percent, of the offspring are expected to have a black coat. What would happen if the homozygous guinea pig were homozygous recessive for coat color? The homozygote would have the genotype bb. Crossing a bb guinea pig with a Bb guinea pig is likely to produce about 50 percent Bb offspring and about 50 percent bb offspring.



**FIGURE 9-9**

The probable results of crossing two rabbits that are heterozygous for black coat color are 50 percent heterozygous black individuals, 25 percent homozygous black individuals, and 25 percent homozygous brown individuals.

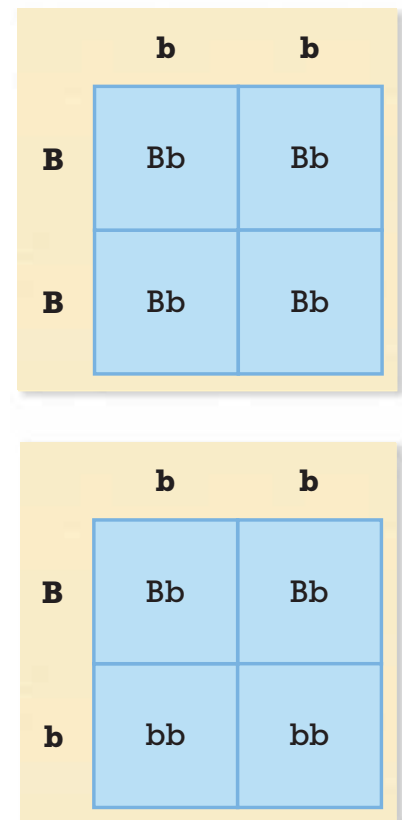
### Example 3: Heterozygous × Heterozygous

In rabbits, the allele for black coat color ( $B$ ) is dominant over the allele for brown coat color ( $b$ ). The Punnett square in Figure 9-9 shows the predicted results of crossing two rabbits that are heterozygous ( $Bb$ ) for coat color. As you can see,  $1/4$  (25 percent) of the offspring are predicted to have the genotype  $BB$ ,  $1/2$  (50 percent) are predicted to have the genotype  $Bb$ , and  $1/4$  (25 percent) are predicted to have the genotype  $bb$ . Thus,  $3/4$  (75 percent) of the offspring resulting from this cross are predicted to have a black coat. One-fourth (25 percent) of the offspring are predicted to have a brown coat.

The ratio of the genotypes that appear in offspring is called the **genotypic ratio**. The probable genotypic ratio of the monohybrid cross represented in Figure 9-9 is  $1 BB : 2 Bb : 1 bb$ . The ratio of the offspring's phenotypes is called the **phenotypic ratio**. The probable phenotypic ratio of the cross represented in Figure 9-9 is 3 black : 1 brown.

### Example 4: Testcross

Recall that in guinea pigs, both  $BB$  and  $Bb$  result in a black coat. How might you determine whether a black guinea pig is homozygous ( $BB$ ) or heterozygous ( $Bb$ )? You could perform a **testcross**, in which an individual of unknown genotype is crossed with a homozygous recessive individual. A testcross can determine the genotype of any individual whose phenotype expresses the dominant trait. You can see from Figure 9-10 that if the black guinea pig of unknown genotype is homozygous black, all offspring will be black. If the individual with the unknown genotype is heterozygous black, about half of the offspring will be black. In reality, if the cross produced any brown offspring, the genotype of the black-coated parent is likely to be heterozygous.



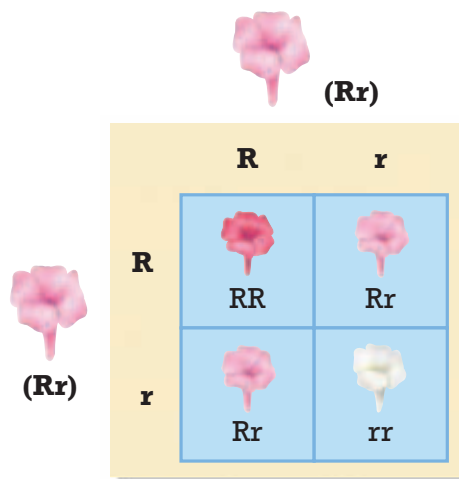
**FIGURE 9-10**

If a black guinea pig is crossed with a brown guinea pig and even one of the offspring is brown, the black guinea pig is heterozygous for coat color.



**FIGURE 9-11**

When red-flowering four o'clocks are crossed with white-flowering four o'clocks, all of the  $F_1$  offspring produce pink flowers, an intermediate between the two parental phenotypes. When the  $F_1$  generation is interbred, red-flowering, pink-flowering, and white-flowering plants are produced because the trait for red flower color has incomplete dominance over the trait for white flower color.



### Example 5: Incomplete Dominance

Recall that in Mendel's pea-plant crosses, one allele was completely dominant over another, a relationship called **complete dominance**. In complete dominance, heterozygous plants and homozygous dominant plants are indistinguishable in phenotype. For example, both pea plants  $PP$  and  $Pp$  for flower color have purple flowers.

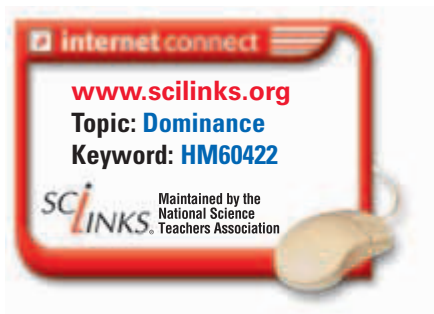
Sometimes, however, the  $F_1$  offspring will have a phenotype in between that of the parents, a relationship called **incomplete dominance**. Incomplete dominance occurs when the phenotype of a heterozygote is intermediate between the phenotypes determined by the dominant and recessive traits. In four o'clocks, for example, both the allele for red flowers ( $R$ ) and the allele for white flowers ( $r$ ) influence the phenotype. Neither allele is completely dominant over the other allele. When four o'clocks self-pollinate, red-flowering plants produce only red-flowering offspring and white-flowering plants produce only white-flowering offspring. However, when red four o'clocks are crossed with white four o'clocks, all of the  $F_1$  offspring have *pink* flowers. One hundred percent of the offspring of this cross have the  $Rr$  genotype, which results in a pink phenotype.

What would be the result of crossing two pink-flowering ( $Rr$ ) four o'clocks? As the Punnett square in Figure 9-11 shows, the probable genotypic ratio is  $1 RR : 2 Rr : 1 rr$ . Given that neither the allele for red flowers ( $R$ ) nor the allele for white flowers ( $r$ ) is completely dominant, the probable phenotypic ratio is 1 red : 2 pink : 1 white.

### Example 6: Codominance

**Codominance** occurs when both alleles for a gene are expressed in a heterozygous offspring. In codominance, neither allele is dominant or recessive, nor do the alleles blend in the phenotype.

For example, the three human MN blood types, M, N, and MN, are determined by two alleles,  $L^M$ , and  $L^N$ . The letters M and N refer to two molecules on the surface of the red blood cell. The genotype of a person with blood type MN is  $L^M L^N$ , and neither allele is dominant over the other. Type MN blood cells carry both M- and N-types of molecules on their surface.



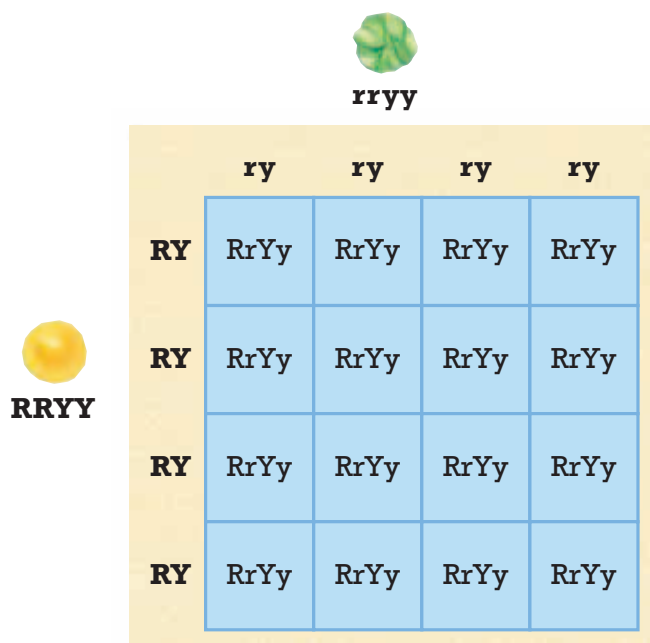
# PREDICTING RESULTS OF DIHYBRID CROSSES

A **dihybrid** (die-HIE-brid) **cross** is a cross in which two characteristics are tracked. The offspring of a dihybrid cross are called *dihybrids*. Predicting the results of a dihybrid cross is more complicated than predicting the results of a monohybrid cross because more combinations of alleles are possible. For example, to predict the results of a cross in which both seed texture and seed color are tracked, you have to consider how four alleles from each parent can combine.

## Homozygous × Homozygous

Suppose that you want to predict the results of a cross between a pea plant that is homozygous for round, yellow seeds and one that is homozygous for wrinkled, green seeds. In pea plants, the allele for round seeds (*R*) is dominant over the allele for wrinkled seeds (*r*), and the allele for yellow seeds (*Y*) is dominant over the allele for green seeds (*y*).

As Figure 9-12 shows, the Punnett square used to predict the results of a cross between a parent of the genotype *RRYY* and a parent of the genotype *rryy* will contain 16 boxes. Alleles are carried by the male and female gametes (pollen and ovule). The independently sorted alleles from one parent—*RY*, *RY*, *RY*, and *RY*—are listed along the left side of the Punnett square. The independently sorted alleles from the other parent—*ry*, *ry*, *ry*, and *ry*—are listed along the top of the Punnett square. Each box is filled with the letters that are above it and to the left of it outside the square. Notice that the genotype of all of the offspring of this cross will be heterozygous for both traits (*RrYy*); therefore, all of the offspring will have round, yellow seed phenotypes.



|                                                                                                    |           |                                                                                                    |             |             |             |
|----------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|-------------|-------------|-------------|
|                                                                                                    |           | <br><b>rryy</b> |             |             |             |
|                                                                                                    |           | <b>ry</b>                                                                                          | <b>ry</b>   | <b>ry</b>   | <b>ry</b>   |
| <br><b>RRYY</b> | <b>RY</b> | <b>RrYy</b>                                                                                        | <b>RrYy</b> | <b>RrYy</b> | <b>RrYy</b> |
|                                                                                                    | <b>RY</b> | <b>RrYy</b>                                                                                        | <b>RrYy</b> | <b>RrYy</b> | <b>RrYy</b> |
|                                                                                                    | <b>RY</b> | <b>RrYy</b>                                                                                        | <b>RrYy</b> | <b>RrYy</b> | <b>RrYy</b> |
|                                                                                                    | <b>RY</b> | <b>RrYy</b>                                                                                        | <b>RrYy</b> | <b>RrYy</b> | <b>RrYy</b> |



## Quick Lab

### Determining Genotypes

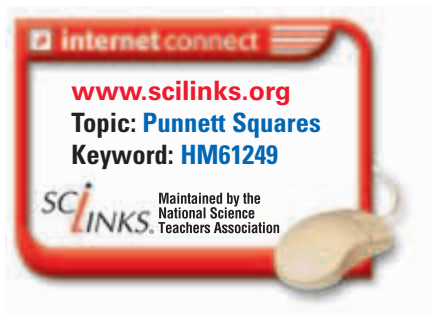
**Materials** pencil and paper

**Procedure** The ability to roll the tongue upward from the sides is a dominant, inherited trait. In one family, both parents and three children are tongue rollers, while one child is not. Determine the genotype and phenotype of each parent.

**Analysis** Are the parents homozygous or heterozygous? Are the children homozygous or heterozygous?

**FIGURE 9-12**

This Punnett square shows a dihybrid cross between a pea plant that is homozygous recessive for wrinkled, green seeds (*rryy*) and a pea plant that is homozygous dominant for round, yellow seeds (*RRYY*).



## Heterozygous × Heterozygous

The same procedure is used to determine the results of crossing two pea plants heterozygous for round, yellow seeds. As Figure 9-13 shows, the offspring of this dihybrid cross are likely to have nine different genotypes. These nine genotypes will result in pea plants that have the following four phenotypes:

- 9/16 that have round, yellow seeds (genotypes  $RRYY$ ,  $RRYy$ ,  $RrYY$ , and  $RrYy$ )
- 3/16 that have round, green seeds (genotypes  $RRyy$  and  $Rryy$ )
- 3/16 that have wrinkled, yellow seeds (genotypes  $rrYY$  and  $rrYy$ )
- 1/16 that have wrinkled, green seeds (genotype  $rryy$ )

**FIGURE 9-13**

A dihybrid cross of two individuals heterozygous for both traits of interest is likely to result in nine different genotypes and four different phenotypes.



**RrYy**



**RrYy**

|           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|
|           | <b>RY</b> | <b>Ry</b> | <b>rY</b> | <b>ry</b> |
| <b>RY</b> | RRYY      | RRYy      | RrYY      | RrYy      |
| <b>Ry</b> | RRYy      | RRyy      | RrYy      | Rryy      |
| <b>rY</b> | RrYY      | RrYy      | rrYY      | rrYy      |
| <b>ry</b> | RrYy      | Rryy      | rrYy      | rryy      |

## SECTION 2 REVIEW

1. Explain why a phenotype might not always indicate genotype.
2. Identify the equation used to determine probability.
3. Explain how you might go about determining the genotype of a purple-flowering plant.
4. Illustrate in the form of a Punnett square the results of crossing a pink-flowering four o'clock with a white-flowering four o'clock.
5. Explain the difference between a monohybrid cross and a dihybrid cross and give an example of each.

### CRITICAL THINKING

6. **Analyzing Concepts** The offspring of two short-tailed cats have a 25 percent chance of having no tail, a 25 percent chance of having a long tail, and a 50 percent chance of having a short tail. Using this information, what can you hypothesize about the genotypes of the parents and the way in which tail length is inherited?
7. **Inferring Relationships** If you crossed two purple-flowering pea plants and all of the  $F_1$  offspring were purple-flowering, what could you say about the genotypes of the parents? If some of the  $F_1$  offspring were white-flowering, what could you say about the genotypes of the parents?

# CHAPTER HIGHLIGHTS

## SECTION 1

## Mendel's Legacy

- The study of how characteristics are transmitted from parents to offspring is called *genetics*.
- Mendel observed seven characteristics of pea plants. Each characteristic occurred in two contrasting traits.
- Self-pollination, in which pollen is transferred from the anthers of a flower to either the stigma of the same flower or the stigma of another flower on the same plant, normally occurs in pea plants. Cross-pollination occurs when pollen is transferred between flowers of two different plants.
- Mendel concluded that inherited characteristics are controlled by factors that occur in pairs. In his experiments on pea plants, one factor in a pair masked the other. The trait that masked the other was called the *dominant trait*. The trait that was masked was called the *recessive trait*.
- The law of segregation states that a pair of factors is segregated, or separated, during the formation of gametes. Two factors for a characteristic are then combined when fertilization occurs and a new offspring is produced.
- The law of independent assortment states that factors for individual characteristics are distributed to gametes independently. The law of independent assortment is observed only for genes that are located on separate chromosomes or are far apart on the same chromosome.
- We now know that the factors that Mendel studied are alleles, or alternative forms of a gene. Each of two or more alternative forms of a gene is called an *allele*. One allele for each trait is passed from each parent to the offspring.

### Vocabulary

genetics (p. 173)  
heredity (p. 173)  
trait (p. 173)  
pollination (p. 174)  
self-pollination (p. 174)

cross-pollination (p. 174)  
true-breeding (p. 175)  
P generation (p. 175)  
F<sub>1</sub> generation (p. 175)  
F<sub>2</sub> generation (p. 175)

dominant (p. 177)  
recessive (p. 177)  
law of segregation (p. 177)  
law of independent assortment (p. 178)

molecular genetics (p. 178)  
allele (p. 178)

## SECTION 2

## Genetic Crosses

- The genotype is the genetic makeup of an organism. The phenotype is the appearance of an organism.
- Probability is the likelihood that a specific event will occur. A probability may be expressed as a decimal, a percentage, or a fraction.
- A Punnett square can be used to predict the outcome of genetic crosses.
- A cross in which one characteristic is tracked is a monohybrid cross. The offspring of a monohybrid cross are called *monohybrids*.
- A testcross, in which an individual of unknown genotype is crossed with a homozygous recessive individual, can be used to determine the genotype of an individual whose phenotype expresses the dominant trait.
- Complete dominance occurs when heterozygous individuals and dominant homozygous individuals are indistinguishable in phenotype.
- Incomplete dominance occurs when two or more alleles influence the phenotype and results in a phenotype intermediate between the dominant trait and the recessive trait.
- Codominance occurs when both alleles for a gene are expressed in a heterozygous offspring. Neither allele is dominant or recessive, nor do the alleles blend in the phenotype as they do in incomplete dominance.
- A cross in which two characteristics are tracked is a dihybrid cross. The offspring of a dihybrid cross are called *dihybrids*.

### Vocabulary

genotype (p. 180)  
phenotype (p. 180)  
homozygous (p. 181)  
heterozygous (p. 181)

probability (p. 181)  
monohybrid cross (p. 182)  
Punnett square (p. 182)  
genotypic ratio (p. 183)

phenotypic ratio (p. 183)  
testcross (p. 183)  
complete dominance (p. 184)

incomplete dominance (p. 184)  
codominance (p. 184)  
dihybrid cross (p. 185)



# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
  - homozygous* and *heterozygous*
  - law of segregation* and *law of independent assortment*
  - genetics* and *heredity*
- Use the following terms in the same sentence: *pollination*, *self-pollination*, and *cross-pollination*.
- For each pair of terms, explain the relationship between the terms.
  - genotype* and *phenotype*
  - monohybrid cross* and *dihybrid cross*
  - allele* and *trait*
- Word Roots and Origins** The word *dominant* comes from the Latin *dominari*, which means “to rule.” Using this information, explain why the term *dominant* is a good name for the genetic phenomenon that the term describes.

## UNDERSTANDING KEY CONCEPTS

- Relate** why Mendel began his experiments by allowing pea plants to self-pollinate for several generations.
- Explain** the difference between dominant and recessive traits.
- Describe** the differences between the P generation, the F<sub>1</sub> generation, and the F<sub>2</sub> generation.
- Relate** why it is not necessary, when the dominant and recessive traits are known, to use the term *homozygous* when referring to the genotype of an individual that has a recessive phenotype.
- State** the difference between a monohybrid cross and a dihybrid cross.
- Propose** how crossing-over during meiosis might affect the segregation of genes that are on the same chromosome.
- Relate** the events of meiosis to the law of segregation.
- Summarize** how a gardener who has a pea plant that produces round seeds can determine whether the plant is homozygous or heterozygous for the allele that determines seed texture. (In pea plants, round seed texture is dominant over wrinkled seed texture.)
- Relate** probability to the study of genetics.
- Predict** the results of a cross between a rabbit homozygous dominant for black coat color (*BB*) and a rabbit homozygous recessive for brown coat color (*bb*).
- 
**CONCEPT MAPPING** Use the following terms to create a concept map that illustrates information related to Mendel's experiments: *pea plants*, *heredity*, *self-pollination*, *cross-pollination*, *F<sub>1</sub> generation*, *F<sub>2</sub> generation*, *trait*, and *true-breeding*.

## CRITICAL THINKING

- Interpreting Graphics** Use the Punnett square below to answer the questions that follow.
  - Does the Punnett square demonstrate a monohybrid cross or a dihybrid cross?
  - List the genotypes of the parents.
  - Give the genotypic ratio predicted by the Punnett square for the cross.

|    | QT   | Qt   | qT   | qt   |
|----|------|------|------|------|
| QT | QQTT | QQTt | QqTT | QqTt |
| Qt | QQTt | QQtt | QqTt | Qqtt |
| qT | QqTT | QqTt | qqTT | qqTt |
| qt | QqTt | Qqtt | qqTt | qqtt |

- Predicting Results** In rabbits, the allele for black coat color (*B*) is dominant over the allele for brown coat color (*b*). Predict the results of a cross between a rabbit homozygous for black coat color (*BB*) and a rabbit homozygous for brown coat color (*bb*).
- 
**Applying Information** Write a report summarizing how an understanding of heredity allows animal breeders to develop animals that have desirable traits. Find out what kinds of animals are bred for special purposes.

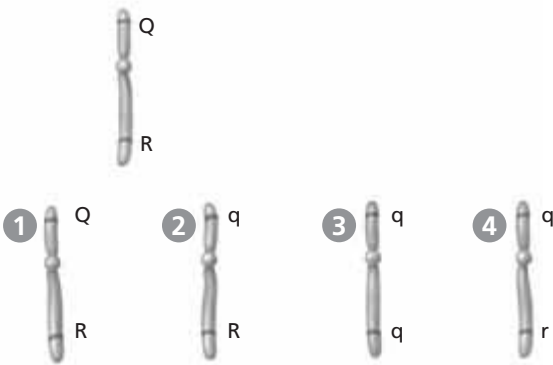


# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- What is a procedure in which an individual of unknown genotype is crossed with a homozygous recessive individual to determine the genotype of the unknown individual called?
  - a monohybrid cross
  - a dihybrid cross
  - a hybrid cross
  - a testcross
- In a monohybrid cross of two heterozygous parents ( $Pp$ ), what would the expected genotypes of the offspring be?
  - $1 PP : 2 Pp : 1 pp$
  - $1 pp : 3 PP$
  - $3 Pp : 1 pp$
  - all  $Pp$
- Which of the following is an example of a genotype of a heterozygous individual?
  - $p$
  - $YY$
  - $Zz$
  - $rr$

**INTERPRETING GRAPHICS:** Use the diagrams of chromosomes below to answer the question that follows. The single chromosome below has two genes, both of which carry a dominant allele  $Q$  and  $R$ .



- Homologous chromosomes are chromosomes that carry genes for the same characteristics, such as eye color or hair color. Which of the chromosomes in the bottom row could not be the homologous chromosome for the single chromosome in the top row?
  - 1
  - 2
  - 3
  - 4

**DIRECTIONS:** Complete the analogy below:

- $Rr$  : genotype :: red :
  - $F_1$  generation
  - heterozygote
  - phenotype
  - dominant

**INTERPRETING GRAPHICS:** Use the diagram of a Punnett square below to answer the question that follows.

|     |     |     |
|-----|-----|-----|
|     | $r$ | $r$ |
| $R$ | 1   | 2   |
| $r$ | 3   | 4   |

The Punnett square above shows the expected results of a cross between two pea plants.  $R$  and  $r$  represent the alleles for round seed and wrinkled seed traits, respectively.

- What would the seed texture phenotype of the plant in box 4 be?
  - round
  - $Rr$
  - wrinkled
  - $rr$

## SHORT RESPONSE

Mendel was able to observe certain traits as they were passed on by carefully controlling how the pea plants were pollinated. Explain why Mendel began his experiments by allowing pea plants to self-pollinate for several generations.

## EXTENDED RESPONSE

A cross between two pea plants that have axial flowers and inflated pods gives the following offspring: 20 that have axial flowers and inflated pods, 7 with axial flowers and constricted pods, and 5 that have terminal flowers and inflated pods.

**Part A** Identify the most probable genotype of the two parents.

**Part B** Use a Punnett square to explain the results.

## Test TIP

Before answering word problems that are genetics questions, write the problem down by using letters to symbolize genotypes.

# Modeling Monohybrid Crosses

## OBJECTIVES

- Predict the genotypic and phenotypic ratios of offspring resulting from the random pairing of gametes.
- Calculate the genotypic ratio and phenotypic ratio among the offspring of a monohybrid cross.

## PROCESS SKILLS

- predicting
- organizing
- analyzing data
- calculating

## MATERIALS

- lentils
- green peas
- 2 Petri dishes

## Background

1. How many traits are involved in a monohybrid cross? How many alleles are involved?
2. What prevents the expression of a recessive allele?
3. When gametes form, what happens to the alleles for each trait?

## PART A Simulating a Monohybrid Cross

1. You will model the random pairing of alleles by choosing lentils and peas from Petri dishes. These dried seeds will represent the alleles for seed color. A green pea will represent *G*, the dominant allele for green seeds, and a lentil will represent *g*, the recessive allele for yellow seeds.
2. The seeds in each Petri dish will represent the alleles from a single parent. Label one Petri dish "female gametes" and the other Petri dish "male gametes." Place one green pea and one lentil in the Petri dish labeled "female gametes," and place one green pea and one lentil in the Petri dish labeled "male gametes."

3. Each parent contributes one allele to each offspring. Model a cross between these two parents by choosing a random pairing of the dried seeds from the two containers. Do so by simultaneously picking one seed from each container *without looking*. Place the pair of seeds together on the lab table. The pair of seeds represents the genotype of one offspring.
4. Record the genotype of the first offspring in your lab report in a table like Table A, shown below.

**TABLE A GAMETE PAIRINGS**

| Trial | Offspring genotype | Offspring phenotype |
|-------|--------------------|---------------------|
| 1     |                    |                     |
| 2     |                    |                     |
| 3     |                    |                     |
| 4     |                    |                     |
| 5     |                    |                     |
| 6     |                    |                     |
| 7     |                    |                     |
| 8     |                    |                     |
| 9     |                    |                     |
| 10    |                    |                     |

5. Return the seeds to their original dishes, and repeat step 3 nine more times. Record the genotype of each offspring in your data table.
6. Based on each offspring's genotype, determine and record each offspring's phenotype. Assume that the allele for green seeds, *G*, is completely dominant over the allele for yellow seeds, *g*.

## PART B Calculating Genotypic and Phenotypic Ratios

- In your lab report, prepare a data table similar to Table B, shown below.
- Determine the genotypic and phenotypic ratios among the offspring. First, count and record the number of homozygous dominant, heterozygous, and homozygous recessive individuals recorded in Table A. Then, record the number of offspring that produce green seeds and the number that produce yellow seeds under "Phenotypes" in your data table.
- Calculate the genotypic ratio for each genotype by using the following equation:  

$$\text{Genotypic ratio} = \frac{\text{number of offspring with a given genotype}}{\text{total number of offspring}}$$
- Calculate the phenotypic ratio for each phenotype by using the following equation:  

$$\text{Phenotypic ratio} = \frac{\text{number of offspring with a given phenotype}}{\text{total number of offspring}}$$
- Now, pool the data for the whole class, and record the data in your lab report in a table like Table C.
- Compare your class's sample with your small sample of 10. Calculate and record in your data table the genotypic and phenotypic ratios for the class data.
- Construct a Punnett square showing the parents and their offspring in your lab report.
-  Clean up your materials before leaving the lab.

**TABLE B OFFSPRING RATIOS**

| Genotypes                          | Total | Genotypic ratio       |
|------------------------------------|-------|-----------------------|
| Homozygous dominant ( <i>GG</i> )  |       | _____ : _____ : _____ |
| Heterozygous ( <i>Gg</i> )         |       |                       |
| Homozygous recessive ( <i>gg</i> ) |       |                       |
| Phenotypes                         |       | Phenotypic ratio      |
| Green seeds                        |       | _____ : _____         |
| Yellow seeds                       |       |                       |

## Analysis and Conclusions

- What characteristic is being studied in this investigation?
- What are the genotypes of the parents? Describe the genotypes of both parents by using the terms *homozygous*, *heterozygous*, or both.
- What does each seed in the Petri dish represent?
- When the seeds were selected and paired, what did the pairs represent?
- Did Tables B and C reflect a classic monohybrid-cross phenotypic ratio of 3:1?
- When the class data were tabulated, did a classic monohybrid-cross phenotypic ratio of 3:1 result?
- If a genotypic ratio of 1:2:1 is observed, what must the genotypes of both parents be?
- Show what the genotypes of the parents would be if 50 percent of the offspring were green and 50 percent of the offspring were yellow.
- Construct a Punnett square for the cross of a heterozygous black guinea pig and an unknown guinea pig whose offspring include a recessive white-furred individual. What are the possible genotypes of the unknown parent?

## Further Inquiry

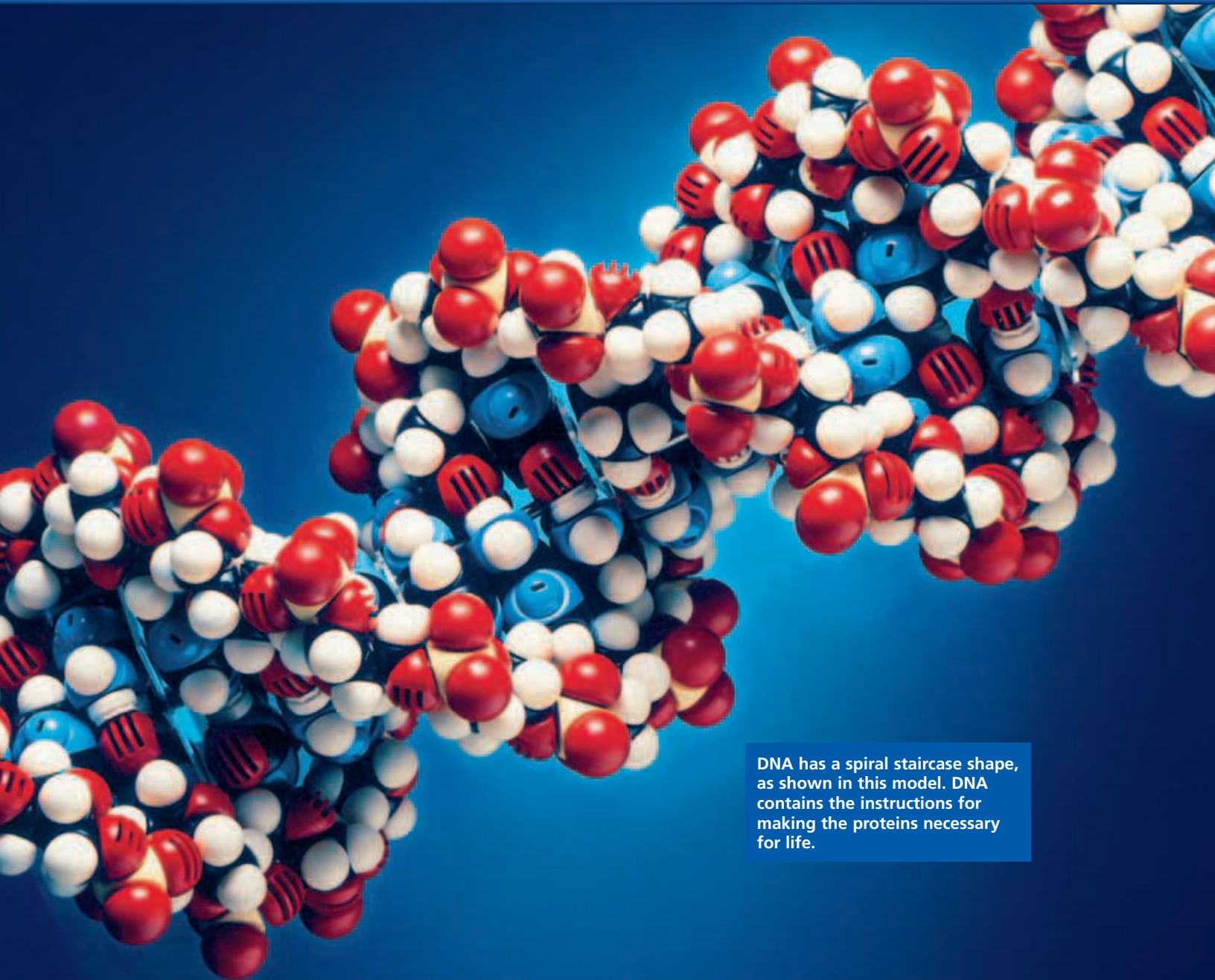
Design a model to demonstrate a dihybrid cross of two parents that are heterozygous for two characteristics. Construct and complete a Punnett square for this cross.

**TABLE C OFFSPRING RATIOS (Entire Class)**

| Genotypes                          | Total | Genotypic ratio       |
|------------------------------------|-------|-----------------------|
| Homozygous dominant ( <i>GG</i> )  |       | _____ : _____ : _____ |
| Heterozygous ( <i>Gg</i> )         |       |                       |
| Homozygous recessive ( <i>gg</i> ) |       |                       |
| Phenotypes                         |       | Phenotypic ratio      |
| Green seeds                        |       | _____ : _____         |
| Yellow seeds                       |       |                       |



# DNA, RNA, AND PROTEIN SYNTHESIS



DNA has a spiral staircase shape, as shown in this model. DNA contains the instructions for making the proteins necessary for life.

SECTION 1 *Discovery of DNA*

SECTION 2 *DNA Structure*

SECTION 3 *DNA Replication*

SECTION 4 *Protein Synthesis*



Unit 6—Gene Expression  
Topics 1–6



For project ideas from  
*Scientific American*, visit  
[go.hrw.com](http://go.hrw.com) and type in  
the keyword **HM6SAB**.

# DISCOVERY OF DNA

*From his studies with pea plants, Mendel concluded that hereditary factors determine many of an organism's traits. But what were these hereditary factors? How did these molecules store hereditary information? Scientists believed that if they could answer these questions, they could understand how cells pass on characteristics to their descendants. The answers to these questions began to emerge during an epidemic of pneumonia in London in the 1920s.*

## GRIFFITH'S EXPERIMENTS

In 1928, British medical officer Frederick Griffith was studying a bacterium called *Streptococcus pneumoniae* (abbreviated *S. pneumoniae*). Some types, or strains, of this bacterium can cause the lung disease pneumonia in mammals. Griffith was trying to develop a vaccine against a disease-causing, or **virulent** (VIR-yoo-luhnt) strain of the bacterium.

As shown in Figure 10-1, each virulent bacterium is surrounded by a capsule made of polysaccharides that protects it from a body's defense systems. The bacteria in a virulent strain grow as smooth-edged colonies when grown in a Petri dish and are called the *S* strain. In contrast, a second strain of *S. pneumoniae* does not cause pneumonia and lacks a capsule. The second strain is called the *R* strain because it grows into rough colonies. The *R* strain is also shown in Figure 10-1.



Colonies of the harmful (*S*) strain



Colonies of the harmless (*R*) strain

## OBJECTIVES

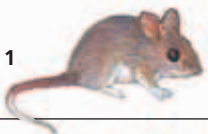
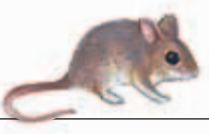
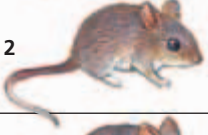
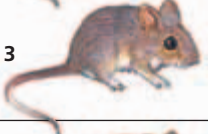
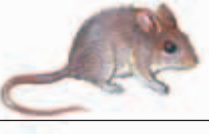
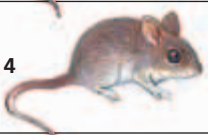
- **Relate** how Griffith's bacterial experiments showed that a hereditary factor was involved in transformation.
- **Summarize** how Avery's experiments led his group to conclude that DNA is responsible for transformation in bacteria.
- **Describe** how Hershey and Chase's experiment led to the conclusion that DNA, not protein, is the hereditary molecule in viruses.

## VOCABULARY

virulent  
transformation  
bacteriophage

**FIGURE 10-1**

Griffith studied *S. pneumoniae* bacteria. The *S* strain can cause pneumonia. The *R* strain does not cause pneumonia.

|              |                                                                                                                                                                                                                                                                   | Result                                                                             | Conclusion                                                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Experiment 1 |  Inject mouse with live <i>R</i> cells.                                                         |  | <i>R</i> cells do not kill the mouse.                                                                                             |
| Experiment 2 |  Inject mouse with live <i>S</i> cells.                                                         |  | <i>S</i> cells kill the mouse.                                                                                                    |
| Experiment 3 | Kill <i>S</i> cells with heat.<br> Inject mouse with heat-killed <i>S</i> cells.                |  | Heat-killed <i>S</i> cells do not kill the mouse.                                                                                 |
| Experiment 4 | Kill <i>S</i> cells with heat.<br>Mix with live <i>R</i> cells.<br> Inject mouse with mixture.  |  | Hereditary material from the heat-killed <i>S</i> cells transforms <i>R</i> cells. The transformed <i>R</i> cells kill the mouse. |

**FIGURE 10-2**

Frederick Griffith used virulent (*S*) and nonvirulent (*R*) bacterial cells to show that the hereditary material can pass from cell to cell.

### Word Roots and Origins

#### transformation

from the Latin *trans*, meaning "across," and *forma*, meaning "a form": to change the condition, character, or function of something

Griffith used the two strains of *S. pneumoniae* bacteria in a series of four experiments, shown in Figure 10-2. These experiments provide insight about the nature of the hereditary material. In Experiments 1 and 2, Griffith injected either live *R* or live *S* cells into mice. He found that only *S* cells killed the mice. In Experiment 3, he injected heat-killed *S* bacteria into mice and found that the mice survived. In his fourth experiment, he injected mice with both heat-killed *S* cells and live *R* cells. He found that the mice died.

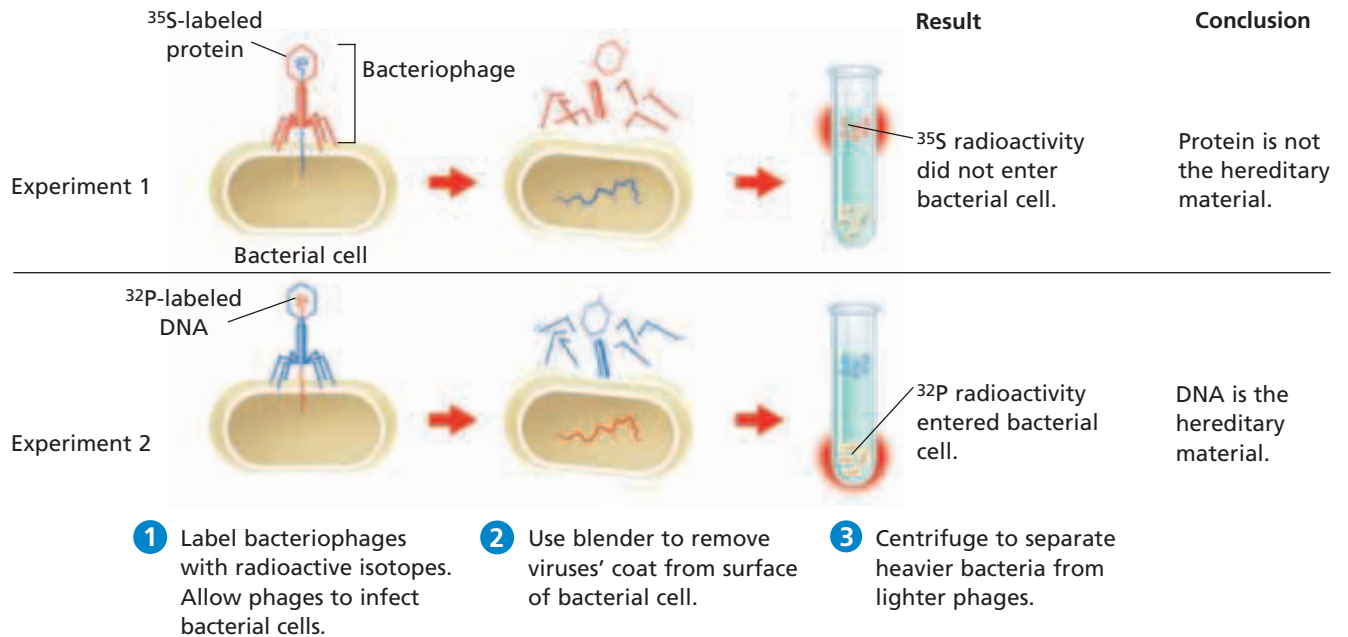
Griffith concluded from his four experiments that heat-killed virulent bacterial cells release a hereditary factor that transfers the disease-causing ability to the live harmless cells. This type of transfer of genetic material from one cell to another cell or from one organism to another organism is called **transformation**.

## AVERY'S EXPERIMENTS

In the early 1940s, American researcher Oswald Avery and his colleagues set out to test whether the transforming agent in Griffith's experiment was protein, RNA, or DNA. The scientists used enzymes to separately destroy each of the three molecules in heat-killed *S* cells. They used a protease enzyme to destroy protein in heat-killed cells in the first experiment, an enzyme called RNase to destroy RNA in the second experiment, and an enzyme called DNase to destroy DNA in the third experiment. Then, they separately mixed the three experimental batches of heat-killed *S* cells with live *R* cells and injected mice with the mixtures.

Avery and his group found that the cells missing protein and RNA were able to transform *R* cells into *S* cells and kill the mice. However, cells missing DNA did not transform *R* cells into *S* cells, and therefore the mice survived. They concluded that DNA is responsible for transformation in bacteria.





## HERSHEY-CHASE EXPERIMENT

In 1952, two American researchers, Martha Chase and Alfred Hershey, set out to test whether DNA or protein was the hereditary material viruses transfer when viruses enter a bacterium. Viruses that infect bacteria are called **bacteriophages**, or just *phages*. As shown in Figure 10-3 in step ❶, Hershey and Chase used radioactive isotopes to label the protein and DNA in the phage. They used radioactive sulfur (<sup>35</sup>S) to label protein and radioactive phosphorus (<sup>32</sup>P) to label DNA. Then, they allowed protein-labeled and DNA-labeled phage to separately infect *Escherichia coli* (abbreviated *E. coli*) bacteria. In step ❷, they removed the phage coats from the cells in a blender. They then used a centrifuge in step ❸ to separate the phage from the *E. coli*. They found that all of the viral DNA and little of the protein had entered *E. coli* cells. They concluded that DNA is the hereditary molecule in viruses.

**FIGURE 10-3**

The experiment of Hershey and Chase showed that DNA carries hereditary information from bacteriophages into the bacteria they infect.

## SECTION 1 REVIEW

1. How did Griffith's experiments show that a hereditary factor was involved in bacterial transformation?
2. Describe how the contributions of Avery and his colleagues revealed that DNA is responsible for transformation in bacteria.
3. How did the Hershey and Chase experiment produce evidence that DNA, and not protein, is the hereditary material in viruses?

### CRITICAL THINKING

4. **Analyzing Methods** Why did heat kill Griffith's *S* bacteria?
5. **Analyzing Results** What were the essential differences between the methods and results of Griffith and Avery's experiments?
6. **Applying Information** What might Hershey and Chase have concluded if they had found both <sup>32</sup>P and <sup>35</sup>S in the bacterial cells?



## SECTION 2

### OBJECTIVES

- **Evaluate** the contributions of Franklin and Wilkins in helping Watson and Crick discover DNA's double helix structure.
- **Describe** the three parts of a nucleotide.
- **Summarize** the role of covalent and hydrogen bonds in the structure of DNA.
- **Relate** the role of the base-pairing rules to the structure of DNA.

### VOCABULARY

nucleotide  
deoxyribose  
nitrogenous base  
purine  
pyrimidine  
base-pairing rules  
complementary base pair  
base sequence

# DNA STRUCTURE

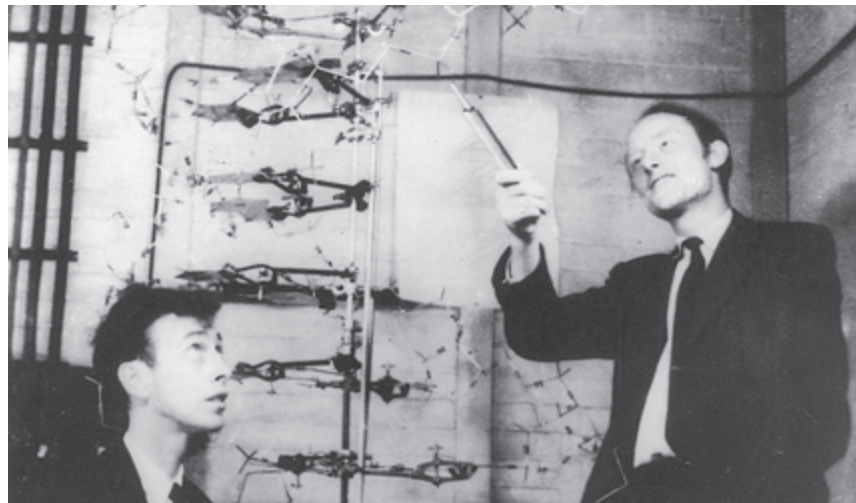
*By the early 1950s, most biologists accepted DNA as the hereditary material. However, they still lacked an understanding of DNA's structure or how this molecule could replicate, store, and transmit hereditary information and direct cell function. These mysteries would soon begin to unravel at Cambridge University in England.*

## DNA DOUBLE HELIX

In the 1950s, a young American biologist, James Watson, teamed up with British graduate student Francis Crick at Cambridge University in England to try to determine the structure of DNA. By 1953, they had put together a model for the structure of DNA as shown in Figure 10-4. They proposed that DNA is made of two chains that wrap around each other in the shape of a double helix, a shape similar to a winding spiral staircase. Their final model was correct and was remarkable because it explained how DNA could replicate.

Watson and Crick relied on other scientists' work to develop their DNA model. Part of that work was X-ray diffraction photographs of DNA crystals, such as the one shown in Figure 10-5. The photographs and crystals were produced by researchers Rosalind Franklin, shown in Figure 10-5, and Maurice Wilkins, at King's College in London.

In 1962, Watson, Crick, and Wilkins received the Nobel Prize in Medicine for their work on DNA. Rosalind Franklin died in 1958 and so could not be named in the award. However, an important genetics institute in Cambridge now bears her name, and her contribution is recognized around the world.



**FIGURE 10-4**

James Watson (left) and Francis Crick stand beside their tin-and-wire model of DNA.

# DNA NUCLEOTIDES

DNA is a nucleic acid made of two long chains (also called strands) of repeating subunits called nucleotides (NOO-klee-oh-TIEDZ). Each **nucleotide** consists of three parts: a five-carbon sugar, a phosphate group, and a nitrogenous base. The three parts of a DNA nucleotide are illustrated in Figure 10-6. The five-carbon sugar in a DNA nucleotide is called **deoxyribose** (dee-AHKS-ee-RIE-bohs). The phosphate group consists of a phosphorus (P) atom bonded to four oxygen (O) atoms. The **nitrogenous** (nie-TRAHJ-uh-nuhs) **base** contains nitrogen (N) atoms and carbon (C) atoms and is a base (accepts hydrogen ions).

## Bonds Hold DNA Together

The DNA double helix is similar to a spiral staircase, as Figure 10-6 shows. The alternating sugar and phosphate molecules form the side “handrails” of the staircase. Nucleotides along each strand are connected by covalent bonds between the sugar of one nucleotide and the phosphate group of the next nucleotide. Each full turn of the DNA helix has 10 nucleotide pairs.

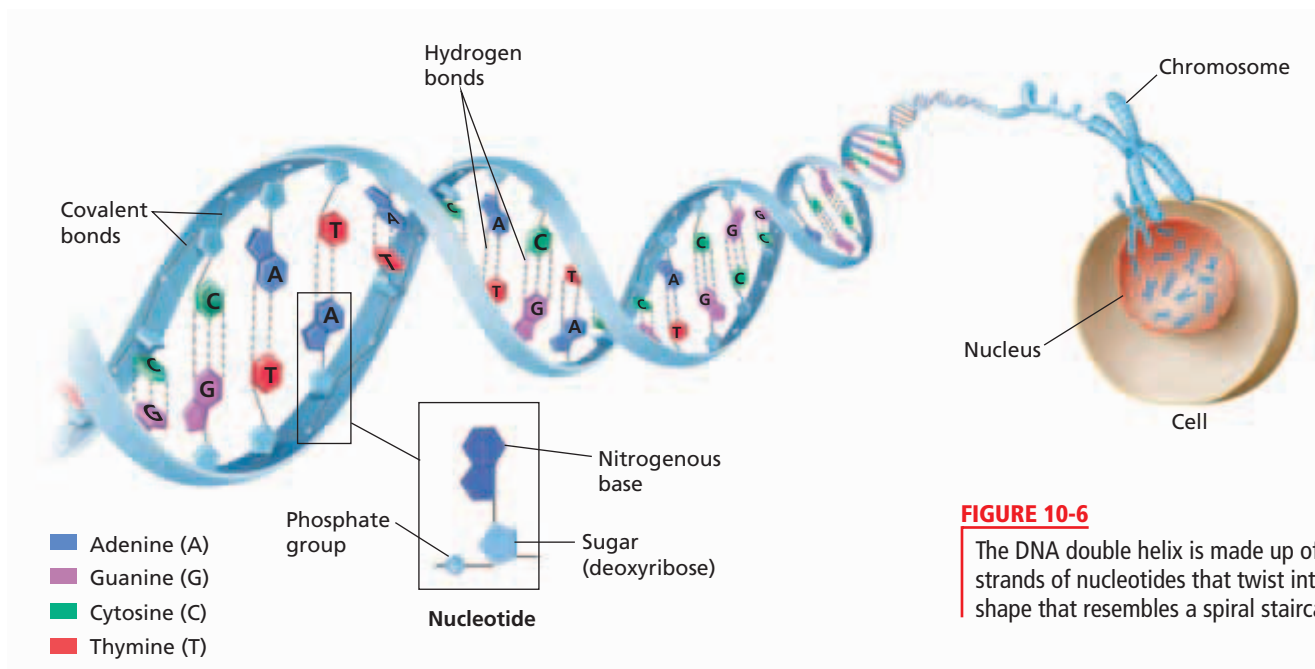
The nitrogenous bases (called “bases” for short) face toward the center of the DNA molecule. The bases on one strand of DNA face—and form bonds called *hydrogen bonds* with—the bases on the other strand. Nitrogenous bases are bonded in pairs between the two strands by two or three hydrogen bonds. The base pairs form the “steps” of the staircase. The base pairs are of uniform width because, in each pair one base has a two-ring structure and the other base has a single-ring structure.

In Figure 10-6, dashed lines indicate the locations of the hydrogen bonds. Hydrogen bonds between the bases help hold the two chains of the DNA double helix together.



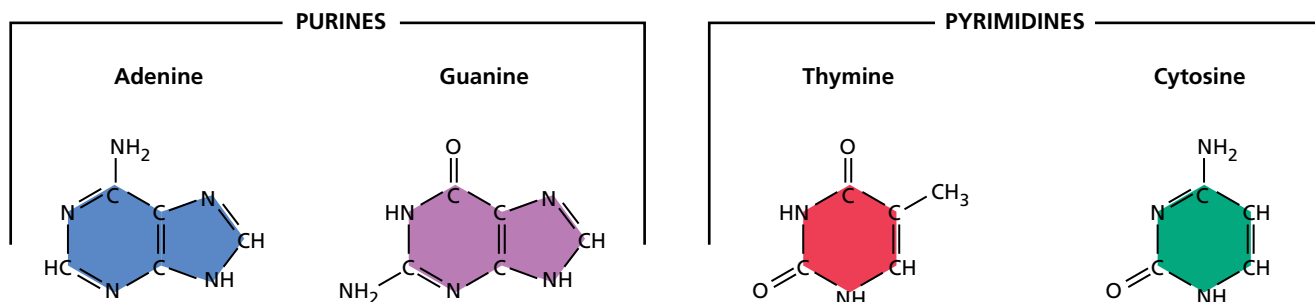
**FIGURE 10-5**

Rosalind Franklin and her X-ray diffraction photo of DNA helped reveal the characteristic shape of a double helix.



**FIGURE 10-6**

The DNA double helix is made up of two strands of nucleotides that twist into a shape that resembles a spiral staircase.



**FIGURE 10-7**

Each nucleotide in a DNA molecule is made of a deoxyribose sugar, a phosphate group, and one of the four nitrogenous bases shown above: thymine, cytosine, adenine, or guanine.

## Nitrogenous Bases

The sugar and phosphate group are identical in all DNA nucleotides. However, the nitrogenous base may be any one of four different kinds—thymine (THIE-MEEN), cytosine (SIET-oh-SEEN), adenine (AD-uh-NEEN), or guanine (GWAH-NEEN). The nitrogenous bases and their chemical structures, called *rings*, are shown above in Figure 10-7. The nitrogenous bases are often represented by the first letter of their name—T (thymine), C (cytosine), A (adenine), and G (guanine).

Nitrogenous bases that have a double ring of carbon and nitrogen atoms, such as adenine and guanine, are called **purines** (PYUR-EENZ). Nitrogenous bases that have a single ring of carbon and nitrogen atoms, such as cytosine and thymine, are called **pyrimidines** (pi-RIM-uh-DEENZ).

### Word Roots and Origins

#### **deoxyribose**

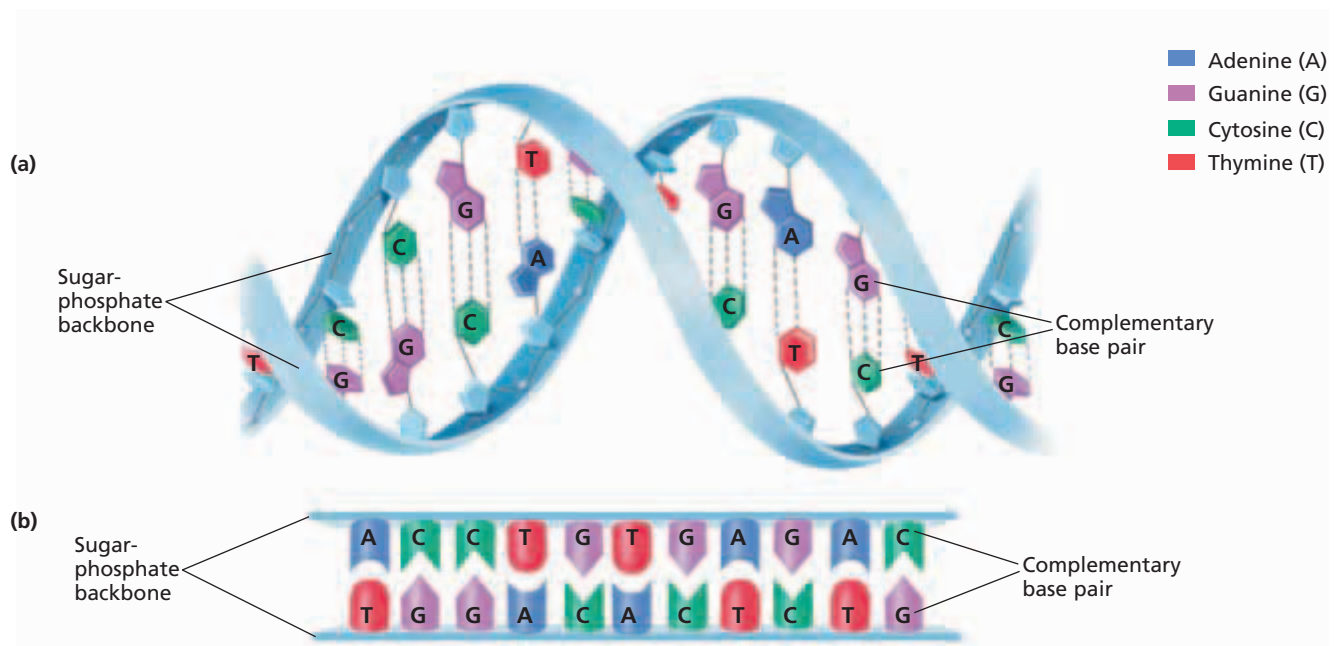
from the Latin *de*, meaning "away from," the Greek *oxys*, meaning "sharp, or acid" (as in oxygen), and *ribose*, a type of sugar

## COMPLEMENTARY BASES

In 1949, American biochemist Erwin Chargaff observed that the percentage of adenine equals the percentage of thymine, and the percentage of cytosine equals that of guanine in the DNA of a variety of organisms. This observation was key to understanding the structure of DNA because it meant bases pair by **base-pairing rules**—in DNA, cytosine on one strand pairs with guanine on the opposite strand, and adenine pairs with thymine, as shown in Figure 10-8. These pairs of bases are called **complementary base pairs**. Notice that each complementary base pair contains one double-ringed purine and one single-ringed pyrimidine.

Because of the base-pairing rules, the order of the nitrogenous bases on the nucleotides in one chain of the DNA molecule is complementary to the order of bases on the opposite chain. For example, if a DNA chain has the sequence ATTC, then the other chain must have the complementary sequence TAAG. The order of nitrogenous bases on a chain of DNA is called its **base sequence**.

Complementary base pairing is important in DNA structure and function for two reasons. First, the hydrogen bonds between the base pairs help hold the two strands of a DNA molecule together. Second, the complementary nature of DNA helps explain how DNA replicates before a cell divides. One strand of a DNA molecule can serve as a template for making a new complementary strand.



## DNA Models

The structure of DNA is often simplified when it is drawn or modeled. For example, the DNA double helix is often illustrated as a straight ladder, as shown at the bottom of Figure 10-8. The sugar-phosphate “handrails” are drawn as a straight line so that the base-pair “steps” between the DNA strands are easier to see. Notice that simplifying the DNA structure highlights the complementary base pairs in each of the DNA nucleotides. In some cases the structure of DNA is simplified even more by just writing the first letter of each of the nitrogenous bases in the DNA nucleotides. For example, the DNA in Figure 10-8b would be represented by

```

A C C T G T G A G A C
T G G A C A C T C T G

```

**FIGURE 10-8**

The DNA double helix resembles a spiral staircase (a), but it is often shown as a straight ladder (b) to more easily show the base pairs.

## SECTION 2 REVIEW

1. What piece of information did Franklin and Wilkins have that helped Watson and Crick determine the double helix structure of DNA?
2. Name the three parts of a nucleotide.
3. Summarize the locations of covalent bonds and hydrogen bonds in a DNA molecule.
4. Describe why the two strands of the double helix are considered to be complementary.
5. State the base-pairing rules in DNA.
6. How do the base-pairing rules relate to the structure of DNA?

### CRITICAL THINKING

7. **Making Predictions** If 2.2 picograms of DNA could be extracted from a certain number of human muscle cells, about how many picograms of DNA could be extracted from the same number of human gamete cells?
8. **Applying Information** Use the base-pairing rules to determine the base sequence that is complementary to the sequence C-G-A-T-T-G.
9. **Making Calculations** A plant's DNA has nucleotides that are 20 percent thymine. What percentage of guanine would be present?



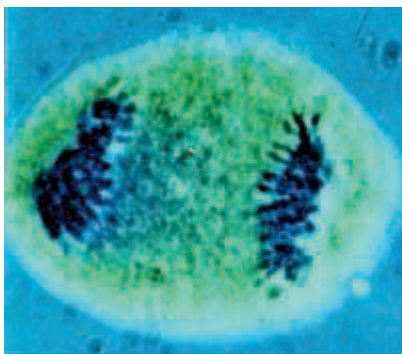
## SECTION 3

### OBJECTIVES

- **Summarize** the process of DNA replication.
- **Identify** the role of enzymes in the replication of DNA.
- **Describe** how complementary base pairing guides DNA replication.
- **Compare** the number of replication forks in prokaryotic and eukaryotic cells during DNA replication.
- **Describe** how errors are corrected during DNA replication.

### VOCABULARY

DNA replication  
helicase  
replication fork  
DNA polymerase  
semi-conservative replication  
mutation



**FIGURE 10-9**

Before a cell divides, its DNA is copied in a process called *DNA replication*. One copy of each chromosome moves to each new cell. In the photo above, the chromosomes (seen as the blue rods on each side of the cell) were in the process of moving during cell division.

# DNA REPLICATION

*Watson and Crick's discovery of the double helix structure of DNA caused great excitement in the scientific community.*

*Scientists realized that this model could explain simply and elegantly how DNA can replicate exactly each time a cell divides, a key feature of hereditary material.*

## HOW DNA REPLICATION OCCURS

**DNA replication** is the process by which DNA is copied in a cell before a cell divides by mitosis, meiosis, or binary fission. During DNA replication, the two nucleotide strands of the original double helix separate along the strands. Because the two strands are complementary, each strand serves as a template to make a new complementary strand. After replication, the two identical double-stranded DNA molecules separate and move to the new cells formed during cell division, as shown in Figure 10-9.

### Steps of DNA Replication

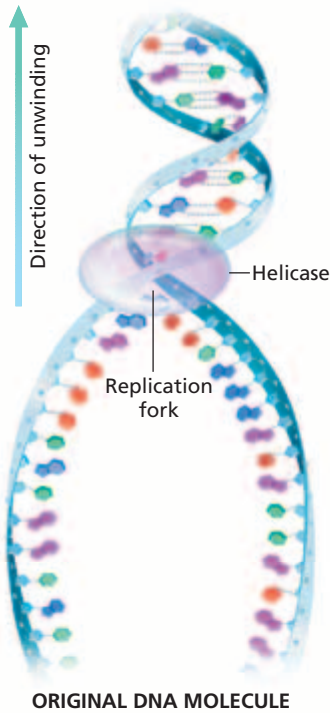
The process of DNA replication is shown in Figure 10-10. In step **1**, enzymes called **helicases** separate the DNA strands. Helicases move along the DNA molecule, breaking the hydrogen bonds between the complementary nitrogenous bases. This action allows the two DNA strands of the double helix to separate from each other. The Y-shaped region that results when the two strands separate is called a **replication fork**.

During step **2**, enzymes called **DNA polymerases** add complementary nucleotides, found floating freely inside the nucleus, to each of the original strands. As the nucleotides on the newly forming strand are added, covalent bonds form between the adjacent nucleotides. Covalent bonds form between the deoxyribose sugar of one nucleotide and the phosphate group of the next nucleotide on the growing strand. Hydrogen bonds form between the complementary nitrogenous bases on the original and new strands.

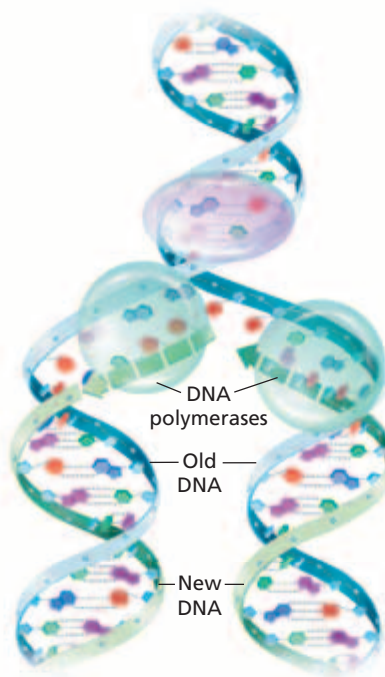
By step **3**, DNA polymerases finish replicating the DNA and fall off. The result is two separate and identical DNA molecules that are ready to move to new cells in cell division.

In each new DNA double helix, one strand is from the original molecule, and one strand is new. This type of replication is called **semi-conservative replication** because each of the new DNA molecules has kept (or conserved) one of the two (or semi) original DNA strands.

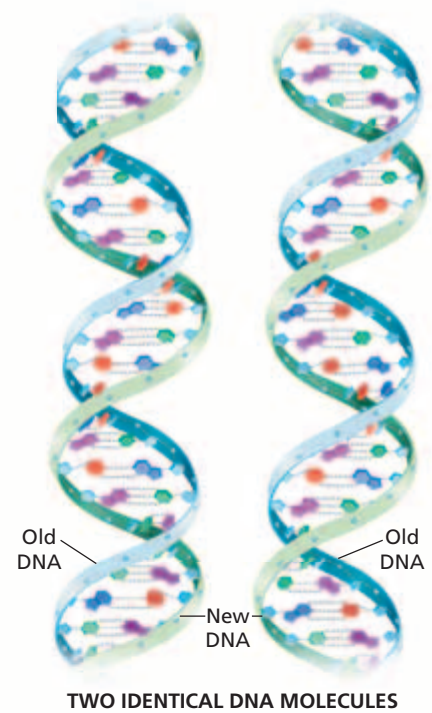
**1** Helicase separates the DNA strands.



**2** DNA polymerases add nucleotides that are complementary to each of the original DNA strands.



**3** DNA polymerases are released. Two DNA molecules identical to the original DNA molecule result.



## Action at the Replication Fork

DNA synthesis occurs in different directions on each strand, as shown by the arrows near the replication fork in step **2** of Figure 10-10. As the replication fork moves along the original DNA, synthesis of one strand follows the movement of the replication fork. Synthesis on the other strand, however, moves in the opposite direction, away from the replication fork, which leaves gaps in the newly synthesized strand. The gaps are later joined together by an enzyme called DNA *ligase*.

## Prokaryotic and Eukaryotic Replication

In prokaryotic cells, which have one circular chromosome, replication begins at one place along the chromosome. Two replication forks are formed and proceed in opposite directions, like two zip-pers opening in opposing directions. Replication continues along each fork until they meet and the entire molecule is copied.

In eukaryotic cells, each chromosome is long, but not circular. At the rate that DNA polymerase adds nucleotides (about 50 nucleotides per second in eukaryotic cells), it would take 53 days to replicate the largest human chromosome. Instead, replication begins at many points or *origins* along the DNA. As with prokaryotes, at each origin, two replication forks move in opposite directions. For example, in a fruit fly chromosome, replication begins simultaneously at about 3,500 sites in a DNA molecule. Only simultaneous replication along chromosomes could allow rapid enough copying of the organism's entire DNA.

**FIGURE 10-10**

During DNA replication, the two strands separate, and each strand serves as a template. DNA polymerases add complementary nucleotides. At the end of replication, there are two identical copies of the original DNA molecule.



**FIGURE 10-11**

Skin cancer can be the result of a DNA mutation in a skin cell that received too much ultraviolet radiation from sunlight. Exposure to harmful radiation from the Sun can be reduced by using sunscreen lotion.

## DNA ERRORS IN REPLICATION

DNA replication usually occurs with great accuracy. Only about one error occurs for every billion paired nucleotides added. That's the equivalent of typing this book 1,000 times and making only one typing error. What accounts for this accuracy? DNA polymerases have repair functions that "proofread" DNA in the same way a friend might check a term paper for spelling errors. For example, if an adenine pairs with a cytosine instead of a thymine, DNA polymerase can repair the error by removing the mispaired cytosine and replacing it with a thymine.

When mistakes in DNA replication do occur, the base sequence of the newly formed DNA differs from the base sequence of the original DNA. A change in the nucleotide sequence of a DNA molecule is called a **mutation**. Mutations can have serious effects on the function of an important gene and disrupt an important cell function.

Some errors escape repair. In addition, chemicals and ultraviolet radiation from the sun can damage DNA. Some mutations can lead to cancer, such as the one shown in Figure 10-11. Thus, an effective mechanism for the repair of damaged DNA is very important to the survival of an organism.

### DNA Replication and Cancer

DNA replication is an elegant process in which genetic information is passed from cell to cell for thousands of generations. It also explains how mutations can arise and lead to altered cells and organisms. Sometimes, the changes allow individuals to survive and reproduce better, so these variations increase in the population over many generations. Sometimes, mutations that are not repaired can cause diseases such as cancer. For example, mutations that affect genes that control how a cell divides can lead to an abnormal mass of cells called a *tumor*. Studying DNA replication is one promising avenue to understanding and treating various types of human cancers.

## SECTION 3 REVIEW

1. Describe what happens at a DNA replication fork during replication.
2. Describe the role of helicases and DNA polymerases during DNA replication.
3. State why DNA replication is a semi-conservative process.
4. Compare the number of replication forks in prokaryotic and eukaryotic DNA during replication.
5. How are replication errors corrected?

### CRITICAL THINKING

6. **Analyzing Concepts** Why are there two DNA polymerases at one replication fork?
7. **Drawing Conclusions** Why are DNA repair enzymes important to an organism's survival?
8. **Evaluating Information** Is a mutation that occurs during the formation of an egg cell or sperm cell more significant than a mutation that occurs in a body cell? Explain.



## DNA REPAIR AND SKIN CANCER

Sometimes, the errors that occur during DNA replication are not fixed by DNA repair enzymes. These unrepaired errors can lead to mutations. Cancer can occur if the mutations happen within genes that control cell growth and cell division. Scientists hope that by studying DNA replication and DNA repair, they can develop treatments or even cure various types of cancers.

### Ultraviolet Light and Skin Cancer

Ultraviolet light, the most energetic part of sunlight, is the



*It is important to guard against skin cancer. For example, apply sunscreen before prolonged exposure to sunlight.*

main cause of mutations that trigger skin cancer. Skin cancer is the most common type of cancer in the United States. Each year, about 1 million Americans get skin cancer.

When UV light reaches the DNA inside a skin cell, thymine bases that are next to each other on the same strand of DNA can become linked by a covalent bond, as shown in the figure below. Linked thymine pairs are called *thymine dimers*. Thymine dimers are usually detected by enzymes moving along the DNA strand because the dimers cause a kink in the DNA, as shown in the figure below. Dimers that are not repaired during DNA replication can cause mutations in genes that control cell division. The mutation can trigger a skin cell to become cancerous.

### DNA Repair Enzymes and Skin Cancer Treatments

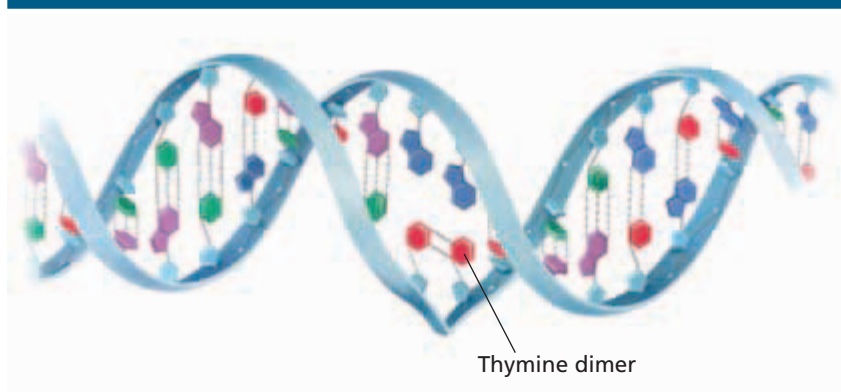
Some organisms do not get skin cancer. One reason is that these organisms have a DNA repair enzyme called *photolyase*

(FOH-toh-LIE-AYS). Photolyase is activated by light and can repair thymine dimers caused by UV radiation. Human skin cells can correct UV-induced dimers by a complex process known as *excision repair* that involves other enzymes. But photolyase uses a more direct and effective mechanism for DNA repair than excision repair. Scientists have already developed a sunscreen containing photolyase to repair the UV-induced DNA damage that occurs when a person is sunburned.

Some researchers want to try to use gene therapy to insert the gene for photolyase in people that are at high risk for skin cancer. Gene therapy is a technique in which a defective gene is replaced with a normal version of the gene. Ongoing studies of DNA repair enzymes may help develop gene therapy and other types of cancer treatments in humans.

### REVIEW

1. How can errors during DNA replication lead to cancer?
2. How does the DNA repair enzyme photolyase prevent skin cancer?
3. **Supporting Reasoned Opinions**  
Would you buy a sunscreen that contains photolyase? Why or why not?



*Unrepaired DNA damage can prevent accurate copying of the DNA and can lead to mutations. One example of DNA damage is covalent cross-linking between two thymine bases, which is called a thymine dimer.*





## SECTION 4

### OBJECTIVES

- **Outline** the flow of genetic information in cells from DNA to protein.
- **Compare** the structure of RNA with that of DNA.
- **Summarize** the process of transcription.
- **Describe** the importance of the genetic code.
- **Compare** the role of mRNA, rRNA, and tRNA in translation.
- **Identify** the importance of learning about the human genome.

### VOCABULARY

ribonucleic acid (RNA)  
transcription  
translation  
protein synthesis  
ribose  
messenger RNA (mRNA)  
ribosomal RNA (rRNA)  
transfer RNA (tRNA)  
RNA polymerase  
promoter  
termination signal  
genetic code  
codon  
anticodon  
genome

**FIGURE 10-12**

DNA contains the instructions for building a protein. DNA transfers the instructions to an RNA molecule in a process called transcription. The RNA moves out into the cytoplasm, where its instructions are read and the protein is assembled in a process called translation.

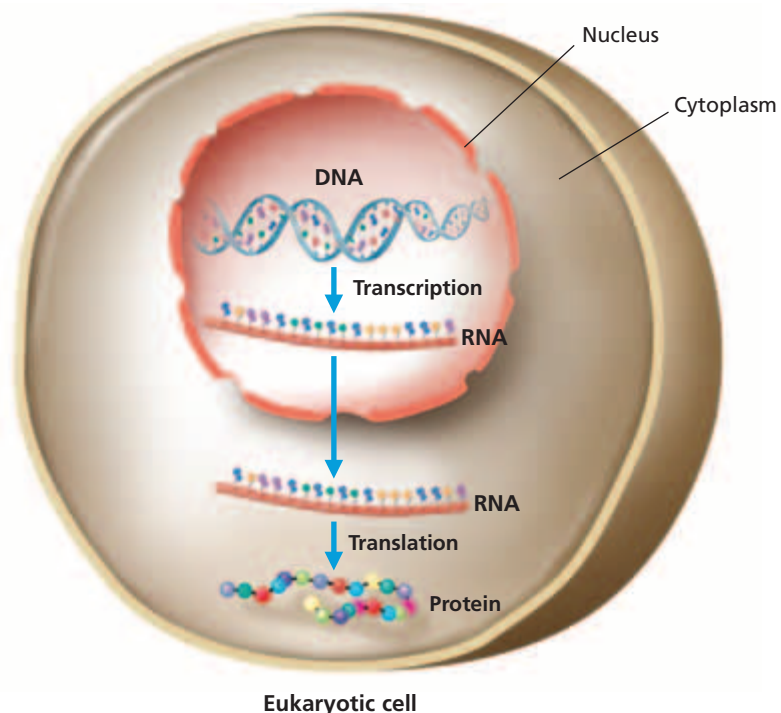
# PROTEIN SYNTHESIS

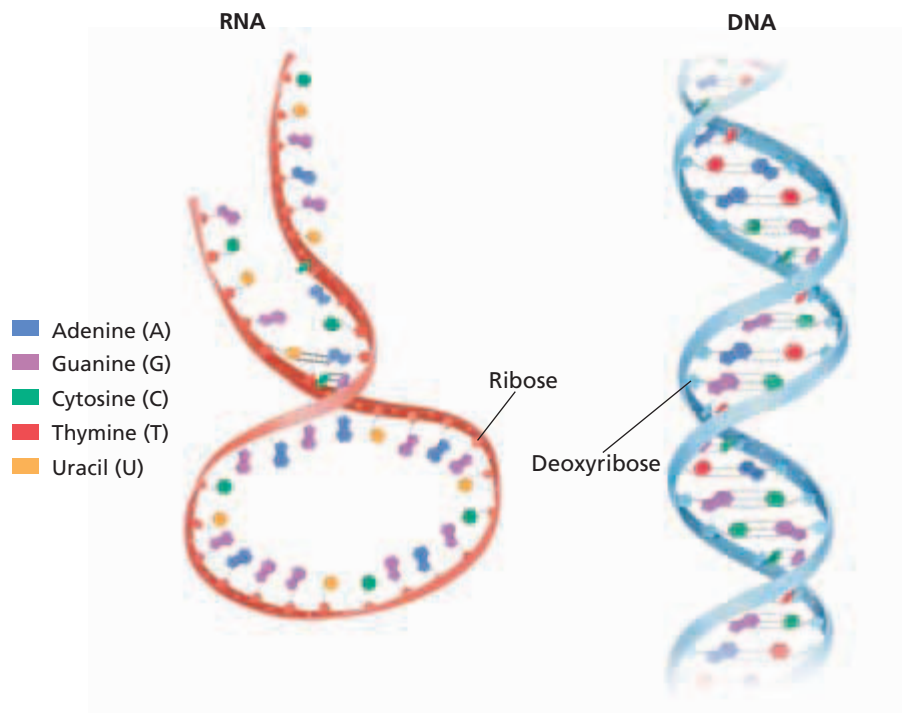
*Characteristics such as hair color are largely determined by genetic factors. But how does inheriting a particular form of a gene result in the appearance of a specific hair color? The structure of DNA helps explain how genes function in making proteins that determine traits in organisms.*

## FLOW OF GENETIC INFORMATION

A gene is a segment of DNA that is located on a chromosome and that codes for a hereditary character. For example, a gene determines a person's hair color. The gene directs the making of the protein called *melanin* (a pigment) in hair follicle cells through an intermediate—the nucleic acid called **ribonucleic acid**, or **RNA**.

Figure 10-12 summarizes the flow of genetic information in a eukaryotic cell. During **transcription**, DNA acts as a template for the synthesis of RNA. In **translation**, RNA directs the assembly of proteins. Forming proteins based on information in DNA and carried out by RNA is called **protein synthesis**, or gene expression. This central concept can be symbolized as  $DNA \rightarrow RNA \rightarrow protein$ . Proteins do important work in cells, such as protecting the body against infections and carrying oxygen in red blood cells.





**FIGURE 10-13**

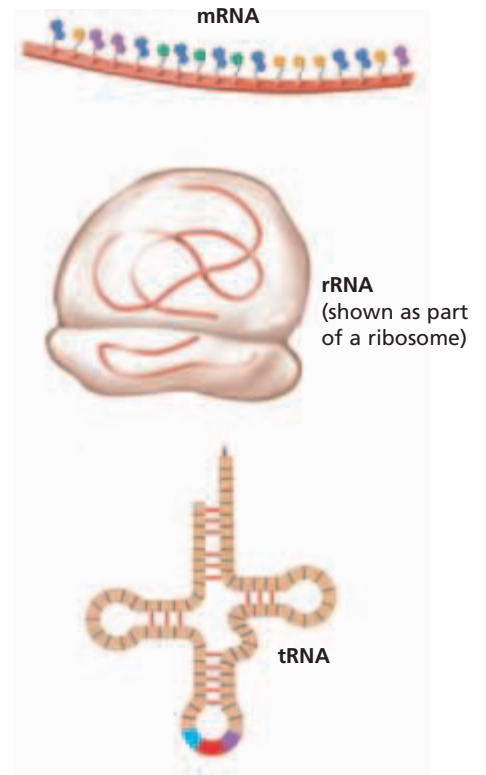
The structure of RNA is different from the structure of DNA. Each of the three major types of RNA—mRNA, tRNA, and rRNA—play a different role during protein synthesis.

## RNA STRUCTURE AND FUNCTION

Like DNA, RNA is a nucleic acid made up of nucleotides. However, as shown in Figure 10-13, the structure of RNA differs from that of DNA in four basic ways. First, RNA contains the sugar **ribose**, not the sugar deoxyribose found in DNA. Second, RNA contains the nitrogenous base **uracil** instead of the nitrogenous base thymine found in DNA. Third, RNA is usually single stranded rather than double stranded like DNA. However, within a single-stranded RNA molecule, some regions fold to form short double-stranded sections. In the double-stranded regions, guanine forms base pairs with cytosine, and uracil forms base pairs with adenine. Fourth, RNA is usually much shorter in length than DNA (about the length of one gene). On the other hand, DNA is usually long, containing hundreds or thousands of genes.

### Types of RNA

Cells have three major types of RNA, as shown in Figure 10-14. Each type of RNA plays a different role in protein synthesis. The first type of RNA is **messenger RNA (mRNA)**, a single-stranded RNA molecule that carries the instructions from a gene to make a protein. In eukaryotic cells, mRNA carries the genetic “message” from DNA in the nucleus to the ribosomes in the cytosol. The second type of RNA is **ribosomal RNA (rRNA)**, which is part of the structure of ribosomes. Ribosomes are organelles in the cell where protein synthesis occurs. Ribosomes are made of rRNAs and many proteins. Figure 10-14 shows a model of a ribosome. The third type of RNA is **transfer RNA (tRNA)**, which transfers amino acids to the ribosome to make a protein. Although the entire tRNA is made of nucleotides linked together, only three are emphasized in Figure 10-14.



**FIGURE 10-14**

Each of the three major types of RNA differs in its structure. Messenger RNA (mRNA) is typically drawn as a relatively straight chain. Ribosomal RNA (rRNA) is shown as part of the structure of a ribosome. Transfer RNA's (tRNA) two-dimensional structure is typically shown with only three of its many nucleotides emphasized.

## Word Roots and Origins

### *transcription*

from the Latin *scribere*, meaning "to write," and *trans*, meaning "across"

# TRANSCRIPTION

Transcription is the process by which the genetic instructions in a specific gene are transcribed or "rewritten" into an RNA molecule. Transcription takes place in the nucleus of eukaryotic cells and in the DNA-containing region in the cytoplasm of prokaryotic cells.

## Steps of Transcription

Transcription occurs in three steps, as shown in Figure 10-15. In step **1**, **RNA polymerase**, an enzyme that catalyzes the formation of RNA on a DNA template, binds to a promoter. A **promoter** is a specific nucleotide sequence of DNA where RNA polymerase binds and initiates transcription. After RNA polymerase binds to the promoter, the DNA strands unwind and separate.

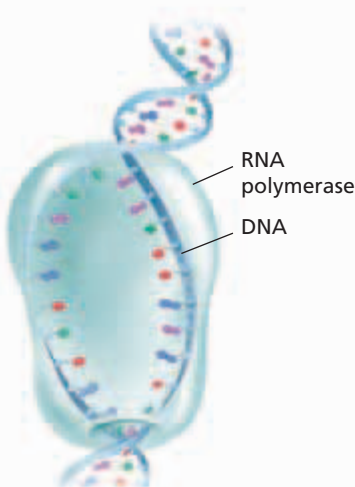
In step **2**, RNA polymerase adds free RNA nucleotides that are complementary to the nucleotides on one of the DNA strands. The resulting chain is an RNA molecule. As in DNA replication, complementary base pairing determines the nucleotide sequence in the newly made RNA. For example, if the bases on the DNA strand was ATCGAC, the bases on the RNA strand would be UAGCUG. Unlike DNA replication, transcription uses only a specific region (a gene) on one of the two DNA strands to serve as the template. As RNA polymerase moves past, the separated DNA strands rewind.

During step **3**, RNA polymerase reaches a **termination signal**, a specific sequence of nucleotides that marks the end of a gene. Upon reaching this "stop" signal, RNA polymerase releases both the DNA and the newly formed RNA. The RNA made during transcription can be one of many types including mRNA, tRNA, or rRNA. The newly made RNA can now perform its job in the cell, and the RNA polymerase can transcribe another gene.

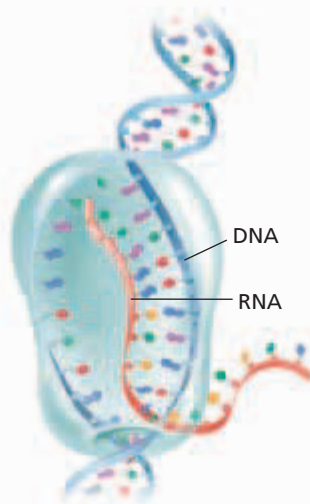
**FIGURE 10-15**

During transcription, the enzyme RNA polymerase "reads" one of the chains, the template strand. RNA polymerase adds and then joins complementary RNA nucleotides, resulting in an RNA strand.

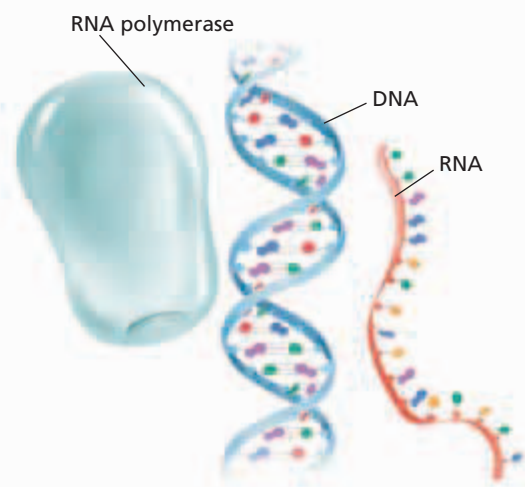
- 1** RNA polymerase binds to the gene's promoter. The two DNA strands unwind and separate.



- 2** Complementary RNA nucleotides are added and then joined.



- 3** When RNA polymerase reaches a termination signal in the DNA, the DNA and new RNA are released by the polymerase.



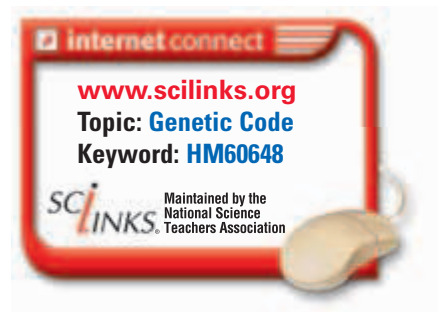
TRANSCRIPTION

# THE GENETIC CODE

During the next process of gene expression, amino acids are assembled based on instructions encoded in the sequence of nucleotides in the mRNA. The **genetic code** is the term for the rules that relate how a sequence of nitrogenous bases in nucleotides corresponds to a particular amino acid. In the genetic code, three adjacent nucleotides (“letters”) in mRNA specify an amino acid (“word”) in a polypeptide. Each three-nucleotide sequence in mRNA that encodes an amino acid or signifies a start or stop signal is called a **codon**.

Table 10-1 lists the 64 mRNA codons and the amino acids they encode in most organisms. For example, the codon *GCU* specifies the amino acid *alanine* in the genetic code. The genetic code is nearly universal to all life on Earth and supports the idea that all organisms share an ancient common ancestor.

Some amino acids are encoded by two, three, or more different codons, as shown in Table 10-1. These codons often differ from one another by only one nucleotide. No codon encodes more than one amino acid. One special codon, AUG, acts as a start codon. A *start codon* is a specific sequence of nucleotides in mRNA that indicates where translation should begin. The start codon encodes the amino acid methionine. Certain sequences of nucleotides in mRNA (UAA, UAG, or UGA), called *stop codons*, do not code for amino acids, but instead signal for translation to end.



**TABLE 10-1 Codons in mRNA**

| First base | Second base              |               |                     |                  | Third base |
|------------|--------------------------|---------------|---------------------|------------------|------------|
|            | U                        | C             | A                   | G                |            |
| <b>U</b>   | UUU } Phenylalanine      | UCU           | UAU } Tyrosine      | UGU } Cysteine   | U          |
|            | UUC } Phenylalanine      | UCC Serine    | UAC } Tyrosine      | UGC } Cysteine   | C          |
|            | UUA } Leucine            | UCA           | UAA } Stop          | UGA } Stop       | A          |
|            | UUG } Leucine            | UCG           | UAG } Stop          | UGG } Tryptophan | G          |
| <b>C</b>   | CUU                      | CCU           | CAU } Histidine     | CGU              | U          |
|            | CUC Leucine              | CCC Proline   | CAC } Histidine     | CGC Arginine     | C          |
|            | CUA                      | CCA           | CAA } Glutamine     | CGA              | A          |
|            | CUG                      | CCG           | CAG } Glutamine     | CGG              | G          |
| <b>A</b>   | AUU } Isoleucine         | ACU           | AAU } Asparagine    | AGU } Serine     | U          |
|            | AUC } Isoleucine         | ACC Threonine | AAC } Asparagine    | AGC } Serine     | C          |
|            | AUA } Methionine (Start) | ACA           | AAA } Lysine        | AGA } Arginine   | A          |
|            | AUG } Methionine (Start) | ACG           | AAG } Lysine        | AGG } Arginine   | G          |
| <b>G</b>   | GUU                      | GCU           | GAU } Aspartic acid | GGU              | U          |
|            | GUC Valine               | GCC Alanine   | GAC } Aspartic acid | GGC Glycine      | C          |
|            | GUA                      | GCA           | GAA } Glutamic acid | GGA              | A          |
|            | GUG                      | GCG           | GAG } Glutamic acid | GGG              | G          |



# TRANSLATION

Although the instructions for making a protein are copied from DNA to mRNA, all three major types of RNA are involved in translation—the making of a protein.

## Protein Structure

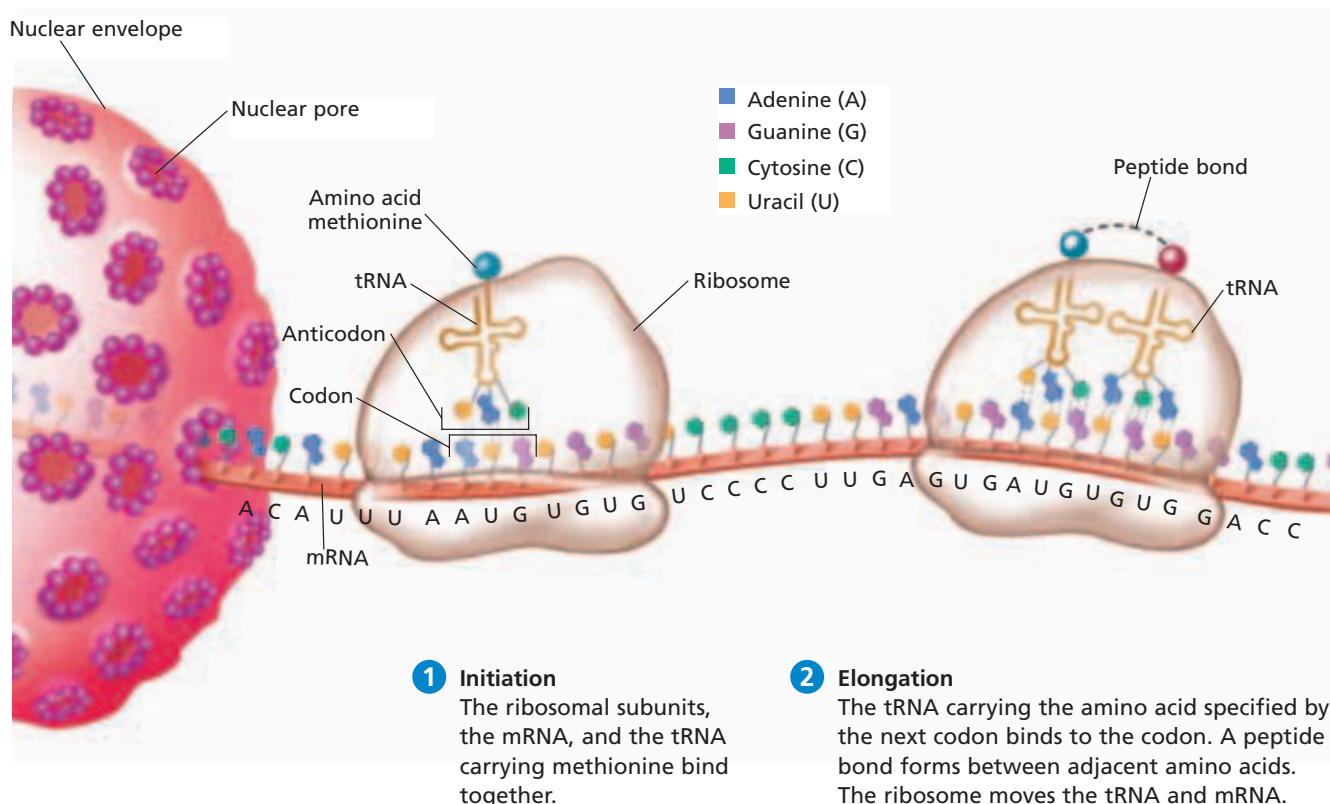
Every protein is made of one or more polypeptides. Polypeptides are chains of amino acids linked by peptide bonds. There are 20 different amino acids found in the proteins of living things. Each polypeptide chain may consist of hundreds or thousands of the 20 different amino acids, arranged in a sequence specific to each protein. The amino acid sequence determines how the polypeptides will twist and fold into the three-dimensional structure of the protein. The shape of the protein is critical to its function.

## Steps of Translation

The translation or decoding of the genetic instructions to form a polypeptide involves five main steps, as shown in Figure 10-16. In step ①, two ribosomal subunits, tRNA, and an mRNA join together. Enzymes first attach a specific amino acid to one end of each tRNA according to the genetic code. The other end of each tRNA contains the **anticodon**, three nucleotides on the RNA that are complementary to the sequence of a codon in mRNA.

**FIGURE 10-16**

During translation, amino acids are assembled from information encoded in mRNA. As the mRNA codons move through the ribosome, tRNAs add specific amino acids to the growing polypeptide chain. The process continues until a stop codon is reached and the newly made protein is released.



A tRNA carrying the amino acid methionine at one end and the anticodon UAC at the other end pairs with the start codon AUG on the mRNA. The first amino acid in nearly all polypeptides is methionine, but this amino acid may be removed later.

In step ②, the polypeptide chain is put together. A tRNA carrying the appropriate amino acid pairs its anticodon with the second codon in the mRNA. The ribosome then detaches methionine from the first tRNA, and a peptide bond forms between methionine and the second amino acid. The first tRNA then exits the ribosome. The ribosome then moves a distance of one codon along the mRNA.

During step ③, the polypeptide chain continues to grow as the mRNA moves along the ribosome. A new tRNA moves in, carrying an amino acid for the next mRNA codon. The growing polypeptide chain moves from one tRNA to the amino acid attached to the next tRNA.

The polypeptide grows one amino acid at a time until step ④. At this step, the ribosome reaches the stop codon. The newly made polypeptide falls off.

During step ⑤, the components of translation come apart. The last tRNA leaves the ribosome, and the ribosome moves away from the mRNA. The translation machinery is now free to translate the same or another mRNA.



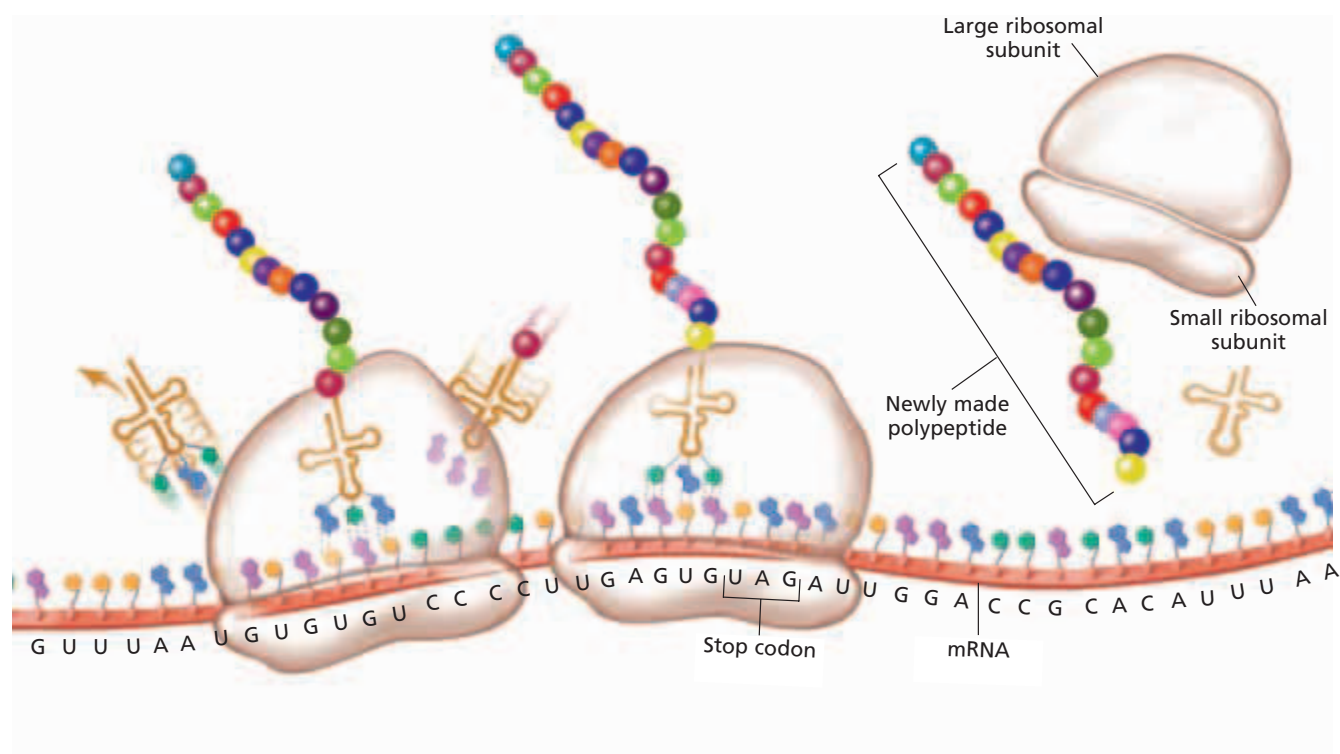
## Quick Lab

### Comparing and Contrasting RNA Types

**Materials** paper and pencil

**Procedure** Create a chart that compares and contrasts the different forms of RNA. Include descriptions of each form's structure and function.

**Analysis** Which types of RNA are alike structurally? What might happen if one type of RNA were missing?



- ③ Elongation (continued)**  
The first tRNA detaches and leaves its amino acid behind. Elongation continues. The polypeptide chain continues to grow.

- ④ Termination**  
The process ends when a stop codon is reached. A stop codon is one for which there is no tRNA that has a complementary anticodon.

- ⑤ Disassembly**  
The ribosome complex falls apart. The newly made polypeptide is released.

## Translating Many Ribosomes at Once

Because a new ribosome begins translating mRNA almost as soon as the preceding ribosome has moved aside, several ribosomes may translate the same mRNA transcript at the same time. In fact, prokaryotes lack a nuclear envelope separating their DNA from ribosomes in the cytosol, thus translation can begin on an mRNA even before transcription of the mRNA has finished. In eukaryotes, translation of an mRNA occurs only after transcription is finished.

## THE HUMAN GENOME

In the years since Watson and Crick discovered the structure of DNA, biologists have achieved a milestone in applying this knowledge to human biology. The entire gene sequence of the human **genome**, the complete genetic content, is now known. Biologists have deciphered the order of the 3.2 billion base pairs in the 23 human chromosomes. The human genome is so large that it would take a person almost 10 years to read the total sequence aloud.

The challenge now is to learn what information the DNA sequences actually encode. An important new field called *bioinformatics* uses computers to compare different DNA sequences. Scientists can program computers to help interpret most DNA sequences and predict where genes lie along the DNA.

To learn where and when human cells use each of the proteins coded for in the approximately 30,000 genes in the human genome will take much more analysis. This information is important because learning which gene sequences control particular biological functions may help diagnose, treat, and prevent genetic disorders, cancer, and infectious diseases in the future.

### SECTION 4 REVIEW

1. Summarize the flow of genetic information.
2. List the four ways in which the structure of RNA differs from that of DNA.
3. Describe the structure and function of each of the three types of RNA.
4. Sequence the main steps of transcription.
5. What is the genetic code?
6. Compare the roles of the three different types of RNA during translation.
7. Describe the significance of identifying the entire sequence of the human genome.

#### CRITICAL THINKING

8. **Making Comparisons** How does the role of RNA polymerase in transcription differ from that of DNA polymerase in DNA replication?
9. **Applying Information** What amino acids would translation of the mRNA with the sequence UAACAAGGAGCAUCC produce?
10. **Analyzing Processes** Discuss why it is important which of the two DNA strands serves as a template during transcription.
11. **Drawing Conclusions** How does the structure of tRNA relate to its function in translation?

# CHAPTER HIGHLIGHTS

## SECTION 1

## Discovery of DNA

- Griffith's experiments showed that hereditary material can pass from one bacterial cell to another. This is called transformation.
- Avery's work showed that DNA is the hereditary material that transfers information between bacterial cells.
- Hershey and Chase confirmed that DNA, and not protein, is the hereditary material.

### Vocabulary

virulent (p. 193)

transformation (p. 194)

bacteriophage (p. 195)

## SECTION 2

## DNA Structure

- Watson and Crick created a model of DNA by using Franklin's and Wilkins's DNA diffraction X-rays.
- DNA is made of two nucleotide strands that wrap around each other in the shape of a double helix.
- A DNA nucleotide is made of a deoxyribose sugar, a phosphate group, and one of four nitrogenous bases: adenine (A), guanine (G), cytosine (C), or thymine (T).
- Nucleotides along each DNA strand are linked by covalent bonds. Complementary nitrogenous bases are bonded by hydrogen bonds.
- Hydrogen bonding between the complementary base pairs, G-C and A-T, holds the two strands of a DNA molecule together.

### Vocabulary

nucleotide (p. 197)

deoxyribose (p. 197)

nitrogenous base (p. 197)

purine (p. 198)

pyrimidine (p. 198)

base-pairing rules (p. 198)

complementary base pair (p. 198)

base sequence (p. 198)

## SECTION 3

## DNA Replication

- DNA replication is the process by which DNA is copied in a cell before a cell divides.
- Replication begins with the separation of the DNA strands by helicases. Then, DNA polymerases form new strands by adding complementary nucleotides to each of the original strands.
- Each new DNA molecule is made of one strand of nucleotides from the original DNA molecule and one new strand.
- Changes in DNA are called mutations. Proofreading and repair prevent many replication errors.

### Vocabulary

DNA replication (p. 200)

helicase (p. 200)

replication fork (p. 200)

DNA polymerase (p. 200)

semi-conservative  
replication (p. 200)

mutation (p. 202)

## SECTION 4

## Protein Synthesis

- The flow of genetic information can be symbolized as DNA → RNA → protein.
- RNA has the sugar ribose instead of deoxyribose and uracil in place of thymine. RNA is single stranded and is shorter than DNA.
- During transcription, DNA acts as a template for directing the synthesis of RNA.
- The genetic code identifies the specific amino acids coded for by each mRNA codon.
- The RNA called mRNA carries the genetic "message" from the nucleus to the cytosol; rRNA is the major component of ribosomes; tRNA carries specific amino acids, helping to form polypeptides.

### Vocabulary

ribonucleic acid (RNA) (p. 204)

transcription (p. 204)

translation (p. 204)

protein synthesis (p. 204)

ribose (p. 205)

messenger RNA

(mRNA) (p. 205)

ribosomal RNA (rRNA) (p. 205)

transfer RNA (tRNA) (p. 205)

RNA polymerase (p. 206)

promoter (p. 206)

termination signal (p. 206)

genetic code (p. 207)

codon (p. 207)

anticodon (p. 208)

genome (p. 210)



# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
  - purine* and *pyrimidine*
  - ribosome* and *ribosomal RNA*
  - messenger RNA* and *transfer RNA*
  - termination signal* and *stop codon*
  - transcription* and *translation*
- Explain the relationship between *codon* and *gene*.
- Use the following terms in the same sentence: *DNA replication*, *replication fork*, *helicase*, and *DNA polymerase*.
- Word Roots and Origins** The word *transcription* is derived from the Latin *scribere*, which means “to write.” The prefix *trans* means “across.” Using this information, explain why the term transcription is a good name for the biological process it describes.

## UNDERSTANDING KEY CONCEPTS

- Summarize** Griffith’s transformation experiments.
- Describe** how Avery’s experiments led to the understanding of DNA as the molecule of heredity in bacteria.
- Describe** the contributions of Hershey and Chase to the understanding that DNA is the hereditary molecule in viruses.
- State** how Watson and Crick were able to build a structural model of DNA.
- Identify** the components of a nucleotide.
- Name** the bonds that link the nucleotides along a DNA strand.
- List** the rules of complementary base pairing.
- Summarize** the major steps that occur during DNA replication.
- Name** the function of DNA polymerase during DNA replication.
- State** how complementary base pairing is important in the replication of DNA.
- Differentiate** the number of replication forks in prokaryotic and eukaryotic DNA.
- Describe** the importance of repair enzymes for the identification of errors during DNA replication.
- Outline** the flow of genetic information in cells.
- Compare** the structure of RNA to that of DNA.
- Summarize** how RNA is formed from a gene during the process of transcription.
- Identify** the function of the genetic code.
- Differentiate** the functions of the three types of RNA involved in protein synthesis.
- Sequence** the major steps of translation.
- Discuss** the importance of learning about the human genome.
- Unit 6—Gene Expression**  
 Write a report summarizing how antibiotics inhibit protein synthesis in bacteria. How do some antibiotics interfere with translation?
-  **CONCEPT MAPPING** Use the following terms to create a concept map that describes the structure of DNA and how it is copied: *nucleotides*, *phosphate group*, *deoxyribose*, *nitrogenous base*, *double helix*, *replication*, *purine*, *pyrimidine*, *DNA polymerases*, and *genes*.

## CRITICAL THINKING

- Analyzing Information** Why is it unlikely that any particular mutation would have any noticeable effect in a population?
- Interpreting Graphics** A segment of DNA has the following sequence:



Write the mRNA transcript from this sequence of DNA. Next, write the tRNA anticodons that would pair with the mRNA transcript. Use Table 10-1 to write the names of the amino acids coded for by the mRNA transcript.

- Analyzing Concepts** A DNA molecule replicates to produce two new DNA molecules. Both of the two new DNA molecules then replicate to form four more new DNA molecules. Are any nucleotide chains from the original DNA present in the last four new DNA molecules? If so, how many?
- Analyzing Current Research** Scientists have determined essentially all of the 3 billion or so nucleotides that spell out the human genome. This genetic information will revolutionize the diagnosis, prevention, and treatment of many human diseases. Propose why this information is important for research on human disease.
- Applying Information** List all codons in the genetic code that could be changed into a stop codon by a single nucleotide mutation.



# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- For which of the following is DNA responsible?
  - directing RNA to make lipids
  - directing RNA to produce glucose
  - encoding information for making proteins
  - encoding information for changing the genetic code
- Where is RNA found?
  - only in proteins
  - only in the nucleus
  - only in the cytoplasm
  - in the nucleus and cytoplasm
- What is the basic unit of DNA called?
  - sugar
  - nucleotide
  - phosphate
  - nucleic acid
- Which of the following nucleic acids is involved in translation?
  - DNA only
  - mRNA only
  - DNA and mRNA
  - mRNA and tRNA

**INTERPRETING GRAPHICS:** The table below shows the percentage of bases in some organisms. Use the table to answer the questions that follow.

**Percentage of Each Nitrogenous Base in Different Organisms**

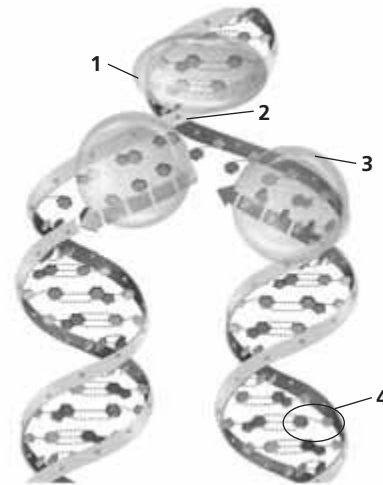
|                | A    | T    | G    | C    |
|----------------|------|------|------|------|
| <i>E. coli</i> | 24.7 | 23.6 | 26.0 | 25.7 |
| Human          | 30.4 | 30.1 | 19.6 | 19.9 |
| Wheat          | 27.3 | 27.1 | 22.7 | 22.8 |

- What is the ratio of purines to pyrimidines for these organisms?
  - about 1:1
  - about 1:2
  - about 1:3
  - about 1:4
- Within each organism, which nucleotides are found in similar percentages?
  - A and T, G and C
  - A and C, G and T
  - A and C, G and U
  - A and G, T and U

**DIRECTIONS:** Complete the following analogy.

7. mRNA : uracil :: DNA :
- guanine
  - thymine
  - adenine
  - cytosine

**DIRECTIONS:** The model below represents a DNA molecule undergoing DNA replication. Use the model to answer the question that follows.



8. Which part of the model represents DNA helicase?
- 1
  - 2
  - 3
  - 4

## SHORT RESPONSE

DNA is made up of two strands of subunits called nucleotides. The two strands are twisted around each other in a double helix shape.

Explain why the structure of a DNA molecule is sometimes described as a zipper.

## EXTENDED RESPONSE

DNA can be damaged by mistakes made during its replication. The mistakes are called mutations.

**Part A** Explain eukaryotic DNA replication.

**Part B** Explain how a mutation during replication can affect a protein that is synthesized.

## Test TIP

Test questions may not be arranged in order of increasing difficulty. If you are unable to answer a question, mark it and move on to other questions.

## Modeling DNA Replication and Protein Synthesis

### OBJECTIVES

- Construct and analyze a model of DNA.
- Use a model to simulate the process of replication.
- Use a model to simulate the process of protein synthesis.

### PROCESS SKILLS

- demonstrating
- identifying
- manipulating a model

### MATERIALS

- plastic soda straws of two different colors, cut into 3 cm sections (54)
- metric ruler
- scissors
- permanent marker
- 54 pushpins (12 red, 12 blue, 12 yellow, 12 green, and 6 white)
- 54 paper clips
- 3 in. × 5 in. note cards
- oval-shaped card

### Background

1. Describe the structure of DNA.
2. State the base-pairing rules.
3. List the steps involved in the copying of DNA before cell division.

4. What are the roles of mRNA, rRNA and tRNA in protein synthesis?
5. Describe the process of transcription and the process of translation.

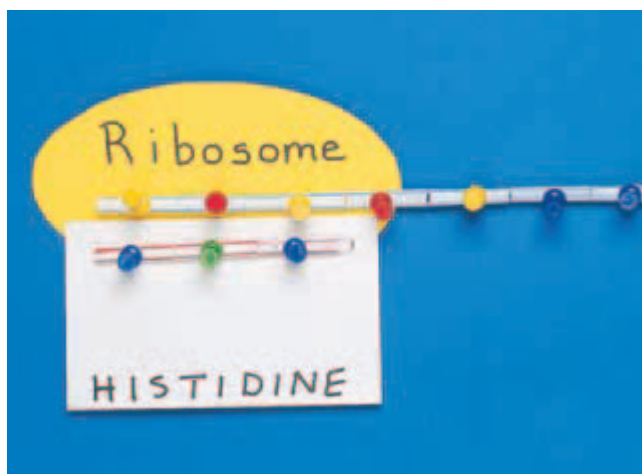
### PART A Making a Model of DNA

1.  **CAUTION** Sharp or pointed objects may cause injury. Handle pushpins carefully. Insert a pushpin midway along the length of each straw segment of one color, as shown in the figure below. Push a paper clip into one end of each straw segment until the clip touches the pin.



2. Keeping the pins in a straight line, insert the paper clip from a blue-pushpin segment into the open end of a red-pushpin segment. Add additional straw segments to the red-segment end in the following order: green, yellow, blue, yellow, blue, yellow, green, red, red, and green. Use the permanent marker to label the blue-segment end "top." This chain of segments is one-half of your first model.
3. Assign nucleotides to the corresponding pushpin colors as follows: red = adenine, blue = guanine, yellow = cytosine, and green = thymine.
4. Construct the other half of your first model. Begin with a yellow segment across from the blue pushpin at the top of your first model. Keep the pins in a straight line. Link segments together in this second strand of DNA according to the base-pairing rules.
5. When you have completed your model of one DNA segment, make a sketch of the model in your lab report. Use colored pencils or pens to designate the pushpin colors. Include a key that indicates which nucleotide each color represents in your sketch.





## PART B Modeling DNA Replication

6. Place the chains parallel to each other on the table. The "top" blue pin of the first chain should face the "top" yellow pin of the second chain.
7. Demonstrate replication by simulating a replication fork at the top pair of pins. Add the remaining straw segments to complete a new DNA model. Be sure to follow the base-pairing rules.
8. Sketch the process of DNA replication in your lab report. Label the replication fork, the segments of original DNA, and the segments of new DNA in your sketch.

## PART C Modeling Protein Synthesis

9. Place the chains of one of the DNA models parallel to each other on the table.
10. Repeat step 1, but use the straw segments of the second color.
11. Assign the uracil nucleotide to the white pushpins. Using the available pushpins and the second set of straw segments, construct a model of an mRNA transcript of the DNA segment. Begin by separating the two chains of DNA and pairing the mRNA nucleotides with the left strand of DNA as you transcribe from the top of the segment to the bottom of the segment.

12. In your lab report, sketch the mRNA model that you transcribed from the DNA segment.
13. Refer to Table 10-1 on page 207 and the photo at the left. Label the note cards with amino acids that you will need to translate your mRNA model. Use the "ribosome" oval cards to model translation.
14. In your lab report, write the sequence of amino acids that resulted from the translation.
15.  Clean up your materials before leaving the lab.

## Analysis and Conclusions

1. Write the base-pair order for the DNA molecule you created by using the following code: red = adenine, blue = guanine, yellow = cytosine, and green = thymine.
2. How does the replicated model of DNA compare with the original model of DNA?
3. Predict what would happen if the nucleotide pairs in the replicated model were not in the same sequence as the pairs in the original model.
4. What is the relationship between the anticodon of a tRNA and the amino acid the tRNA carries?
5. Write the mRNA transcript of the DNA sequence presented below.

CTG TTC ATA ATT

Next, write the tRNA anticodons that would pair with the mRNA transcript. Use Table 10-1 to write the amino acids coded for by the mRNA transcript.

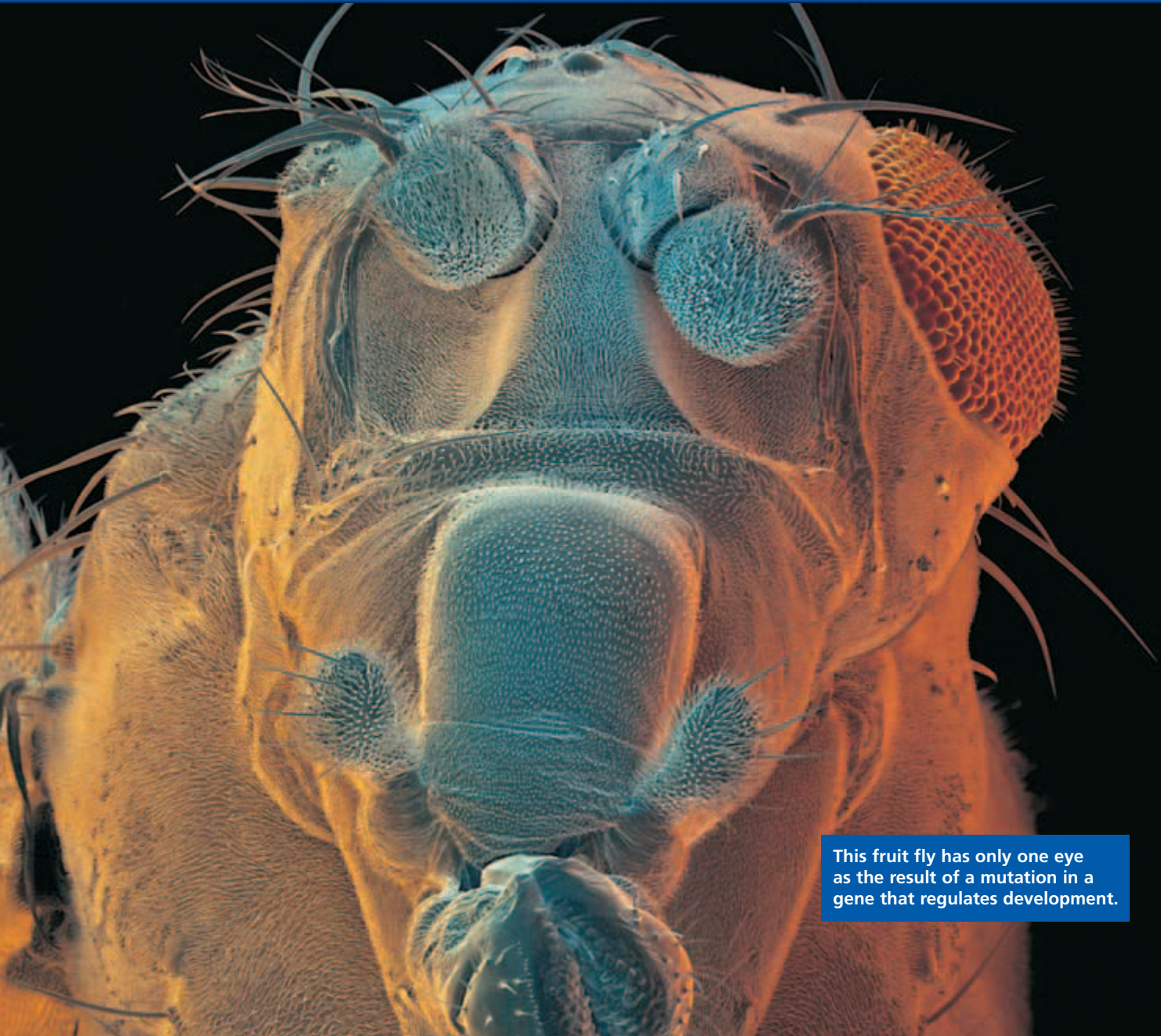
6. If you transcribed the "wrong" side of the DNA molecule, what would the result be? How might the proteins that the organism produced be affected?
7. What are the advantages of having DNA remain in the nucleus of eukaryotic cells?

## Further Inquiry

Design models to represent a eukaryotic and a prokaryotic cell. Use these models along with the models you constructed in this investigation to demonstrate where replication, transcription, and the steps of protein synthesis occur.



# GENE EXPRESSION



This fruit fly has only one eye as the result of a mutation in a gene that regulates development.

**SECTION 1** *Control of Gene Expression*

**SECTION 2** *Gene Expression in Development and Cell Division*



Unit 6—Gene Expression  
Topics 1–6

# CONTROL OF GENE EXPRESSION

*Cells use information in genes to build several thousands of different proteins, each with a unique function. But not all proteins are required by the cell at any one time. By regulating gene expression, cells are able to control when each protein is made.*

## ROLE OF GENE EXPRESSION

**Gene expression** is the activation or “turning on” of a gene that results in transcription and the production of mRNA. Most of the mRNA produced in cells is translated into proteins. But cells do not always need to produce all of the proteins for which their genes contain instructions. Recall that proteins have many different functions. Some proteins play a structural role. Others are enzymes that act as catalysts in chemical reactions. Mechanisms to control gene expression have evolved so that each protein is produced only when it is needed.

The complete genetic material contained in an individual is called the **genome** (JEE-NOHM). By regulating gene expression, cells are able to control which portion of the genome will be expressed and when. Most gene expression occurs at two steps, transcription and translation. Gene expression begins when the enzyme RNA polymerase transcribes the DNA nucleotide sequence of a gene into a specific mRNA. During translation, this mRNA then migrates to a ribosome, where it is translated into a specific protein.

## GENE EXPRESSION IN PROKARYOTES

Scientists first studied gene expression in prokaryotes. Much of our initial knowledge of gene expression came from the work of French scientists François Jacob (1920–) and Jacques Monod (1910–1976) at the Pasteur Institute in Paris. In the early 1960s, Jacob and Monod discovered how genes control the metabolism of the sugar lactose in *Escherichia coli*, a bacterium that lives in the intestines of mammals. Jacob and Monod won the Nobel Prize in 1965 for their discoveries.

## OBJECTIVES

- **Explain** why cells regulate gene expression.
- **Discuss** the role of operons in prokaryotic gene expression.
- **Determine** how repressor proteins and inducers affect transcription in prokaryotes.
- **Describe** the structure of a eukaryotic gene.
- **Compare** the two ways gene expression is controlled in eukaryotes.

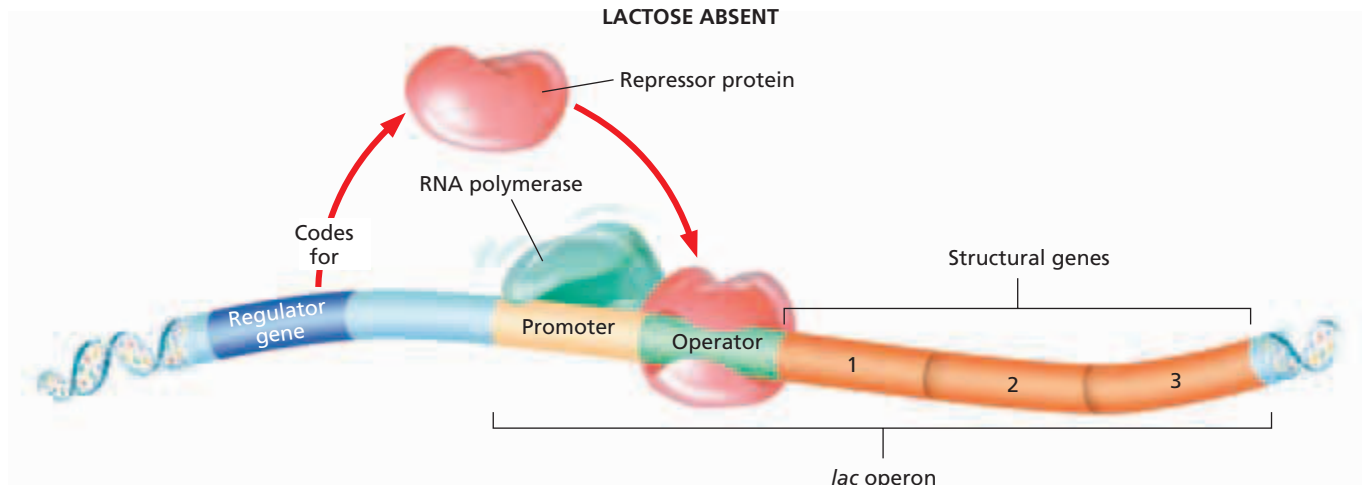
## VOCABULARY

gene expression  
genome  
structural gene  
operator  
operon  
*lac* operon  
repressor protein  
regulator gene  
inducer  
euchromatin  
intron  
exon  
pre-mRNA  
transcription factor  
enhancer

## Word Roots and Origins

**genome**

from the words *gene* and  
*chromosome*



**FIGURE 11-1**

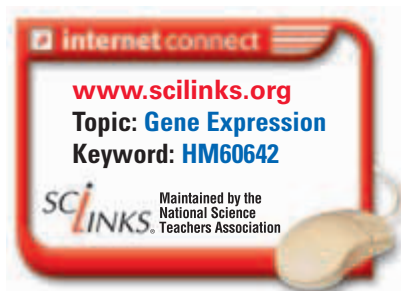
In the *lac* operon of *E. coli*, three structural genes code for the enzymes needed to utilize lactose. When lactose is absent, a repressor protein attaches to the operator. The presence of the repressor protein on the operator blocks the advancement of RNA polymerase.

Lactose is a disaccharide that is composed of the two monosaccharides glucose and galactose. When *E. coli* bacteria are in the presence of lactose, the lactose induces *E. coli* to produce three enzymes. These three enzymes control metabolism of lactose. The production of these enzymes is regulated by three elements found within the DNA of *E. coli*:

- **Structural Genes** Genes that code for polypeptides are called **structural genes**. The structural genes studied by Jacob and Monod code for enzymes that allow *E. coli* to metabolize lactose. The three structural genes that code for these three enzymes are located next to each other on the chromosome.
- **Promoter** Recall that a promoter is a DNA segment that is recognized by the enzyme RNA polymerase. This enzyme then initiates transcription.
- **Operator** An **operator** is a DNA segment that serves as a kind of “switch” by controlling the access of RNA polymerase to the promoter. Thus, the operator controls the ability of RNA polymerase to move along the structural genes.

The structural genes, the promoter, and the operator collectively form an operon. An **operon** (AHP-uhr-AHN) is a series of genes that code for specific products and the regulatory elements that control these genes. Researchers have found that the clustered arrangement of genes that form an operon is a pattern that occurs commonly among bacteria. Jacob and Monod named the operon that they studied the **lac operon** because its structural genes coded for the enzymes that regulate lactose metabolism. The *lac* operon, shown in Figure 11-1 above, includes the entire segment of DNA required to produce the enzymes involved in lactose metabolism.

Jacob and Monod found that the genes for the enzymes for lactose utilization were expressed only when lactose was present. How were the bacteria able to shut off these genes when lactose was absent? Their research showed that gene activation in the *lac* operon depends on whether the operon is “turned off” or “turned on.”





## Operon “Turned Off”

In the absence of lactose, a protein called a *repressor* attaches to the operator. A **repressor protein** is a protein that inhibits genes from being expressed. Repressor proteins are coded for by **regulator genes**, which are located some distance from the operators they affect. The attachment of the repressor protein to the operator physically blocks the advancement of RNA polymerase toward the structural genes and thus inhibits transcription. Figure 11-1 shows how the attachment of the repressor protein to the operator (the “switch”) causes the *lac* operon to “turn off.”

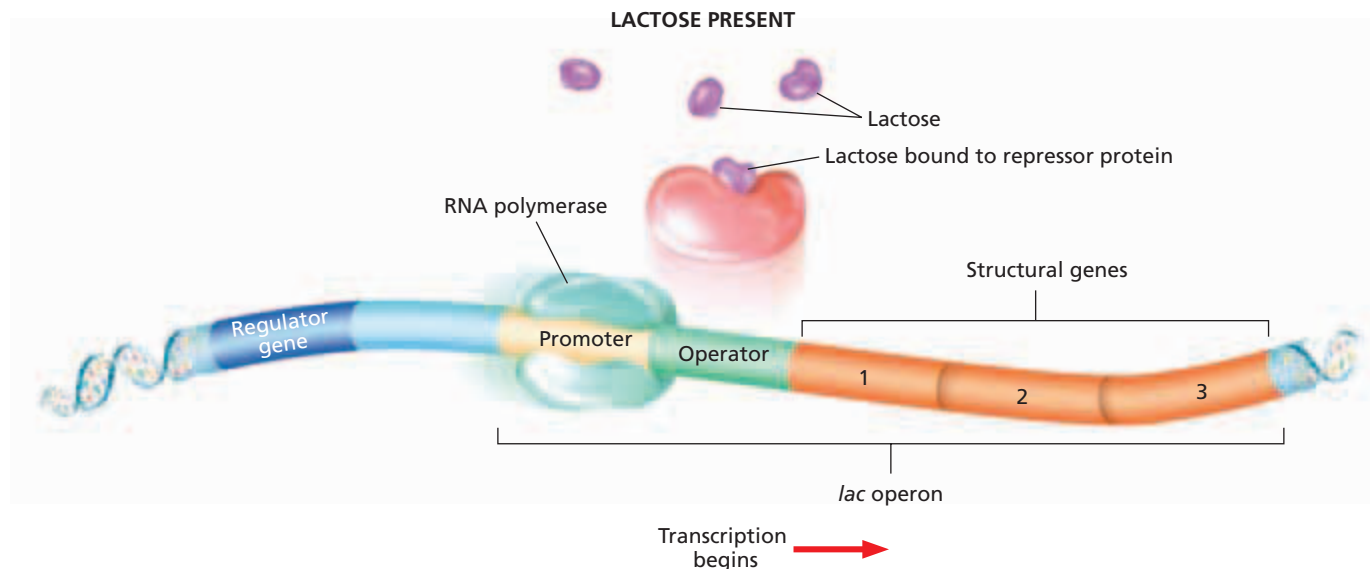
## Operon “Turned On”

When lactose is present in the *E. coli* cell, lactose binds to the repressor protein. This binding changes the shape of the repressor protein. The change in shape causes the repressor protein to detach from the operator (“the switch”), as shown in Figure 11-2 below. RNA polymerase is no longer blocked from transcribing the structural genes of the *lac* operon. The operon—including the three structural genes—is now “turned on,” so all three enzymes required for lactose metabolism are produced. Because it activates, or induces, transcription, lactose acts as an inducer. An **inducer** is a molecule that initiates gene expression.

The *lac* operon illustrates in simple terms the great advantage of regulating gene expression. Cells of *E. coli* are able to “turn off” or “turn on” lactose metabolism depending on whether lactose is present. Because lactose acts as an inducer, the *lac* operon is “turned on” only in the presence of lactose. As a result, lactose induces its own metabolism. When the level of lactose drops, the repressor protein again attaches to the operator, which “turns off” the *lac* operon. Therefore, the three enzymes used in lactose metabolism are not produced when lactose is not present. By controlling gene expression, *E. coli* bacteria conserve resources and produce only those proteins that are needed.

**FIGURE 11-2**

When lactose is present in an *E. coli* cell, lactose acts as an inducer by binding to the repressor protein. The repressor protein then changes shape and detaches. The detachment of the repressor protein allows the transcription of the three structural genes to proceed, and mRNA is produced.





# GENE EXPRESSION IN EUKARYOTES

Eukaryotes are vastly different from prokaryotes. Their genomes are much larger than those of prokaryotes. In addition, the DNA of eukaryotic cells is located in several individual chromosomes instead of in the single circular chromosome that occurs in prokaryotes. Finally, most eukaryotes are multicellular organisms made of specialized cells. Although each cell type contains a complete set of the organism's genes, only some of these genes are expressed at a given time. Different cell types produce different proteins. Not surprisingly, the control of gene expression in eukaryotes is far more complex than it is in prokaryotes. Although operons are common in prokaryotes, they have not been found often in eukaryotes.

## Structure of a Eukaryotic Gene

Much of the control of gene expression in eukaryotes occurs at the level of the individual chromosome. In eukaryotes, gene expression is partly related to the coiling and uncoiling of DNA within each chromosome. Recall that eukaryotic DNA is organized as fibers of chromatin wrapped around small specialized proteins called *histones*. Prior to mitosis or meiosis, the DNA and histones coil tightly, forming the structures we recognize as chromosomes. After mitosis or meiosis, certain regions of the DNA coils relax, thus making transcription possible. This uncoiled form, known as **euchromatin** (yoo-KROH-muh-tin), is the site of active transcription of DNA into RNA. However, some portions of the chromatin in specific cells remain permanently coiled, so their genes can never be transcribed. Thus, the degree to which DNA is uncoiled indicates the degree of gene expression.

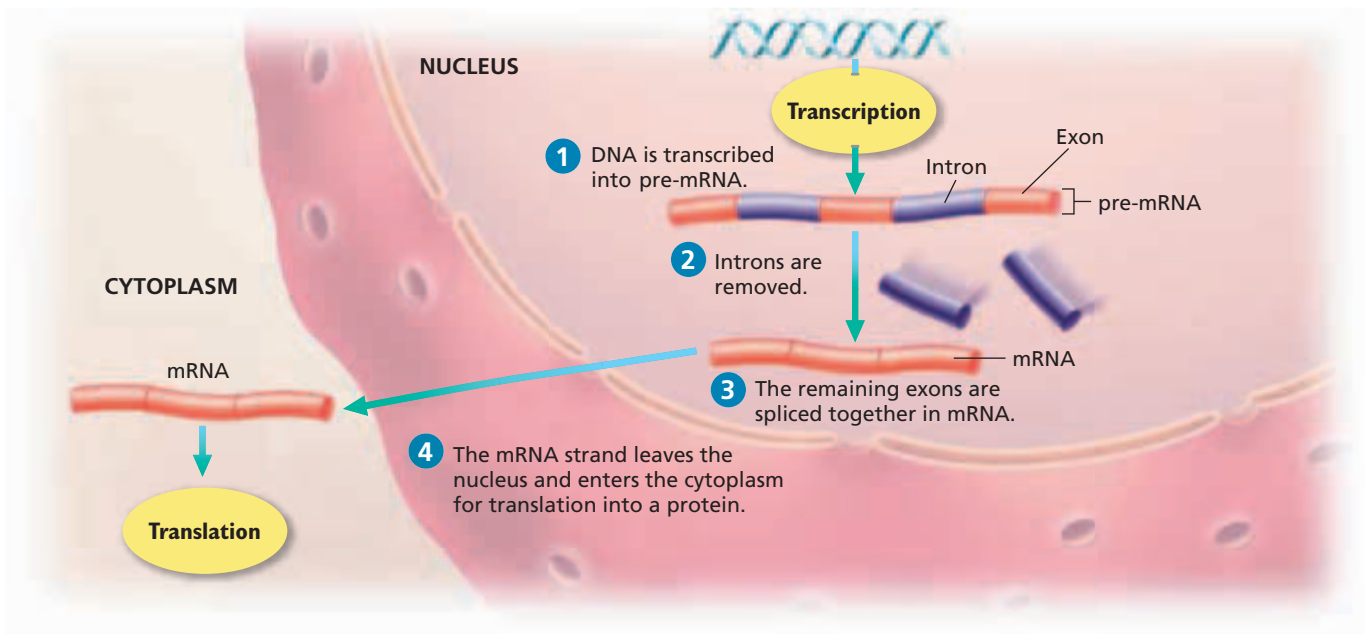
As in prokaryotes, the promoter is the binding site of RNA polymerase in eukaryotes. In the eukaryotic gene, there are two kinds of segments beyond the promoter: introns and exons. **Introns** (IN-trahnz) are sections of a structural gene that are transcribed but are not translated. **Exons** (EK-sahnz) are the sections of a structural gene that, when expressed, are transcribed and translated.

The benefits of the intron-exon pattern of gene organization are not yet fully understood. For many years, scientists were uncertain of the role of introns in the cell. However, recent research suggests that the noncoding RNA transcribed from introns performs important functions even though it is not translated. Some of these functions include regulating RNA that *is* translated, interacting with this coding RNA to influence gene expression, and acting as “switches” that allow protein production only when “turned on” by the presence of certain chemical targets. Scientists continue to explore the role of introns and noncoding RNA. For example, some researchers are investigating medicines that work by affecting the actions of introns and noncoding RNA.

### Word Roots and Origins

#### *intron* and *exon*

The “int” in the word *intron* comes from the “int” in the word *intervening*. The “ex” in the word *exon* comes from the “ex” in the word *expressed*.



## Control After Transcription

In prokaryotes, transcription and translation occur within the cytoplasm. In eukaryotes, however, transcription occurs in the nucleus, and then mRNA passes through the nuclear envelope and into the cytoplasm, where translation occurs. The physical separation of transcription and translation by the nuclear envelope gives eukaryotes more opportunities to regulate gene expression.

Unlike prokaryotes, eukaryotes can control gene expression by modifying RNA after transcription. When transcription occurs, both introns and exons are transcribed, as shown in step 1 of Figure 11-3. The result is a large molecule known as pre-mRNA. **Pre-mRNA** is a form of messenger RNA (mRNA) that contains both introns and exons. (Note that the terms *intron* and *exon* can be used to describe both DNA segments and the RNA segments that are transcribed.) A molecule of mRNA is formed when introns are removed (step 2) from pre-mRNA and the remaining exons are spliced (joined) to one another (step 3). Complex assemblies of RNA and protein called *spliceosomes* split the pre-mRNA at each end of an intron and join the exons. The end result is an mRNA molecule containing only the exons. The mRNA strand leaves the nucleus and enters the cytoplasm to begin the manufacture of a protein on the ribosomes (step 4). The nucleotides in the removed introns can be used again during the transcription of additional pre-mRNA. Similar RNA splicing occurs following the transcription of transfer RNA and ribosomal RNA.

The removal of introns and splicing of an mRNA molecule have also been found to occur in another way. Scientists have discovered that RNA molecules can act as biological catalysts. RNA itself can act as a catalyst to remove introns from mRNA molecules as they form in the nucleus. Until this discovery, it was thought that all enzymes were proteins. RNA molecules that act as enzymes are called *ribozymes*.

**FIGURE 11-3**

Both introns and exons are transcribed to form pre-mRNA. Spliceosomes cut out the introns and join the remaining exons together, forming mRNA.



## Quick Lab

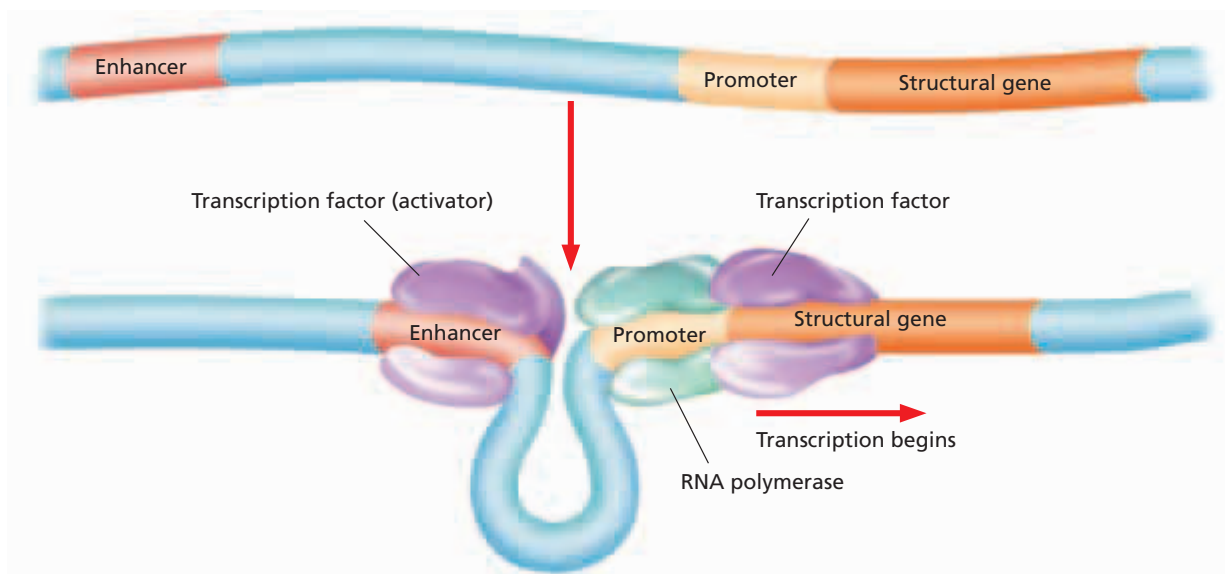
### Modeling Post-Transcription Control

**Materials** felt-tip markers, paper, scissors, tape

#### Procedure

1. Write a sentence that contains only three-letter words and makes sense.
2. Hide the words in random places in a long sequence of letters. This sequence should contain random letters and other three-letter words that make no sense in the sentence you are hiding. Print the sequence of letters all the same size, equally spaced, and with no breaks between them.
3. Trade papers with another team. Use scissors to cut out the "introns." Find the message, and reassemble it with tape.

**Analysis** What represents pre-mRNA in this activity? What represents mRNA?



**FIGURE 11-4**

Many enhancers are located far (thousands of nucleotide bases) away from the genes they activate. Transcription factors facilitate transcription by binding to the enhancer and to the promoter.

## Control at the Onset of Transcription

Most gene regulation in eukaryotes occurs when RNA polymerase binds to a gene—the onset of transcription. Eukaryotic cells, like prokaryotic cells, have regulatory genes. But eukaryotic gene regulation involves more proteins, and the interactions are more complicated. Regulatory proteins in eukaryotes are known as **transcription factors**.

Transcription factors help in the placement of RNA polymerase at the correct area on the promoter, as shown in Figure 11-4. Many different transcription factors may influence one gene.

Transcription factors may also bind sequences of DNA called **enhancers**. In general, enhancers are located at a position far—thousands of nucleotide bases away—from the promoter. A loop in the DNA may bring the enhancer and its *activator* (the attached transcription factor) into contact with the RNA polymerase and transcription factors at the promoter. Transcription factors bound to enhancers can activate transcription factors bound to promoters, as shown in Figure 11-4.

## SECTION 1 REVIEW

1. Why is it beneficial for organisms to control gene expression?
2. Describe the role of operons in prokaryotic organisms.
3. How does lactose affect the functioning of the *lac* operon?
4. Name the sections of eukaryotic genes that are transcribed and translated.
5. Distinguish between pre-mRNA and mRNA.

### CRITICAL THINKING

6. **Making Comparisons** What region of a prokaryotic gene is analogous to the enhancer region of a eukaryotic gene?
7. **Predicting Results** How would RNA polymerase be affected if the repressor protein were not bound to the proper site on a gene?
8. **Relating Concepts** How might the absence of a nuclear envelope in prokaryotes prevent prokaryotes from controlling gene expression by modifying RNA after transcription?

# GENE EXPRESSION IN DEVELOPMENT AND CELL DIVISION

*The control of gene expression plays an important role in the growth of eukaryotes as different cells become specialized to perform different tasks. When the expression of genes is altered—by mutations, for example—abnormalities and even cancer can result.*

## GENE EXPRESSION IN DEVELOPMENT

All multicellular, sexually reproducing organisms begin life as a fertilized egg, or zygote. Although every cell in the developing zygote contains all of the organism's genes, only a small number of the genes are expressed. Certain genes are turned on and off as various proteins are needed at different times during the organism's life. For example, as eukaryotes grow, cells become specialized to perform different tasks. Muscle cells specialize in movement, and liver cells specialize in making enzymes that break down fat. The development of cells that have specialized functions is known as **cell differentiation** (DIF-uh-EN-shee-AY-shuhn). As organisms grow and develop, organs and tissues develop to produce a characteristic form. This development of form in an organism is called **morphogenesis** (MOR-foh-JEN-uh-sis).

### Homeotic Genes

**Homeotic** (HOH-mee-AH-tik) **genes** are regulatory genes that determine where certain anatomical structures, such as appendages, will develop in an organism during morphogenesis. Homeotic genes seem to be master genes of development that determine the overall body organization of multicellular organisms.

When a homeotic gene is transcribed and translated, regulatory proteins are formed. It is thought that these proteins regulate development by switching groups of developmental genes on or off. Such control of gene expression increases or decreases the rates of cell division in various areas of the developing organism. The resultant variation in growth rates in specific areas of the organism produces specific patterns of structural development.

## OBJECTIVES

- **Summarize** the role of gene expression in an organism's development.
- **Describe** the influence of homeotic genes in eukaryotic development.
- **State** the role of the homeobox in eukaryotic development.
- **Summarize** the effects of mutations in causing cancer.
- **Compare** the characteristics of cancer cells with those of normal cells.

## VOCABULARY

cell differentiation  
homeotic gene  
homeobox  
proto-oncogene  
oncogene  
tumor  
cancer  
tumor-suppressor gene  
metastasis  
carcinogen  
carcinoma  
sarcoma  
lymphoma  
leukemia

## Word Roots and Origins

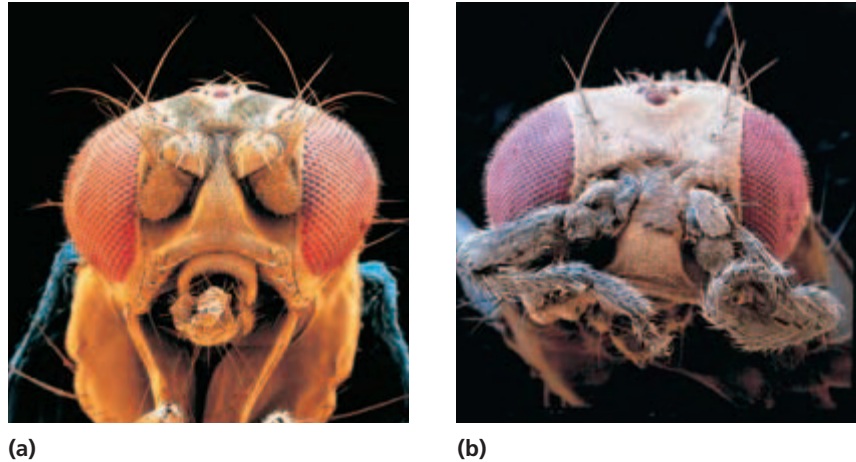
### homeotic

from the Greek *homoion*, meaning  
"to make like"



**FIGURE 11-5**

(a) Homeotic genes are expressed normally in this fruit fly. (b) This fruit fly has legs growing out of its head. This abnormality is caused by a homeotic mutation.

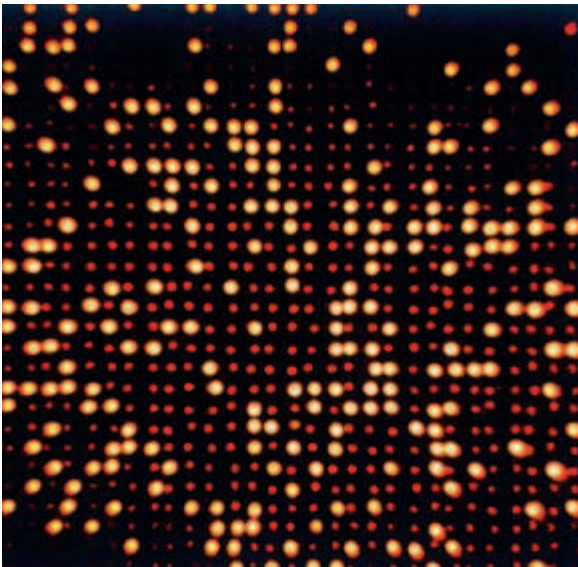


## Homeobox Sequences

One of the best-known examples of homeotic genes is found in fruit flies of the genus *Drosophila*, shown in Figure 11-5a. Each homeotic gene of this fruit fly shares a common DNA sequence of 180 nucleotide pairs. This specific DNA sequence within a homeotic gene is called a **homeobox**, and the homeobox codes for proteins that regulate patterns of development. As the fruit fly embryo becomes an elongated larva, specific homeoboxes control the morphogenesis of specific regions in the larva. Each of these homeoboxes will also control a specific part of the adult fruit fly. As Figure 11-5b shows, a mutation in a homeotic gene can lead to abnormalities. The same or very similar homeobox sequences have been found in homeotic genes of many eukaryotic organisms. It is thought that all organisms may have similar homeoboxes that code for their anatomy.

**FIGURE 11-6**

The lighted spots on this grid of a DNA chip indicate to scientists which genes are being expressed in the cells being studied.



## Tracking Changes in Gene Expression

The control of gene expression is important not only in the development of an organism but throughout the organism's life. Only a fraction of an organism's genes are expressed in any one cell. And cells constantly switch genes on and off. In the 1990s, researchers developed a tool for tracking gene expression called a *DNA chip*.

DNA chips contain a microscopic grid with thousands of known DNA fragments that are "tagged" with a fluorescent compound. A sample of mRNA from the organism being studied is spread over the grid. When spots on the grid light up, as shown in Figure 11-6, mRNA segments from the sample have linked with complementary sequences of DNA on the chip. Scientists can use this information to determine at once which genes are being expressed.

DNA chips have many practical applications but will likely have a significant impact in medicine. In fact, the technology is already being used to better understand gene expression in cancer.

# GENE EXPRESSION, CELL DIVISION, AND CANCER

The division of cells is regulated by many genes, including genes called **proto-oncogenes** (PROHT-oh-AHNG-kuh-JEENZ), which regulate cell growth, cell division, and the ability of cells to adhere to one another. These genes code for regulatory proteins that ensure that the events of cell division occur in the proper sequence and at the correct rate.

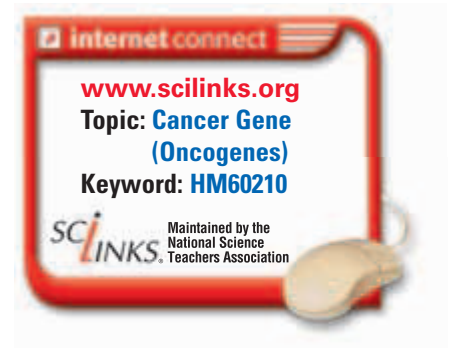
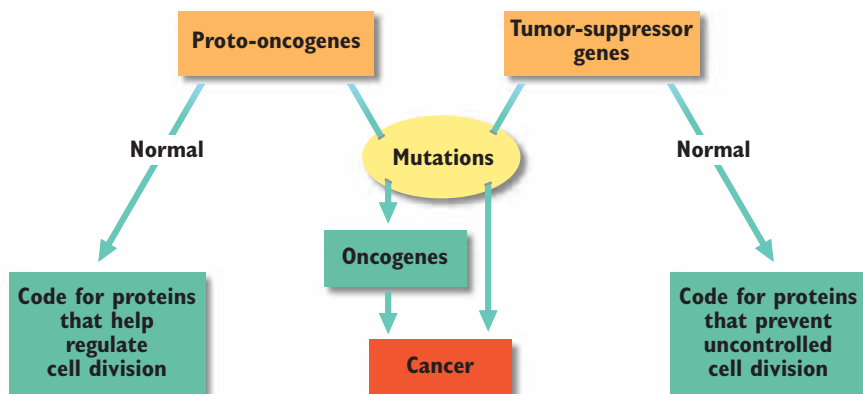
A mutation in a proto-oncogene can change the gene into an **oncogene**, a gene that can cause uncontrolled cell proliferation. The mutation may lead to the overexpression of proteins that initiate cell division or to the expression of such proteins at inappropriate times during the cell cycle. These conditions can lead to uncontrolled cell division.

## Tumor Development

A **tumor** is an abnormal proliferation of cells that results from uncontrolled, abnormal cell division. The cells that make up a *benign* (bi-NIEN) *tumor* remain within a mass. Benign tumors generally pose no threat to life unless they are allowed to grow until they compress vital organs. Examples of benign tumors are the fibroid cysts that can occur in a woman's breasts or uterus. Most benign tumors can be removed by surgery if necessary.

In a *malignant* (muh-LIG-nuhnt) *tumor*, the uncontrolled dividing cells may invade and destroy healthy tissues elsewhere in the body. This uncontrolled growth of cells that can invade other parts of the body is called **cancer**.

Some genes act as “brakes” to suppress tumor formation. **Tumor-suppressor genes** code for proteins that prevent cell division from occurring too often. In cancer, these tumor-suppressor genes are damaged, and a decrease in the activity of tumor-suppressing proteins can increase the rate of cell division. Cells have three types of tumor-suppressing genes, all of which must be damaged before cancer can occur. Figure 11-7 illustrates how mutations in proto-oncogenes and tumor-suppressor genes may lead to cancer.



**FIGURE 11-7**

Mutations in proto-oncogenes or tumor-suppressor genes can destroy normal gene functioning, possibly resulting in cancer. A mutation in a proto-oncogene may cause the gene to become an oncogene, a gene that can trigger cancer.

## Eco Connection

### Secondhand Tobacco Smoke

In 1992, the Environmental Protection Agency (EPA) declared secondhand tobacco smoke, also called *environmental tobacco smoke*, to be a human carcinogen. According to the EPA standards, secondhand smoke contains more than 3,000 chemical compounds, including four known human carcinogens and several other probable human carcinogens. In fact, the air in an enclosed room of smokers could contain up to six times the air pollution of a busy highway.

Thousands of nonsmokers die of lung cancer each year as a result of breathing secondhand tobacco smoke. There is no safe level of tobacco smoke. As more is learned about the contents and effects of tobacco smoke, regulations to protect people from secondhand smoke are being enacted across the United States. In fact, smoking indoors is now prohibited in many public places.

## Gene Expression in Cancer

The expression of oncogenes and mutated tumor-suppressor genes causes cancer cells to behave differently than normal cells. A normal cell must be attached to other cells, to a membrane, or to fibers between cells in order to divide. Also, normal mammalian cells will stop dividing after about 40 cell divisions—or sooner if they become too crowded. Cancer cells, however, continue to divide indefinitely, even when they are very densely packed, seemingly ignoring the normal cellular message to stop dividing. They also continue dividing even after they are no longer attached to other cells, a trait that facilitates the spread of cancer cells throughout the body. The spread of cancer cells beyond their original site is called **metastasis** (muh-TAS-tuh-sis). When metastasis occurs, cancer cells can invade healthy tissues and begin forming new tumors.

## Causes of Cancer

The mutations that alter the expression of genes coding for cell-regulating proteins can occur spontaneously but are more likely to occur as a result of the organism's exposure to a carcinogen (kahr-SIN-uh-juhn). A **carcinogen** is any substance that can induce or promote cancer. Most carcinogens are *mutagens* (MYOOT-uh-JUHNZ), agents that cause mutations to occur within a cell.

Well-known carcinogens include the chemicals in tobacco smoke, asbestos, and ionizing radiation, such as X rays or ultraviolet light from the sun. For example, tobacco smoke, shown in Figure 11-8, has been found to be the cause of more than 85 percent of all lung cancers. Certain viruses can cause cancer. Many viral genes are actually oncogenes. Viruses can also stimulate uncontrolled growth in host cells by causing mutations in proto-oncogenes or tumor-suppressor genes, thus accelerating the rate of cell division in the host cell. Or they may activate the cell's own oncogenes. Viruses have been found to cause some cancers in blood-forming tissues, and the human papilloma virus has been shown to cause cervical cancer.



**FIGURE 11-8**

Tobacco smoke contains more than 60 known carcinogens, including cyanide, formaldehyde, and lead.



# MILESTONES

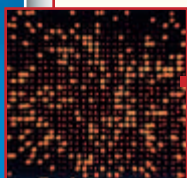
IN

## Recent DNA Research

### Timeline

**1983** The polymerase chain reaction is invented.

**1990** The Human Genome Project is launched.



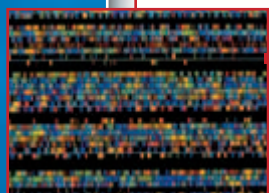
**1992** The first DNA chip patent is issued.

**1997** Dolly, a successful clone of an adult sheep, is born.



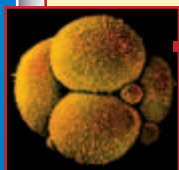
**1998** The first genome sequence of a multicellular organism, the roundworm, is completed.

**2002** Mouse genome sequencing is completed, one year ahead of schedule.



**2003** Sequencing of the human genome is completed.

**2003** Gene linked to heart attacks is discovered.



**2004** Korean scientists report the cloning of human embryos.

*Watson and Crick's model of the DNA double helix, established in 1953, has served as a foundation for the ever-growing body of DNA research. In the relatively brief time since the structure of DNA was first determined, several groundbreaking discoveries—many of them very recent—have expanded greatly our knowledge of DNA.*

One area in which many developments have been made in recent years is genetic technology. One of the most important of these was the development of the polymerase chain reaction (PCR) in 1983. PCR allows researchers to quickly make billions of copies of a specific segment of DNA. The ability to copy DNA in mass quantities has made the study of DNA much easier. Another important technological development is the DNA chip, patented in 1992. The DNA chip can be used to track gene expression in organisms. Other recent technological developments include fluorescence in-situ hybridization (FISH), a method used to identify specific parts of a chromosome, and spectral karyotyping (SKY), a form of FISH used to study complex changes in genetic material.

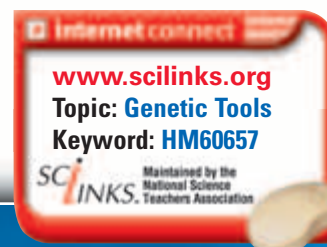
An enormous amount of research has been done in recent years to map the genomes of various organisms, including humans. The Human Genome Project was a cooperative, 13-year effort to map the human genome. The Human Genome Project was funded by the United States Government, contributions from several other countries, and private corporations. By the completion of sequencing in 2003, the U.S. government had spent over 437 million dollars. The project determined the sequence of the 3 billion base pairs that make up human DNA. Remarkably, all goals of the project were completed on time, and at a significantly lower cost than projected. As a result of the Human Genome Project, more than 1,400 genes related to disease have been identified.

Since 1995, scientists have sequenced the genomes of more than 150 organisms. In 1998, the first full sequencing of a genome in a multicellular organism—the roundworm—was completed. In 2002, sequencing of the mouse genome was completed. The mouse genome has its own version of nearly every human gene.

Along with all this recent DNA research come many difficult ethical questions. In 1997, the first successful cloning of an organism from differentiated cells resulted in the birth of Dolly the sheep. Since then many animals have been cloned. But Dolly began to suffer early from conditions normally found only in older sheep, raising questions about possible problems of premature aging in clones. Dolly was euthanized in 2003.

### Review

1. Name a practical application of the Human Genome Project.
2. **Critical Thinking** Why might it be useful to study the genomes of other organisms besides humans?
3. **Critical Thinking** Why might clones such as Dolly the sheep suffer from premature aging?





**FIGURE 11-9**

The top photo shows a healthy lung. The bottom photo shows carcinomas in a diseased lung. Lung cancer is one of the deadliest forms of cancer; 87 percent of lung cancer patients die within five years of diagnosis.



## Risks of Developing Cancer

Whether a person actually develops cancer seems to depend on many factors. Some families exhibit higher-than-average rates of certain cancers, leading researchers to determine that some people have a genetic predisposition to these types of cancer. With regard to cancers caused by mutagens, the number of exposures to the carcinogen and the amount of carcinogen in each exposure are significant factors. Mutations in gametes (egg or sperm cells) are especially important because these mutations are passed along to offspring.

Usually, more than one mutation is needed to produce a cancer cell. Perhaps this helps to explain why the cancer risk increases with the number of exposures to carcinogens and with the age of the individual. The longer an individual lives, the more mutations he or she will accumulate. But according to the National Cancer Institute in 2003, heightened awareness of the causes of cancer, combined with improved detection and treatment of the disease, has resulted in a decline in the number of deaths in the United States caused by the four most common cancers. The death rate for all cancers combined has also stabilized.

## Kinds of Cancer

Malignant tumors can be categorized according to the types of tissues affected. **Carcinomas** (KAHR-suh-NOH-muhz) grow in the skin and the tissues that line the organs of the body. Lung cancer, shown in Figure 11-9, and breast cancer are examples of carcinomas. **Sarcomas** (sahr-KOH-muhz) grow in bone and muscle tissue. **Lymphomas** (lim-FOH-muhz) are solid tumors that grow in the tissues of the lymphatic system. Tumors in blood-forming tissues may cause **leukemia** (loo-KEE-mee-uh), the uncontrolled production of white blood cells. Usually, it takes several years for cancer to develop. However, when a vital organ, such as the liver or pancreas, is involved, the symptoms caused by organ dysfunction due to cancer may develop more rapidly.

## SECTION 2 REVIEW

1. How can morphogenesis be affected by the control of gene expression?
2. What is the role of homeotic genes in fruit flies of the genus *Drosophila*?
3. Explain the relationship between a homeobox and a homeotic gene.
4. Describe how mutations in proto-oncogenes or tumor-suppressor genes can lead to cancer.
5. List three ways in which cancer cells differ from normal cells.

### CRITICAL THINKING

6. **Relating Concepts** Why might X rays be more dangerous to an ovary than to muscle tissue?
7. **Predicting Patterns** Tobacco products were first introduced in Europe in the late 1500s. Draw a graph showing a possible pattern of lung cancer rates in Europe over the past 1000 years.
8. **Inferring Relationships** What does the presence of similar homeobox sequences among many eukaryotic organisms suggest about the possible evolutionary relationships between these organisms?

# CHAPTER HIGHLIGHTS

## SECTION 1

## Control of Gene Expression

- Gene expression is the activation of a gene that results in transcription and the production of mRNA. Only a fraction of any cell's genes are expressed at any one time.
- A promoter and an operator regulate the transcription of structural genes. In prokaryotes, the structural genes, the promoter, and the operator collectively form an operon.
- A promoter is the segment of DNA that is recognized by the enzyme RNA polymerase, which then initiates transcription. An operator is the segment of DNA that acts as a "switch" by controlling the access of RNA polymerase to the promoter.
- A repressor protein can inhibit genes from being expressed. Repressor proteins are coded for by regulator genes. A repressor protein attaches to the operator, physically blocking the advancement of RNA polymerase.
- An inducer is a molecule that initiates gene expression. In *E. coli*, lactose serves as an inducer. An inducer binds to the repressor protein. As a result, the shape of the repressor protein changes, and the repressor protein detaches from the operator. RNA polymerase can then advance to the structural genes.
- Eukaryotes do not have operons. The genomes of eukaryotes are larger and more complex than those of prokaryotes.
- Eukaryotic genes are organized into noncoding sections, called *introns*, and coding sections, called *exons*.
- In eukaryotes, gene expression can be controlled after transcription—through the removal of introns from pre-mRNA—or at the onset of transcription—through the action of transcription factors.

### Vocabulary

gene expression (p. 217)  
genome (p. 217)  
structural gene (p. 218)  
operator (p. 218)

operon (p. 218)  
*lac* operon (p. 218)  
repressor protein (p. 219)  
regulator gene (p. 219)

inducer (p. 219)  
euchromatin (p. 220)  
intron (p. 220)  
exon (p. 220)

pre-mRNA (p. 221)  
transcription factor (p. 222)  
enhancer (p. 222)

## SECTION 2

## Gene Expression in Development and Cell Division

- The development of specialized cells is called *cell differentiation*. The development of form in an organism is called *morphogenesis*. Both cell differentiation and morphogenesis are governed by gene expression.
- Homeotic genes are regulatory genes that determine where anatomical structures will be placed during development.
- Within each homeotic gene, a specific DNA sequence known as the *homeobox* regulates patterns of development. The homeoboxes of many eukaryotic organisms appear to be very similar.
- Mutations of proto-oncogenes or tumor-suppressor genes may lead to cancer. Cancer is the uncontrolled growth of abnormal cells.
- A carcinogen is any substance that can induce or promote cancer. Most carcinogens are mutagens, substances that cause mutations.
- Unlike normal cells, cancer cells continue to divide indefinitely, even if they become densely packed. Cancer cells will also continue dividing even if they are no longer attached to other cells.

### Vocabulary

cell differentiation (p. 223)  
homeotic gene (p. 223)  
homeobox (p. 224)  
proto-oncogene (p. 225)

oncogene (p. 225)  
tumor (p. 225)  
cancer (p. 225)

tumor-suppressor gene (p. 225)  
metastasis (p. 226)  
carcinogen (p. 226)

carcinoma (p. 228)  
sarcoma (p. 228)  
lymphoma (p. 228)  
leukemia (p. 228)

# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
  - operator* and *operon*
  - proto-oncogene* and *oncogene*
  - intron* and *exon*
  - homeotic gene* and *homeobox*
- Explain the relationships between carcinogen and mutagen.
- Use the following terms in the same sentence: *gene*, *gene expression*, *regulator gene*, and *repressor protein*.
- Word Roots and Origins** The word *morphogenesis* is derived from the Greek *morphe*, which means “shape,” and the Latin *genus*, which means “birth.” Using this information, explain why the term *morphogenesis* is a good name for the biological process that the term describes.

## UNDERSTANDING KEY CONCEPTS

- Identify** the term that describes the activation of a gene that results in transcription and the production of mRNA.
- Name** the kind of organism in which gene expression was first observed.
- Describe** how *E. coli* benefit by making enzymes to utilize lactose only when lactose is in the cellular environment.
- Explain** what causes the *lac* operon to “turn off” and “turn on.”
- Compare** pre-mRNA with mRNA.
- Describe** the role of enhancers in the control of gene expression.
- Evaluate** the relationship between gene expression and morphogenesis.
- Identify** the role of homeoboxes in morphogenesis.
- Compare** the roles of proto-oncogenes and tumor-suppression genes.

- State** the unusual characteristic of cancer cells that can lead to metastasis.

- Define** *carcinogen*.

- Unit 6—Gene Expression**

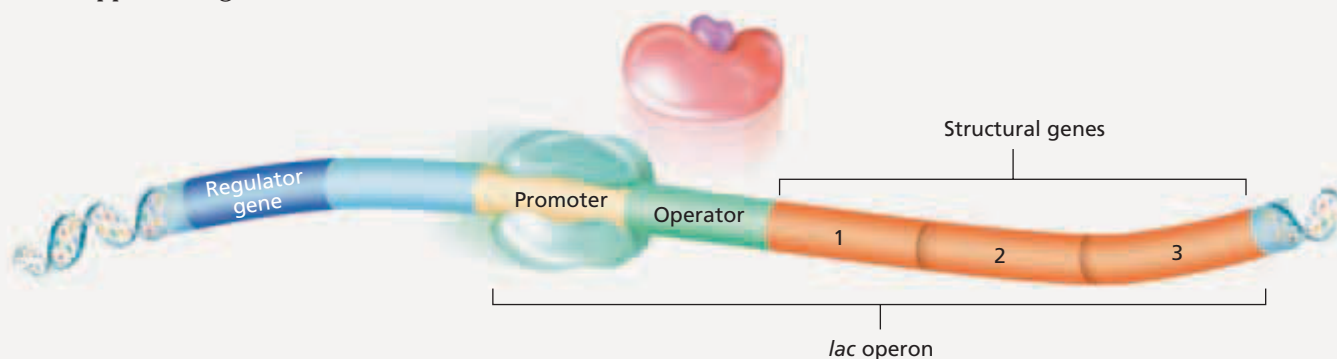


Write a report describing the influence of homeotic genes on an organism's development.

- CONCEPT MAPPING** Use the following terms to create a concept map that shows how a mutated gene can lead to cancer: *exon*, *gene expression*, *intron*, *mutagen*, *oncogene*, *proto-oncogene*, *tumor-suppressor gene*, and *tumor*.

## CRITICAL THINKING

- Analyzing Information** Kwashiorkor is a disease in children caused by a diet high in carbohydrates but lacking in complete protein. When children with kwashiorkor are put on a diet rich in protein, they may become very ill with ammonia poisoning, and some even die. The high level of ammonia in their blood is due to the inadequate metabolism of protein. What does this tell you about the enzymes that metabolize protein?
- Relating Concepts** Fruit flies of the genus *Drosophila* feed on fermenting fruit, which often contains a large amount of alcohol. If these fruit flies are fed a diet that has a high alcohol content, the amount of the enzyme that metabolizes alcohol in the digestive tract increases. What does this increase tell you about the enzyme?
- Interpreting Graphics** Study the diagram of the *lac* operon shown below.
  - Describe the role of the following elements shown in the diagram: promoter, operator, and structural genes.
  - What does it mean to say that an operon is “turned on”?
  - Is the operon “turned on” in the diagram shown below?



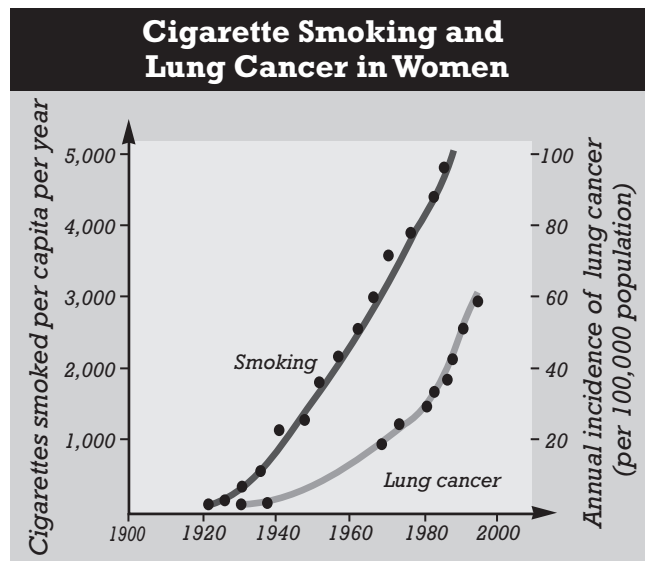


# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- Which of the following codes for a repressor protein?
  - enhancer
  - promoter
  - regulator gene
  - structural gene
- Which component of an operon controls the advancement of RNA polymerase?
  - exon
  - operator
  - promoter
  - structural gene
- Pre-mRNA contains which of the following?
  - exons only
  - introns only
  - both introns and exons
  - neither introns nor exons

**INTERPRETING GRAPHICS:** The graph below shows the number of cigarettes smoked per capita per year between 1920 and 2000 and the annual incidence of lung cancer among women. Use the graph to answer the question that follows.

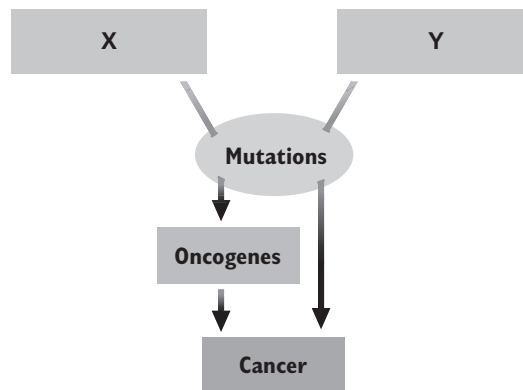


- What was the relationship between number of cigarettes smoked and incidence of lung cancer?
  - There was no relationship between cigarette smoking and lung cancer.
  - As the number of cigarettes smoked decreased, the incidence of lung cancer increased.
  - As the number of cigarettes smoked increased, the incidence of lung cancer increased.
  - As the number of cigarettes smoked increased, the incidence of lung cancer decreased.

**DIRECTIONS:** Complete the following analogy.

- skin : carcinoma :: blood-forming tissue :
  - sarcoma
  - leukemia
  - lymphoma
  - carcinogen

**INTERPRETING GRAPHICS:** The diagram below shows how mutations in certain genes can lead to cancer. Use the diagram to answer the questions that follow.



- What does *X* represent?
  - mutagens
  - carcinogens
  - proto-oncogenes
  - tumor-suppressor genes
- What does *Y* represent?
  - mutagens
  - carcinogens
  - proto-oncogenes
  - tumor-suppressor genes

## SHORT RESPONSE

A biologist isolates mRNA from a mouse brain and liver and finds that the two types of mRNA differ.

Can these results be correct, or has the biologist made an error? Explain your answer.

## EXTENDED RESPONSE

Mutations may occur in gametes or in body cells.

**Part A** In which cell type could a mutation cause genetic variation in a population?

**Part B** Explain how genetic variation could result from a mutation in this cell type.

## Test TIP

When using a graph to answer a question, make sure you know what variables are represented on the *x*- and *y*-axes before answering.



# Modeling Gene Expression in the *lac* Operon

## OBJECTIVES

- Make a model of the *lac* operon.
- Demonstrate the mechanisms that regulate gene expression in the *lac* operon of *Escherichia coli*.
- Simulate the transcription of the structural genes in the *lac* operon.

## PROCESS SKILLS

- comparing and contrasting
- identifying
- demonstrating
- manipulating a model

## MATERIALS

- pipe cleaner
- large colored beads, 6
- colored modeling clay in three colors
- labeling tape
- marking pen
- pencil
- paper

## Background

1. Define *gene*.
2. What is the role of RNA polymerase in protein synthesis?
3. Where does protein synthesis occur? What is the function of mRNA in protein synthesis?
4. What is the role of ribosomes during protein synthesis?
5. What are the roles of the operator, promoter, and structural genes within the *lac* operon?
6. How does the presence or absence of lactose affect the *lac* operon?
7. What is a regulator gene?

## PART A Making a Model of the *lac* Operon

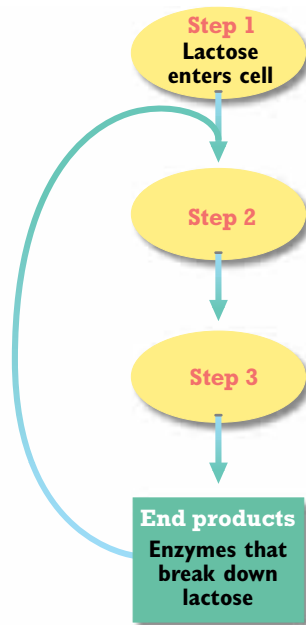
1. In this investigation, you will use the materials provided to make a model of a *lac* operon. Allow the pipe cleaner to represent the portion of DNA that constitutes the *lac* operon.
2. Thread the pipe cleaner through three beads of similar size, shape, and color. These three beads represent the structural genes of the *lac* operon.
3. Add one bead to represent the operator portion of the *lac* operon. Also, add beads to represent the promoter and the regulator gene, respectively.
4. Using labeling tape and a marking pen, label each of the beads you have placed on the pipe cleaner. This represents a model of the *lac* operon.
5. Compare the sequence of the labeled beads on the pipe cleaner with the sequence of segments in the diagram of the *lac* operon in the chapter. When your model of the *lac* operon correctly reflects the parts of the *lac* operon in the figure, proceed to Part B.

## PART B Modeling the *lac* Operon When It Is “Turned Off”

6. Choose one color of modeling clay to represent the enzyme RNA polymerase, and choose another color to represent the repressor molecule. Use the modeling clay to mold an RNA polymerase molecule and a repressor molecule.
7. Using the molecules you made out of clay in step 6, modify your model of the *lac* operon so that it shows the *lac* operon when it is “turned off.”
8. In your lab report, draw your model of the *lac* operon when it is “turned off.” Label all parts of your drawing. How does the presence of the repressor molecule prevent transcription of the structural genes?

### PART C Modeling the *lac* Operon When It Is “Turned On”

9. Choose a third color of modeling clay to represent the inducer molecule. Use the modeling clay to form an inducer molecule.
10. Using the inducer molecule you made out of clay, modify your model of the *lac* operon so that it shows the *lac* operon when it is “turned on.”
11. Simulate the activation of the *lac* operon and the transcription of the structural genes.
12. In your lab report, prepare a diagram of your model that shows the expression of the structural genes in the *lac* operon. Include ribosomes and mRNA in your diagram. Label all parts of your diagram.
13. The graphic organizer below shows the sequence of steps that occurs after lactose enters *E. coli* cells. Copy this graphic organizer in your lab report. Complete the graphic organizer by describing what takes place during step 2 and step 3. Explain how the end product affects the events shown in the graphic organizer.
14.  Clean up your materials before leaving the lab.



### Analysis and Conclusions

1. What substance serves as an inducer in the *lac* operon?
2. How might a mutation in the regulator gene affect the *lac* operon?
3. Look at the diagram you made in step 12. Refer to your diagram, and predict what will happen when the inducer is no longer present.
4. How would the loss of the promoter site from the operon affect the production of the enzymes needed to utilize lactose?
5. In homes and apartments, a consistent temperature is maintained by means of a thermostat, which regulates when heating (or air conditioning) is turned on or off. In what way does the *lac* operon function like a thermostat?
6. Biological processes often take place in a series of sequential steps called a *biochemical pathway*. Many biochemical pathways are controlled by feedback inhibition. In feedback inhibition, a pathway's end product affects an earlier step in the pathway and causes the pathway to stop. Explain how the function of the *lac* operon is similar to the process of feedback inhibition.

### Further Inquiry

1. Use classroom or library references to find examples of feedback inhibition in biology. Describe why models of feedback inhibition are sometimes called *feedback loops*.
2. The products of the *lac* operon are produced when lactose is present. In this way, the presence of a specific molecule stimulates transcription of the structural genes. In contrast, some operons are repressed when a specific molecule is present. Use classroom or library references to find out how the *trp* operon functions in *E. coli*. Then, compare the function of the *trp* operon with the function of the *lac* operon.

# INHERITANCE PATTERNS AND HUMAN GENETICS

Almost every human body cell except a sperm or an egg has 23 pairs of chromosomes. Each chromosome contains thousands of genes that play an important role in how a person develops, functions, and grows.

## SECTION 1 *Chromosomes and Inheritance*

## SECTION 2 *Human Genetics*



Unit 5—Heredity  
Topics 1–6

# CHROMOSOMES AND INHERITANCE

*Francis Collins and his lab group discovered the gene responsible for cystic fibrosis (CF). Cystic fibrosis often is a fatal genetic disorder. Thick, sticky mucus builds up and blocks ducts in the pancreas and intestines and causes difficulty in breathing. In this chapter, you will learn how diseases, such as CF, and characteristics, such as eye color, are inherited and expressed.*

## CHROMOSOMES

Jeff Pinard, a student in Collins's lab, studied more about how CF is inherited. Pinard, shown in Figure 12-1, has CF. Pinard and the rest of Collins's group were able to study the CF gene in part because of work carried out by geneticists in the early 1900s.

### Early Work

In the early 1900s, researcher Thomas Hunt Morgan began experimenting with the small fruit fly *Drosophila melanogaster*. Morgan observed that the flies have four pairs of chromosomes. He also observed that three of the pairs were identical in both females and males, but one pair differed in size and shape. In females, the fourth pair had two identical chromosomes, now called *X chromosomes*. In males, the fourth pair had one X chromosome, but also a shorter chromosome, now called a *Y chromosome*. Today, geneticists call the X and Y chromosomes *sex chromosomes*.



**FIGURE 12-1**

Student Jeff Pinard uses molecular techniques to learn about the genetic variation that causes his symptoms of cystic fibrosis (CF).

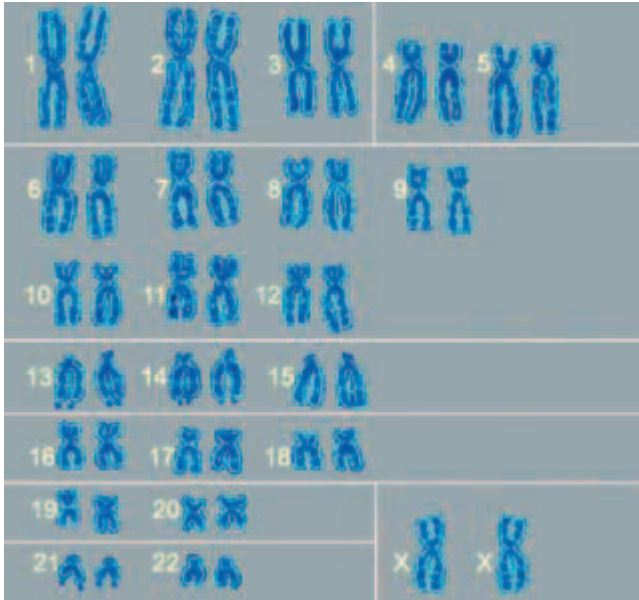
## OBJECTIVES

- **Distinguish** between sex chromosomes and autosomes.
- **Explain** the role of sex chromosomes in sex determination.
- **Describe** how an X- or Y-linked gene affects the inheritance of traits.
- **Explain** the effect of crossing-over on the inheritance of genes in linkage groups.
- **Distinguish** between chromosome mutations and gene mutations.

## VOCABULARY

sex chromosome  
autosome  
sex-linked trait  
linked gene  
chromosome map  
map unit  
germ-cell mutation  
somatic-cell mutation  
lethal mutation  
deletion  
inversion  
translocation  
nondisjunction  
point mutation  
substitution  
frameshift mutation  
insertion mutation





(a) Female karyotype



(b) Male karyotype

### FIGURE 12-2

Human female and male karyotypes have in common 22 chromosome pairs, called *autosomes*. The karyotype of females (a) differs from that of males (b) in only the 23rd pair, which consists of the sex chromosomes. The 23rd chromosome pair has two X chromosomes in females and one X chromosome and one Y chromosome in males.

## Sex Chromosomes and Autosomes

The **sex chromosomes** contain genes that determine the sex (gender) of an individual. The remaining chromosomes that are not directly involved in determining the sex of an individual are called **autosomes**. As in fruit flies, human males have an X and a Y chromosome, and human females have two X chromosomes. Figure 12-2a shows an example of the 23 pairs of human chromosomes from a female; Figure 12-2b shows an example of chromosomes from a male. In certain organisms, such as chickens and moths, males have two identical sex chromosomes, and females have two different sex chromosomes. Most plants and some fish lack sex chromosomes entirely.

## Sex Determination

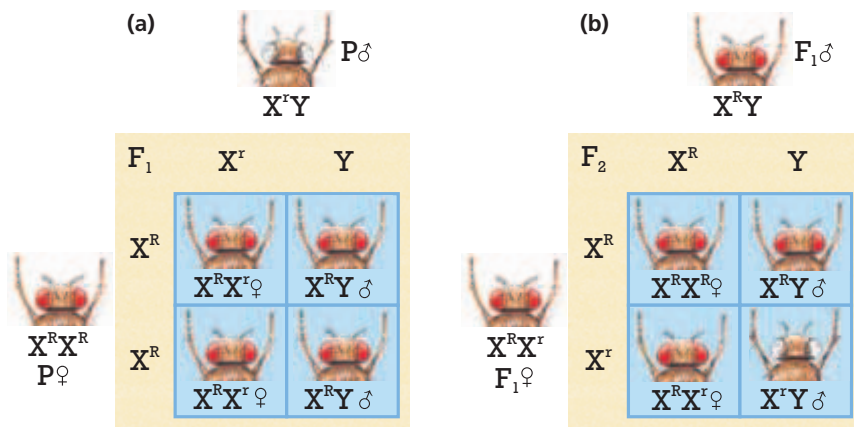
Like other homologous chromosomes, sex chromosomes pair during meiosis I. As meiosis proceeds, the paired chromosomes separate and move to different cells. As a result, a sperm cell has an equal chance of receiving an X chromosome or a Y chromosome. Each egg, however, receives only an X chromosome. This system of sex determination results in a one-to-one ratio of males to females. Each egg and sperm cell also receives a single copy of each autosome.

In mammals, when an egg that carries an X chromosome is fertilized by a sperm carrying a Y chromosome, the offspring has an XY pair and is male. Likewise, when an egg is fertilized by a sperm cell carrying an X chromosome, the offspring has an XX pair and is female. In a male mammal, the Y chromosome contains a gene called *SRY* for **Sex-determining Region Y**. This gene codes for a protein that causes the gonads of an embryo to develop as testes. Because female embryos do not have the *SRY* gene, the gonads develop as ovaries.

### Word Roots and Origins

#### **autosome**

from the Greek *autos*, meaning "self," and *soma*, meaning "a body"



**FIGURE 12-3**

Eye color is a sex-linked trait in fruit flies, as these two Punnett squares show. (a) A male ( $\delta$ ) with white eyes mated to a female ( $\text{♀}$ ) with red eyes yields all red-eyed F<sub>1</sub> offspring. (b) An F<sub>1</sub> cross results in an F<sub>2</sub> generation in which all females are red-eyed, half of the males are red-eyed, and half of the males are white-eyed.

## EFFECTS OF GENE LOCATION

When Morgan was doing his research with fruit flies, one of the lab members noticed that a single male fruit fly had white eyes instead of the red eyes that are normally found in the flies. Morgan crossed this white-eyed male with a normal red-eyed female, and found that all the F<sub>1</sub> offspring had red eyes, as shown in Figure 12-3a. This demonstrated that the red-eye trait is dominant to the white-eye trait. Morgan next crossed F<sub>1</sub> males with F<sub>1</sub> females, as shown in Figure 12-3b. The resulting F<sub>2</sub> generation showed the expected ratio of three red-eyed flies to one white-eyed fly. Unexpectedly, however, all of the white-eyed flies were male.

### Sex-Linked Genes and Traits

Based on this surprising observation, Morgan hypothesized that the gene for eye color is carried on the X chromosome and that the Y chromosome lacks an allele for the eye-color gene. An X chromosome carries a gene for eye color, either  $X^R$  (red-eye allele) or  $X^r$  (white-eye allele). In a cross of an  $X^R X^R$  female (red-eyed) with an  $X^r Y$  male (white-eyed), all of the F<sub>1</sub> females will be  $X^R X^r$  (red-eyed), and all of the F<sub>1</sub> males will be  $X^R Y$  (red-eyed).

In the F<sub>2</sub> generation, half of the females will be  $X^R X^R$ , and the other half will be  $X^R X^r$ . Because all have the dominant allele  $R$ , all will be red-eyed. In the F<sub>2</sub> males, however, half will be  $X^R Y$  (red-eyed), but the other half will be  $X^r Y$  (white-eyed).

The results of these experiments showed Morgan not only that genes reside on chromosomes but also that the red eye-color gene resides on the X chromosome. Morgan called genes located on the X chromosome *X-linked genes*. He called genes on the Y chromosome, such as *SRY* in humans, *Y-linked genes*. The term **sex-linked trait** refers to a trait that is coded for by an allele on a sex chromosome. The X chromosome is much larger than the Y chromosome, so there are more X-linked than Y-linked traits. Most X-linked alleles have no homologous counterpart on the Y chromosome. Because males have only one X chromosome, a male who carries a recessive allele on the X chromosome will exhibit the sex-linked trait.



### Quick Lab

#### Modeling Linkage

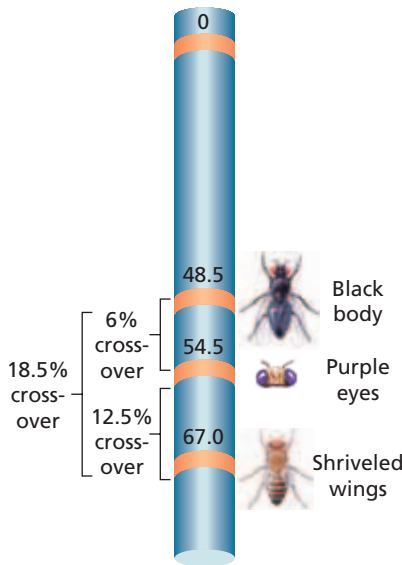
**Materials** Two kinds of candy, toothpicks, pencil, paper

**Procedure** Use two kinds of candy, each kind of which has two colors, to represent genes for two traits. Long noses are dominant over short noses. Large ears are dominant over small ears. One color of candy will represent the dominant allele, and a different color candy will represent the recessive allele. Use these materials to determine the outcome of a cross between two individuals, each heterozygous for both traits. Your teacher will tell you if the genes are linked or not.

1. Draw a Punnett square. Use the appropriate alleles to make gametes for each individual. Then place the allele combinations in each square representing the possible zygotes from that cross.
2. If your genes are linked, you must use toothpicks to link the genes together before you arrange gametes on your Punnett square.

**Analysis** What is the phenotypic ratio in the offspring when the genes are not linked? What is the phenotypic ratio when the genes are linked? Explain the difference.

### GENE LOCATIONS ON A FRUIT FLY CHROMOSOME MAP



**FIGURE 12-4**

The cross-over frequency between the gene for black body and the gene for purple eyes is 6 percent, so these two genes are 6 map units apart. The cross-over frequency between the purple-eye gene and the shriveled-wing gene is 12.5 percent, which is equivalent to 12.5 map units. The black-body gene and the shriveled-wing gene are 18.5 map units apart.

## Linked Genes

Morgan and other geneticists hypothesized that if genes are inherited together, the reason is that they occur on the same chromosome. For example, Morgan studied two fly genes—one for body color and one for wing length—located on the same autosome. Gray body, *G*, was dominant to black body, *g*, and long wings, *L*, were dominant to short wings, *l*. Morgan crossed gray-bodied, long-winged (*GGLL*) flies with black-bodied, short-winged (*ggll*) flies. All the  $F_1$  offspring had the genotype *GgLl* and were gray with long wings.

Morgan then crossed members of the  $F_1$  generation with one another (*GgLl*  $\times$  *GgLl*) to produce an  $F_2$  generation. The flies in the  $F_2$  generation occurred in a phenotypic ratio of three gray, long-winged flies to one black, short-winged fly. If the alleles of the two genes had been located on different chromosomes, they would have assorted independently and produced an  $F_2$  generation with a phenotypic ratio of 9:3:3:1 as in Mendel's peas. Morgan called pairs of genes that tend to be inherited together **linked genes**, and he called a set of linked genes a *linkage group*.

Morgan hypothesized that genes are linked because they are found on the same chromosome. An unexpected observation helped confirm this hypothesis. His  $F_2$  crosses produced a few offspring unlike either parent, with gray bodies and short wings (*Ggll*) or black bodies and long wings (*ggLl*). Morgan realized mutations are too rare to explain all the exceptions he saw. Morgan thus inferred that the natural rearrangement process during crossing-over must be responsible. Recall that *crossing-over* is the exchange of pieces of DNA between homologous chromosomes. Crossing-over during the first division of meiosis does not create new genes or delete old ones. Instead, it rearranges allele combinations.

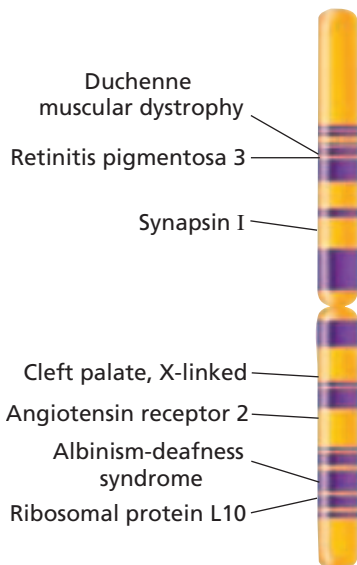
## Chromosome Mapping

The farther apart two genes are located on a chromosome, the more likely a cross-over will occur. The greater the percentage of  $F_2$  offspring showing recombinant traits, the farther apart the genes for those traits must lie on a chromosome.

Researchers conduct breeding experiments and use the resulting data to prepare a chromosome map. A **chromosome map** is a diagram that shows the linear order of genes on a chromosome. Alfred H. Sturtevant, one of Morgan's students, made the first chromosome map for flies, as shown in Figure 12-4. To prepare his map, Sturtevant compared the frequency of crossing-over for several genes. The percentage of crossing-over for two traits is proportional to the distance between them on a chromosome. Sturtevant defined one **map unit** as a frequency of crossing-over of 1 percent.

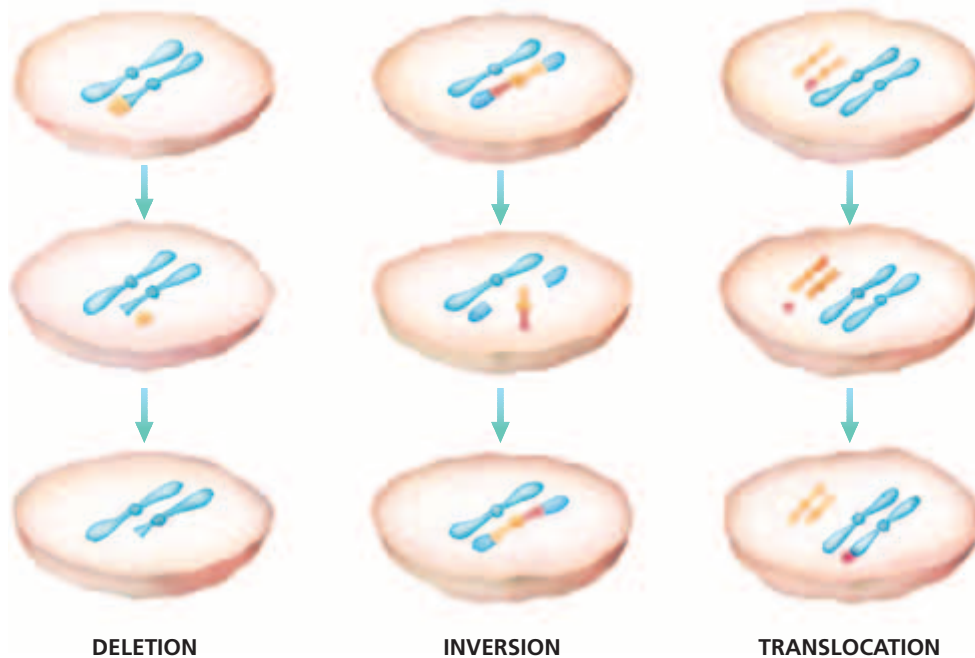
Today, researchers have new techniques to map genes. A simplified map of the human X chromosome, made by using these new techniques, is shown in Figure 12-5.

### HUMAN X CHROMOSOME



**FIGURE 12-5**

The locations of a few genes on the X chromosome are shown here.



## MUTATIONS

Cystic fibrosis results from a mutation. A *mutation* is a change in the nucleotide-base sequence of a gene or DNA molecule. **Germ-cell mutations** occur in an organism's gametes. Germ-cell mutations do not affect the organism itself, but they can be passed on to offspring. **Somatic-cell** (soh-MAT-ik SEL) **mutations** take place in an organism's body cells and can therefore affect the organism. For example, certain types of human skin cancer and leukemia result from somatic-cell mutations. Somatic-cell mutations cannot be inherited.

**Lethal mutations** cause death, often before birth. Some mutations, however, result in phenotypes that are beneficial to the individual. Organisms with beneficial mutations have a better chance of surviving and reproducing, and therefore have an evolutionary advantage. Mutations provide the variation upon which natural selection acts. Mutations can involve an entire chromosome or a single DNA nucleotide.

### Chromosome Mutations

Chromosome mutations involve changes in the structure of a chromosome or the loss or gain of a chromosome. Three types of chromosome mutations are shown in Figure 12-6. A **deletion** is the loss of a piece of a chromosome due to breakage. In an **inversion**, a chromosomal segment breaks off, flips around backward, and reattaches. In a **translocation**, a piece of one chromosome breaks off and reattaches to a nonhomologous chromosome. In **nondisjunction** (NAHN-dis-JUNGK-shuhn), a chromosome fails to separate from its homologue during meiosis. One gamete receives an extra copy of a chromosome, and another gamete receives no copies. An example of nondisjunction that results in Down syndrome is shown in Figure 12-7.

**FIGURE 12-6**

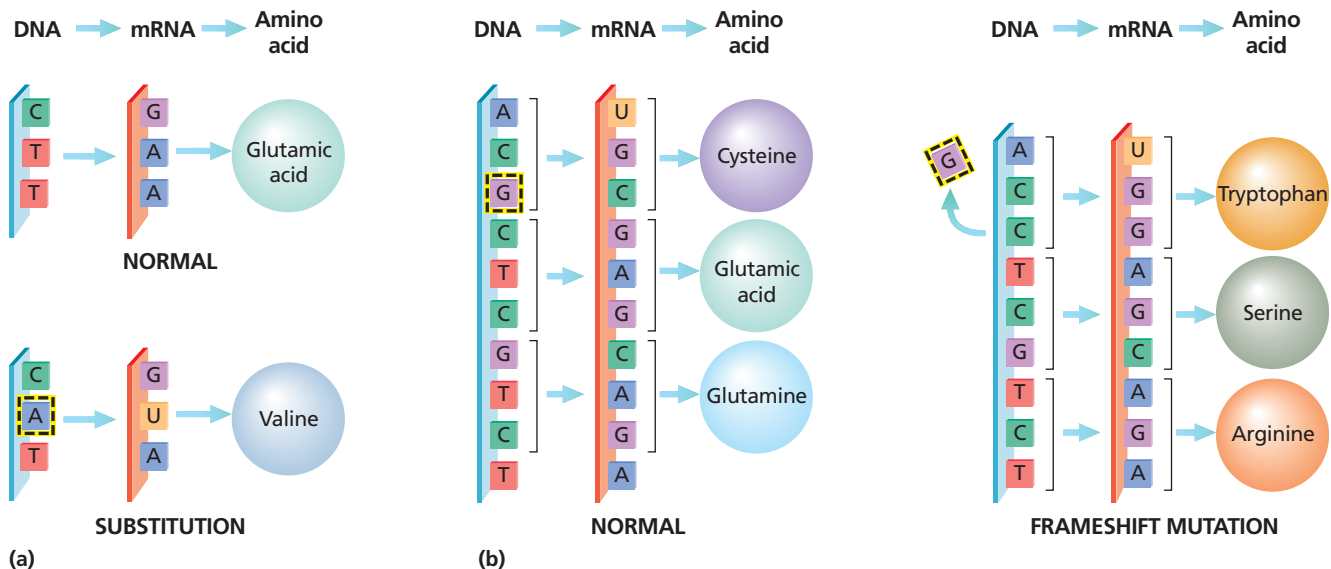
In some types of mutations, chromosomes break. In a deletion, a piece of a chromosome is lost. In an inversion, a piece flips and reattaches. In a translocation, a broken piece attaches to a nonhomologous chromosome.



**FIGURE 12-7**

Some chromosome mutations are the loss or gain of entire chromosomes. The mutation that gives a person three copies of chromosome 21 results in Down Syndrome.





**FIGURE 12-8**

(a) In a substitution mutation, one nucleotide replaces another, forming a new codon that may signal the insertion of the wrong amino acid. (b) Deleting a nucleotide causes all subsequent codons to be incorrectly read, resulting in a frameshift mutation. Adding a nucleotide shifts the codon grouping too, and causes misreading.

## Gene Mutations

The substitution, addition, or removal of a single nucleotide is a **point mutation**, which is a change that occurs within a single gene or other segment of DNA on a chromosome. In a **substitution**, one nucleotide replaces another, as shown in Figure 12-8a. If this substitution occurs in a codon, the amino acid can be changed. In a deletion mutation, one or more nucleotides in a gene are lost. This loss can cause incorrect grouping of the remaining codons, called a **frameshift mutation**, making all amino acids downstream change. A frameshift mutation is shown in Figure 12-8b. This mutation, in turn, can have a disastrous effect on the protein's function. In **insertion mutations**, one or more nucleotides are added to a gene, which can also result in a frameshift mutation.

## SECTION 1 REVIEW

1. Compare sex chromosomes with autosomes.
2. How does the inheritance of sex chromosomes result in approximately equal numbers of males and females among the offspring of fruit flies?
3. Explain why Morgan did not find white-eyed female *D. melanogaster* in the  $F_2$  generation when he crossed white-eyed males with red-eyed females.
4. How do sex linkage and crossing-over show that genes are found on chromosomes?
5. How can crossing-over between two alleles be used to map their locations on chromosomes?
6. Describe how nondisjunction can change chromosome number.

### CRITICAL THINKING

7. **Predicting Results** What cross could Morgan have performed to make the first white-eyed female fruit fly?
8. **Analyzing Concepts** Three genes (A, B, and C) are on the same chromosome. However, only C and A are usually expressed together in the same phenotype. Explain this result and describe the relative positions of A, B, and C on the chromosome.
9. **Relating Concepts** Chromosome mutations most often occur during nuclear division. Explain why this is a true statement.

# HUMAN GENETICS

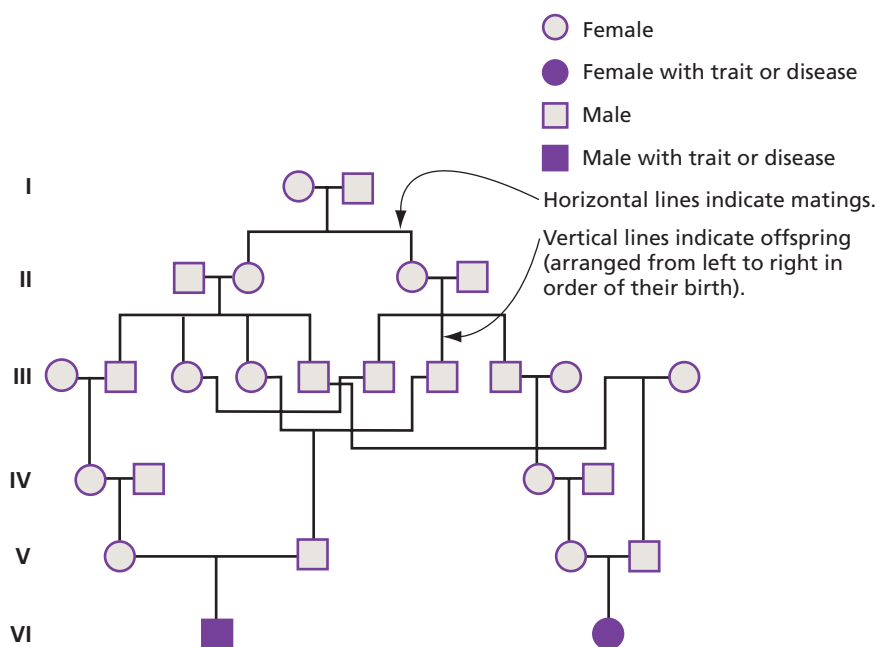
*This section investigates how geneticists analyze genetic data from families to track the inheritance of human genes. It also explores the genetic and environmental factors that influence human genetic traits and disorders, and discusses how geneticists detect and treat human genetic disorders.*

## INHERITANCE OF TRAITS

Geneticists can study human genetic traits and trace genetic diseases from one generation to the next by studying the phenotypes of family members in a pedigree.

### Pedigrees

A **pedigree** is a diagram that shows how a trait is inherited over several generations. Figure 12-9 is a pedigree of a family that has several cases of cystic fibrosis. In a pedigree, squares stand for males and circles stand for females. A filled symbol means that the person has the trait or condition. An empty symbol means that the person does not have the trait or condition. A horizontal line joining a male and female indicates a mating. A vertical line indicates offspring arranged from left to right in order of their birth. Roman numerals label different generations.



## OBJECTIVES

- **Analyze** pedigrees to determine how genetic traits and genetic disorders are inherited.
- **Summarize** the different patterns of inheritance seen in genetic traits and genetic disorders.
- **Explain** the inheritance of ABO blood groups.
- **Compare** sex-linked traits with sex-influenced traits.
- **Explain** how geneticists can detect and treat genetic disorders.

## VOCABULARY

pedigree  
carrier  
genetic disorder  
polygenic  
complex character  
multiple allele  
codominance  
incomplete dominance  
sex-influenced trait  
Huntington's disease  
amniocentesis  
chorionic villi sampling  
genetic counseling  
gene therapy

**FIGURE 12-9**

This pedigree for cystic fibrosis (CF) shows that each of the two affected individuals in the sixth generation has unaffected parents. Note that a cystic fibrosis allele from the first generation passed unexpressed through the next four generations. Marriages within the family during those four generations resulted in affected individuals who had two copies of the recessive disease allele for the CF gene.

## Patterns of Inheritance

Biologists learn about genetic diseases by analyzing *patterns of inheritance*, the expression of genes over generations. Pedigrees help to interpret patterns of inheritance. For example, if a trait is autosomal, it will appear in both sexes equally. If a trait is sex-linked, it is usually seen only in males. Most sex-linked traits are recessive.

If a trait is autosomal dominant, every individual with the trait will have a parent with the trait. If the trait is recessive, an individual with the trait can have one, two, or neither parent exhibit the trait.

If individuals with autosomal traits are homozygous dominant or heterozygous, their phenotype will show the dominant characteristic. If individuals are homozygous recessive, their phenotype will show the recessive characteristic. Two people who are heterozygous carriers of a recessive mutation will not show the mutation, but they can produce children who are homozygous for the recessive allele.

The pedigree in Figure 12-9 shows that the condition of cystic fibrosis is inherited as an autosomal recessive. Individuals such as the four people in the fifth generation in the pedigree are called **carriers** because they have one copy of the recessive allele but do not have the disease. Although carriers do not express the recessive allele, they can pass it to their offspring.

### Word Roots and Origins

#### *polygenic*

from the Greek *poly*, meaning "many," and the Greek *genesis*, meaning "origin"

## GENETIC TRAITS AND DISORDERS

Genes controlling human traits show many patterns of inheritance. Some of these genes cause genetic disorders. **Genetic disorders** are diseases or disabling conditions that have a genetic basis.

### Polygenic Inheritance

Single genes having two or more alleles can determine traits, such as blood type or cystic fibrosis. Geneticists have learned, however, that most human characteristics are **polygenic** (PAHL-ee-JEHN-ik) characters: they are influenced by several genes. Polygenic characters show

many degrees of variation, as seen in Figure 12-10. Skin color, for example, results from the additive effects of three to six genes. These genes control the amount of the brownish-black pigment called *melanin* in the skin. The more melanin skin cells produce, the darker the skin. Each of three to six genes has an allele that produces low amounts of melanin and another allele that makes high amounts of melanin. The final amount of melanin in a person's skin that is not exposed to sunlight comes from the number of high-melanin alleles among these few skin-color genes. Eye color, height, and hair color are also polygenic characters.

**FIGURE 12-10**

Many characters, such as height, weight, hair color, and skin color, are polygenic. Often, the environment strongly influences polygenic characters.



## Complex Characters

Many human conditions are **complex characters**—characters that are influenced strongly both by the environment and by genes. Skin color is both polygenic and complex. Exposure to sunlight generally causes the skin to become darker, no matter what the skin-color genotype is. Human height is another polygenic character that is controlled by an unknown number of genes that influence the growth of the skeleton. Height, however, is also influenced by environmental factors, such as nutrition and disease.

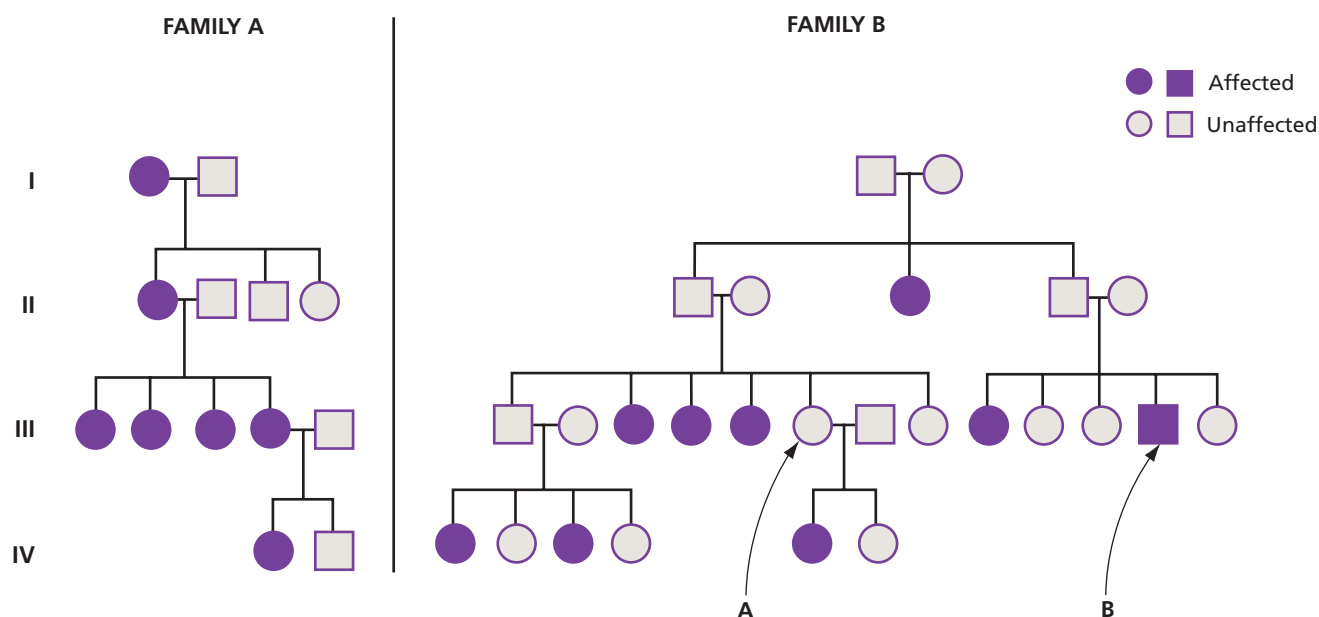
Other complex characters play a role in diseases or conditions such as breast cancer, diabetes, heart disease, stroke, and schizophrenia. Most breast cancer, for example, occurs in people with no familial history of the disease. But breast cancer also runs in some families.

Geneticist Marie-Claire King studied the genetics of breast cancer in families in which several individuals had the disease at younger ages than the average breast cancer patient does. Figure 12-11 shows the pedigrees of two of the families she studied. In Family A, each affected person has an affected parent, which is the inheritance pattern of a dominant trait. Family B demonstrates that additional genetic and environmental factors can influence whether or not a person expresses a trait. In Family B, a female, individual III-A, does not develop breast cancer herself but has a child who goes on to develop breast cancer. Notice also, individual III-B in Family B is a male who has breast cancer. This again shows that many factors in addition to the central gene influence the onset of a genetic disease.

Biologists hope that by identifying the environmental components that contribute to a disease, they can educate people in ways that minimize their risk of developing the disease. For breast cancer, for example, non-genetic risk factors include a diet high in saturated fat.

**FIGURE 12-11**

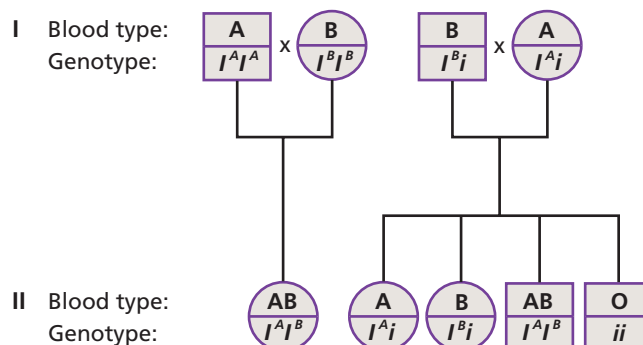
These diagrams show the pedigrees of families in which there is hereditary breast cancer.





| Phenotype  |                                                                                   |                      |
|------------|-----------------------------------------------------------------------------------|----------------------|
| Blood type | Molecule on red blood cell surface                                                | Genotype             |
| A          |  | $I^A I^A$ or $I^A i$ |
| B          |  | $I^B I^B$ or $I^B i$ |
| AB         |  | $I^A I^B$            |
| O          | Neither A nor B                                                                   | $ii$                 |

(a) Blood Types



(b) Inheritance of Blood Type Alleles

FIGURE 12-12

The ABO gene has three alleles (a). Allele  $I^A$  causes a type A sugar to appear on the surface of red blood cells. Allele  $I^B$  causes red blood cells to display a type B sugar. The third allele,  $i$ , does not cause the display of any sugars. In (b), a pedigree shows two examples of how blood type can be inherited.

## Multiple Alleles

Many genes have more than three alleles. Genes with three or more alleles are said to have **multiple alleles**. For example, in humans, the ABO blood groups (blood types) are controlled by the three alleles  $I^A$ ,  $I^B$ , and  $i$ . The alleles  $I^A$  and  $I^B$  are codominant. In **codominance**, both alleles are expressed in the phenotype of a heterozygote. Both  $I^A$  and  $I^B$  are dominant to the recessive  $i$  allele. The  $I^A$  and  $I^B$  alleles encode variants of an enzyme that cause two different sugar molecules to appear on the surface of red blood cells. The  $i$  allele lacks the activity of the enzyme entirely, so neither of the sugars appear on the red blood cell surface. Figure 12-12a shows how combinations of the three different alleles can produce four different blood types—A, B, AB, and O. Notice that a person who inherits two  $i$  alleles has type O blood. Figure 12-12b shows how blood type can be inherited.

## Incomplete Dominance

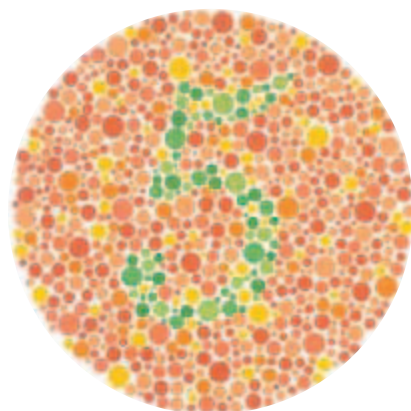
Sometimes, an individual displays a trait that is intermediate between the two parents, a condition known as **incomplete dominance**. For example, in Caucasians, the child of a straight-haired parent and a curly-haired parent would have wavy hair. Straight hair and curly hair are homozygous traits. Wavy hair is heterozygous and is intermediate between straight and curly hair.

## X-Linked Traits

Some complex characters are determined by X-linked genes, and a pedigree will usually reveal many affected males and no affected females. A male inherits his X chromosome from his mother. One form of **colorblindness** is a recessive X-linked disorder in which an individual cannot distinguish certain colors, such as red and green. Several X-linked genes encode proteins that absorb red or green light in the eye. Red-green colorblindness occurs because mutations disrupt these genes, so the eye cannot absorb some colors of light. Eye doctors often test for colorblindness by using a chart similar to the one in Figure 12-13.

FIGURE 12-13

A person who has a red-green colorblindness might not be able to see the number 5 in the center of the circle in this color-vision test chart.



## Sex-Influenced Traits

**Sex-influenced traits** are involved in other complex characters. Males and females can show different phenotypes even when they share the same genotype. Sex-influenced traits are usually autosomal. For example, an allele that is dominant in males but recessive in females controls pattern baldness, the type of baldness usually found in men. The difference is due to higher levels of the hormone testosterone in men, which interacts with the genotype to produce pattern baldness.

## Single-Allele Traits

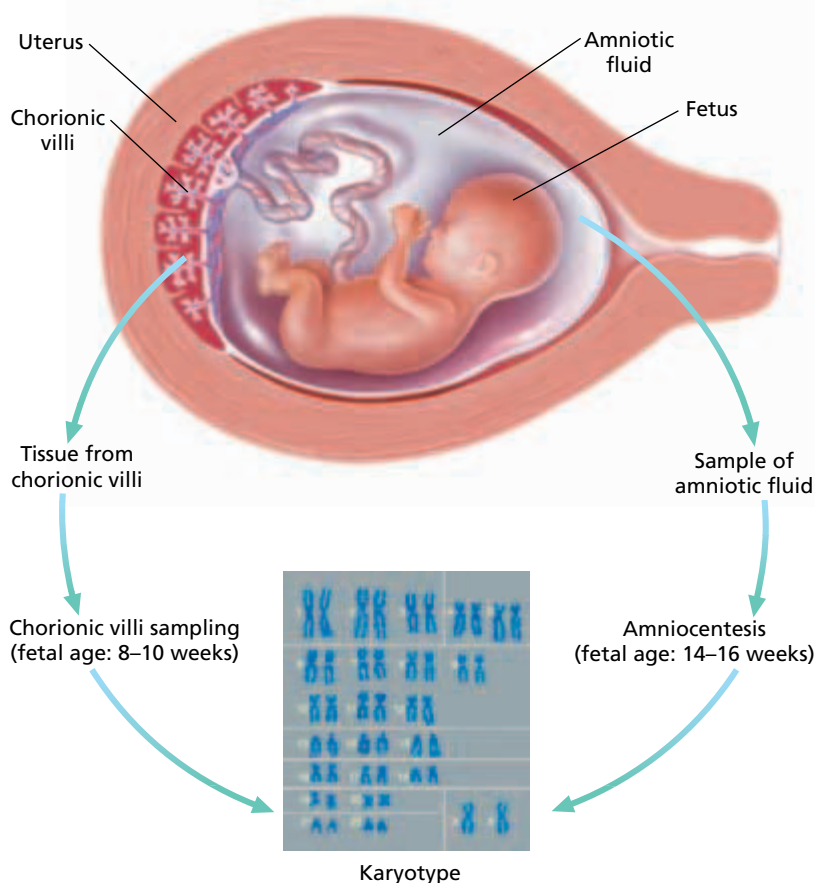
A single allele of a gene controls single-allele traits. Geneticists have discovered that more than 200 human traits are governed by single dominant alleles. **Huntington's disease** (HD) is an autosomal dominant condition characterized by forgetfulness and irritability. It develops as an affected person reaches 30 or 40 years of age and progresses to muscle spasms, severe mental illness, and, finally, death. Because a dominant gene exists in every HD heterozygote, each affected person has at least one affected parent. Unfortunately, many HD patients have already had children by the time their symptoms appear. Direct DNA testing is beginning to allow for earlier diagnosis of the HD allele.

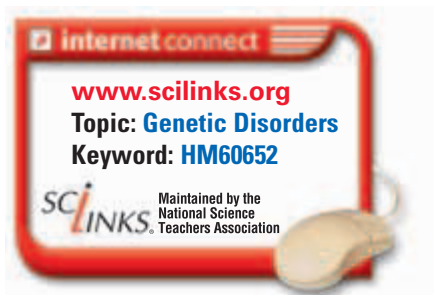
## DETECTING GENETIC DISEASE

Many people with a family history of genetic disease seek genetic screening before having children. *Genetic screening* is an examination of a person's genetic makeup. It may involve karyotypes, blood tests for certain proteins, or direct tests of DNA. Physicians can now also detect more than 200 genetic disorders in the fetus. The technique called **amniocentesis** (AM-nee-oh-sen-TEE-sis), shown in Figure 12-14, allows a physician to remove some amniotic fluid from the amnion, the sac that surrounds the fetus, between the 14th and 16th week of pregnancy. Geneticists can analyze fetal cells for genetic disease by examining chromosomes and proteins in the fluid.

**FIGURE 12-14**

Geneticists can use fetal cells obtained by amniocentesis or by chorionic villi sampling to prepare fetal karyotypes that might display chromosome mutations. This allows physicians to diagnose chromosomal abnormalities before a child's birth.





In **chorionic villi** (KAWR-ee-AHN-ik VIL-ie) **sampling**, also shown in Figure 12-14, the physician takes a sample of the chorionic villi—cells derived from the zygote that grow between the mother’s uterus and the placenta—between the 8th and 10th week. Both procedures allow technicians to analyze fetal cells, chromosomes, proteins and detect genetic disease.

Table 12-1 lists some important genetic disorders, symptoms, and their patterns of inheritance.

**TABLE 12-1** *Some Important Genetic Disorders*

| Disorder (Gene)                | Symptom                                                                                       | Normal protein; function; effect                                                                                                           | Pattern of inheritance and chromosome location | Frequency among human births                         |
|--------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------|
| Huntington’s disease (gene HD) | gradual deterioration of brain tissue in middle age; shortened life expectancy                | huntingtin protein; involved in movement of vesicles in nerve cells; mutation causes extra copies of the codon CAG in the gene             | autosomal dominant on chromosome 4             | 1 in 10,000                                          |
| Cystic fibrosis (gene CFTR)    | mucus clogs lungs and pancreas; victims today live to early adulthood or longer               | cystic fibrosis transmembrane conductance regulator; regulates the transport of chloride ions in epithelial cells                          | autosomal recessive on chromosome 7            | Up to 1 in 900 French Canadians; 1 in 2000 Europeans |
| Sickle cell anemia (gene HBB)  | organ damage due to impaired blood flow                                                       | beta globin; carries oxygen in blood; mutation causes red blood cells to change shape and clog capillaries                                 | autosomal recessive on chromosome 11           | 1 in 500 African Americans                           |
| Tay-Sachs disease (gene HEXA)  | deterioration of central nervous system in infancy; death in early childhood                  | hexosaminidase A; breaks down cellular wastes in lysosome; mutation allows waste buildup, causing nerve-cell death                         | autosomal recessive on chromosome 15           | 1 in 600 Jews of European descent                    |
| Phenylketonuria (gene PAH)     | infant brain fails to develop normally; death in childhood                                    | phenylalanine hydroxylase; catalyzes change of the amino acid phenylalanine to tyrosine; without the enzyme, a toxic substance accumulates | autosomal recessive on chromosome 12           | 1 in 18,000 Americans                                |
| Marfan syndrome (gene FBN1)    | long limbs, loose joints, deformed vertebral column, crowded teeth, rupture of large arteries | fibrillin-1; a major component of connective tissue; lack of this protein causes weakness of ligaments and blood-vessel sheaths            | autosomal dominant on chromosome 15            | 1 in 20,000 Americans                                |
| Breast cancer (gene BRCA1)     | malignant tumors in breast tissue                                                             | breast cancer-1; inhibits growth of breast and ovarian tumors, probably by encouraging repair of DNA damage                                | autosomal dominant on chromosome 17            | About 8 percent of breast cancer patients            |
| Hemophilia (gene F8)           | prolonged bleeding due to ineffective blood clotting                                          | coagulation factor 8; helps cause blood to clot; the mutant protein does not function in clotting                                          | X-linked recessive on chromosome X             | 1 in 7,000                                           |

## Genetic Counselor

### Job Description

A genetic counselor is a health professional who has a specialized graduate degree and experience in the areas of medical genetics and counseling.

### Focus On a Genetic Counselor

Robin Bennett is a senior genetic counselor and clinic manager at the Medical Genetics Clinic of the University of Washington. Individuals and couples who seek genetic counseling include people whose family members have birth defects or genetic disorders and people who may be at risk for an inherited condition. Clients may be expecting a child or considering parenthood, or they may

have been recently diagnosed with a disease. "The whole family becomes your patient," says Bennett. At the first meeting with a client, Bennett usually draws up a pedigree, obtains medical records of affected relatives, and, in some cases, orders blood tests. "After the appointment, I send the client a written report, which helps patients and their families and healthcare providers to understand the information."

### Education and Skills

- **High school**—a focus on science and math courses
- **College**—bachelor's degree, including course work in genetics, chemistry, statistics, psychology, and



developmental biology; M.S. degree in genetic counseling; national certification is available

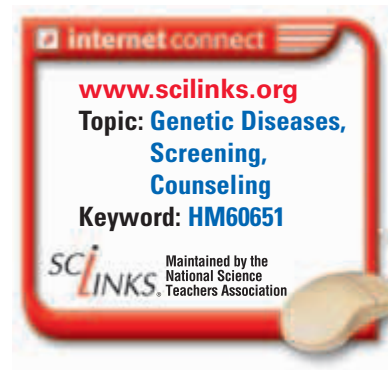
- **Skills**—strong verbal skills, emotional stability, and strong writing skills



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## Genetic Counseling

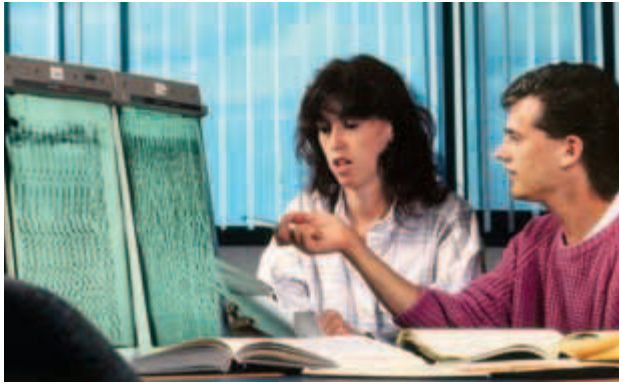
Many people with a family history of a genetic disease also undergo **genetic counseling**, the process of informing a person or couple about their genetic makeup. Genetic counseling is a form of medical guidance that informs individuals about problems that might affect their offspring. By studying the data from genetic screening tests and the family's pedigree, a genetic counselor can predict the likelihood that a couple will produce an affected child. For diseases that have both genetic and environmental influences, such as diabetes, physicians and counselors can advise families on how to lower risk factors.



## TREATING GENETIC DISEASE

Physicians can treat genetic diseases in several ways. For many diseases they can treat just the symptoms. For example, an individual with the genetic disease phenylketonuria (PKU) lacks an enzyme that converts the amino acid phenylalanine into the amino acid tyrosine. Phenylalanine builds up in the body and causes severe mental retardation. Physicians prescribe strict food regimens for phenylketonuria (PKU) patients to eliminate the amino acid phenylalanine from their diets. PKU can be detected by means of a blood test administered to infants during the first few days of life.





**FIGURE 12-15**

Jeff Pinard discusses with a colleague the results of his DNA analyses of cystic fibrosis.

For cystic fibrosis patients, physicians prescribe 45-minute-long sessions of pounding on the back and chest to dislodge sticky mucus.

For some diseases, physicians can implement symptom-prevention measures. For example, a physician might prescribe insulin injections to patients with diabetes. For patients with hemophilia, a doctor might prescribe injections of a missing blood-clotting protein. Physicians can even do some types of surgery to correct genetic defects in a fetus before birth.

## Gene Therapy

Another level of treatment currently in development involves replacing the defective gene. This type of therapy, called **gene therapy**, is a technique that places a healthy copy of a gene into the cells of a person whose copy of the gene is defective. Gene therapy relies on knowing gene sequences like the one Pinard is viewing in Figure 12-15. Medical researchers place a functional allele of the gene, such as the *CFTR* gene, into the DNA of a virus. They then introduce the modified virus into a patient's lungs where the virus infects the cells and brings along the functional gene. This improves the patient's symptoms, but only until the infected cells slough off. Then the patient must undergo the procedure again. Researchers are working to increase the effectiveness of gene therapy.

Gene therapy, in which only body cells are altered, is called *somatic cell gene therapy*. This contrasts with *germ cell gene therapy*, the attempt to alter eggs or sperm. Bioethicists, who study ethical issues in biological research, generally view somatic cell gene therapy as an extension of normal medicine to improve patients' health. Germ cell gene therapy, however, poses more risks and ethical issues because future generations could be affected in unpredictable ways.

## SECTION 2 REVIEW

1. A husband and wife have a son with cystic fibrosis. Their second child, a daughter, does not. Prepare a pedigree for this family.
2. Explain the difference between a polygenic character and a complex character.
3. A husband and wife have the ABO blood group genotypes  $I^A I^B$  and  $ii$ . What ABO blood types can their children have?
4. Use Table 12-1 to compare Huntington's disease with sickle cell anemia.
5. Describe the methods physicians can use to detect genetic diseases in an unborn fetus.

### CRITICAL THINKING

6. **Predicting Patterns** A woman with cystic fibrosis marries a man who is heterozygous for cystic fibrosis. What is the likelihood that their children will have cystic fibrosis?
7. **Applying Information** Why is colorblindness less common among females?
8. **Analyzing Patterns** A man with blood type B marries a woman with blood type A. Their first child is blood type O. What is the probability their next child will be blood type AB? blood type B?

# CHAPTER HIGHLIGHTS

## SECTION 1

## Chromosomes and Inheritance

- Genes reside on chromosomes. Sex chromosomes contain genes that determine an organism's sex. The remaining chromosomes that are not directly involved in determining the sex of an individual are called autosomes.
- In mammals, an individual carrying two X chromosomes is female. An individual carrying an X and a Y chromosome is male.
- Genes found on the X chromosome are X-linked genes. A sex-linked trait is a trait whose allele is located on a sex chromosome. Because males have only one X chromosome, a male who carries a recessive allele on the X or Y chromosome will exhibit the sex-linked condition.
- Pairs of genes that tend to be inherited together are called linked genes. They occur close to each other on the same chromosome. The farther apart two genes are located on a chromosome, the more likely a cross-over will occur. Researchers use recombinant percentages to construct chromosome maps showing relative gene positions.
- Germ-cell mutations occur in gametes and can be passed on to offspring. Somatic-cell mutations occur in body cells and affect only the individual organism.
- Chromosome mutations are changes in the structure of a chromosome or the loss or gain of an entire chromosome. Gene mutations are changes in one or more of the nucleotides in a gene.

### Vocabulary

sex chromosome (p. 236)  
autosome (p. 236)  
sex-linked trait (p. 237)  
linked gene (p. 238)  
chromosome map (p. 238)

map unit (p. 238)  
germ-cell mutation (p. 239)  
somatic-cell  
mutation (p. 239)  
lethal mutation (p. 239)

deletion (p. 239)  
inversion (p. 239)  
translocation (p. 239)  
nondisjunction (p. 239)  
point mutation (p. 240)

substitution (p. 240)  
frameshift mutation (p. 240)  
insertion mutation (p. 240)

## SECTION 2

## Human Genetics

- Geneticists use pedigrees to trace diseases or traits through families. Pedigrees reveal inheritance patterns of genes.
- A carrier has one copy of a recessive allele but does not express the trait.
- Polygenic characters, such as skin color, are controlled by two or more genes.
- Complex characters, such as height, are influenced by both genes and environment.
- Multiple-allele characters, such as ABO blood groups, are controlled by three or more alleles of a gene.
- The gene for colorblindness, an X-linked recessive, is found on the X chromosome.
- A sex-influenced trait, such as pattern baldness, is expressed differently in men than in women even if it is on an autosome and both sexes have the same genotype.
- Genetic screening examines a person's genetic makeup and potential risks of passing disorders to offspring. Amniocentesis and chorionic villi sampling help physicians test a fetus for the presence of genetic disorders.
- Genetic counseling informs screened individuals about problems that might affect their offspring.
- Genetic disorders are treated in various ways. Among the treatments are symptom-relieving treatments and symptom-prevention measures, such as insulin injections for diabetes.
- Gene therapy is a type of treatment under development. In gene therapy, a defective gene is replaced with a copy of a healthy gene.
- Somatic cell gene therapy alters only body cells. Germ cell gene therapy attempts to alter eggs or sperm.

### Vocabulary

pedigree (p. 241)  
carrier (p. 242)  
genetic disorder (p. 242)  
polygenic (p. 242)

complex character (p. 243)  
multiple allele (p. 244)  
codominance (p. 244)

incomplete  
dominance (p. 244)  
sex-influenced trait (p. 245)  
Huntington's disease (p. 245)

amniocentesis (p. 245)  
chorionic villi sampling (p. 246)  
genetic counseling (p. 247)  
gene therapy (p. 248)

# CHAPTER REVIEW

## USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
  - germ-cell mutation* and *somatic-cell mutation*
  - multiple-allele* and *polygenic*
  - sex-linked trait* and *sex-influenced trait*
  - amniocentesis* and *chorionic villi sampling*
- Explain the relationship between a *chromosome map* and a *map unit*.
- Use the following terms in the same sentence: *point mutation*, *substitution*, and *frameshift mutation*.
- Word Roots and Origins** The word *somatic* is derived from the Greek *somatikos*, which means “body.” Using this information, explain why the term *somatic-cell mutation* is a good name for what the term describes.

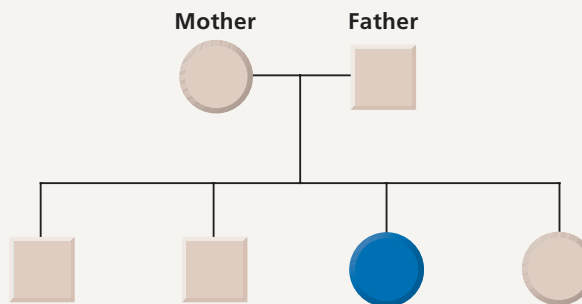
## UNDERSTANDING KEY CONCEPTS

- Compare** sex chromosomes and autosomes.
- Identify** the evidence that led Morgan to hypothesize that the gene for eye color in *Drosophila melanogaster* is carried on the X chromosome.
- Describe** where the human *SRY* gene resides and its significance.
- Relate** how X-linked genes affect the inheritance of sex-linked traits.
- Evaluate** the relationship between crossing over, recombinant types, and chromosome mapping.
- Differentiate** between a chromosome mutation and a point mutation.
- Differentiate** between nondisjunction and translocation mutations.
- State** the type of information that is obtained by analyzing a pedigree.
- Describe** the pattern of inheritance in Huntington’s disease.
- Predict** the possible genotypes for a person whose ABO blood group is type A.
- Summarize** two ways in which geneticists can detect genetic disorders.
- Describe** how gene therapy is used to treat genetic disorders.
- Unit 5—Heredity**  
 Write a report summarizing the latest findings on treating a genetic disorder with gene therapy. What obstacles still need to be overcome?

-  **CONCEPT MAPPING** Use the following terms to create a concept map that describes the ways that changes in DNA occur: *mutation*, *chromosome mutation*, *substitution*, *deletion*, *point mutation*, *death*, *inversion*, *translocation*, *nondisjunction*, *germ-cell mutation*, *lethal mutation*, and *gametes*.

## CRITICAL THINKING

- Analyzing Data** In *Drosophila* the genes for body color and wing length are on the same chromosome. Gray body (*G*) is dominant to black body (*g*), and long wings (*L*) are dominant to short wings (*l*). Assume that both dominant alleles are on the same chromosome. Draw a Punnett square representing the cross  $Gg Ll \times Gg Ll$ . Write the phenotypic and genotypic ratios that would be expected among the offspring, assuming that crossing-over does not occur.
- Relating Concepts** Individuals who are heterozygous for sickle cell anemia should avoid extreme conditions that severely reduce the amount of oxygen available to the body, such as playing vigorous sports at high elevations. Explain why this would be advisable.
- Inferring Relationships** A 20-year-old man with cystic fibrosis has a sister who is soon to be married. If you were the man, how would you explain your sister’s likelihood of having children with cystic fibrosis?
- Interpreting Graphics** The individual shown in blue (third from the left, bottom row) in the pedigree below is affected by a genetic disorder. State the pattern of inheritance for the disorder and whether the disorder is autosomal or sex-linked. Explain your answer.



- What advice might a genetic counselor give to the unaffected brothers and sister shown in the pedigree in question 22?



# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- Which can a chromosomal map show?
  - the sex of the individual
  - the presence of mutant alleles
  - the positions of genes on a chromosome
  - whether a gene is autosomal or recessive
- Which can result from the deletion of a single nucleotide?
  - trisomy
  - a translocation
  - nondisjunction
  - a frameshift mutation
- At the present time amniocentesis cannot reveal which of the following?
  - eye color
  - genetic disease
  - sex of the fetus
  - chromosomal abnormalities
- A geneticist working with the fruit fly *Drosophila melanogaster* discovers a mutant phenotype that appears only in males who are offspring of males of the same phenotype. What does this information suggest about the mutant phenotype?
  - The trait is X-linked.
  - The trait is Y-linked.
  - The trait is autosomal dominant.
  - The trait is autosomal recessive.

**INTERPRETING GRAPHICS:** The table below shows the genotypes and phenotypes of pattern baldness. Use the table to answer the question that follows.

**Genotypes and Phenotypes of Pattern Baldness**

| Genotype | Phenotype |          |
|----------|-----------|----------|
|          | Female    | Male     |
| BB       | Bald      | Bald     |
| Bb       | Not bald  | Bald     |
| bb       | Not bald  | Not bald |

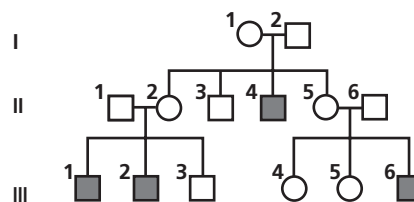
- Which statement best explains why men and women express the *Bb* genotype differently?
  - The trait is polygenic.
  - The trait has multiple alleles.
  - Pattern baldness is a sex-linked trait.
  - Pattern baldness is a sex-influenced trait.

**DIRECTIONS:** Complete the following analogy.

6. translocation : chromosome mutation ::  
substitution :
- gene mutation
  - point mutation
  - germ-cell mutation
  - somatic-cell mutation

**INTERPRETING GRAPHICS:** The image below is a pedigree showing the inheritance of hemophilia in a family. Use the pedigree to answer the question that follows.

**A Pedigree of Hemophilia**



- Which type of inheritance pattern is associated with hemophilia?
  - autosomal recessive
  - sex-linked dominant
  - sex-linked recessive
  - autosomal dominant

## SHORT RESPONSE

Consider a couple about to get married. The woman has cystic fibrosis, but the man does not.

What benefit would they gain by seeing a genetic counselor?

## EXTENDED RESPONSE

Colorblindness is a recessive, sex-linked trait. A woman and a man, both with normal vision, have three daughters with normal vision. One of the daughters marries a man with normal vision, and they have a son who is colorblind.

**Part A** Which parent of the son is the carrier of the trait? Explain your answer.

**Part B** What is the likelihood that the children of a woman heterozygous for colorblindness and a colorblind man will express the trait? Explain your answer.

## Test TIP

Test questions are not necessarily arranged in order of increasing difficulty. If you are unable to answer a question, mark it and move on to another question.



## Analyzing Karyotypes

### OBJECTIVES

- **Make** a human karyotype by arranging chromosomes in order by length, centromere position, and banding pattern.
- **Identify** a karyotype as normal or abnormal.
- **Identify** any genetic disorder that is present and describe the effect of the genetic disorder on the individual.

### PROCESS SKILLS

- Identifying
- Describing
- Analyzing
- Modeling

### MATERIALS

- chromosome spread
- human karyotyping form
- metric ruler
- photomicrograph of chromosomes
- scissors
- transparent tape

### Background

1. Define and describe a human karyotype.
2. Compare the two procedures used to obtain the cells needed for preparing a fetal karyotype.
3. List four types of chromosomal mutations that can be identified by preparing a fetal karyotype.
4. Name the disorder that occurs in people who have an extra copy of chromosome 21.

### Procedure

1. Obtain a photomicrograph and note the letter identifying which person the cells were taken from.
2.  **CAUTION** Always cut in a direction away from your face and body. Carefully cut apart the chromosomes on each photomicrograph. Be sure to leave a slight margin around each chromosome.

3. Arrange the chromosomes in homologous pairs. The members of each pair will be the same length and will have their centromeres located in the same area. Use the ruler to measure the length of the chromosome and the position of the centromere. The banding patterns of homologous chromosomes will be similar and may also help you pair the chromosomes.

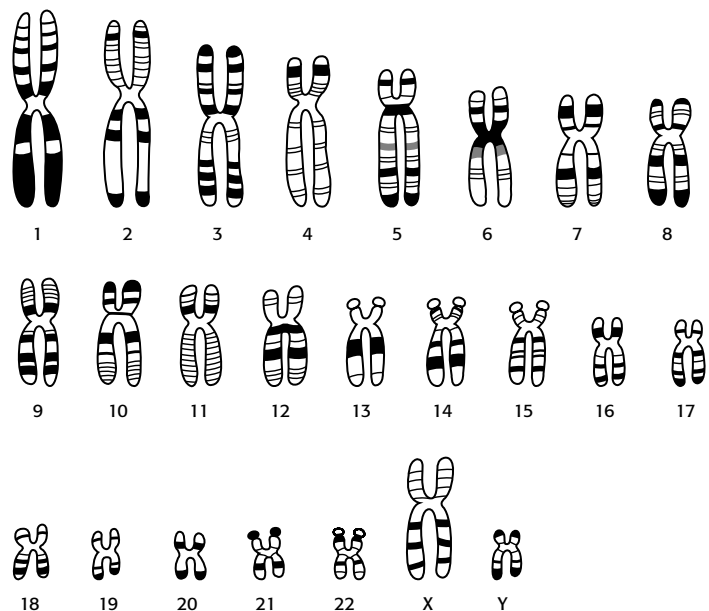


Figure 1 Normal Human Karyotype

4. Arrange the pairs according to their length. Begin with the largest chromosomes and move to the smallest.
5. Tape each pair of homologous chromosomes to a human karyotyping form. Place the centromeres on the lines provided. Place the longest chromosome at position 1, and the shortest at position 22. Place the two sex chromosomes at position 23.
6. The diagram you have made is a karyotype, as in Figure 1. Analyze your karyotype to determine the sex of the individual. Use the information in Table 1 to guide your analysis.

**TABLE 1 GENETIC DISORDERS CAUSED BY AN ABNORMAL CHROMOSOME NUMBER**

| Name of abnormality              | Chromosome affected     | Description of abnormality                                                                                                                      |
|----------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Down syndrome, or Trisomy 21     | #21                     | 47 chromosomes; mental retardation with specific characteristic features; may have heart defects and respiratory problems                       |
| Edwards' syndrome, or Trisomy 18 | #18                     | 47 chromosomes; severe mental retardation; very characteristic malformations of the skull, pelvis, and feet, among others; die in early infancy |
| Patau syndrome, or Trisomy 13    | #13                     | 47 chromosomes; abnormal brain function that is very severe; many facial malformations; usually die in early infancy                            |
| Turner's syndrome                | Single X in female (XO) | 45 chromosomes; in females only; missing an X chromosome; do not develop secondary sex characteristics; are infertile                           |
| Klinefelter's syndrome           | Extra X in male (XXY)   | 47 chromosomes; in males only; sterile, small testicles; otherwise normal appearance                                                            |
| XYY syndrome                     | Extra Y in male (XYY)   | 47 chromosomes; in males only; low mental ability; otherwise normal appearance                                                                  |
| Triple X syndrome                | Extra X in female (XXX) | 47 chromosomes; sterility sometimes occurs; normal mental ability                                                                               |

7. Create a data table like the example below, and record your results. Pool your data with that from the rest of the class.

**TABLE 2 POOLED CLASS DATA**

| Letter identifier | Sex | Condition | Chromosome abnormality |
|-------------------|-----|-----------|------------------------|
| A                 |     |           |                        |
| B                 |     |           |                        |
| C                 |     |           |                        |
| D                 |     |           |                        |
| E                 |     |           |                        |
| F                 |     |           |                        |
| G                 |     |           |                        |
| H                 |     |           |                        |
| I                 |     |           |                        |

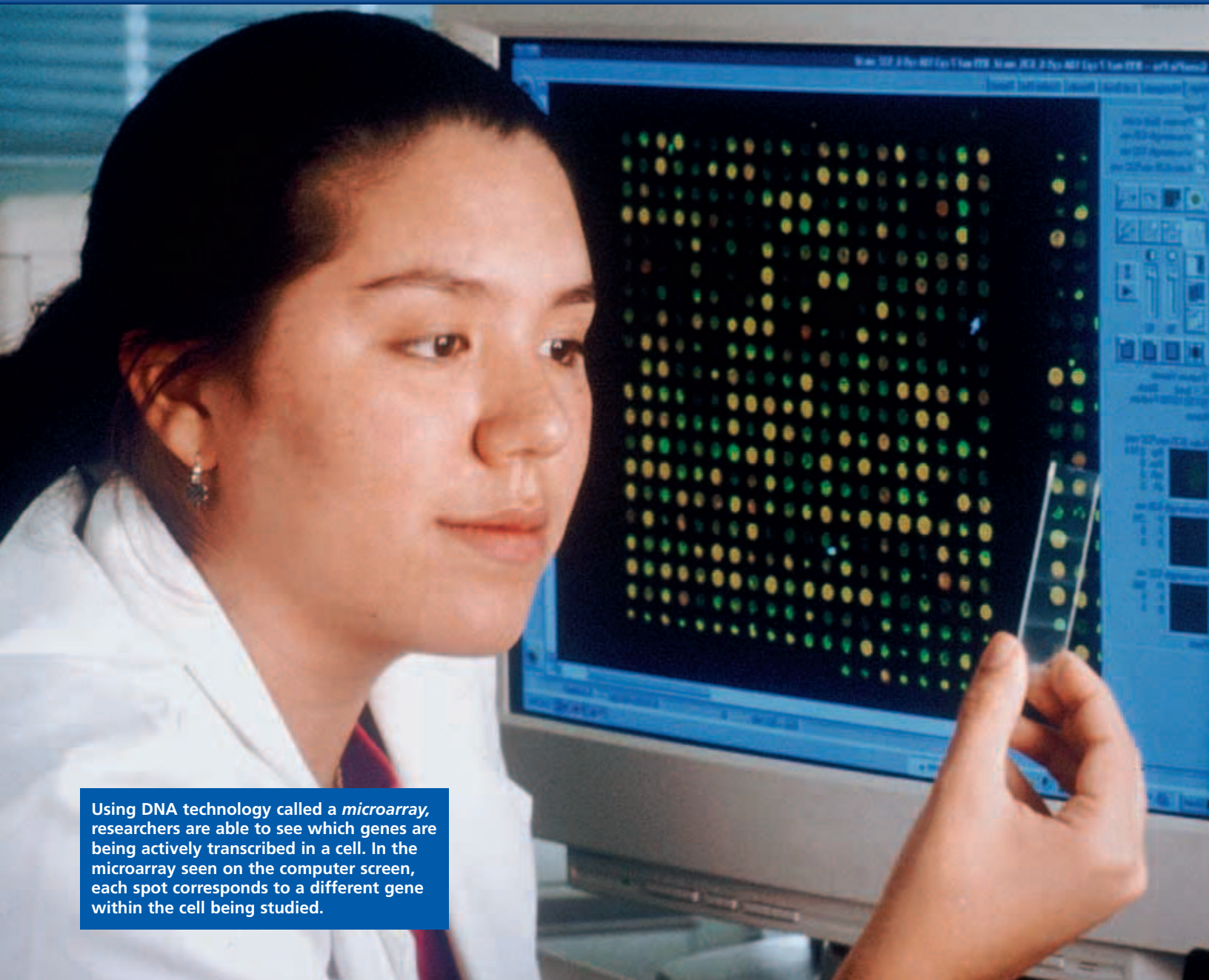
8.  Clean up your materials and wash your hands before leaving the lab.

## Analysis and Conclusions

1. Is the fetus represented by your karyotype male or female? How do you know?
2. Does the fetus have a genetic disorder? Explain.
3. Assume that two students started with the same photomicrograph. One student concluded that the individual had Down syndrome. The other student concluded that the individual had Edwards' syndrome. Explain how this could happen.
4. How is sex determined in a person who has more than two sex chromosomes? Explain your answer.
5. In this lab, you examined karyotypes for the presence of abnormal chromosome numbers in both autosomes and sex chromosomes. Which condition seems to have a greater influence on a person's health—trisomy of an autosome or trisomy of a sex chromosome?
6. Assume that an individual has a deletion mutation in one of their chromosomes. What would the karyotype look like in this situation?
7. How might banding patterns be important to detecting an inversion mutation?
8. Some medical labs make karyotypes from several of an individual's cells before drawing conclusions about the individual's health. Do you think this is necessary? Why or why not?

## Further Inquiry

Trisomy occurs when an individual has three copies of the same chromosome. Monosomy occurs when an individual has only one copy of a chromosome. In this lab, you examined a fetal karyotype for the presence of three different trisomies. Find out why monosomies are rarely detected.



Using DNA technology called a *microarray*, researchers are able to see which genes are being actively transcribed in a cell. In the microarray seen on the computer screen, each spot corresponds to a different gene within the cell being studied.

## SECTION 1 DNA Technology

## SECTION 2 The Human Genome Project

## SECTION 3 Genetic Engineering



Unit 6—Gene Expression  
Topics 1–6



For project ideas from *Scientific American*, visit [go.hrw.com](http://go.hrw.com) and type in the keyword **HM6SAC**.

# DNA TECHNOLOGY

*Today, scientists manipulate DNA for many practical purposes by using techniques collectively called DNA technology.*

*For example, DNA technology can be used as evidence in a criminal case by providing positive identification of DNA left at the scene of a crime. Scientists also use DNA technology to improve food crops, to determine if a person has the genetic information for certain diseases before symptoms appear, and to do research on treatments and cures for genetic diseases.*

*This chapter discusses the tools of DNA technology, how scientists apply these tools to study entire genomes, and how they use them to improve human lives.*

## DNA IDENTIFICATION

Except for identical twins, no two people in the world are genetically alike. A majority of DNA is actually the same in all humans, but about 0.10 percent of the human genome (the complete genetic material in an individual) varies from person to person. Because of the 0.10 percent variation, scientists are able to identify people based on their DNA. DNA identification compares DNA samples in regions of a chromosome that differ. DNA identification is useful for many purposes, including determining a person's paternity, identifying human remains, tracing human origins, and providing evidence in criminal cases.

### Noncoding DNA

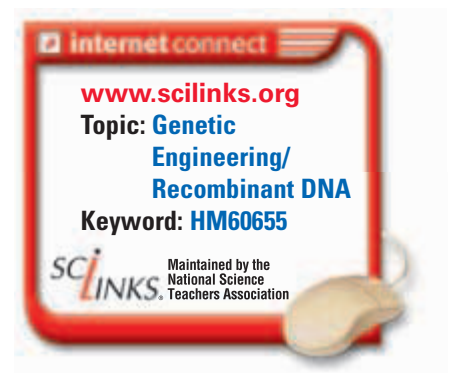
Amazingly, about 98 percent of our genetic makeup does not code for any protein. The regions of DNA that do not code for proteins are called *noncoding DNA*. Noncoding DNA contains many **length polymorphisms**, or variations in the length of the DNA molecule between known genes. Some length polymorphisms in noncoding regions come from short, repeating sequences of DNA. For example, a repeating nucleotide sequence might be CACACA, and so on. These sequences can repeat a few or many times in tandem (one "behind" another) and so are called **variable number tandem repeats (VNTR)**. The number of tandem repeats at specific places (loci) in DNA varies among individuals. For each of the many VNTR loci in a person's DNA, he or she will have a certain number of repeats. Geneticists have calculated how frequently VNTR used in DNA identification occur in the general population. With these numbers, they can determine how rare a particular DNA profile is.

## OBJECTIVES

- **Explain** the significance of non-coding DNA to DNA identification.
- **Describe** four major steps commonly used in DNA identification.
- **Explain** the use of restriction enzymes, cloning vectors, and probes in making recombinant DNA.
- **Summarize** several applications of DNA identification.

## VOCABULARY

length polymorphism  
variable number tandem repeats (VNTR)  
polymerase chain reaction (PCR)  
primer  
restriction enzyme  
gel electrophoresis  
DNA fingerprint  
genetic engineering  
recombinant DNA  
clone  
vector  
plasmid  
probe





## STEPS IN DNA IDENTIFICATION

The main steps involved in DNA identification are (1) isolate the DNA in a sample and, if needed, make copies, (2) cut the DNA into shorter fragments that contain known VNTR areas, (3) sort the DNA by size, and (4) compare the size fragments in the unknown sample of DNA to those of known samples of DNA. If a match occurs between the unknown sample and a known sample, then a person's identity can be confirmed.

### Copying DNA: Polymerase Chain Reaction

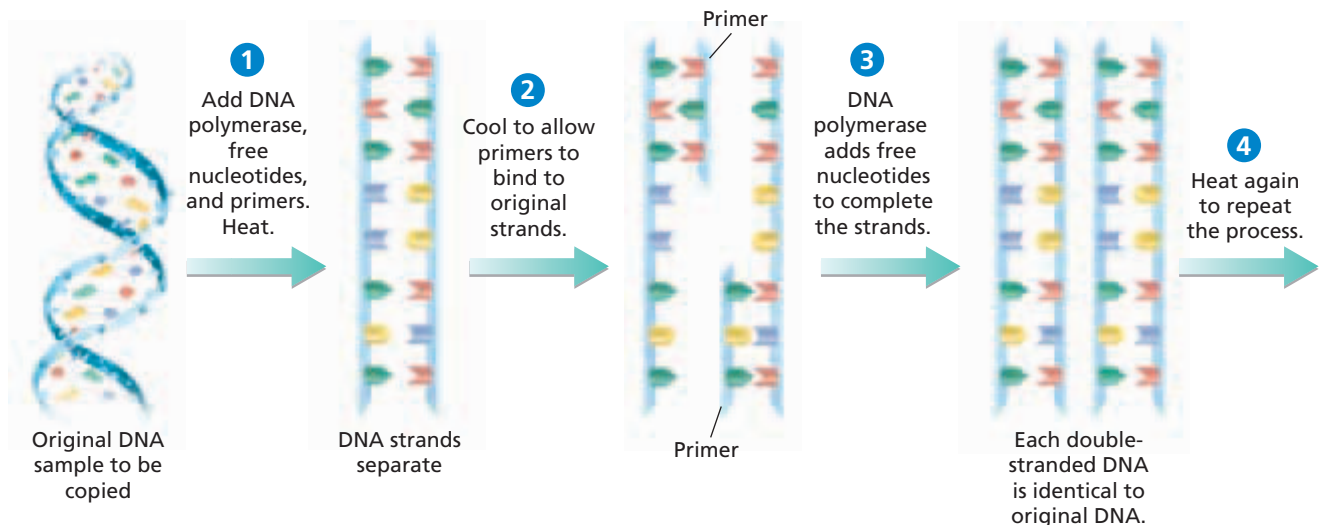
Often, DNA that is recovered from a crime scene or from human tissues is present in very small amounts. In these cases, scientists need to make copies in order to have enough DNA to use for DNA identification. The **polymerase chain reaction (PCR)** is a technique that quickly produces many copies of a DNA fragment.

As shown in step ① of Figure 13-1, PCR requires a *template*, a DNA fragment containing the sequence the scientist wants to copy. PCR also requires a supply of the four DNA nucleotides, heat-tolerant DNA polymerase, and primers. **Primers** are artificially made pieces of single-stranded DNA that are 20 to 30 nucleotides long that must be present for DNA polymerase to initiate replication. Primers are complementary to the ends of the DNA fragment that is to be copied.

When all the ingredients are combined, copying can begin. Primers bind to DNA, and DNA polymerase makes a copy of both DNA strands. Heating breaks the bonds holding the template DNA to the newly made strands. After cooling, as shown in step ②, primers can once more bind to the DNA. Then, in step ③, DNA polymerase can copy again. In step ④, the cycle is repeated. With each new cycle, the DNA between the two primers doubles.

**FIGURE 13-1**

In the polymerase chain reaction (PCR), a scientist chooses a DNA fragment to copy and designs primers that will bind to both ends of the fragment. DNA polymerase copies the segment between the two primers. Repeating the procedure through about 30 cycles generates millions of copies of a single piece of DNA fragment.

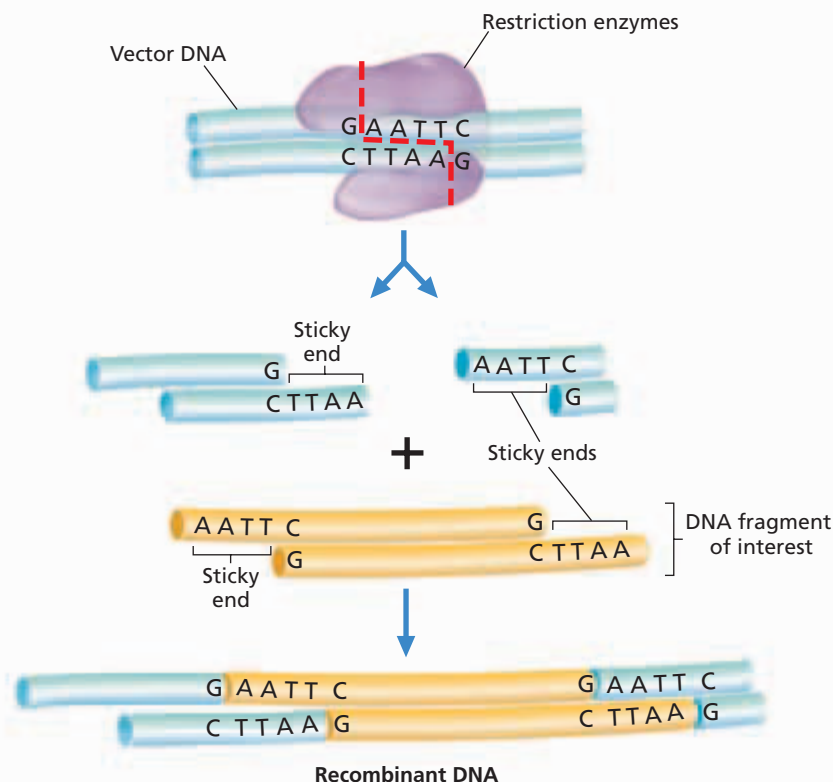


## Cutting DNA: Restriction Enzymes

To cut long DNA molecules into shorter pieces, biologists use bacterial proteins called **restriction enzymes**. Restriction enzymes recognize specific short DNA sequences, and cut the DNA in or near the sequence, as Figure 13-2 shows. Some restriction enzymes leave DNA overhangs that function like “sticky ends” so that other pieces of DNA with complementary sequences can bind to them.

## Sorting DNA by Size: Gel Electrophoresis

DNA fragments can be studied using a technique called **gel electrophoresis**. Gel electrophoresis separates nucleic acids or proteins according to their size and charge. As shown in step ① of Figure 13-3, the DNA samples are cut with a restriction enzyme. The DNA is then placed in wells made on a thick gel, as shown in step ②. An electric current runs through the gel for a given period of time. Negatively charged DNA fragments migrate toward the positively charged end of the gel. Smaller DNA fragments migrate faster and farther than longer fragments, and this separates the fragments by size. In step ③, the DNA is transferred to a nylon membrane and radioactive probes are added. The probes bind to complementary DNA. In step ④, an X-ray film is exposed to the radiolabeled membrane. The resulting pattern of bands is called a **DNA fingerprint**.

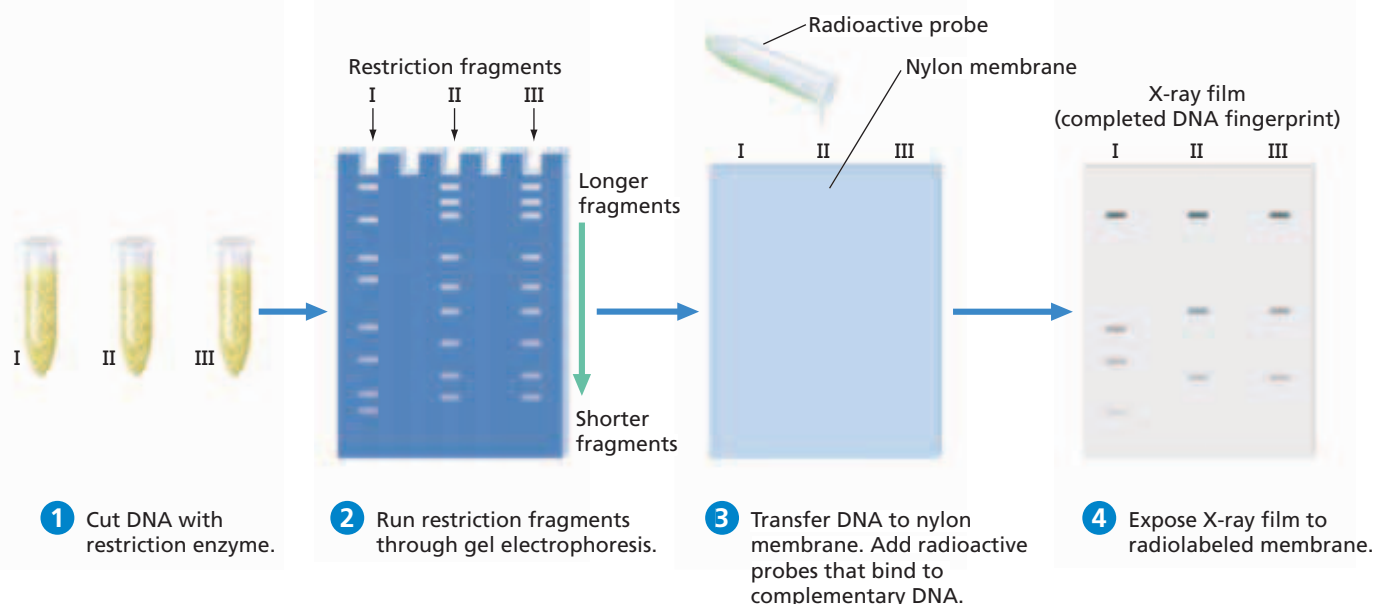


**FIGURE 13-2**

The restriction enzyme in this figure recognizes the sequence GAATTC on each DNA and cuts each chain between the G and A nucleotides. DNA fragments with sticky ends result.

**FIGURE 13-3**

In a DNA fingerprint, DNA samples are cut, transferred to a nylon membrane, and exposed to radioactive probes. Exposing X-ray film to the membrane makes the fingerprint visible.





**FIGURE 13-4**

A DNA fingerprint can pinpoint subtle genetic differences that identify each individual uniquely. Identical twins share the same DNA fingerprint.

## Comparing DNA: DNA Fingerprints

To permanently preserve the DNA fingerprint, a technician can place a positively charged nylon membrane over the gel and transfer the negatively charged DNA to the membrane. To visualize specific DNA pieces, biologists can prepare a nucleic acid complementary to the DNA of interest and label it radioactively. A sheet of X-ray film placed over the gel will be exposed only where the desired DNA is on the gel, providing a DNA fingerprint. An example of a DNA fingerprint is shown in Figure 13-4.

## Accuracy of DNA Fingerprints

What makes a DNA fingerprint so powerful as an identification tool is the combined analysis of many VNTR loci. Analyzing only one VNTR locus is like having only one digit in a person's telephone number. DNA fingerprinting typically compares from five to thirteen VNTR loci. Thirteen loci, as often used by some crime labs in their criminal profiles, produce the odds that two people will share a DNA profile at about one in 100 billion. There are about 6.5 billion people on Earth, so this should be enough to remove any doubt.

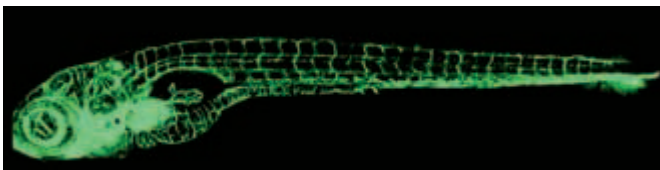
# RECOMBINANT DNA

The techniques of DNA technology are sometimes used to modify the genome of a living cell or organism. The process of altering the genetic material of cells or organisms to allow them to make new substances is called **genetic engineering**. **Recombinant DNA** results when DNA from two different organisms is joined.

An organism with recombinant DNA is shown in Figure 13-5. To study blood vessel growth, researchers combined a jellyfish gene encoding green fluorescent protein (GFP) that glows under ultraviolet light with a zebrafish gene that is involved in blood-vessel development. They inserted the GFP/blood-vessel gene into zebrafish embryos. The fish's blood-vessel cells transcribed the recombinant DNA and made the green fluorescent proteins. As the zebrafish grew, their blood vessels glowed green, and researchers could more easily study their growth.

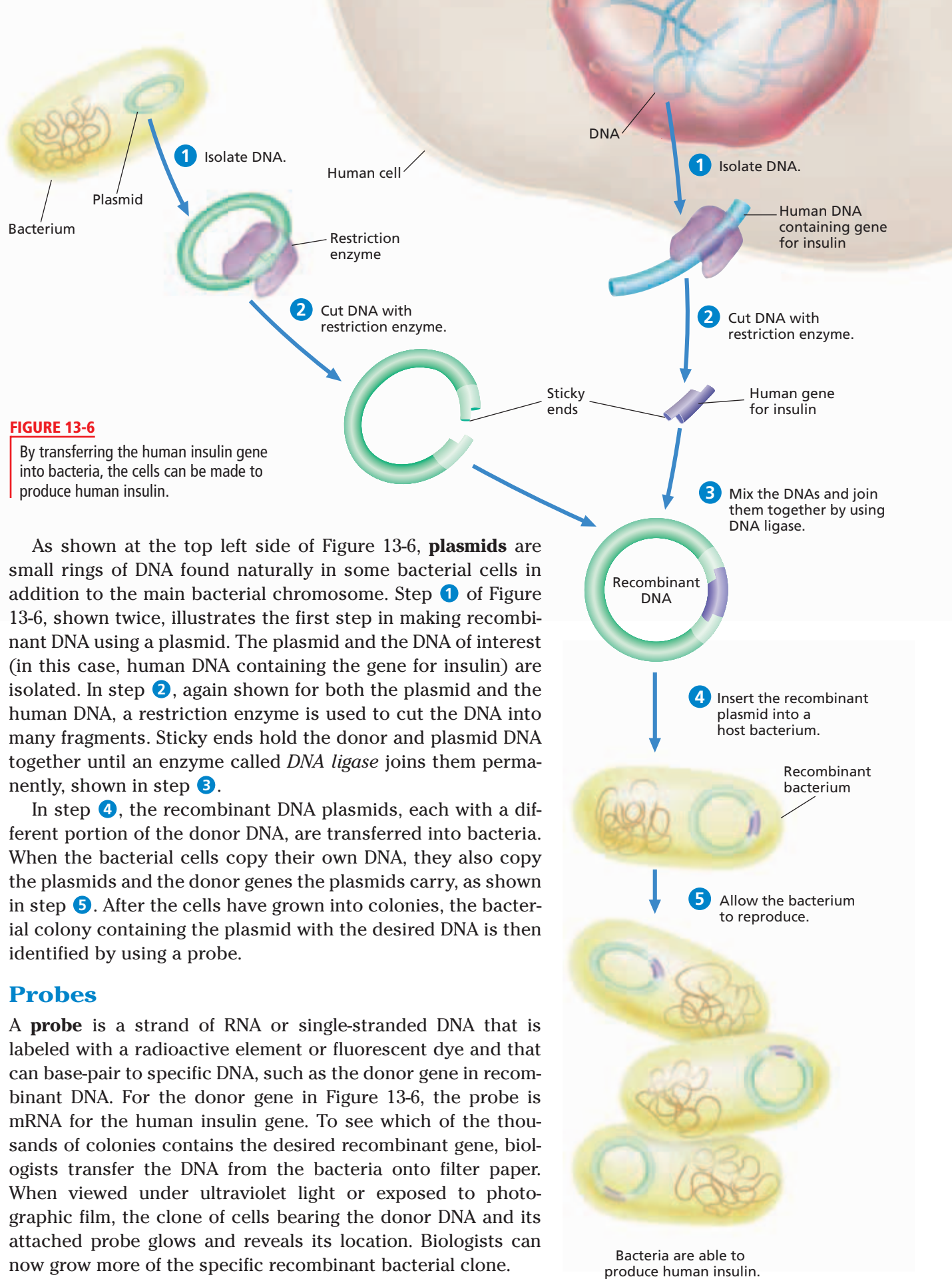
**FIGURE 13-5**

Researchers had questions about which molecules cause blood vessels to grow. To answer these questions, they used genetic engineering to make the proteins that control blood-vessel growth in zebrafish glow green.



## Cloning Vectors

A **clone** is an exact copy of a DNA segment, a whole cell, or a complete organism. Biologists also use the term *clone* as a verb, meaning to make a genetic duplicate. Researchers can clone DNA fragments by inserting them into **vectors**, DNAs that can replicate within a cell, usually a bacterium or yeast, and that can carry foreign DNA. When the vector carrying the foreign DNA enters bacteria and the bacteria reproduce, they grow a colony of cloned cells that includes the foreign DNA. Cloning vectors include viruses that infect bacteria and plasmids.







## Quick Lab

### Comparing Unique Characteristics

**Materials** ink pad, paper, pencil, scissors

#### Procedure

1. Cut four 3-in. squares of plain paper.
2. Draw lines on a full-sized sheet of paper to divide it into four equal squares.
3. Each team member will press a right thumb on the ink pad and then will quickly press the inked thumb in a square on the large sheet of paper and again on one of the small squares of paper.
4. Examine each of the thumbprints, and list the differences and similarities of the prints. Describe how each thumbprint is unique. Then, shuffle the four thumbprints on the small squares, and try to match each with its duplicate on the large sheet of paper.

**Analysis** What characteristics are common to all thumbprints? What characteristics made each thumbprint unique? How do a person's fingerprints relate to his or her DNA fingerprints?

## APPLICATIONS FOR DNA TECHNOLOGY

DNA technology has been used for many purposes. For example, DNA identification has been used in forensics to identify criminals and to free the wrongly convicted. DNA has also been used to identify human remains. For example, DNA identification techniques were used to identify Czar Nicholas II of Russia and his family, who were executed by the Bolsheviks in 1918.

In addition to forensic applications, DNA technology is of great importance in other scientific efforts. Anthropologists use DNA identification techniques to trace human origins and migrations. Environmental conservationists use the same techniques to trace migrations and movements of threatened or endangered organisms in the effort to preserve their species.

Recombinant DNA techniques give microorganisms new capabilities that have useful applications. The first recombinant DNA product to be used commercially was human insulin (for the treatment of diabetes) in 1982. A recombinant DNA molecule was made by inserting a human gene for insulin into a bacterial plasmid. These bacteria are now grown in vats, from which large amounts of human insulin are extracted and used to treat diabetes.

Since 1982, more than 30 products made using DNA technology have been approved and are being used around the world. These proteins are preferred to conventional drugs because they are highly specific and have fewer side effects. Medically important proteins include factors to treat immune-system deficiencies and anemia. Clotting factors for people with hemophilia, human growth hormone for people with growth defects, interferons for viral infections and cancer, and proteins, such as growth factors to treat burns and ulcers, are just a few other genetically engineered medicines in use today.

## SECTION 1 REVIEW

1. Summarize the significance of noncoding DNA to DNA identification.
2. Describe the steps in a polymerase chain reaction.
3. What role do restriction enzymes play in DNA technology?
4. What are "sticky ends," and in what way do they function in making recombinant DNA?
5. Explain the role cloning vectors play in making recombinant DNA.
6. List three ways that DNA technology could be used to improve the lives of humans.

### CRITICAL THINKING

7. **Relating Concepts** Explain why the process of genetic fingerprinting uses only specific, short segments of DNA rather than the entire genome.
8. **Evaluating Conclusions** A student performing electrophoresis on a DNA sample believes that her smallest DNA fragment is the band nearest the negative pole of the gel. Do you agree with her conclusion? Explain your answer.
9. **Applying Information** Explain the statement that the genetic code is universal.

# THE HUMAN GENOME PROJECT

*One exciting application of DNA technology has been the sequencing of the entire human genome. This section discusses how researchers used modern genetic tools to sequence the human genome and what their findings mean for 21st century biology and society.*

## MAPPING THE HUMAN GENOME

In 1990, geneticists around the world tackled one of the most ambitious projects in scientific history—the Human Genome Project. The **Human Genome Project** is a research effort undertaken to sequence all of our DNA and locate within it all of the functionally important sequences, such as genes. That is, the project seeks to determine the sequence of all 3.3 billion nucleotides of the human genome and to map the location of every gene on each chromosome. Information from the project will provide insight into our evolutionary past, genome organization, gene expression, and cell growth.

The Human Genome Project linked more than 20 scientific laboratories in six countries. By 2001, the draft sequence of the human genome appeared in two landmark papers in the science journals *Science* and *Nature*. The high-quality sequence was completed in 2003—two years ahead of schedule. Figure 13-7 shows an example of how the sequence of nucleotides in a piece of DNA is displayed on a computer screen.

### Important Insights

Scientists with the Human Genome Project were surprised by some of the discoveries they made, including the following:

1. Only about 2 percent of the human genome codes for proteins.
2. Chromosomes have unequal distribution of exons—sequences of nucleotides that are transcribed and translated.
3. The human genome is smaller than previously estimated. The human genome has only about 20,000 to 25,000 protein-coding genes—far fewer than the 100,000 originally estimated. Scientists now know that RNAs are not used only for translating DNA into proteins. Many RNAs are involved in regulating gene expression.

## OBJECTIVES

- **Discuss** two major goals of the Human Genome Project.
- **Summarize** important insights gained from the Human Genome Project.
- **Explain** why animal model species are useful to study genes.
- **State** how information from the Human Genome Project will be applied to future projects.
- **Relate** bioinformatics, proteomics, and microarrays to the Human Genome Project.

## VOCABULARY

Human Genome Project  
proteome  
single nucleotide  
polymorphism (SNP)  
bioinformatics  
proteomics  
two-dimensional gel  
electrophoresis

**FIGURE 13-7**

Researchers with the Human Genome Project developed automated DNA sequencing machines that could determine the order of millions of base pairs per day. The four bases are represented by different colors.



## Word Roots and Origins

### proteome

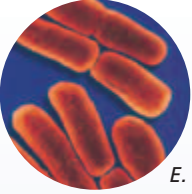
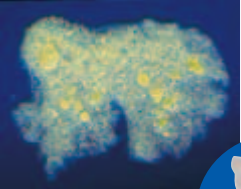
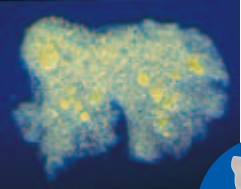
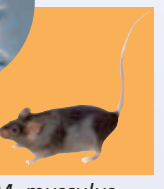
a new, combined word from *prote* in the word *protein* and *ome* in the word *genome*

- The exons of human genes are spliced in many ways, allowing the same gene to encode different versions of a protein. An organism's complete set of proteins is called its **proteome**. The human proteome is quite complex.
- About half of the human genome arises from the shuffling of transposons, pieces of DNA that move from one chromosome location to another. Transposons appear to have no specific role in development or physiology.
- There are about 8 million **single nucleotide polymorphisms (SNP)**, unique spots where individuals differ by a single nucleotide. SNPs are important for mapping the genome in more detail, and in the identification of human disease genes.

## Model Species

In order to better understand how human genes control development and health and to explain how genes affect behaviors, biologists want to map similar genes in *model species*. Since the Human Genome Project, many other genome sequence projects have been completed, and more are underway. Some of the model species for genome sequencing, as seen in Table 13-1, include a bacterium, a roundworm, a fruit fly, a zebrafish, and a mouse. Researchers can induce mutations in each of these species to look for gene actions. Because they represent such a broad phylogenetic base, it is possible to make valid generalizations that relate to larger groups.

**TABLE 13-1** Genome Sizes of Some Species

|                                                                                                              | Domain/<br>Kingdom | Organism                        | Common<br>name    | Genome<br>size (million<br>bases) | Number<br>of genes |
|--------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------|-------------------|-----------------------------------|--------------------|
| <br><i>E. coli</i>        | Archaea            |                                 |                   |                                   |                    |
|                                                                                                              | Archaeobacteria    | <i>Pyrococcus furiosus</i>      | <i>Pyrococcus</i> | 1.9                               | 2,065              |
|                            | Bacteria           |                                 |                   |                                   |                    |
|                                                                                                              | Eubacteria         | <i>Chlamydia trachomatis</i>    | Chlamydia         | 1.0                               | 894                |
|                            |                    | <i>Escherichia coli</i>         | <i>E. coli</i>    | 4.6                               | 4,289              |
| <br><i>D. discoideum</i>  | Eukarya            |                                 |                   |                                   |                    |
|                                                                                                              | Protista           | <i>Dictyostelium discoideum</i> | Amoeba            | 34                                | ~9,000             |
| <br><i>L. longiflorum</i> | Fungi              | <i>Saccharomyces cerevisiae</i> | Yeast             | 12                                | 6,000              |
|                                                                                                              | Plantae            | <i>Arabidopsis thaliana</i>     | Mustard           | 125                               | 23,174             |
| <br><i>S. cerevisiae</i>  |                    | <i>Lilium longiflorum</i>       | Easter lily       | 100,000                           | ~25,000            |
|                                                                                                              | Animalia           | <i>Drosophila melanogaster</i>  | Fruit fly         | 120                               | 13,600             |
| <br><i>M. musculus</i>    |                    | <i>Caenorhabditis elegans</i>   | Roundworm         | 97                                | 19,049             |
|                                                                                                              |                    | <i>Xenopus tropicalis</i>       | Frog              | 1,700                             | ~30,000            |
|                           |                    | <i>Homo sapiens</i>             | Human             | 3,300                             | ~20,000            |
|                                                                                                              |                    | <i>Mus musculus</i>             | Mouse             | 3,630                             | ~30,000            |
|                                                                                                              |                    | <i>Danio rerio</i>              | Zebrafish         | 1,700                             | ~3,000             |

## Applications

Information from the Human Genome Project has been and will continue to be applied to different medical, industrial, commercial, and scientific purposes. For example, scientists have already discovered specific genes responsible for several genetic disorders, including cystic fibrosis, Duchenne muscular dystrophy, and colon cancer. Researchers may improve diagnoses, treatments, and therapies for the more than 4,000 human genetic disorders.

John Carpten of the National Institutes of Health, pictured in Figure 13-8, is a researcher taking such an approach to prostate cancer. Carpten knew that prostate cancer rates in men are second only to skin cancer rates and that some forms of the disease run in families. By studying male relatives with and without prostate cancer, he and his team were able to map a gene for susceptibility to prostate cancer. The gene, called *RNASEL*, occurs near a gene sequenced during the Human Genome Project. *RNASEL* encodes an RNA-digesting enzyme. Carpten isolated the *RNASEL* gene from patients, and found mutations. His discoveries may one day lead to new prostate cancer treatments.



**FIGURE 13-8**

Dr. John Carpten is a researcher at the National Human Genome Research Institute. He uses the results of the Human Genome Project to study prostate cancer.

## THE FUTURE OF GENOMICS

It is clear that although the more than 3 billion letters of the human genetic code have been sequenced, this is only the beginning of the quest to understand the human genome. New tools and fields of scientific research have arisen that enable the acquisition, analysis, storage, modeling, and distribution of the information contained within DNA and protein sequence data. Geneticists will now be able to apply their new tools to benefit human health.

### Bioinformatics

Keeping track of the billions of nucleotide base pairs in a complex genome requires substantial computer power. Much of the progress in studying genomes comes from advances in the new scientific field of bioinformatics. **Bioinformatics** combines biological science, computer science, and information technology to enable the discovery of new biological insights and unifying principles. Bioinformatics utilizes databases to store and integrate the data coming out of genomic research. One such database called the BLAST search allows rapid comparisons between a gene or protein sequence from one organism and similar sequences from all other organisms in a database at the National Center for Biotechnology Information. The BLAST tool allows a worker to find corresponding genes in different organisms. By knowing the function of a gene from a related model species such as a mouse, the researcher can then infer the function of a previously unstudied human gene.





**FIGURE 13-9**

Two-dimensional gel electrophoresis separates proteins in two steps. The first dimension separates proteins according to their isoelectric points. The second dimension separates proteins according to their molecular weights.

## Proteomics

As important as genomes are, it is the proteins they encode that actually carry out the work of cells. To understand how genes work, biologists must understand proteins. **Proteomics** is the study of all of an organism's proteins, including their identities, structures, interactions, and abundances. A key tool in proteomics is **two-dimensional gel electrophoresis**, a method that separates the proteins in a sample into individual spots, as shown in Figure 13-9. A researcher can cut out a protein spot from the gel, then use special methods to determine the amino acid sequence in part of the protein.

Using bioinformatics, he or she can then search the DNA of a sequenced genome and match an individual gene to that unique protein. Proteomics and bioinformatics will allow medical researchers to identify new targets for therapeutic drugs and develop new markers for diagnosing disease.

## Microarrays

An important tool of the genomic revolution is a technique called *DNA microarray*, a two-dimensional arrangement of DNA molecules representing thousands of cloned genes. A DNA microarray can show which genes are active in a cell. To prepare the DNA microarray, robotic machines arrange tiny amounts of sequences from thousands of genes on a single microscope slide. To investigate, for example, how tumor cells differ from normal cells, mRNA from the tumor is labeled with fluorescent dyes and poured onto the microarray slides. The more mRNA that binds to its complement at a particular spot on the slide, the brighter the color, indicating that this specific gene is highly active. Physicians use DNA microarray analysis to classify patients' cancers. Such classification can lead to better-informed decisions about which type of treatment is best. A DNA microarray is shown on the first page of this chapter.

## SECTION 2 REVIEW

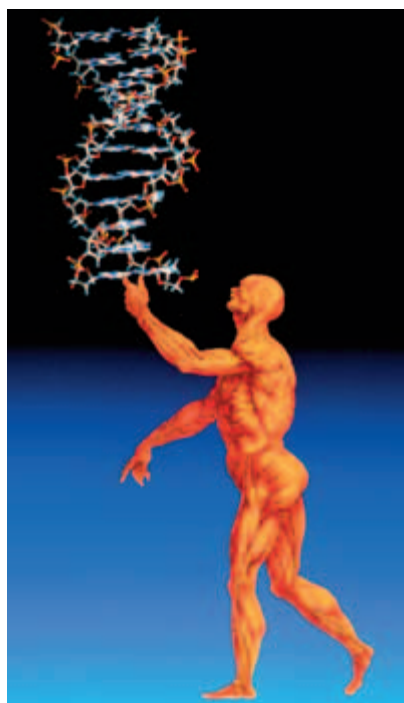
1. Describe two major goals of the Human Genome Project.
2. Summarize four insights from the initial analysis of the complete sequence of the human genome.
3. Why are model species useful to study genes?
4. List and describe two genetic disorders that might be treated using DNA technology information gained in the Human Genome Project.
5. Distinguish between proteomics and bioinformatics.

### CRITICAL THINKING

6. **Relating Concepts** Describe how the rapid growth of the computer technology industry in the 1990s relates to the Human Genome Project.
7. **Analyzing Models** Some people might assume that the more nucleotides in a genome, the more genes it contains. Make a graph of the data in Table 13-1. Explain why you agree or disagree.
8. **Evaluating Choices** If you were the scientist who developed the model organism list, which organisms would you include? Justify your choices.

## WHO OWNS GENES?

The very large U.S. government investment in the Human Genome Project has spurred research in genetics. Within months of the genetic map's release, researchers around the country applied for patents on thousands of different genes and parts of genes. By the end of 2004, more than three million gene-related patent applications had been filed. Most gene-related patents have been issued to the government, the University of California, and three top biotechnology companies. Some observers criticize the idea of patenting human genes. Others support it.



*Today the public is debating whether or not patents should be awarded for human genes.*

### The Patent Process

The United States Commerce Department issues patents on inventions. By proving that he or she is the first to invent something, an inventor can get exclusive rights to develop and sell products based on the invention for 20 years.

The patent office must determine that an invention is practically useful, new, clever, and creative in design, and useful to a skilled technician for the practical purpose intended.

### Gene Patents

The patent office has ruled that whole genes as they occur in nature on chromosomes are not patentable. If a person isolates a gene, however, or modifies the gene, it becomes a candidate for a patent. Most patent applications are for gene fragments that contain one or more changed base pairs and may represent gene sequences that lead to disease or new drugs.

### Arguments Against Gene Patents

One critic of human gene patents, Dr. Jonathan King, points out that the human genome mapping project came from 50 years of public investment and therefore shouldn't be patentable by private companies. Others object to patenting any part of nature. Still others argue that patenting gene fragment base-pair sequences rewards the easy part. The cost of identifying a gene is only a small fraction of the total cost of developing

useful applications from genetic information. It is argued that gene patents penalize anyone (such as pharmaceutical companies) who would invest in the research for drugs or therapies derived from the gene sequence. Most critics agree that gene patenting could slow the study of the human genome.

### Arguments in Favor of Gene Patents

Some companies say patents are necessary because of high research and development costs. Companies fear they will waste millions of dollars on duplicated efforts because only patented information is made public. Unpatented information is kept secret. They point out that owning a patent draws investors, which helps fund research. The public nature of a patent reduces secrecy and allows researchers to share techniques.

## REVIEW

1. What characteristics of an invention must be present for a gene to be patented?
2. What are the arguments for and against gene patents?
3. **Critical Thinking** Are you in favor of gene patenting? Why or why not?



## SECTION 3

### OBJECTIVES

- **Discuss** the uses of genetic engineering in medicine.
- **Summarize** how gene therapy is being used to try to cure genetic disorders.
- **Discuss** cloning and its technology.
- **Describe** two ways genetic engineering has been used to improve crop plants.
- **Discuss** environmental and ethical issues associated with genetic engineering.

### VOCABULARY

gene therapy  
cloning by nuclear transfer  
telomere  
DNA vaccine  
bioethics

# GENETIC ENGINEERING

*In addition to DNA fingerprints and genomics, genetic engineering techniques are being used in medical, industrial, commercial, and agriculture settings. This section discusses some of these applications and the ethical issues the techniques raise.*

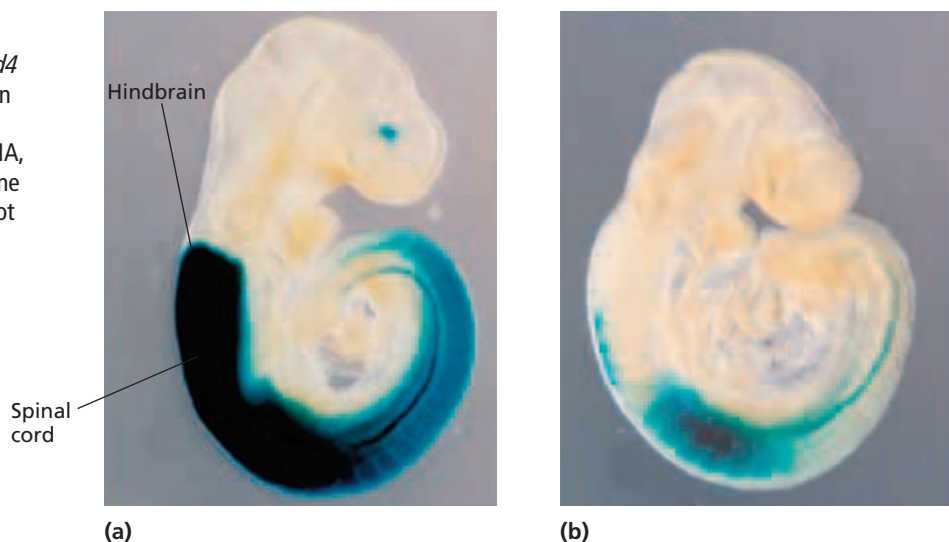
## MEDICAL APPLICATIONS

Genetic engineering has allowed biologists to study how genes function. For example, researchers in Montreal used genetic engineering to study brain development in mice. They wanted to determine what activates the gene *Hoxd4* as the hindbrain develops in an embryo. This is important because abnormal hindbrain development may contribute to autism, a disorder that disrupts a child's ability to socialize and communicate.

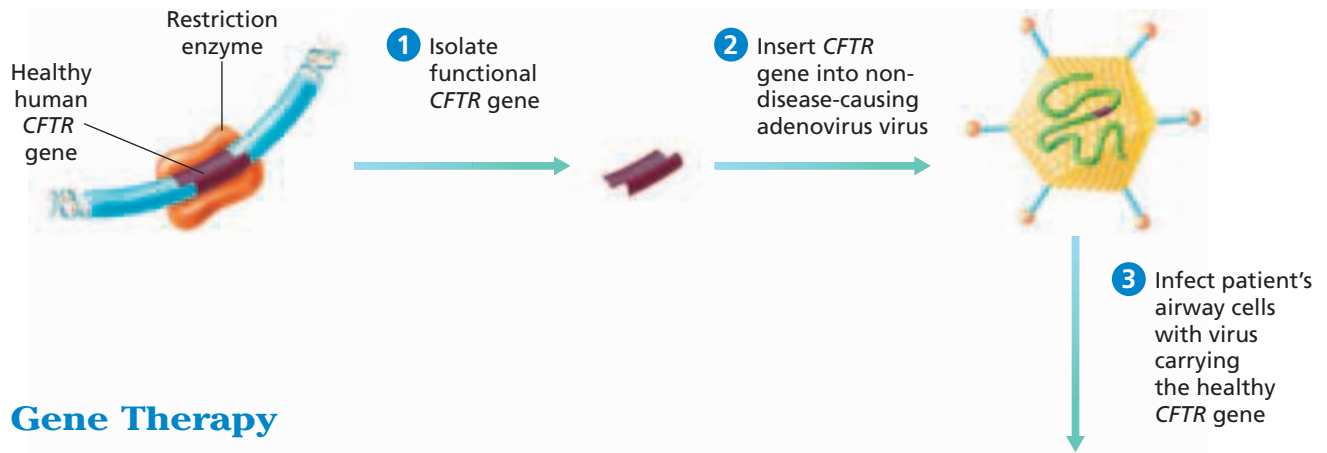
The researchers combined the *Hoxd4* gene and a region adjacent to the gene with a "reporter gene." The reporter gene encodes an enzyme that can make a blue-colored product. They inserted the recombinant DNA into mouse cells, grew embryos, and found that the region adjacent to the *Hoxd4* gene could turn on the reporter gene and its blue product, as shown in Figure 13-10a. When they mutated the adjacent region, they discovered (by the lack of blue color) that it was expressed in the spinal cord, but not in the embryo's hindbrain, as shown in Figure 13-10b. They concluded that the DNA sequence adjacent to *Hoxd4* helps control hindbrain development. Experiments such as these are unraveling the mysteries of gene function during development and may eventually provide therapies for disease.

**FIGURE 13-10**

(a) DNA adjacent to the mouse *Hoxd4* gene participates in normal hindbrain development. (b) After a researcher deliberately mutates the flanking DNA, he or she can see that the *Hoxd4* gene is expressed in the spinal cord but not in the hindbrain.







## Gene Therapy

Genetic engineering has also allowed biologists to try to treat genetic disorders in different ways. One method is a technique called gene therapy. In **gene therapy** a genetic disorder is treated by introducing a gene into a patient's cells. Gene therapy works best for disorders that result from the loss of a single protein. For example, the lung disease *cystic fibrosis* results from the lack of a functional gene called the *CFTR* gene. When functional, the gene encodes a protein that helps transport ions into and out of cells in the breathing passages. Without that gene, poor ion exchange causes the symptoms of cystic fibrosis, including the buildup of sticky mucus that blocks the airways.

Figure 13-11 summarizes the steps involved in gene therapy. In step 1, researchers isolate the functional gene (such as the *CFTR* gene). In step 2, they insert the healthy gene into a viral vector. In step 3, they introduce the recombinant virus to the patient by infecting the patient's airway by means of a nasal spray. The healthy copy of the *CFTR* gene temporarily produces the missing protein and improves ion exchange. The traditional treatment for cystic fibrosis involves thumping sessions—clapping on the back and chest for half-hour periods several times a day to dislodge mucus.

Cystic fibrosis research has accelerated since the discovery of the *CFTR* gene in 1989. In the laboratory, researchers were able to add a healthy copy of *CFTR* into the DNA of cystic fibrosis cells. The result was an immediate return to a normal ion transport mechanism. However, trials in the laboratory are different from trials on living humans. Apparently, the cells that express the highest levels of *CFTR* are deeper in the lungs than the surface cells that current forms of gene therapy can reach. Because the cells that line the airway slough off periodically, the treatment must be repeated. In addition, patients may suffer immune reactions to the treatment. Researchers hope to overcome these obstacles and to one day provide a permanent cure.

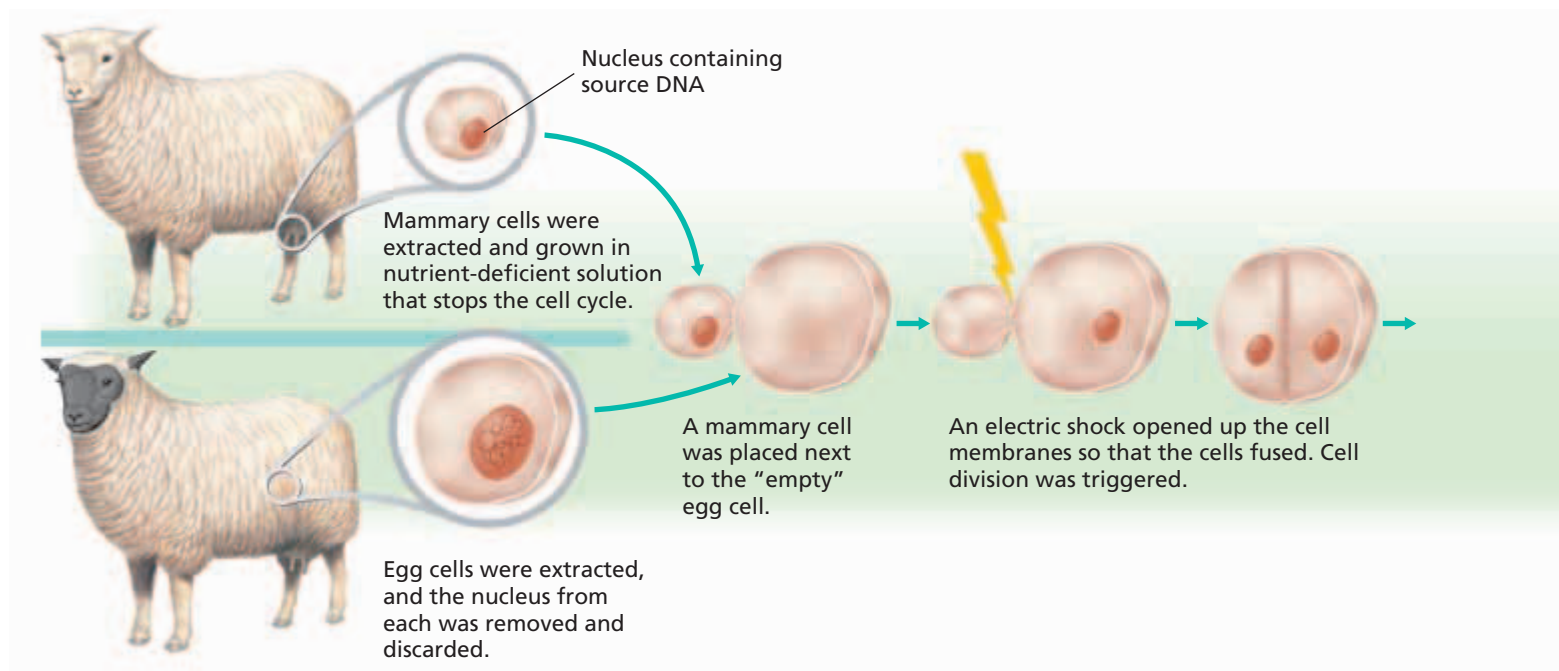
People with certain kinds of hemophilia, acquired immunodeficiency syndrome (AIDS), or some cancers are future candidates for gene therapy. Until recombinant DNAs can be inserted into the correct cells, however, and immune reactions can be prevented, gene therapy may continue to be a short-term solution.



**FIGURE 13-11**

The steps in gene therapy for cystic fibrosis are summarized. The patient in the photo is receiving gene therapy for cystic fibrosis. A healthy copy of the gene responsible for cystic fibrosis is being administered through a nasal spray.





**FIGURE 13-12**

Biologists cloned a sheep by transferring a somatic cell nucleus from one animal into the egg cell of another. They implanted the resulting embryo into a surrogate mother, and the offspring that developed was a clone of the original nucleus donor.

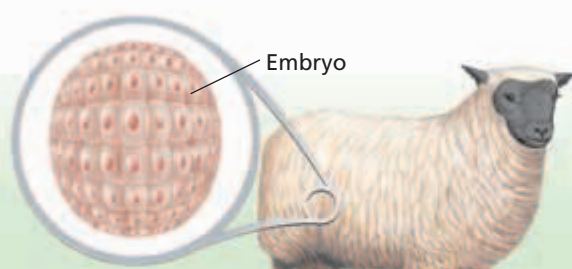
## Cloning

In the 1990s, biologists began cloning whole organisms, such as sheep and mice. The name for this procedure is **cloning by nuclear transfer**, the introduction of a nucleus from a body cell into an egg cell to generate an organism identical to the nucleus donor. The first animal successfully cloned from an adult tissue was a sheep named Dolly in 1996.

As shown in Figure 13-12, scientists in Scotland isolated a mature, functioning mammary cell nucleus from an adult sheep. They also isolated an egg cell from a second sheep and removed the nucleus. They then fused the mammary cell with the "empty" egg cell. The egg was stimulated to divide and grew into an embryo. The researchers implanted this embryo into the uterus of a surrogate mother who gave birth to a lamb, which they called Dolly. Dolly's nuclear DNA was identical to the original donor of the mammary gland cell.

Despite the successful cloning, Dolly suffered premature aging and disease and died at age 6, only half of a normal sheep's lifespan. Researchers found that Dolly had short **telomeres**, or repeated DNA sequences at the ends of chromosomes that shorten with each round of cell division. Short telomeres may be associated with premature aging. Other cloned species, however, have not experienced similar telomere shortening.

The goal of most animal cloning is to alter the genome in some useful way. For example, researchers have altered and cloned goats so that they secrete human blood clotting factors into their milk. Cloned pigs have been altered in the hope that pig livers, hearts, and other organs might not trigger organ rejection if transplanted into human recipients. Some researchers are cloning animals as models for the study of human disease, such as cystic fibrosis.



The embryo developed *in vitro* and was later implanted into a surrogate mother.

After a 5-month pregnancy, a lamb was born that was genetically identical to the sheep from which the mammary cell was extracted.



## Vaccines

A vaccine is a substance containing all or part of a harmless version of a pathogen that physicians introduce into the body to produce immunity to disease. The immune system recognizes the pathogen's surface proteins and responds by making defensive proteins called *antibodies*. A **DNA vaccine** is a vaccine made from the DNA of a pathogen but does not have disease-causing capability. The DNA vaccine is injected into a patient where it directs the synthesis of a protein. The immune system mounts a defense against the protein. If the vaccinated person contacts the disease agent in the future, his or her new immunity should provide protection. Researchers are currently working on developing DNA vaccines to prevent AIDS, malaria, and certain cancers.



## AGRICULTURAL APPLICATIONS

Plant researchers are using genetic engineering to develop new strains of plants called *genetically modified (GM) crops*. In a world of exponentially increasing human population, the need for more food with better nutritional value presents a challenge to plant biologists.

### Increasing Yields and Improving Nutrition

To feed the planet's hungry population, biologists have made crop plants that are more tolerant to environmental conditions. They have also added genes to strains of wheat, cotton, and soybeans that make the plants resistant to weed-controlling chemicals called *herbicides*. To increase the amount of food a crop will yield, researchers have transferred genes for proteins that are harmful to insects and other pests into crop plants. The plants are protected from serious damage and yield more food. Similar techniques have been used to make plants resistant to certain diseases.

### Word Roots and Origins

#### herbicide

from the Latin *herba*, meaning "plant," and *cida*, meaning "to kill"

Genetic engineers have also been able to improve the nutritional value of many crop plants. For example, in Asia many people use rice as a major food source, yet rice has low levels of iron and beta carotene, which the body uses to make vitamin A. As a result, millions suffer from iron and vitamin A deficiencies. Genetic engineers have added genes to rice to overcome these deficiencies.

## ETHICAL ISSUES

**Bioethics** is the study of ethical issues related to DNA technology. Many scientists and nonscientists are involved in identifying and addressing any ethical, legal, and social issues that may arise as genetic engineering techniques continue to be developed. They want to make sure that none of the tools turn out to be dangerous or have unwanted results and that any technology and data that arise are carefully used. Almost all scientists agree that continued restraint and oversight are needed.

For example, some people are concerned that GM food crops might harm the environment in unusual ways. What would happen if introduced genes for herbicide resistance jumped to the wild, weedy relative of a GM crop? To this end, in the mid-1970s, government agencies set standards for safety procedures and required permits and labels for certain GM products. Most biologists agree that rigorous testing should be conducted and safeguards required before farmers release GM organisms into the environment.

Most scientists currently consider gene therapy to be unethical if it involves reproductive cells that would affect future generations. Most people consider cloning of human embryos for reproduction unethical.

The welfare of each patient is most important. Confidentiality of each individual's genetic make up is vital to prohibit discrimination in the workplace. Decisions about ethical issues must be made not just by scientists but by the involvement of an informed public.

### SECTION 3 REVIEW

1. List two types of medical products that can be produced using DNA technology.
2. How have medical researchers used gene therapy to help people with cystic fibrosis?
3. What are the main steps in cloning a sheep?
4. Describe a potential safety and environmental concern with regard to genetically modified (GM) crops.
5. Relate bioethics to the continued development of genetic engineering techniques.

#### CRITICAL THINKING

6. **Forming Reasoned Opinions** Should genetically engineered food products require special labels? Why or why not?
7. **Applying Information** If you were to genetically engineer a crop, what would it be, and how would you improve it?
8. **Evaluating Information** In what way might employers discriminate against a person if his or her genome were known to them?

# CHAPTER HIGHLIGHTS

## SECTION 1

## DNA Technology

- DNA technology provides the tools to manipulate DNA molecules for practical purposes.
- The repeating sequences in noncoding DNA vary between individuals and thus can be used to identify an individual.
- To identify a DNA sample, scientists isolate the DNA and copy it using the polymerase chain reaction (PCR). The DNA is then cut into fragments using restriction enzymes. The fragments are separated by size using gel electrophoresis.
- The resulting pattern is compared to the pattern from a known sample of DNA treated in the same way.
- Restriction enzymes recognize and cut specific nucleotide sequences. This process creates single chains called sticky ends on the ends of each piece of DNA. The enzyme DNA ligase can rejoin sticky ends and connect DNA fragments.
- Researchers use restriction enzymes to insert DNA fragments into vectors. The resulting DNA from two different organisms is called recombinant DNA.

### Vocabulary

length polymorphism (p. 255)  
variable number tandem repeats (VNTR) (p. 255)

polymerase chain reaction (PCR) (p. 256)  
primer (p. 256)  
restriction enzyme (p. 257)

gel electrophoresis (p. 257)  
DNA fingerprint (p. 257)  
genetic engineering (p. 258)  
recombinant DNA (p. 258)

clone (p. 258)  
vector (p. 258)  
plasmid (p. 259)  
probe (p. 259)

## SECTION 2

## The Human Genome Project

- The goals of the Human Genome Project were to determine the nucleotide sequence of the entire human genome and map the location of every gene on each chromosome. This information will advance the diagnosis, treatment, and prevention of human genetic disorders.
- The Human Genome Project yielded important information about human genes and proteins. For example, there are far fewer protein-encoding human genes than once believed but far more proteins because of the complex way they are encoded.
- The Human Genome Project included sequencing the genes of many model species to provide insights into gene function.
- Information from the Human Genome Project has been applied to medical, commercial, and scientific purposes.
- Bioinformatics uses computers to catalog and analyze genomes. Microarrays, two-dimensional arrangements of cloned genes, allow researchers to compare specific genes such as those that cause cancer. Proteomics studies the identities, structures, interactions, and abundances of an organism's proteins.

### Vocabulary

Human Genome Project (p. 261)  
proteome (p. 262)

single nucleotide polymorphism (SNP) (p. 262)

bioinformatics (p. 263)  
proteomics (p. 264)

two-dimensional gel electrophoresis (p. 264)

## SECTION 3

## Genetic Engineering

- Genetic engineering is being used to provide therapies for certain genetic diseases.
- Gene therapy refers to treating genetic disorders by correcting a defect in a gene or by providing a normal form of a gene. Researchers hope that gene therapy can be used to cure genetic disorders in the future.
- In cloning by nuclear transfer, a nucleus from a body cell of one individual is introduced into an egg cell (without its nucleus) from another individual. An organism identical to the nucleus donor results.
- Genetic engineering is used to produce disease-resistant, pest-resistant, and herbicide-resistant crops in an effort to improve the yields and nutrition of the human food supply.
- Some people fear that the release of genetically modified organisms would pose a separate environmental risk. Many safety, environmental, and ethical issues involved in genetic engineering have not been resolved.

### Vocabulary

gene therapy (p. 267)

cloning by nuclear transfer (p. 268)

telomere (p. 268)  
DNA vaccine (p. 269)

bioethics (p. 270)



# CHAPTER REVIEW

## USING VOCABULARY

1. Explain the relationship between *length polymorphisms* and *variable number tandem repeats (VNTR)*.
2. For each pair of terms, explain how the meanings of the terms differ.
  - a. *gel electrophoresis* and *DNA fingerprint*
  - b. *restriction enzyme* and *recombinant DNA*
  - c. *vector* and *plasmid*
3. Use the following terms in the same sentence: *cloning by nuclear transfer* and *telomeres*.
4. **Word Roots and Origins** The word *herbicide* derives from the Latin *herba*, meaning “plant” and *cida*, meaning “to kill.” Using this information, explain how the term *herbicide* relates to an increase in food crop yields.

## UNDERSTANDING KEY CONCEPTS

5. **Describe** how variable number tandem repeats (VNTR) are used to identify DNA.
6. **Name** the four major steps used in DNA identification.
7. **Identify** the purpose for using the polymerase chain reaction (PCR).
8. **Explain** how restriction enzymes are used.
9. **Describe** how a DNA fingerprint is prepared.
10. **Relate** cloning vectors to recombinant DNA.
11. **Name** two applications of DNA identification.
12. **State** the two major goals of the Human Genome Project.
13. **Discuss** three insights gained from the Human Genome Project.
14. **Describe** the importance of sequencing the genomes of model species.
15. **State** how information from the Human Genome Project will be applied to future projects.
16. **Discuss** the role of bioinformatics in the Human Genome Project.
17. **Explain** how gene therapy has been used to treat cystic fibrosis.
18. **Describe** how biologists used cloning by nuclear transfer to clone animals.
19. **List** two ways that genetic engineering may increase crop yields.
20. **Describe** the steps scientists are taking to ensure that genetic engineering techniques are carefully used in society.

21.  **CONCEPT MAPPING** Use the following terms to create a concept map about genetic engineering: *DNA of interest*, *vectors*, *recombinant DNA*, *plasmids*, *restriction enzymes*, and *sticky ends*.

## CRITICAL THINKING

22. **Applying Information** In the past, breeders developed new plant and animal varieties by selecting organisms with desirable traits and breeding them. Name an advantage and a disadvantage of genetic engineering techniques over this method.

### 23. Unit 6—Gene Expression

 The United States has regulations regarding the release of genetically engineered organisms into the environment. Write a report about reasons why people have concerns regarding the release of such organisms.

24. **Interpreting Graphics** The photograph below shows eight lanes (columns) in a gel. Several of these lanes contain DNA fingerprints of samples taken from a crime scene, a victim, and four suspects. State which suspect's DNA fingerprint matches the blood found at the crime scene. Is it likely that the blood found at the crime scene belongs to the suspect? Explain.



#### KEY

- |            |             |
|------------|-------------|
| 1 Control  | 5 Suspect 1 |
| 2 Blood    | 6 Suspect 2 |
| 3 Victim   | 7 Suspect 3 |
| 4 Standard | 8 Suspect 4 |

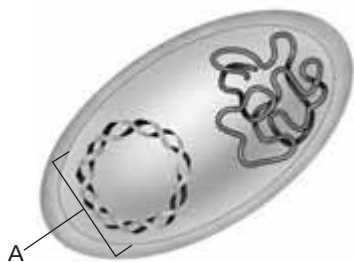


# Standardized Test Preparation

**DIRECTIONS:** Choose the letter of the answer choice that best answers the question.

- Which is a molecule containing DNA from two different organisms?  
A. vector DNA  
B. a DNA clone  
C. plasmid DNA  
D. recombinant DNA
- Which of the following is used to cut DNA molecules in specific locations?  
F. cloning vectors  
G. cloning enzymes  
H. restriction enzymes  
J. polymerase chain reaction
- What is the term used for inserting a healthy copy of a gene into a person who has a defective gene?  
A. cloning vector  
B. gene therapy  
C. recombinant DNA  
D. polymerase chain reaction (PCR)
- Which is the process used in animal cloning?  
F. DNA cloning  
G. recombinant DNA  
H. polymerase chain reaction  
J. cloning by nuclear transfer

**INTERPRETING GRAPHICS:** The graphic below shows a bacterial cell. Use the graphic to answer the questions that follow.

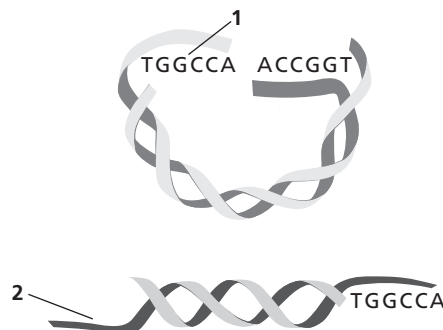


- Which best describes molecule A?  
A. It is an insulin gene.  
B. It is recombinant DNA.  
C. It is a bacterial plasmid.  
D. It is a disease-causing virus.
- How is a bacterial plasmid described after donor DNA is inserted into the bacterium's DNA?  
F. vector DNA  
G. cloned DNA  
H. plasmid DNA  
J. recombinant DNA

**DIRECTIONS:** Complete the following analogy.

7. Proteomics : proteins :: genomics :  
A. lipids  
B. genes  
C. proteins  
D. carbohydrates

**INTERPRETING GRAPHICS:** The diagram below is of two pieces of DNA that were cut with the same restriction enzyme. Use the diagram to answer the question that follows.



8. Which nucleotide sequence must the sticky end labeled 2 have if it is to bond with the sticky end labeled 1?  
F. UGGCCU  
G. TCCGGA  
H. ACCGGT  
J. CTTAAG

## SHORT RESPONSE

A probe is a strand of RNA or single-stranded DNA that is labeled with a radioactive element or fluorescent dye.

How do biologists use a probe to find cloned DNA?

## EXTENDED RESPONSE

One concern about genetic engineering involves confidentiality and insurance.

*Part A* How could the human genome be misused, relative to confidentiality issues?

*Part B* What might people's concern be about health insurance, and why?

## Test TIP

Check for careless mistakes such as marking two answers for a single question.

# Analyzing DNA Using Gel Electrophoresis

## OBJECTIVES

- Use restriction enzymes to cut DNA.
- Separate DNA fragments of different sizes.

## PROCESS SKILLS

- experimenting
- analyzing
- calculating

## MATERIALS

- |                                                      |                                |
|------------------------------------------------------|--------------------------------|
| ■ protective clothing                                | ■ water bath (37°C)            |
| ■ ice, crushed                                       | ■ gel-casting tray             |
| ■ ice bucket                                         | ■ gel comb, 6-well             |
| ■ microtube rack                                     | ■ hot-water bath (65°C)        |
| ■ microtube A—Uncut DNA                              | ■ agarose (0.8%)               |
| ■ microtube B— <i>Hind</i> III                       | ■ hot mitt                     |
| ■ microtube C— <i>Bam</i> HI                         | ■ zipper-lock plastic bag      |
| ■ microtube D— <i>Eco</i> RI                         | ■ graduated cylinder (10 mL)   |
| ■ microtube E—Unknown                                | ■ 1× TBE buffer                |
| ■ permanent marker                                   | ■ freezer                      |
| ■ micropipetter (0.5–10 µL)                          | ■ loading dye (5 µL)           |
| ■ micropipetter tips                                 | ■ gel chamber and power supply |
| ■ 10× restriction buffer for each restriction enzyme | ■ WARD'S DNA stain             |
| ■ Lambda virus DNA (34 µL)                           | ■ staining trays               |
|                                                      | ■ water, distilled             |
|                                                      | ■ ruler, metric                |

## Background

1. DNA has a negative charge and flows toward the positive end of a gel during electrophoresis. Small DNA fragments move faster than larger fragments.
2. The distance that each DNA fragment moves is used to calculate the  $R_f$ , or relative mobility, of a fragment. The  $R_f$  is used to calculate the number of base pairs in the fragment.

## PART A Cutting DNA

1.    Wear safety goggles, gloves, and a lab apron at all times.

2. Fill an ice bucket with ice. Obtain one each of the following microtubes: A—Uncut DNA, B—*Hind*III, C—*Bam*HI, D—*Eco*RI, and E—Unknown. Microtubes B–D contain 1 µL of the indicated restriction enzyme. Place all microtubes in the ice. *Restriction enzymes MUST be kept on ice until step 6.*
3. With a permanent marker, write the initials for everyone in your group *on the top* of microtubes A–E.
4.  **CAUTION** If you get a chemical on your skin or clothing, wash it off at the sink while calling to your teacher. Set a micropipetter to 1 µL, and put a tip on the end of the micropipetter, as shown in (a) below. Using a new tip for each microtube, add 1 µL of the corresponding 10× restriction buffer to each of microtubes B–D. Place the buffer on the side of the tube. *Do not touch the micropipetter tips to the solutions in the microtubes.*



(a) MICROPIPETTER



(b) CASTING TRAY

5. Reset the micropipetter to 8 µL. Using the micropipetter and a new tip for each microtube, add 8 µL of the Lambda virus DNA to the side of each of microtubes B–D. Gently flick the bottom of each microtube with your finger to mix the solutions. *Do not shake the microtubes!* Reset the micropipetter to 10 µL, and add 10 µL of Lambda DNA to microtube A.
6. Place all of the microtubes into a 37°C water bath. After 50–60 minutes, remove the microtubes from the water bath, and immediately put them into a freezer. If the class period ends before 50 minutes have passed, your teacher will give you further directions. While the restriction enzymes are working, go to Part B.

## PART B Preparing an Agarose Gel

7. Set up a gel-casting tray, as shown in (b). Place a gel comb in the grooves of the gel-casting tray. Make sure that the comb does not touch the bottom of the tray. If it does, get another comb from your teacher.
8. Write the names of the members of your group on a paper towel. Carry your tray to the table with the melted agarose, and place your tray on the paper towel.
9.  Using a hot mitt, pour melted 0.8% agarose into your gel-casting tray until the agarose reaches a depth of 3 mm. Make sure that the agarose spreads evenly throughout the tray. *Do not move your gel tray before the agarose solidifies.*
10. Let the gel cool (about 20–30 minutes) until the agarose solidifies.
11. While the gel is cooling, write your name, the date, and your class period on a zipper-lock plastic bag. Pour 5 mL of 1× TBE buffer into the bag.
12. When the gel has solidified, carefully remove the gel comb by pulling it straight up. If the comb does not come up easily, pour a little 1× TBE buffer on the comb area. After removing the gel comb, open the plastic bag and carefully slide the gel tray into the bag. *Do not remove the gel from the gel-casting tray.* Store the gel according to your teacher's instructions.

## PART C Running a Gel

13. Retrieve your microtubes (A–E) and your gel. If the materials in the microtubes are frozen, hold each tube in your hand until the solutions thaw.
14. Set a micropipetter to 1 µL, and place a tip on the end. Add 1 µL of loading dye to each microtube. *Use a new tip for each microtube.* Gently tap each microtube on your lab table to thoroughly mix the solutions. *Do not shake the microtubes.*
15. Remove your gel (still in the gel-casting tray) from the plastic bag, and place it in a gel chamber. Orient the gel so that the wells are closest to the black wire, or anode.
16. Set a micropipetter to 10 µL, and place a new tip on the end. Open microtube A, and remove 10 µL of solution. Carefully place the solution into the well in lane 1, the left-most lane. To do this, place both elbows on the lab table, lean over the gel, and slowly lower the micropipetter tip into the opening of the well before depressing the plunger. *Do not jab the micropipetter tip through the bottom of the well.*

17. *Using a new micropipetter tip for each tube, repeat step 16 for each of the remaining microtubes.* Use lane 2 for microtube B, lane 3 for microtube C, lane 4 for microtube D, and lane 5 for microtube E.
18. Very slowly fill the gel chamber with 1× TBE buffer until the level of the buffer is approximately 1–2 mm above the surface of the gel.
19.  **CAUTION** Follow all of the manufacturer's precautions regarding the use of this equipment. Close the gel chamber, and connect it to a power supply according to your teacher's instructions.
20. Allow an electric current to flow through the gel. You will see a blue line moving away from the wells. When the blue line is approximately 5 mm from the end of the gel, disconnect the power supply and remove the gel. Store the gel overnight in the plastic bag.

## PART D Analyzing a Gel

21. To stain a gel, carefully place the gel (wells up) into a staining tray. Pour WARD'S DNA stain into the staining tray until the gel is completely covered. Cover the staining tray, and label it with your initials. Allow the stain to sit for at least 2 hours. Next, carefully pour the stain into the sink drain, and flush it down the drain with water. *Do not let the stained gel slip out of the staining tray.*
22. To destain a gel, cover the gel with distilled water by pouring water to one side of the gel. Let the gel sit overnight (or at least 8–12 hours). The bands of DNA will appear as purple lines against a light background.
23. Calculate the  $R_f$  for each fragment using the following equation:
$$R_f = \frac{\text{distance in mm that DNA fragment migrated}}{\text{distance in mm from well to the dye}}$$
24.   Dispose of your materials according to your teacher's directions, and wash your hands before leaving the lab.

## Analysis and Conclusions

1. Which two samples appear to have the same pattern of DNA bands?
2. Which restriction enzyme cut the DNA in the unknown sample? Justify your answer.
3. What are some measures that you took to prevent contamination of your DNA samples during this lab?



# UNIT 4

# EVOLUTION

“The past, the finite greatness of the past! For what is the present, after all, but a growth out of the past.”

Walt Whitman

## CHAPTERS

- 14 *History of Life*
- 15 *Theory of Evolution*
- 16 *Population Genetics and Speciation*
- 17 *Classification of Organisms*



Charles Darwin studied the giant tortoises (above) on the Galápagos Islands. Observations that Darwin made during his voyage on the HMS Beagle (right) led him to look for a mechanism by which evolution occurs.



### Internet connect

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Internet resources are located throughout this unit.

**sciLINKS**  
Maintained by the  
National Science  
Teachers Association



Archeologists (above) scrutinize fossilized remains of human ancestors for clues to human evolution. Darwin observed a number of finches (right) on the Galápagos Islands and found that each species was adapted to a different food source.



**Australopithecus  
robustus**

# HISTORY OF LIFE

Molten rock breaks through Earth's crust in the form of a volcano. Conditions in a volcano are similar to those thought to have been present on early Earth.



**SECTION 1** *Biogenesis*

**SECTION 2** *Earth's History*

**SECTION 3** *The First Life-Forms*



# BIOGENESIS

The principle of **biogenesis** (BIE-oh-JEN-uh-sis), which states that all living things come from other living things, seems very reasonable to us today. Before the seventeenth century, however, it was widely thought that living things could also arise from nonliving things in a process called **spontaneous generation**. This seemed to explain why maggots appeared on rotting meat and why fish appeared in ponds that had been dry the previous season—people thought mud might have given rise to the fish. In attempting to learn more about the process of spontaneous generation, scientists performed controlled experiments. As you read about these experiments, refer to the figures that show the experimental design.

## OBJECTIVES

- **Compare** the principle of biogenesis with the idea of spontaneous generation.
- **Summarize** the results of experiments by Redi and by Spallanzani that tested the hypothesis of spontaneous generation.
- **Describe** how Pasteur's experiment disproved the hypothesis of spontaneous generation.

## VOCABULARY

biogenesis  
spontaneous generation

## REDI'S EXPERIMENT

Flies have often been viewed as pesky creatures. Most people are too busy trying to get rid of flies to even think about studying them. In the middle of the 17th century, however, the Italian scientist Francesco Redi (1626–1697) noticed and described the different developmental forms of flies. Redi observed that tiny wormlike maggots turned into sturdy oval cases, from which flies eventually emerge. He also observed that maggots seemed to appear where adult flies had previously landed.

These observations led Redi to question the commonly held belief that flies were generated spontaneously from rotting meat. Figure 14-1 shows an experiment that Redi conducted in 1668 to test his hypothesis that meat kept away from adult flies would remain free of maggots.

**FIGURE 14-1**

In Redi's experiment, maggots were found only in the control jars because that was the only place where adult flies could reach the meat to lay eggs.





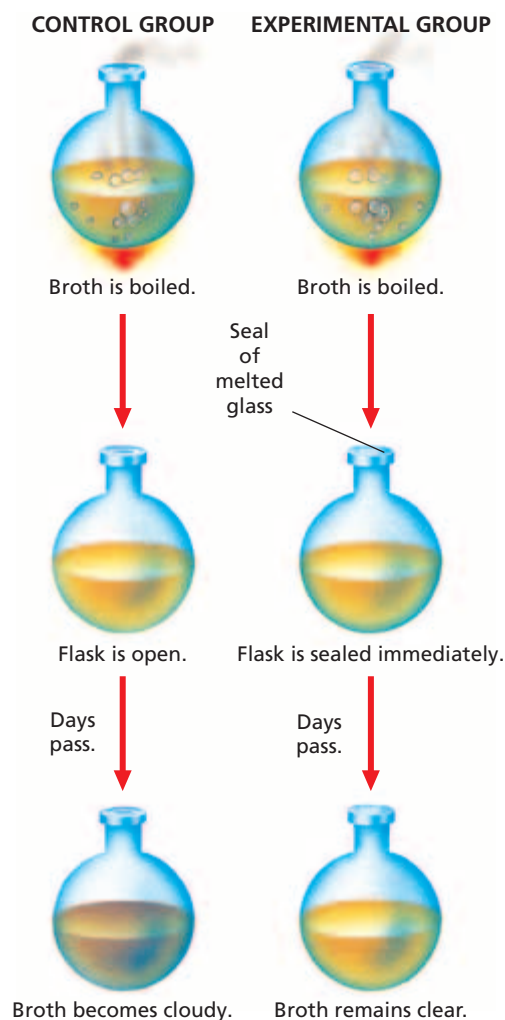


Redi's experimental group consisted of netting-covered jars that contained meat. The control group consisted of uncovered jars that also contained meat. The netting allowed air to enter and prevented flies from landing on the meat. After a few days, maggots were living in the meat in the open jars, but the net-covered jars remained free of maggots. Redi's experiment showed convincingly that flies come only from eggs laid by other flies. Redi's hypothesis was confirmed, and a major blow was struck against the hypothesis of spontaneous generation.

## SPALLANZANI'S EXPERIMENT

**FIGURE 14-2**

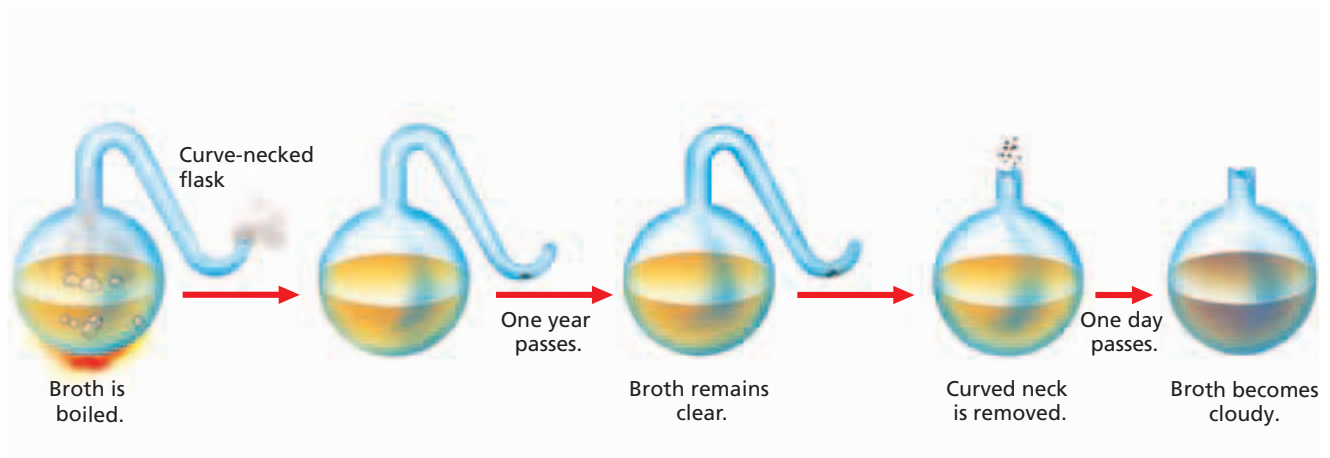
In Spallanzani's experiment, he boiled meat broth in open flasks. Then, he sealed the flasks of the experimental group by melting the glass necks of the flasks closed. The broth inside remained uncontaminated by microorganisms.



At about the same time that Redi carried out his experiment, other scientists began using a new tool—the microscope. Their observations with the microscope revealed that the world is teeming with tiny creatures. They discovered that microorganisms are simple in structure and amazingly numerous and widespread. Many investigators at the time thus concluded that microorganisms arise spontaneously from a “vital force” in the air.

In the 1700s, another Italian scientist, Lazzaro Spallanzani (1729–1799), designed an experiment to test the hypothesis of spontaneous generation of microorganisms, as shown in Figure 14-2. Spallanzani hypothesized that microorganisms formed not from air but from other microorganisms. He knew that microorganisms grew easily in food, such as broth made from boiled meat. Spallanzani reasoned that boiling broth in a flask would kill all the microorganisms in the broth, on the inside of the glass, and in the air in the flask. For his experimental group, Spallanzani boiled clear, fresh broth until the flasks filled with steam. While the broth was hot, he sealed the flasks by melting their glass necks. The control-group flasks of broth were left open. The broth in the sealed flasks remained clear and free of microorganisms, while the broth in the open flasks became cloudy because it was contaminated with microorganisms.

Spallanzani concluded that the boiled broth became contaminated only when microorganisms from the air entered the flask. Spallanzani's opponents, however, objected to his method and disagreed with his conclusions. They claimed that Spallanzani had heated the experimental flasks too long, destroying the “vital force” in the air inside them. Air lacking this “vital force,” they claimed, could not generate life. Thus, those who believed in spontaneous generation of microorganisms kept the idea alive for another century.



**FIGURE 14-3**

In Pasteur's experiment, a flask with a curved but open neck prevented microorganisms from entering. Broth boiled in the flasks became contaminated by microorganisms only when the curved necks were removed from the flasks.

## PASTEUR'S EXPERIMENT

By the mid-1800s, the controversy over spontaneous generation had grown fierce. The Paris Academy of Science offered a prize to anyone who could clear up the issue once and for all. The winner of the prize was the French scientist Louis Pasteur (1822–1895).

Figure 14-3 shows how Pasteur set up his prize-winning experiment. To answer objections to Spallanzani's experiment, Pasteur made a curve-necked flask that allowed the air inside the flask to mix with air outside the flask. The curve in the neck of the flask prevented solid particles, such as microorganisms, from entering the body of the flask. Broth boiled inside the experimental curve-necked flasks remained clear for up to a year. But when Pasteur broke off the curved necks, the broth became cloudy and contaminated with microorganisms within a day. Pasteur reasoned that the contamination was due to microorganisms in the air.

Those who had believed in the spontaneous generation of microorganisms gave up their fight. With Pasteur's experiment, the principle of biogenesis became a cornerstone of biology.

### Word Roots and Origins

#### *biogenesis*

from the Greek *bioun*, meaning "to live," and *gignesthai*, meaning "to be born"

## SECTION 1 REVIEW

1. What does the term *spontaneous generation* mean?
2. Explain how Redi's experiment disproved the hypothesis that flies formed in food by spontaneous generation.
3. What caused people to think the air contained a "vital force" that produced living organisms?
4. Describe the argument that Spallanzani's experiment failed to disprove the occurrence of spontaneous generation. Explain how Pasteur's experiment addressed these criticisms.

### CRITICAL THINKING

5. **Making Comparisons** Spallanzani and Pasteur both used a technique that is now widely used to preserve food. What was this technique?
6. **Analyzing Methods** What would have happened if Pasteur had tipped one of his flasks so that the broth in the flask had come into contact with the curve of the neck?
7. **Analyzing Ideas** If spontaneous generation does not occur and the principle of biogenesis is true, what scientific question remains?

## SECTION 2

### OBJECTIVES

- **Outline** the modern scientific understanding of the formation of Earth.
- **Summarize** the concept of half-life.
- **Describe** the production of organic compounds in the Miller-Urey apparatus.
- **Summarize** the possible importance of cell-like structures produced in the laboratory.

### VOCABULARY

radiometric dating  
isotope  
mass number  
radioactive decay  
radioactive isotope  
half-life  
microsphere  
coacervate

# EARTH'S HISTORY

*If spontaneous generation does not happen on Earth today, then the question remains: How did cell-based life arise in the first place? The key to answering this question lies in scientific hypotheses that conditions on early Earth were very different from present conditions. Scientists continue to form and test these hypotheses by modeling conditions and processes that could have given rise to the first cellular life on Earth.*

## THE FORMATION OF EARTH

Evidence from computer models of the sun suggests that about 5 billion years ago, our solar system was a swirling mass of gas and dust, as shown in Figure 14-4. Over time, most of this material was pulled together by gravity and formed the sun. The remaining gas, dust, and debris circled the young sun. Scientists think that the planets formed through repeated collisions of this space debris. During a 400 million-year period, Earth grew larger as gravity pulled in more debris. The collisions between Earth and space debris also released a great deal of thermal energy. Some collisions would have released enough energy to melt large portions of Earth's surface.

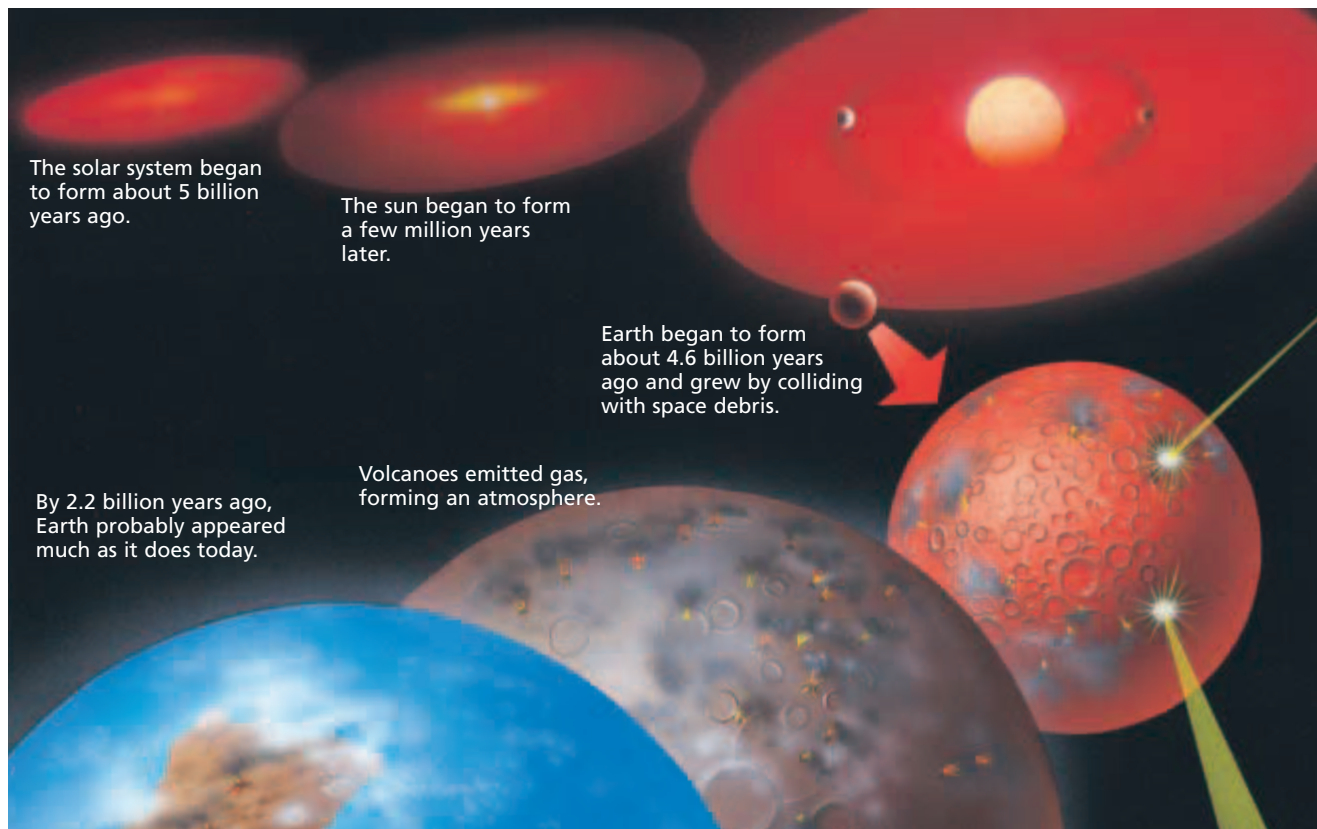
### Earth's Age

The estimated age of Earth, more than 4 billion years, is about 700,000 times as long as the period of recorded history. It is about 50 million times as long as an average human life span. How can we determine what happened so long ago? Scientists have explored Earth's surface and examined its many layers to establish a fairly complete picture of its geologic history. Early estimates of Earth's age were made from studying layers of sedimentary rock in Earth's crust. The age of Earth could not be estimated accurately, however, until the middle of the twentieth century, when modern methods of establishing the age of materials were developed.

### Radiometric Dating

Methods of establishing the age of materials include the techniques known as **radiometric dating**. Recall that the atomic number of an element is the number of protons in the nucleus. All atoms of an element have the same atomic number, but their number of neutrons can vary. Atoms of the same element that differ in the number of neutrons they contain are called **isotopes** (IE-suh-TOHPs). Most elements have several isotopes.





The **mass number** of an isotope is the total number of protons and neutrons in the nucleus. The mass number of the most common carbon isotope is 12. If you recall that the atomic number of carbon is 6, you can calculate that this carbon isotope has six protons and six neutrons. Isotopes are designated by their chemical name followed by their mass number; for example, carbon exists as both *carbon-12* and *carbon-14*.

Some isotopes have unstable nuclei, which undergo **radioactive decay**; that is, their nuclei release particles or radiant energy, or both, until the nuclei become stable. Such isotopes are called **radioactive isotopes**. Rates of decay of radioactive isotopes have been determined for many isotopes. The length of time it takes for one-half of any size sample of an isotope to decay to a stable form is called the **half-life** of the isotope. Depending on the isotope, half-lives vary from a fraction of a second to billions of years.

The quantity of a particular radioactive isotope in a material can be measured to determine the material's age. This amount is compared with that of some other substance whose quantity in the material remains constant over time. For example, organic materials can be dated by comparing the amount of carbon-14, a radioactive isotope, with the amount of carbon-12, a stable isotope. Living things take carbon into their bodies constantly. Most of the carbon is in the form of carbon-12. A very small proportion of it, however, is in the form of carbon-14, which undergoes decay. This ratio of carbon-14 to carbon-12 is a known quantity for living organisms.

**FIGURE 14-4**

It took about one-half billion years for Earth to form from a swirling mass of gases. Sometime after Earth formed, living organisms began to affect the atmosphere.

### Word Roots and Origins

#### **isotope**

from the Greek *isos*, meaning "equal," and *topos*, meaning "place"





## Quick Lab

### Modeling Radioactive Decay

**Materials** shoebox, 50 pennies, 50 paper clips

**Procedure** Place the pennies in the shoebox. Close the lid, shake the box, and then set the box down. Replace each face-up penny with a paper clip. Record the number of pennies and paper clips in the box. Repeat this process five times.

**Analysis** If the pennies in the box represent an unstable isotope, such as carbon-14, what do the paper clips represent? What does each shake of the box represent?

**FIGURE 14-5**

This deerskin quiver, with a wooden bow and arrows, is about 3,000 years old. Carbon-14 dating methods can be used for organic materials less than 60,000 years old.



**TABLE 14-1** Some Isotopes Used in Radiometric Dating

| Isotope      | Half-life (years) |
|--------------|-------------------|
| Carbon-14    | 5,730             |
| Uranium-235  | 704,000,000       |
| Potassium-40 | 1,250,000,000     |
| Uranium-238  | 4,500,000,000     |

When an organism dies, its uptake of carbon stops, and decay of the existing carbon-14 continues. Thus, over time, the amount of carbon-14 declines with respect to the original amount of the stable carbon-12. After 5,730 years, half of the carbon-14 in a sample will have decayed. After another 5,730 years, half of the remaining carbon-14 in the sample likewise will have decayed. Use of carbon-14 dating is limited to organic remains less than about 60,000 years old, such as the leather quiver and wooden bow and arrows shown in Figure 14-5. Isotopes with longer half-lives are used to date older rocks.

Radioactive isotopes occur naturally in all matter. Some of the isotopes commonly used in radiometric dating procedures appear in Table 14-1. Radiometric dating is accurate within certain limits. The techniques depend on making careful measurements and obtaining samples that are uncontaminated with more recent material. Scientists compare several types of independent measurements to determine a range of plausible dates.

Scientists have estimated Earth's age by using a dating method that is based on the decay of uranium and thorium isotopes in rock crystals. Collisions between Earth and large pieces of space debris probably caused the surface of Earth to melt many times as the planet was formed. Therefore, the age of the oldest unmelted surface rock should tell us when these collisions stopped and the cooling of Earth's surface began. The oldest known rocks and crystals are about 4 billion years old. So, scientists infer that organic molecules began to accumulate about 4 billion years ago.

## FIRST ORGANIC COMPOUNDS

All of the elements found in organic compounds are thought to have existed on Earth and in the rest of the solar system when the Earth formed. But how and where were these elements assembled into organic compounds? An important hypothesis to solve this puzzle was proposed in the 1920s by two scientists: Soviet Alexander I. Oparin (1894–1980) and American John B. S. Haldane (1892–1964). They thought that the early atmosphere contained ammonia,  $\text{NH}_3$ ; hydrogen gas,  $\text{H}_2$ ; water vapor,  $\text{H}_2\text{O}$ ; and compounds made of hydrogen and carbon, such as methane,  $\text{CH}_4$ .

According to Oparin, at high temperatures, these gases might have formed simple organic compounds, such as amino acids. When Earth cooled and water vapor condensed to form lakes and seas, these simple organic compounds would have collected in the water. Over time, these compounds could have entered complex chemical reactions, fueled by energy from lightning and ultraviolet radiation. These reactions, Oparin reasoned, ultimately would have resulted in the macromolecules essential to life, such as proteins.

## Synthesis of Organic Compounds

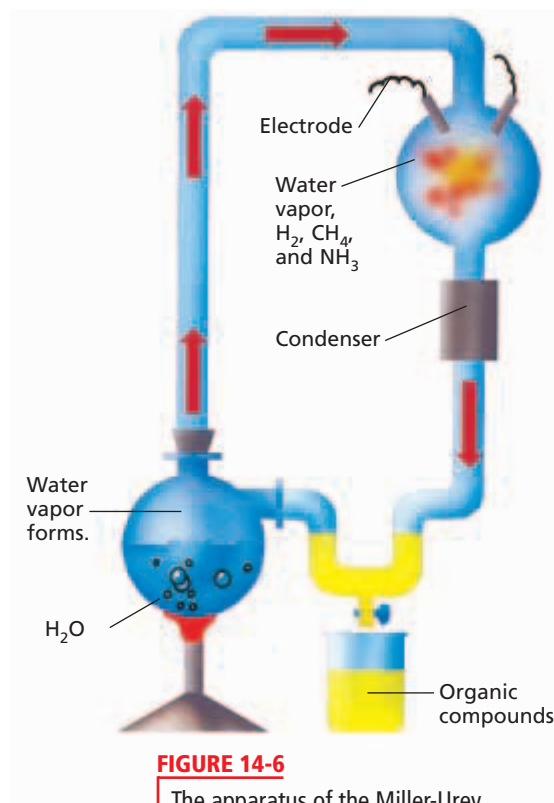
Oparin carefully developed his hypotheses, but he did not perform experiments to test them. So, in 1953, an American graduate student, Stanley L. Miller (1930–), and his professor, Harold C. Urey (1893–1981), set up an experiment using Oparin's hypotheses as a starting point. Their apparatus, illustrated in Figure 14-6, included a chamber containing the gases Oparin assumed were present in the young Earth's atmosphere. As the gases circulated in the chamber, electric sparks, substituting for lightning, supplied energy to drive chemical reactions. The Miller-Urey experiment, and other variations that have followed, produced a variety of organic compounds, including amino acids.

Since the 1950s, scientists have used similar experiments to test and revise hypotheses about the origin of simple organic compounds. In such experiments, scientists have combined a variety of chemicals and energy sources to produce an assortment of organic compounds, including amino acids, ATP, and nucleotides. Scientists are convinced that basic organic compounds could have formed on early Earth in many ways.

Furthermore, scientists who study planet formation have proposed new hypotheses about early Earth's atmosphere. For example, one hypothesis holds that the atmosphere of early Earth was composed largely of carbon dioxide,  $\text{CO}_2$ ; nitrogen,  $\text{N}_2$ ; and water vapor,  $\text{H}_2\text{O}$ . Laboratory simulations of these atmospheric conditions have shown that both carbon dioxide and oxygen gas interfere with the production of organic compounds. Therefore, the production of organic compounds might only have been possible in areas protected from the atmosphere, such as those that exist in undersea hot springs.

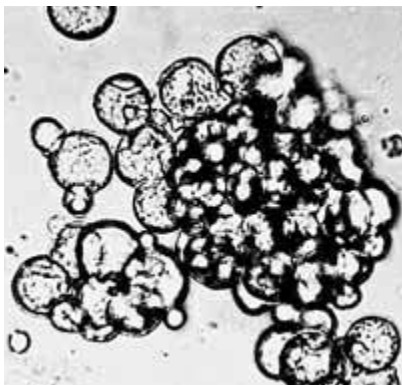
## Organic Compounds from Beyond Earth

Some scientists hypothesize that organic compounds could have been carried to Earth by debris from space. In 1970, a broad mixture of organic compounds was found in a newly fallen meteorite. Because the meteorite was recovered before it was contaminated with organic compounds from Earth, these compounds must have formed in space. So, organic compounds from space could have accumulated on the surface of early Earth.



**FIGURE 14-6**

The apparatus of the Miller-Urey experiment was intended to test Oparin's hypothesis about the conditions for formation of organic molecules on early Earth. Although similar experiments and hypotheses continue to be revised and tested, the experiment was an important milestone in this area of scientific investigation.



**FIGURE 14-7**

Membrane-bound structures, such as these, have been formed in the laboratory under conditions that may have existed on early Earth. Structures such as these may have enclosed replicating molecules of RNA and may have been the forerunners of the first cells.

## FROM MOLECULES TO CELL-LIKE STRUCTURES

Sidney Fox (1912–1998) and others have done extensive research on the physical structures that may have given rise to the first cells. These cell-like structures, such as the ones shown in Figure 14-7, form spontaneously in the laboratory from solutions of simple organic chemicals. The structures include **microspheres**, which are spherical in shape and are composed of many protein molecules that are organized as a membrane, and **coacervates** (coh-AS-uhr-VAYTS), which are collections of droplets that are composed of molecules of different types, including lipids, amino acids and sugars.

For many years, it had been assumed that all cell structures and the chemical reactions of life required enzymes that were specified by the genetic information of the cell. Both coacervates and microspheres, however, can form spontaneously under certain conditions. For example, the polymers that form microspheres can arise when solutions of simple organic chemicals are dripped onto the surface of hot clay. The heat vaporizes the water, encouraging polymerization. Coacervates and microspheres have a number of cell-like properties, including the ability to take up certain substances from their surroundings. Coacervates can grow, and microspheres can bud to form smaller microspheres. These properties of coacervates and microspheres show that some important aspects of cellular life can arise without direction from genes. Thus, these studies suggest that the gap between the nonliving chemical compounds and cellular life may not be quite as wide as previously thought.

However, microspheres and coacervates do not have all of the properties of life. These cell-like structures do not have hereditary characteristics. Thus, these structures cannot respond to natural selection. Scientists are still investigating hypotheses about how living cells may have formed from simpler ingredients.

## SECTION 2 REVIEW

1. Outline the major steps in the formation of Earth, as reconstructed by modern scientists.
2. If 1.0 g of a radioactive isotope had a half-life of 1 billion years, how much of it would be left after each of the following intervals of time: 1 billion years, 2 billion years, 3 billion years, and 4 billion years?
3. What are two possible sources of simple organic compounds on early Earth?
4. What properties do microspheres and coacervates share with cells?

### CRITICAL THINKING

5. **Evaluating Conclusions** Which parts of Oparin's hypothesis were tested by the Miller-Urey experiment? Which parts were not tested?
6. **Making Inferences** Some radioactive isotopes that are used in medicine have half-lives of a few years. Would these isotopes also be useful in dating rocks? Why or why not?
7. **Designing an Experiment** Form your own hypothesis about a stage in the formation or development of early life on Earth. Describe how scientists might test this hypothesis.

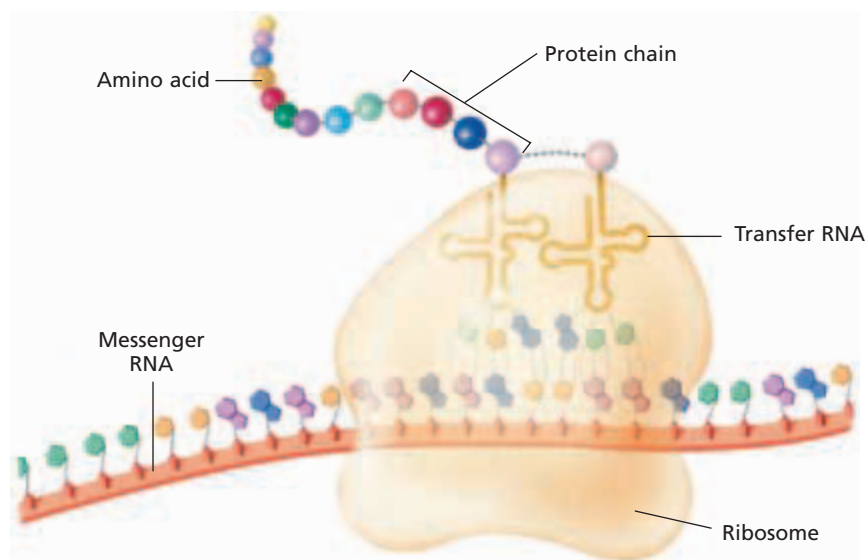
# THE FIRST LIFE-FORMS

*Scientists continue to investigate many competing hypotheses about possible transitions from simple organic molecules to cellular life. A critical part of each such hypothesis is to explain how molecules could be organized into self-replicating systems, in other words, to explain the origin of heredity.*

## THE ORIGIN OF HEREDITY

Chapter 10 provides a detailed explanation of how hereditary information affects the phenotype of cells. Recall that the hereditary information contained in a DNA molecule is first transcribed into an RNA message, and then the RNA message is translated into a protein, as shown in Figure 14-8. Thus, DNA serves as the template for RNA, which in turn serves as the template for specific proteins.

In recent years, scientists have taken a closer look at the DNA-RNA-protein sequence. Why is RNA necessary for this process? Why doesn't DNA, which is a template itself, carry out protein synthesis directly? The clues to a more complete understanding of RNA function may be found in its structure. RNA molecules take on a greater variety of shapes than DNA molecules do. An example is the unique shape of transfer RNA, shown in Figure 14-8. These shapes are dictated by hydrogen bonds between particular nucleotides in an RNA molecule, much as the shapes of proteins depend on hydrogen bonds between particular amino acids. These questions and observations led to the speculation that some RNA molecules might actually behave like proteins and catalyze chemical reactions.



## OBJECTIVES

- **Explain** the importance of the chemistry of RNA in relation to the origin of life.
- **List** three inferred characteristics that describe the first forms of cellular life on Earth.
- **Compare** the two types of autotrophy used by early cells.
- **Relate** the development of photosynthesis to the development of aerobic respiration in early cells.
- **Explain** the theory of endosymbiosis.

## VOCABULARY

ribozyme  
archaea  
chemosynthesis  
cyanobacteria  
ozone  
endosymbiosis

**FIGURE 14-8**

Messenger RNA is transcribed from a DNA template. Transfer RNA translates the three-base codons in the mRNA, assembling a protein from the specified amino acids.



## Eco Connection

### Archaea

Some species of archaea, such as *Methanosarcina barkeri* pictured in Figure 14-9, are referred to as methanogens. Within these curious prokaryotic cells, hydrogen gas reacts with carbon dioxide to produce methane, CH<sub>4</sub>, a simple carbon compound. Methanogens are poisoned by oxygen, but they can live in watery environments where other organisms have consumed all free oxygen, such as in swamps and even the intestines of animals.

Methanogens may prove useful to humans in two significant ways: they are currently used in the cleanup of organic waste, such as sewage, and they may eventually be harnessed for large-scale production of methane for use as a fuel source.

Other species of archaea are being used in the cleanup of petroleum spills into soil, such as occur when underground gasoline tanks develop leaks. This technique, called *bioremediation*, often relies on archaeal species that are already present in the soil. These cells are stimulated by application of nutrient-rich solutions formulated to their taste. As the cells multiply, they metabolize petroleum, releasing harmless byproducts.

## THE ROLES OF RNA

In the early 1980s, researcher Thomas Cech (1947–) found that a type of RNA found in some unicellular eukaryotes is able to act as a chemical catalyst, similar to the way an enzyme acts. Cech used the term **ribozyme** (RIE-buh-ZIEM) for an RNA molecule that can act as a catalyst and promote a specific chemical reaction.

Later studies based on Cech's discovery indicated that ribozymes could act as catalysts for their own replication. In fact, self-replicating systems of RNA molecules have been created in the laboratory. These findings support the hypothesis that life could have started with self-replicating molecules of RNA.

Furthermore, other roles for RNA have been discovered. RNA plays a vital role in DNA replication, protein synthesis, and other basic biochemistry. Perhaps most or all of the chemistry and genetics of early cells were based on RNA. This model of the beginnings of life on Earth is sometimes called *the RNA world*.

Self-replicating RNA molecules might have been not only the first case of heredity but also the first case of competition. For an RNA molecule, self-replication might involve competing with other RNA molecules for a limited number of available nucleotides. A particular RNA molecule with a slightly different structure than other RNA molecules might be more successful in getting nucleotides from its environment and thus more likely to be replicated. Thus, over time, the more-efficient RNA molecules would “survive,” and the “world” of RNA would slowly change.

There are several competing hypotheses about how RNA or other simple replicating systems could have evolved into modern cellular life. Some hypotheses propose that certain kinds of minerals formed a template on which organic molecules could line up and form polymers. Another hypothesis is that self-replicating RNA started to evolve inside cell-like structures such as microspheres or coacervates. The self-replicating RNA could have provided the hereditary information that the cell-like structures lack. If the RNA molecules were able to direct the assembly of the structures that carried them, a cell-like system would be formed.

## THE FIRST CELLS

Although lacking direct evidence of the first cells, scientists can make some inferences. First, many scientists think that little or no oxygen gas existed on early Earth. Second, the oldest fossils that are thought to be cells are the size and shape of some living prokaryotes. Finally, the first cells might have developed in an environment filled with organic molecules for food. Thus, the first cells were probably anaerobic, heterotrophic prokaryotes.

### Word Roots and Origins

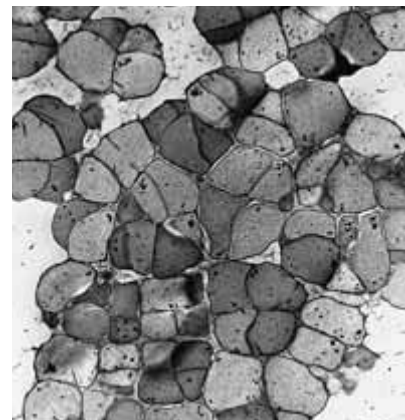
#### *archaea*

from the Greek *archein*,  
meaning “to begin”

We can reason that a growing population of heterotrophs that depended on spontaneously formed organic molecules for food eventually would have removed most of these molecules from the environment. At this point, autotrophs would have begun to have an advantage. The first autotrophs, however, probably did not depend on photosynthesis the way that most autotrophs do today.

## Chemosynthesis

If we look for living organisms that may be similar to these early organisms, we find the archaea (ahr-KEE-uh). The **archaea** are a related group of unicellular organisms, many of which thrive under extremely harsh environmental conditions. *Methanosarcina barkeri*, the archaeon shown in Figure 14-9, lives in anaerobic environments. Many species of archaea are autotrophs that obtain energy by chemosynthesis (KEE-moh-SIN-thuh-sis) instead of photosynthesis. In the process of **chemosynthesis**,  $\text{CO}_2$  serves as a carbon source for the assembly of organic molecules. Energy is obtained from the oxidation of various inorganic substances, such as sulfur.



**FIGURE 14-9**

This archaeal species, *Methanosarcina barkeri*, produces methane during metabolism. Archaea are thought to be similar to the types of cellular life that first populated Earth about 4 billion years ago.

## Photosynthesis and Aerobic Respiration

Some forms of life had become photosynthetic by 3 billion years ago. Scientists infer this from a variety of geologic evidence, such as the chemical traces of photosynthetic activity. Also, most of the oldest known fossils of cells are similar to modern **cyanobacteria** (SIE-uh-no-bak-TIR-ee-uh)—a group of photosynthetic, unicellular prokaryotes. *Lyngbya*, a genus of modern cyanobacteria, is shown in Figure 14-10a. *Lyngbya* cells often grow in colonies and form layered structures called *stromatolites*, shown in Figure 14-10b. Fossils of stromatolites as old as 3.5 billion years are known.

Oxygen, a byproduct of photosynthesis, was damaging to many early unicellular organisms. Oxygen could destroy some coenzymes essential to cell function. Within some organisms, however, oxygen bonded to other compounds, thereby preventing the oxygen from doing damage. This bonding was one of the first steps in aerobic respiration. Thus, an early function of aerobic respiration may have been to prevent the destruction of essential organic compounds by oxygen.

Many scientists think that it took a billion years or more for oxygen gas levels to reach today's levels. The oxygen gas,  $\text{O}_2$ , eventually reached the upper part of the atmosphere, where it was bombarded with sunlight. Some wavelengths of sunlight can split  $\text{O}_2$  to form highly reactive single oxygen atoms,  $\text{O}$ . These  $\text{O}$  atoms react with  $\text{O}_2$  and form **ozone**,  $\text{O}_3$ . Ozone is poisonous to both plant and animal life, but in the upper atmosphere, a layer of ozone absorbs much of the ultraviolet radiation from the sun. Ultraviolet radiation damages DNA, and without the protection of the ozone layer, life could not have come to exist on land.

**FIGURE 14-10**

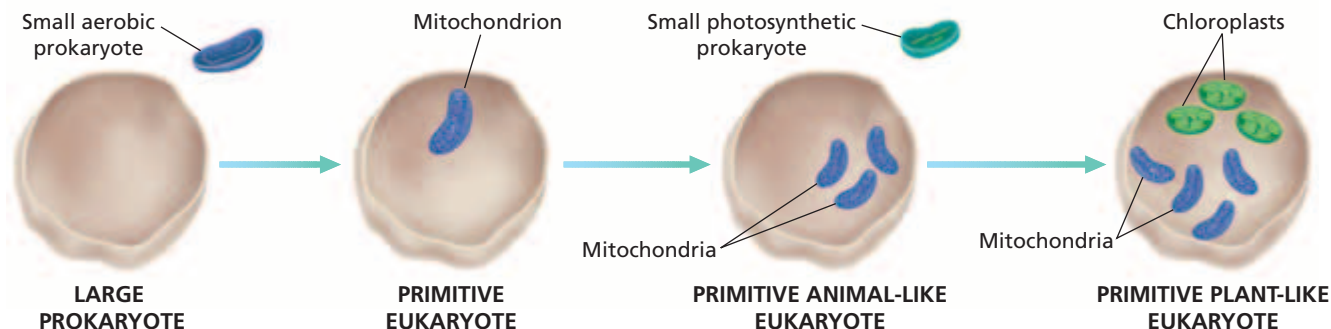
Most of the oldest known fossils of cells are similar to modern cyanobacteria, such as members of the genus *Lyngbya* (a). Cyanobacteria such as these sometimes form large colonies that grow in layers and form stromatolites (b).



(a)



(b)



**FIGURE 14-11**

According to the theory of endosymbiosis, large prokaryotic, unicellular organisms engulfed smaller prokaryotic, unicellular organisms. Engulfed prokaryotes eventually gave rise to modern mitochondria and chloroplasts, which carry their own DNA and replicate independently from the rest of the cell that contains them.

## THE FIRST EUKARYOTES

Recall that eukaryotic cells differ from prokaryotic cells in several ways. Eukaryotic cells are larger, their DNA is organized into chromosomes in a cell nucleus, and they contain membrane-bound organelles. How did eukaryotic cells evolve? Researcher Lynn Margulis (1938–) proposed that early prokaryotic cells may have developed a mutually beneficial relationship. A large body of evidence suggests that between about 2.0 billion and 1.5 billion years ago, a type of small aerobic prokaryote was engulfed by and began to live and reproduce *inside* of a larger, anaerobic prokaryote. This theory is called **endosymbiosis** (EN-doh-SIM-bie-OH-sis) and is modeled in Figure 14-11. The eukaryotes provided a beneficial environment, and the prokaryotes provided a method of energy synthesis.

Scientists infer that endosymbiotic aerobic prokaryotes evolved into modern mitochondria, which perform aerobic respiration in eukaryotic cells. In a later case of endosymbiosis, photosynthetic cyanobacteria may have evolved into chloroplasts, which perform photosynthesis in modern eukaryotic plant and algae cells. There is compelling evidence to support these hypotheses for the origin of these organelles. Both chloroplasts and mitochondria replicate independently from the replication cycle of the cell that contains them. Moreover, chloroplasts and mitochondria contain some of their own genetic material, which differs from that of the rest of the cell. Finally, the DNA of these organelles is found in a circular arrangement that is characteristic of prokaryotic cells.

### Word Roots and Origins

#### **endosymbiosis**

from the Greek *endon*, meaning "within," *syn* meaning "together," and *biosis*, meaning "way of life"

## SECTION 3 REVIEW

1. How does RNA differ from DNA?
2. Describe three major scientific inferences about the first living cells on Earth.
3. What traits make archaeobacteria likely relatives of Earth's earliest organisms?
4. Explain the difference between chemosynthesis and photosynthesis.
5. Explain the theory of endosymbiosis.

### CRITICAL THINKING

6. **Evaluating Conclusions** What evidence supports the hypothesis that mitochondria were once free-living prokaryotic cells?
7. **Identifying Relationships** How did anaerobic cells influence the development of aerobic cells?
8. **Predicting Consequences** Some forms of air pollution reduce the thickness of Earth's ozone layer. How might this change affect modern life?



# CHAPTER HIGHLIGHTS

## SECTION 1

### Biogenesis

- Before the 1600s, it was generally thought that organisms could arise from nonliving material by spontaneous generation.
- Redi showed in 1668 that rotting meat kept away from flies would not produce new flies. Maggots appeared only on meat that had been exposed to flies.
- Spallanzani showed in the 1700s that microorganisms would not grow in broth when its container was heated and then sealed. He inferred that microorganisms do not arise spontaneously but, rather, are carried in the air.
- Pasteur in the 1800s used a variation of Spallanzani's design to prove that microorganisms are carried in the air and do not arise by spontaneous generation.

#### Vocabulary

biogenesis (p. 279)

spontaneous generation (p. 279)

## SECTION 2

### Earth's History

- Scientists think that Earth formed by the gravitational accumulation of dust and debris moving through space.
- Isotopes are atoms with varying numbers of neutrons. The ages of rocks and other materials can be determined by measuring the amount of radioactive decay that has occurred in radioactive isotopes found in samples of those materials. An isotope's half-life is the time that one-half of a sample of the isotope takes to decay.
- The first simple organic compounds on early Earth may have formed under conditions of high energy and in an atmosphere very different from that of today's Earth.
- Meteorites may have brought organic compounds to Earth.
- Further chemical reactions may have converted simple organic compounds into the macromolecules important to life. Lightning, ultraviolet radiation, or heat from within the Earth could have provided the energy for these reactions. These conditions have been experimentally modeled.
- Cell-like structures, including microspheres and coacervates, form spontaneously in certain kinds of solutions. These structures could have been a step in the formation of modern cells but lack hereditary material.
- Scientists continue to investigate many hypotheses about the origins of organic molecules and cells in Earth's history.

#### Vocabulary

radiometric dating (p. 282)  
isotope (p. 282)

mass number (p. 283)  
radioactive decay (p. 283)

radioactive isotope (p. 283)  
half-life (p. 283)

microsphere (p. 286)  
coacervate (p. 286)

## SECTION 3

### The First Life-Forms

- In addition to serving as a template for protein assembly, some RNA molecules can act as enzymes. Like proteins, RNA molecules can assume different shapes. These shapes depend on areas of attraction between the RNA nucleotides. For these reasons, the first molecule that held hereditary information may have been RNA rather than DNA.
- The first cells that formed on Earth were probably heterotrophic prokaryotes.
- The first autotrophic cells probably used chemosynthesis to make food. Chemosynthesis produces energy through the oxidation of inorganic substances.
- Most modern autotrophic cells use photosynthesis to make food. An important byproduct of photosynthesis is oxygen.
- Once oxygen began to accumulate on Earth, cells would need to bind oxygen to other compounds in order to prevent damage to cell enzymes. This binding function may have been a first step toward aerobic respiration in cells.
- Eukaryotic cells may have evolved from large prokaryotic cells that engulfed smaller prokaryotic cells. The engulfed prokaryotic cells may have become the ancestors of organelles such as mitochondria and chloroplasts.

#### Vocabulary

ribozyme (p. 288)  
archaea (p. 289)

chemosynthesis (p. 289)  
cyanobacteria (p. 289)

ozone (p. 289)  
endosymbiosis (p. 290)



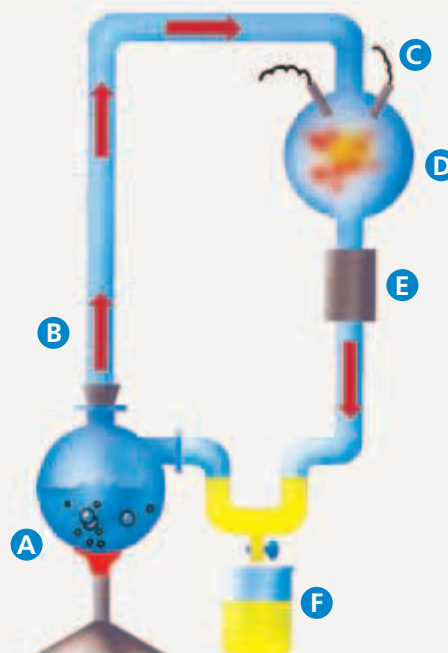
# CHAPTER REVIEW

## USING VOCABULARY

- For each of the following pairs of terms, explain how the meanings of the terms differ.
  - biogenesis* and *spontaneous generation*
  - ribozyme* and *enzyme*
  - photosynthesis* and *chemosynthesis*
  - archaebacteria* and *cyanobacteria*
- Explain the relationships between radioactive decay, isotopes, and half-life.
- \_\_\_\_ and \_\_\_\_ are examples of nonliving, cell-like structures that can form in certain solutions.
- Word Roots and Origins** The word *biogenesis* is derived from the Greek word *gignesthai*, which means “origin.” The prefix *bio* means “to be born.” Using this information, explain how the meaning of *biogenesis* relates to this chapter.

## UNDERSTANDING KEY CONCEPTS

- Predict** what would have happened if Redi had used jars covered with netting that had very large openings in his experiment.
- Evaluate** Spallanzani’s experiment, and explain why his results did not conclusively disprove the theory of spontaneous generation.
- Describe** Pasteur’s experiment, and explain how it disproved spontaneous generation.
- Relate** the role of gravity to the early formation of Earth.
- Calculate** the age of a sample containing thorium-230 (whose half-life is 75,000 years) after three-fourths of the sample has decayed.
- Identify** the natural phenomenon that the electric spark in the Miller-Urey experiment was intended to model.
- Compare** the hypothesis tested in the Miller-Urey experiment with subsequent hypotheses about the conditions under which life may have first formed on Earth.
- Compare** microspheres and coacervates with modern, living cells.
- Explain** why many scientists who investigate the origin of life have focused on RNA chemistry.
- Describe** a hypothesis about the role of RNA in the original development of cells on early Earth.
- Identify** which modern organisms are thought to be most like the first forms of life on Earth.
- Infer** which environmental factors would probably favor the evolution of autotrophs.
- List** and describe two types of autotrophs. State how each makes its own food.
- Describe** the conditions under which the ability to bind oxygen gas is an advantage to a cell.
- Explain** how the formation of the ozone layer permitted organisms to colonize land.
- Explain** the theory of endosymbiosis.
- CONCEPT MAPPING** Use the following terms to create a concept map illustrating the theory of endosymbiosis: *anaerobic prokaryotes*, *eukaryotes*, *cyanobacteria*, *aerobic prokaryotes*, *endosymbiosis*, *mitochondria*, and *chloroplasts*.
- Comparing Concepts** How does the principle of biogenesis pose a scientific question regarding the origin of life on Earth?
- Evaluating Conclusions** People once believed fish could form from the mud in a pond that sometimes dried up. How could you demonstrate that this conclusion is false?
- Evaluating Methods** Why did the Miller-Urey experiment not include oxygen gas,  $O_2$ , in the mixture of gases simulating Earth’s first atmosphere?
- Interpreting Graphics** The apparatus shown below is an example of the Miller-Urey experiment modeling conditions on early Earth. Explain the function of each part of the apparatus.



## CRITICAL THINKING



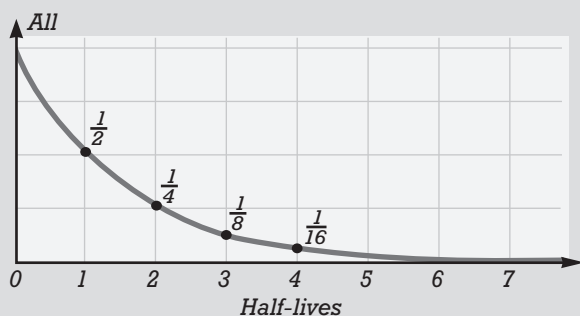
# Standardized Test Preparation

**DIRECTIONS:** Choose the letter that best answers the question.

- In the 17th and 18th centuries, which of the following was the hypothesis of spontaneous generation used to explain?
  - how new life started
  - how eukaryotes evolved
  - how simple organic compounds formed
  - how coacervates and microspheres formed
- Coacervates are similar to cells but lack which of the following?
  - genetic information
  - interior fluid
  - complex organic molecules
  - the chemical properties of cells
- The planets of our solar system gained mass for a half-billion years after their formation as a result of which of the following?
  - flames from the sun
  - collisions with space debris
  - tidal forces generated by moons
  - the synthesis of organic molecules

**INTERPRETING GRAPHICS:** The graph below shows the decay of a radioactive isotope over time. Use the graph to answer the question that follows.

**Amount Remaining of Radioactive Sample**



- If the half-life of carbon-14 is 5,730 years, how many years would it take for  $\frac{7}{8}$  of the original amount of carbon-14 in a sample to decay?
  - 5,014 years
  - 11,460 years
  - 17,190 years
  - 22,920 years

**DIRECTIONS:** Complete the following analogy.

- chloroplasts : cyanobacteria :: mitochondria :
  - archaea
  - aerobic prokaryotes
  - anaerobic eukaryotes
  - chemosynthetic bacteria

**INTERPRETING GRAPHICS:** Use the table below to answer the questions that follow.

**Estimated Abundance of Some Elements on Earth and in Meteorites**

| Element   | Earth<br>(% of total mass) | Meteorites<br>(% of total mass) |
|-----------|----------------------------|---------------------------------|
| Iron      | 36.0                       | 27.2                            |
| Oxygen    | 28.7                       | 33.2                            |
| Magnesium | 13.6                       | 17.1                            |
| Silicon   | 14.8                       | 14.3                            |
| Sulfur    | 1.7                        | 1.9                             |

- According to the table, which element is found on Earth in a greater abundance than the element's abundance in meteorites?
  - iron
  - sulphur
  - oxygen
  - magnesium
- If this table is typical of the relative abundance of all elements on Earth, in meteorites, and on other planets, which of the following statements would be supported?
  - Earth and meteorites have similar origins.
  - Earth and meteorites have different origins.
  - All meteorites formed from parts of Earth.
  - All elements on Earth come from meteorites.

## SHORT RESPONSE

Describe the prevailing scientific model of the original formation of Earth.

## EXTENDED RESPONSE

Little direct evidence has been found of any life that existed in the first few billion years of Earth's history.

**Part A** Describe the major steps that may have led to the development of eukaryotic organisms.

**Part B** Describe evidence that supports hypotheses about some of these steps.

## Test TIP

For short-response and essay questions, be sure to answer the prompt as fully as possible. Include major steps, important facts, descriptive examples, and supporting details in your response.

# Making Microspheres

## OBJECTIVES

- Make microspheres from amino acids by simulating the conditions found on early Earth.
- Compare the structure of microspheres with the structure of living cells.

## PROCESS SKILLS

- observing
- comparing and contrasting
- modeling
- relating

## MATERIALS

- safety goggles
- lab apron
- heat-protective gloves
- 500 mL beaker
- hot plate
- 125 mL Erlenmeyer flasks, 2
- ring stand with clamp
- balance
- amino acid mixture (supplied by your teacher)
- glass stirring rod
- tongs
- clock or timer
- 1% sodium chloride (NaCl) solution
- 50 mL graduated cylinder
- dropper
- microscope slide
- coverslip
- compound light microscope
- 1% sodium hydroxide (NaOH) solution

## Background

1. Microspheres are very small, spherical vessels that are bounded by a membranelike layer of amino acids. Microspheres can be created in the laboratory under controlled conditions.
2. How do microspheres differ from living cells?
3. How do microspheres resemble living cells?
4. What role might have been played by microspheres or similar structures before life began on Earth?

## Procedure

1.   Put on safety goggles, a lab apron, and heat-protective gloves before beginning this investigation.
2.   **CAUTION** Do not plug in or unplug the hot plate with wet hands. Use care to avoid burns when working with the hot plate. Do not touch the hot plate. Use tongs to move heated objects. Turn off the hot plate when not in use. Fill a 500 mL beaker half full with water, and heat it on a hot plate. You will use the beaker as a hot-water bath. Leave space on the hot plate for a 125 mL Erlenmeyer flask, to be added later.
3.  While waiting for the water to boil, clamp a 125 mL Erlenmeyer flask to a ring stand. Add 6 g of the amino acid mixture to the flask.
4. When the water in the beaker begins to boil, move the ring stand carefully so that the flask of amino acids sits in the hot-water bath.
5. When the amino acids have heated for 20 minutes, measure 10 mL of NaCl solution in a graduated cylinder, and pour the solution into a second Erlenmeyer flask. Place the second flask on the hot plate beside the hot-water bath.



6. When the NaCl solution begins to boil, use tongs to remove the flask containing the NaCl solution from the hot plate. Then, while holding the flask with tongs, slowly add the NaCl solution to the hot amino acids while stirring.
7. Let this NaCl–amino acid solution boil for 30 seconds.
8. Remove the solution from the water bath, and allow it to cool for 10 minutes.
9.  **CAUTION** Slides break easily. Use caution when handling them. Use a dropper to place a drop of the solution on a microscope slide, and cover the drop with a coverslip.
10. Place the slide on the microscope stage. Examine the slide under low power for tiny spherical structures. Then, examine the structures under high power. These tiny sphere-shaped objects are microspheres.
11.  **CAUTION** If you get the sodium hydroxide (NaOH) solution on your skin or clothing, wash it off at the sink while calling to your teacher. If you get the sodium hydroxide solution in your eyes, immediately flush your eyes at the eyewash station while calling to your teacher. Place a drop of 1% NaOH solution at the edge of the coverslip to raise the pH as you observe the microspheres. What happens?
12. In your lab report, make a table similar to the one shown below. Based on your observations of microspheres and cells, complete your table. Consider the appearance of microspheres and cells, their method

of reproduction, their interaction with their environment, and any other characteristics that you observe.

13.   Clean up your lab materials, and wash your hands before leaving the lab.

## Analysis and Conclusions

1. Suggest how the microspheres in step 10 were formed.
2. What did you observe when the pH was raised in step 11?
3. What does this suggest about the relationship of pH to microsphere formation?
4. Compare and contrast microspheres with living cells.
5. What characteristics would microspheres have to exhibit in order to be considered living?
6. How might the conditions you created in the lab be similar to those that are thought to have existed when life first evolved on Earth?
7. Predict what would happen to microspheres if they were placed in hypotonic and hypertonic solutions.

## Further Inquiry

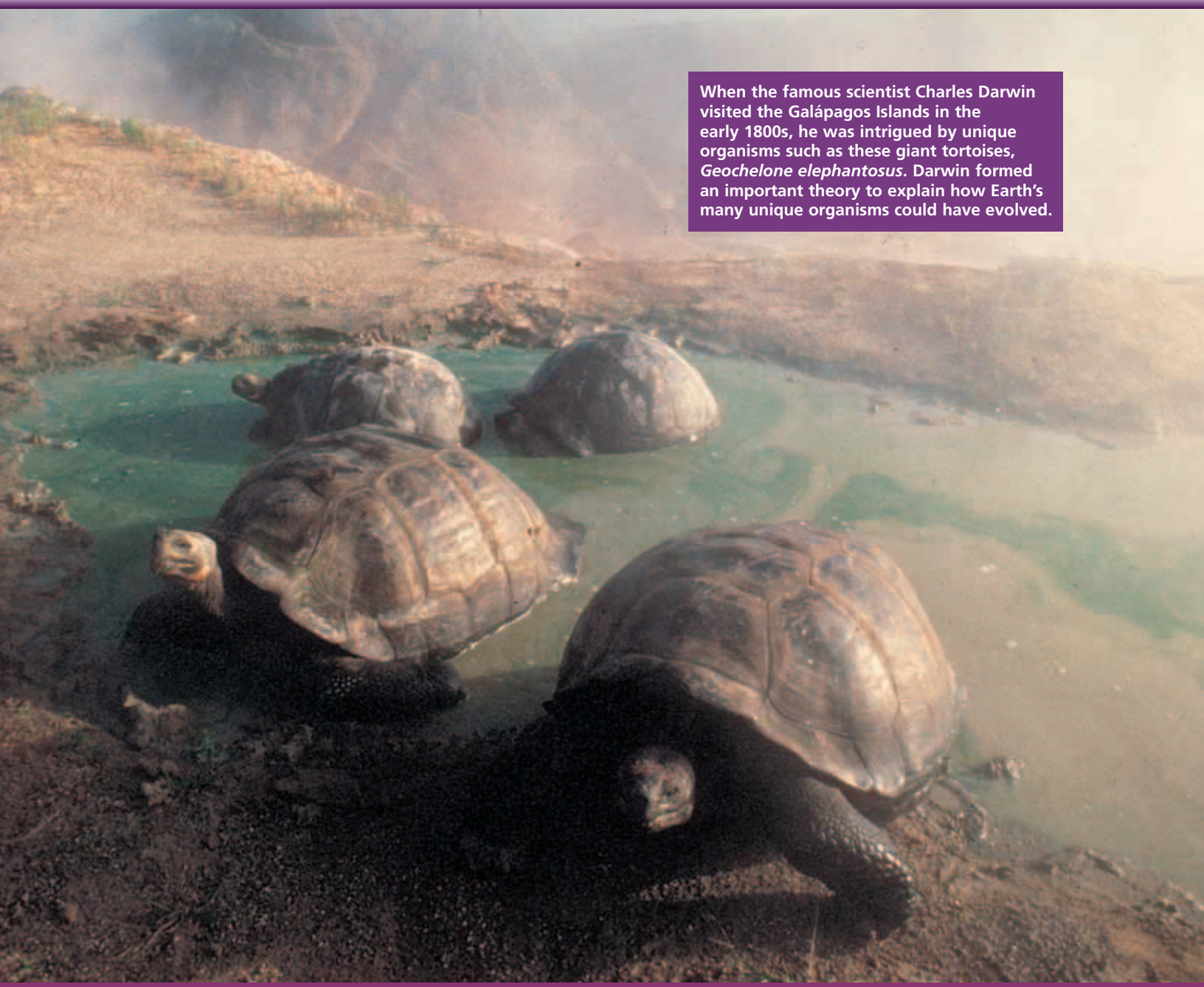
1. What do you think would happen if you added too much or too little heat? What happens to proteins at high temperatures? How can you test for the right amount of heat to use?
2. Do you think your microsphere experiment would have worked if you had substituted other amino acids? How can you test your hypothesis?

### COMPARING MICROSPHERES WITH CELLS

| Cell-like characteristics | Characteristics that are not cell-like |
|---------------------------|----------------------------------------|
|                           |                                        |
|                           |                                        |
|                           |                                        |
|                           |                                        |
|                           |                                        |



# THEORY OF EVOLUTION



When the famous scientist Charles Darwin visited the Galápagos Islands in the early 1800s, he was intrigued by unique organisms such as these giant tortoises, *Geochelone elefantosus*. Darwin formed an important theory to explain how Earth's many unique organisms could have evolved.

**SECTION 1** *History of Evolutionary Thought*

**SECTION 2** *Evidence of Evolution*

**SECTION 3** *Evolution in Action*

# HISTORY OF EVOLUTIONARY THOUGHT

*In the 1830s, the young English naturalist Charles Darwin took a trip around the world on a ship called HMS Beagle. He was fascinated by diverse and unique organisms, such as the giant tortoises of the Galápagos Islands. Darwin went on to form one of the most important theories in biology.*

## THE IDEA OF EVOLUTION

After visiting the Galápagos Islands, Charles Darwin (1809–1882), shown in Figure 15-1, noted that groups of animals varied from island to island. For example, the tortoises on the same island resembled each other closely, but those from neighboring islands were different. Darwin noticed similarities and differences among many organisms as he traveled around the world. He became convinced that organisms had changed over time, and he wanted to understand why.

The development of new types of organisms from preexisting types of organisms over time is called **evolution**. Modern scientists also define evolution as a heritable change in the characteristics within a population from one generation to the next. Darwin sought to present the evidence that the evolution of new organisms occurs. Furthermore, he formed a theory to explain how evolution could occur. Others had tried to form such theories earlier, but Darwin's theory became the basis for modern explanations of evolution. Remember that in science, a *theory* is a well-supported explanation for some aspect of the natural world that incorporates many observations, inferences, and tested hypotheses. In forming his theory, Darwin took years to put together data from many sources and to take account of the ideas of other scientists of his time.

### Ideas Of Darwin's Time

In Europe in the 18th century, most scientists thought that all species were permanent and unchanging. Furthermore, they thought that the Earth was only thousands—and not billions—of years old. However, scientists began to present evidence that the species on Earth have changed over time, and that the Earth is much older than anyone had thought.

## OBJECTIVES

- **Define** the biological process of evolution.
- **Summarize** the history of scientific ideas about evolution.
- **Describe** Charles Darwin's contributions to scientific thinking about evolution.
- **Analyze** the reasoning in Darwin's theory of evolution by natural selection.
- **Relate** the concepts of adaptation and fitness to the theory of natural selection.

## VOCABULARY

evolution  
strata  
natural selection  
adaptation  
fitness

**FIGURE 15-1**

Charles Darwin first studied to be a doctor and then a minister, but was also interested in nature. At the age of 22, he set off on a five-year voyage that became an important part of the history of science.







**FIGURE 15-2**

The strata of the Grand Canyon contain evidence of the history of the area over millions of years of geologic time.

### Word Roots and Origins

#### evolution

from the Latin prefix *e-*, meaning "out," and *volvere*, meaning "to roll"

## Ideas About Geology

By the 1800s, scientists in Europe had begun to study rock layers—called **strata**—such as those shown in Figure 15-2. They found that strata are formed as new layers of rock are deposited over time. They inferred that, in general, lower strata were formed first and are thus the oldest. The scientists also found that different rock strata hold fossils of different kinds of organisms.

French anatomist Georges Cuvier (coo-VYAY) (1769–1832) spent years reconstructing the appearance of unique organisms from fossil bones. Cuvier gave convincing evidence that some organisms in the past differed greatly from any living species and that some organisms had become *extinct*, meaning the species had ceased to live after a point in time. Cuvier also found that deeper and older strata hold fossils that are increasingly different from living species. Finally, Cuvier found

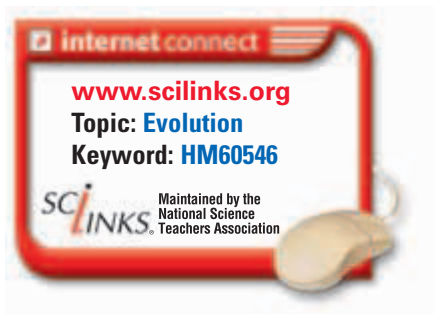
many "sudden" changes in the kinds of organisms found in one rock stratum compared to the next.

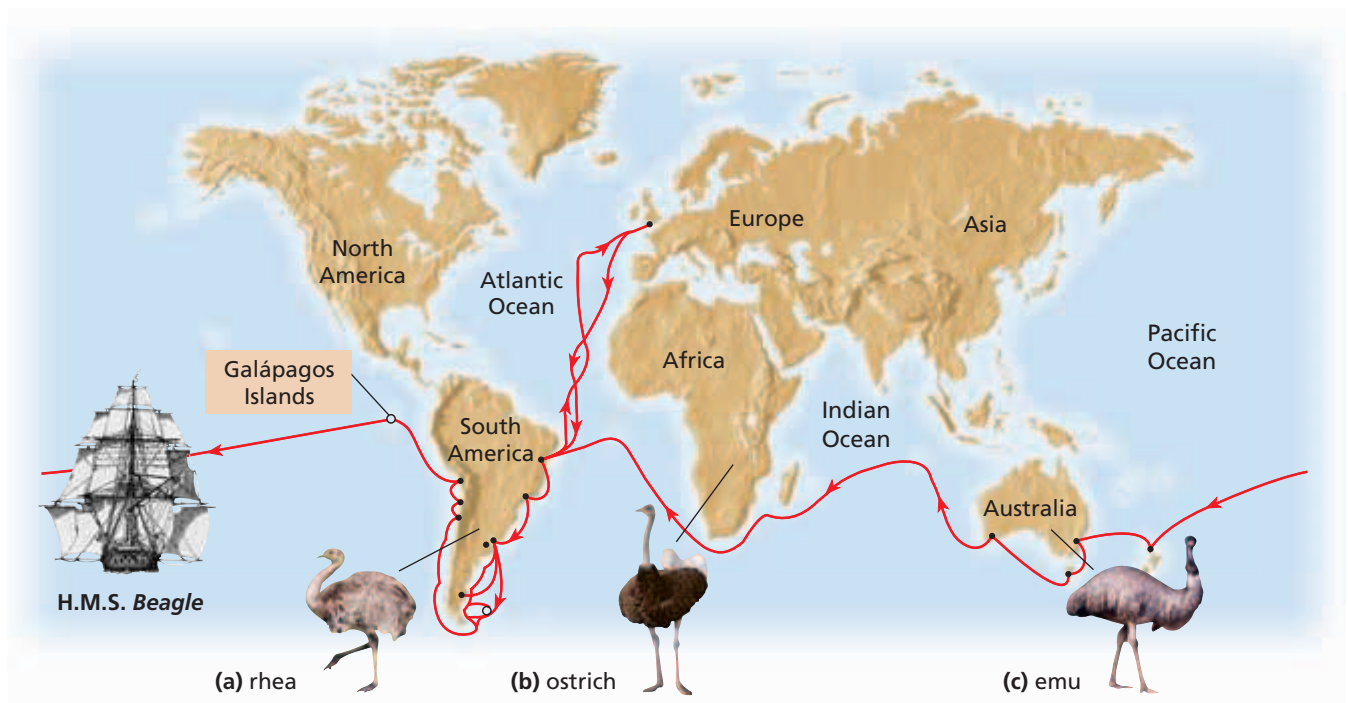
To explain these observations, Cuvier promoted the idea of *catastrophism* (kuh-TAS-truh-FIZ-uhm)—the idea that sudden geologic catastrophes caused the extinction of large groups of organisms at certain points in the past. Even though scientists no longer accept all of his explanations, Cuvier contributed to scientific acceptance that geologic change and extinction had occurred.

The English geologist Charles Lyell (1797–1875) shared some of Cuvier's ideas but thought that the geologic processes that have changed the shape of Earth's surface in the past continue to work in the same ways. Lyell's idea is called *uniformitarianism* (YOON-uh-FAWRM-uh-TER-ee-uhn-iz-um). Charles Darwin read Lyell's writings while on his trip around the world. He was excited to find how well Lyell's ideas fit with his own observations and ideas. Lyell had shown evidence from geology that fit with Darwin's evidence from biology. Darwin referred to Lyell's work in many of his writings.

## Lamarck's Ideas on Evolution

The French biologist Jean Baptiste Lamarck (1744–1829) also supported the idea that populations of organisms change over time. Lamarck put forward a new idea to explain how evolution could happen, though this idea is no longer accepted among scientists. Lamarck thought that simple organisms could arise from nonliving matter. He also thought that simple forms of life inevitably develop into more complex forms. He proposed that individuals could acquire traits during their lifetimes as a result of experience or behavior, then could pass on those traits to offspring. Lamarck's idea is called the *inheritance of acquired characteristics*. Even though Darwin himself once accepted Lamarck's idea, it was rejected by many scientists of their time and has not been supported by modern scientific study of the mechanisms of inheritance.





## DARWIN'S IDEAS

At about the same time in the mid-1800s, both Charles Darwin and the English naturalist Alfred Russel Wallace (1823–1913) formed a new theory to explain how evolution may take place. Both Darwin and Wallace had been on sea voyages around the world; Darwin's voyage is shown in Figure 15-3. In 1858, the ideas of Darwin and Wallace were presented to a prestigious group of scientists in London. The following year, Darwin published a book entitled *On the Origin of Species by Means of Natural Selection*.

Darwin had two goals in his book: first, he wanted to present the large amount of evidence that evolution occurs; and second, he wanted to explain the variety and distribution of organisms on Earth in terms of natural processes that are observable every day.

### Descent with Modification

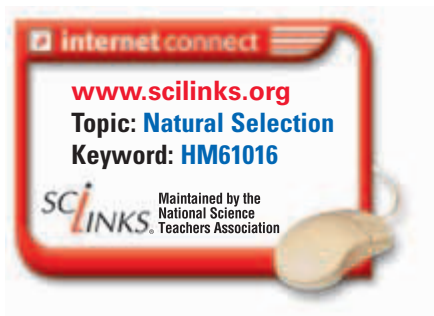
Darwin used the phrase *descent with modification* to describe the process of evolution. He carefully reviewed the evidence that every species—living or extinct—must have descended by reproduction from preexisting species and that species must be able to change over time. Darwin was not the first person to put forward the idea of descent with modification, but he was the first to argue that *all* species had descended from only one or a few original kinds of life.

Darwin saw the animals of the Galápagos Islands as evidence of descent with modification. For example, the islands are home to 13 similar species of finches. Each of these bird species has a beak that is best adapted for a certain kind of food. But Darwin suspected that all 13 species descended from and diverged from just a few ancestral finches. These ancestors could have flown to the Galápagos Islands from elsewhere sometime after the islands were formed.

**FIGURE 15-3**

On his voyage, Darwin noticed the locations of similar organisms around the world. The rhea (a), ostrich (b), and emu (c), for example, are clearly related species that have similar body forms and occupy similar habitats. They occur on different continents, however. Darwin wondered what causes such distributions in nature.





## Natural Selection

Darwin proposed the theory of **natural selection** as the mechanism for descent with modification. In forming his theory, Darwin thought carefully about the forces that could cause changes in organisms over time. The following summary explains the four main parts of Darwin's reasoning, as shown in Figure 15-4:

**1 Overproduction** More offspring can be produced than can survive to maturity. For example, each female deer has one or more offspring per year for many years in a lifetime. This multiplication could increase the total number of deer in a short time. But each new deer needs food and is vulnerable to predators and disease, so not all of the deer live for very long.

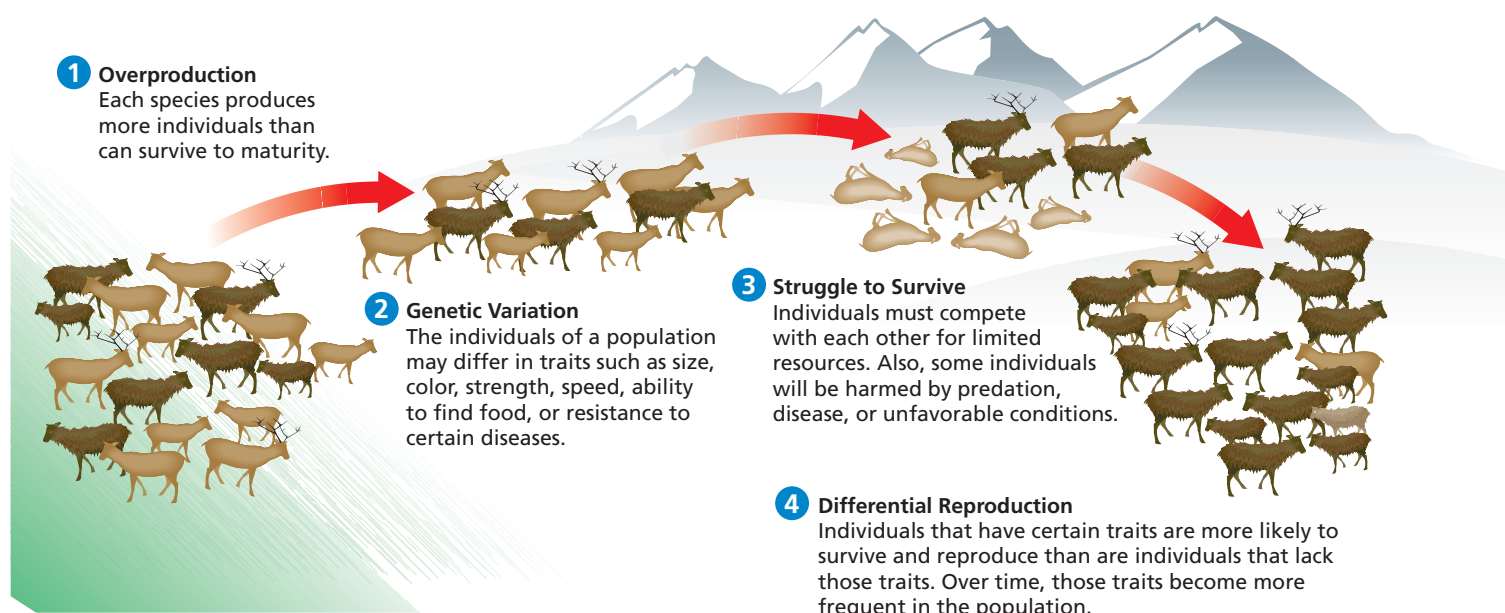
Darwin drew this part of his reasoning from a popular book about human social problems by English clergyman and economist Thomas Malthus (1766–1834). Malthus pointed out that human populations can increase more quickly than food supplies and that populations are often limited by conditions such as war, disease, or lack of food. Darwin realized that the environment limits the populations of *all* organisms by causing deaths or by limiting births.

**2 Genetic Variation** Within a population, individuals have different traits. For example, some deer may have thicker fur or longer legs than others. Also, some of this variation can be inherited. For example, deer that have thick fur tend to have offspring with thick fur. Occasionally, new traits may appear in a population.

**3 Struggle to Survive** Individuals must compete with each other in what Darwin called a “struggle for existence.” Some variations improve an individual's chance to survive and reproduce, but some variations reduce this chance. For example, deer that have thick fur may survive in the cold better than deer that have thin fur. A trait that makes an individual successful in its environment, such as thick fur in cold climates, is called an **adaptation**.

**FIGURE 15-4**

This diagram represents the process of natural selection explained by Charles Darwin as the mechanism for evolution.



- 4 **Differential Reproduction** Darwin concluded that organisms with the best adaptations are most likely to survive and reproduce. And through inheritance, the adaptations will become more frequent in the population. So, populations may begin to differ as they become adapted to different environments, even if they descended from the same ancestors. This conclusion is the core idea of Darwin's theory.

Darwin explained that natural selection could account for descent with modification as species become better adapted to different environments. That is, the theory of natural selection proposes that nature changes species by selecting traits. The environment "selects" the traits that may increase in a population by selecting the parents for each new generation.

Darwin sometimes used the phrase "survival of the fittest" to describe natural selection. In evolutionary terms, **fitness** is a measure of an individual's hereditary contribution to the next generation. This kind of fitness is more than simply living a long time. A fit individual is one that has offspring that also live long enough to reproduce in a given environment. For example, if thick fur is an advantage for a deer living in the mountains, then deer that have thick fur are more likely to live long enough to reproduce and pass on the genes for thick fur. If a certain trait increases an individual's fitness, the proportion of individuals with that trait is likely to increase over time. So, adaptations are those traits that increase the fitness of individuals, and populations tend to be well adapted to survive the conditions in which they live.

In evolutionary theory, the term *adaptation* is also used to describe changes in traits in populations over time. This meaning is different from that of short-term adaptation by an individual to a temporary condition. Also, long-term adaptation in populations is not the same as acclimatization in individuals. *Acclimatization* is a short-term process in which physiological changes take place in a single being in its own lifetime. An example of acclimatization is an animal adjusting to a new climate by growing thicker fur.

## Eco Connection

### Galápagos Islands

The unique ecosystems of the Galápagos Islands have been in danger since the discovery of the islands by Europeans in 1535. For 400 years, the islands were a favorite stopping place for pirates and whalers. Sailors valued the large native tortoises as a meat source because the tortoises could live for long periods aboard a ship with little or no food or water. Over the years, the number of native tortoises was severely reduced. Three of the 14 original subspecies of tortoises are now extinct. However, since 1936, the government of Ecuador and scientists from many countries have taken steps to preserve and restore the islands' native species and habitats. In 1965, the Charles Darwin Research Station began to breed and reintroduce the tortoises and other species.

## SECTION 1 REVIEW

1. Explain Darwin's use of the phrase *descent with modification* to describe the process of evolution.
2. Describe two scientists' ideas about geology that influenced ideas about evolution in the 1800s.
3. Explain the difference between an acquired characteristic and an inherited characteristic.
4. In what ways was Darwin an important scientist?
5. Describe the four parts of reasoning in Darwin's theory of evolution by natural selection. Use examples in your answer.

### CRITICAL THINKING

6. **Inferring Meaning** Explain why some biologists say that "fitness is measured in grandchildren."
7. **Analyzing Processes** Suppose that an individual has a new trait that makes it live longer than others in its population. Does this individual have greater fitness? Explain your answer.
8. **Applying Information** What have you learned about heredity and genetics that could support Darwin's theory of natural selection?

## SECTION 2

### OBJECTIVES

- **Relate** several inferences about the history of life that are supported by evidence from fossils and rocks.
- **Explain** how biogeography provides evidence that species evolve adaptations to their environments.
- **Explain** how the anatomy and development of organisms provide evidence of shared ancestry.
- **Compare** the use of biological molecules with other types of analysis of evolutionary relationships.
- **Describe** the ongoing development of evolutionary theory.

### VOCABULARY

fossil  
superposition  
relative age  
absolute age  
biogeography  
homologous structure  
analogous structure  
vestigial structure  
phylogeny

**FIGURE 15-5**

These fossils of pterosaur bones (a), fern leaves (b), and trilobite exoskeletons (c) were buried in ancient sediment. The insect (d) was trapped in ancient tree sap. In each case, the surrounding material later became rock.



(a)



(b)



(c)



(d)

# EVIDENCE OF EVOLUTION

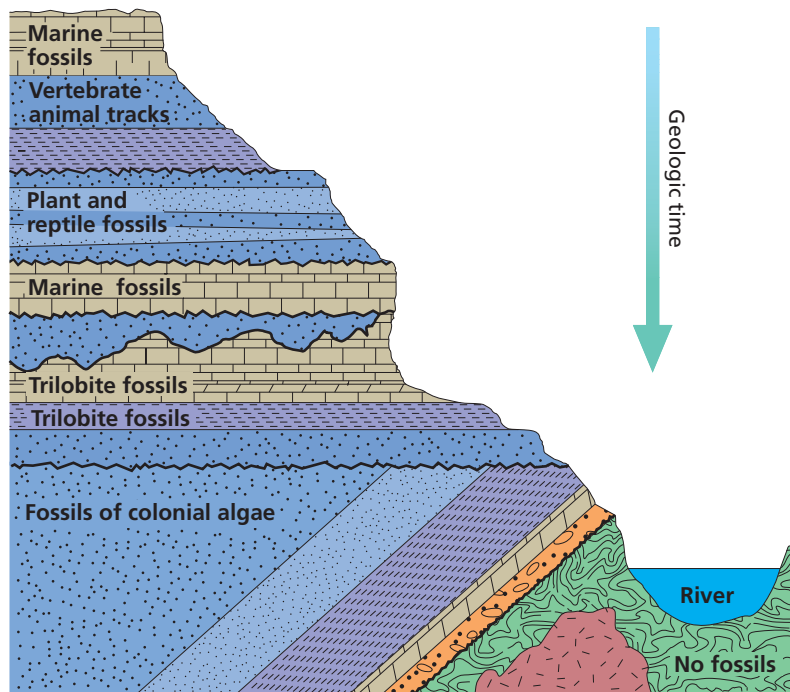
*Many kinds of evidence give insight into the history of life on Earth and the patterns of change among organisms. Fossils that are different from organisms living today are strong evidence that organisms on Earth can change over time. But evidence of evolution is also found inside living organisms.*

## THE FOSSIL RECORD

A **fossil** is the remains or traces of an organism that died long ago. Fossils of many kinds of organisms can be formed under a number of different conditions, as shown in Figure 15-5. Fossils show that different types of organisms appeared at different times and places on Earth. Some fossils are of organisms that have become *extinct*, meaning the species is no longer alive. Fossils are among the most powerful evidence of evolution.

### The Age of Fossils

In 1669, Danish scientist Nicolaus Steno (1638–1686) proposed the principle of **superposition**. This principle states that if the rock strata at a location have not been disturbed, the lowest stratum was formed before the strata above it. Successive strata are newer, and the most recent stratum is on the top. Geologists in the 1700s and 1800s built on Steno's ideas by comparing strata in different places and by comparing the fossils found in different strata. The geologists began to put together a timeline for the order in which different groups of rocks and fossils were formed. This timeline is commonly known as the *geologic time scale* (a simple version of this table appears in the Appendix). Today, geologists often can tell a fossil's **relative age**—its age compared to that of other fossils—by referring to the geologic time scale and to records of known fossils.



**FIGURE 15-6**

Erosion has exposed layer after layer of sedimentary rock in the Grand Canyon. Fossils trapped in the layers show that organisms changed over time as conditions on the surface changed over time. Deeper and older layers often contain fossils of organisms, such as trilobites, that no longer live on Earth.

For certain rocks, scientists can estimate the time since formation, or the **absolute age** of the rock, by techniques such as *radio-metric dating*. Scientists use both relative and absolute dating as they compare fossils around the world. In this way, scientists try to make their history of life on Earth as precise as possible. However, the fossil record is an incomplete history of life because not all organisms have left fossil evidence. Most fossils are formed only by rare events. For example, a fossil could form if an organism died in a place where it did not fully decay and where rock later formed over the remains. Then, if the fossil escapes destruction over time, humans will discover it only by chance. Nevertheless, Earth is rich with fossil evidence of organisms that lived in the past.

## The Distribution of Fossils

A number of additional inferences can be made from the fossil record. First, we can infer that different organisms lived at different times. For example, the rock strata shown in Figure 15-6 hold different kinds of fossils in successive layers. Second, we can infer that today's organisms are different from those of the past. Trilobites, for example, were unlike any organism alive today. Third, we can infer that fossils found in adjacent layers are more like each other than to fossils found in deeper or higher layers. In other words, organisms that lived during closer time periods are more alike than organisms that lived in widely separated time periods. Different species of trilobites appear in nearby strata, whereas deeper and older strata do not hold trilobites but hold other kinds of organisms. Fourth, by comparing fossils and rocks from around the planet, we can infer when and where different organisms existed. Each fossil provides evidence about the environment in which the organism existed and for which the organism had become adapted.





## Transitional Species

Finally, we can infer from the fossil record that species have differed in a gradual sequence of forms over time. This inference is based on *transitional species*, which have features that are intermediate between those of hypothesized ancestors and later descendant species. For example, evolutionary scientists have hypothesized that modern marine mammals, such as whales, evolved from early mammals in the fossil record that walked with four legs on land. This hypothesis predicts that there should be fossils that share characteristics of both ancient land-dwelling mammals and modern whales.

Indeed, scientists have found several fossils, such as those in Figure 15-7, that form a sequence of transitional forms. Comparing these fossils with each other and with modern whales reveals a sequence of differences in the structure of the hind limbs, forelimbs, vertebrae, and skull of each species. Scientists explain these differences as increasing adaptations for life in water. For example, hind limbs were smaller in later species and are absent in modern whales except for tiny pelvic bones. The hypothesis of whale evolution from land mammals is strongly supported by these fossil finds. However, there are other groups of organisms for which fossils of hypothesized transitional species have not been found.

**FIGURE 15-7**

The fossil skeletons below form a sequence of transitional forms that support the hypothesis that whales evolved from four-legged, land-dwelling mammals.



**1** *Pakicetus* (pak-uh-SEE-tuhs)

Scientists think that whales evolved from land-dwelling mammals. One of these ancestors may have belonged to the genus *Pakicetus*, which lived about 50 million years ago. The fossil skeleton of a pakicetid is shown here.



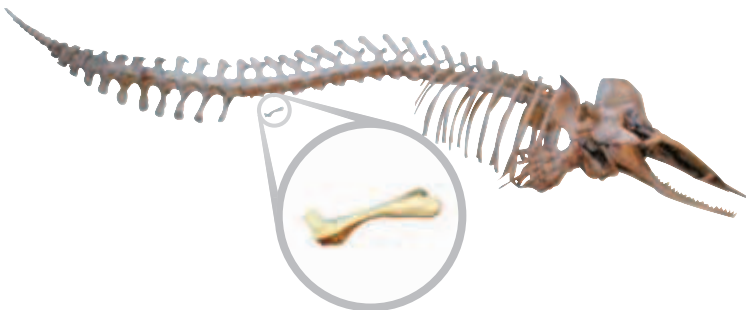
**2** *Ambulocetus* (am-byoo-loh-SEE-tuhs)

This genus of mammal lived in coastal waters about 49 million years ago. It could swim by kicking its legs and using its tail for balance. It could also waddle on land with its short legs.



**3** *Dorudon* (DOH-roo-don)

This genus of mammal lived in the oceans about 40 million years ago. It resembled a giant dolphin and propelled itself with a massive tail. It had forelimbs that were flippers and tiny hind limbs that could not have been used for walking or swimming.



**4** **Modern toothed whales**

Modern whales have forelimbs that are flippers. They also have tiny, nonfunctioning hip bones at the rear of their bodies.

# BIOGEOGRAPHY

**Biogeography** is the study of the locations of organisms around the world. When traveling, Charles Darwin and Alfred Russel Wallace saw evidence of evolution in the distribution of organisms. Both Darwin and Wallace observed animals that seemed closely related yet were adapted to different environments in nearby regions. However, they also observed animals that seemed unrelated but that had similar adaptations to similar environments in regions that were far apart. Again, the model of descent with modification provides an explanation for these patterns of distribution.

The mammals of the continent of Australia provide a striking example of biogeographic evidence of evolution. There are native Australian animals that resemble wolves, cats, mice, moles, or anteaters. However, most Australian mammals are marsupials, mammals that have pouches in their bodies for carrying their young. A possible explanation for this pattern is that these animals evolved in isolation on the Australian continent.

## ANATOMY AND EMBRYOLOGY

Descent with modification also predicts the findings of *anatomy*—the study of the body structure of organisms—and *embryology*—the study of how organisms develop. Look at the bones in the forelimbs of humans, penguins, alligators, and bats shown in Figure 15-8. These forelimbs are used in different ways in each animal, yet each limb has a similar bone structure. One explanation for the commonalities among the forelimb bones of the four animals is that an early ancestor shared by all these vertebrates had a forelimb with a similar bone structure. As generations passed, different populations of descendants adapted to different environments. Bones inherited from ancestors may have become modified for different tasks.

Biologists define **homologous structures** as anatomical structures that occur in different species and that originated by heredity from a structure in the most recent common ancestor of the species. Homologous organs often have a related structure even if their functions differ between species. On the other hand, **analogous structures** have closely related functions but do not derive from the same ancestral structure. For example, even though birds, bats, and moths have wings, their wings have very different underlying structures. Scientists think that wings evolved independently in each of these groups of animals.

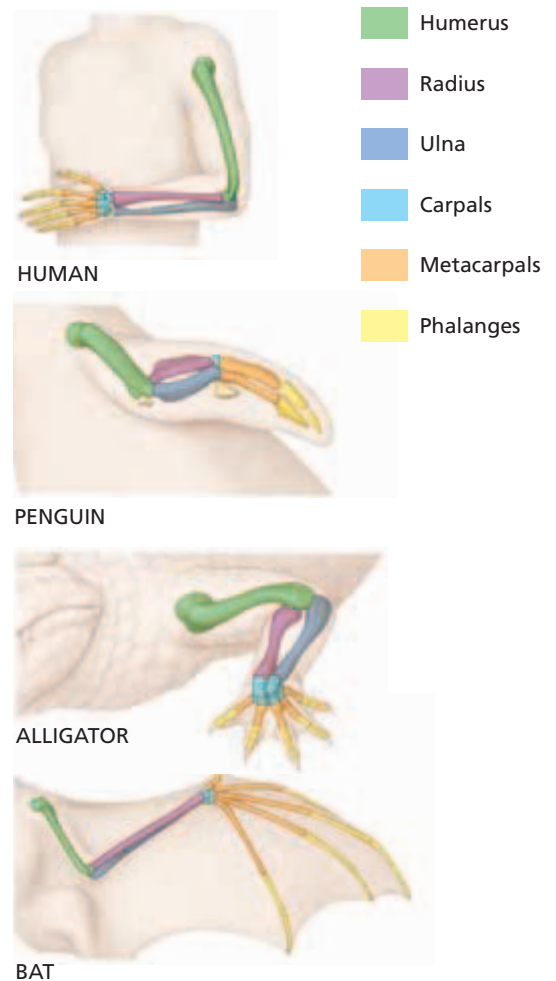
### Word Roots and Origins

#### *homologous*

from the Greek *homo*, meaning “equivalent to” and *logos*, meaning “relation” or “reason”

**FIGURE 15-8**

Humans, penguins, alligators, and bats each have forelimbs with homologous parts.





(a) dogfish, *Squalus acanthias*



(b) chicken, *Gallus gallus*



(c) cat, *Felis catus*

#### FIGURE 15-9

Modern analysis of vertebrate embryos shows that fish, chick, and cat embryos are remarkably similar at certain stages. This pattern would be expected if all were descended from a common ancestor. Processes occurring later in development then modify the ancestral body structures.

Further evidence of evolution is found in structures called **vestigial structures** that seem to serve no function but that resemble structures with functional roles in related organisms. For example, the human tailbone, or coccyx, is made up of four fused vertebrae that resemble the bones in an animal's tail. Other examples of vestigial structures are the pelvic bones of modern whales and the human appendix.

The development of animal embryos is also evidence of descent with modification. As shown in Figure 15-9, some stages of vertebrate embryo development are very alike, although the similarities fade as development proceeds. One possible explanation for these similarities is that vertebrates share a common ancestor and have inherited similar stages of development.

## BIOLOGICAL MOLECULES

Organisms that share many traits should have a more recent common ancestor than organisms that share fewer traits. Darwin made this prediction by studying anatomy, but modern studies of biological molecules also support this prediction. In all species, DNA and RNA are the molecular basis for inheritance of traits. Furthermore, DNA affects traits encoding the amino acid sequences that form proteins. Biologists can compare the DNA, RNA, proteins, and other biological molecules from many different organisms. Then, they can look for similarities and differences among the data for each species. The greater the number of similarities between any given species, the more closely the species are related through a common ancestor. However, because the study of biological molecules is still relatively new, scientists continue to debate how to interpret this kind of evidence.

## DEVELOPING THEORY

Not until the mid-1900s did scientists begin to integrate the theory of natural selection with new understandings of genetics. This blending has been called the *modern synthesis* of evolutionary theory. As in all areas of science, hypotheses and theories about evolution continue to be formed, challenged, and revised. Many aspects of evolution remain poorly understood, and some observations remain unexplained. Although modern evolutionary theory, like all theories in science, can never be “proven,” it is widely accepted and applied by scientists because it explains the broadest range of observations and makes useful predictions.